

Goldbach Programme

Master File

Research knowledge base for future models

Factorization-graph programme for the binary Goldbach conjecture: the canonical selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$, the stopped traversal, Canonical Root Dynamics (CRD), the complete theorem registry, the current proof frontier, and the full failure log with all 239 live entries reproduced.

This is not a publication draft. It is a project bible. It contains no marketing language and no suggestion that Goldbach is proved anywhere. Binary Goldbach is **OPEN**. Universal success of the canonical root is **OPEN**. The p_0 -centred proof programme is **SUPERSEDED**.

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Chapter 1

AI Navigation Map

HOW TO USE THIS DOCUMENT

If you are a model receiving only this file, read in this order.

1. **Chapter 2** (three pages) — what is true right now. Do not skip it; several beliefs that were live in earlier versions of the corpus are dead.
2. **Chapter 3** — the notation is fixed. One symbol has one meaning throughout. Do not silently re-define $h, m, Q, R, T, U, J, \eta, \nu, \gamma$.
3. **Chapter 4** — the theorem registry. Before using any result, look up its status and dependencies here.
4. **Chapter 9** — the only place where new work should start.
5. **Chapters 11 and 12** — read *before* proposing anything. 239 recorded ways to waste a run.

Three rules that are violated most often.

1. Failure of the canonical traversal is *not* global Goldbach failure. Never invoke a theorem whose hypothesis is $\text{Good}_N = \emptyset$ under p_0 -failure alone. FAIL-UPPERHALF-SCOPE-001
2. A successful-world witness does not refute a statement conditioned on a closed failed world; a noncanonical root does not refute a canonical statement. FAIL-F1-DOMAIN-MIX-001
3. Never quote a count without its domain. The unit-complement row $N - r = 1$ is a separate class and is excluded everywhere. FAIL-UNIT-COMPLEMENT-001, FAIL-CENSUS-DOMAIN-001

Never raise a status. **OPEN**, **CONJECTURE**, **EMPIRICAL PATTERN**, **VERIFIED COMPUTATION** and **PROVED** are different things and stay different. A finite scan with zero counterexamples is never promoted to a theorem. If you prove something, add a new registry row; do not edit an old one.

IF YOU NEED X → GO TO SECTION Y

If you need ...	Go to ...
definitions (all of them)	Chapter 3: notation registry §3.1, objects §3.2, valuation layer §3.7
p_0 — definition, geometry, why it matters	§3.6 (definition), Chapter 6 (whole programme), §6.5 (what is and is not special)
BFS / stopped traversal — exact algorithm	§3.3 (definition), §3.4 (runnable reference implementation), §3.5 (historical variants)
EAC — exceptional autonomous component	§3.8; orientation warning in §3.8; prior-art boundary §14.2

If you need ...	Go to ...
canonical failed world	§3.9 (definition $R_N(r)$, closure, unique source); fixed point B_N^{low} in §3.10
CRD — Canonical Root Dynamics	Chapter 8; epistemic convention §8.1; architecture wall §8.2
C0 / C0a — small-gap canonical obligation	§8.3
C0b — canonical reflection-defect ledger	§8.4; branch targets TI/TD/TS/TF in §8.5
P1 / P1BR — branchwise production upgrade	§9.4
L1 — actual layer realisation	§9.5
E1 — escape or coupling	§9.6
W1 — simultaneous clean witnesses	§9.7
K1 — source-sensitive capacity	§9.8
X1 — canonical failure exclusion	§9.9
RPD — repeated prime-power redundancy	§9.10
computational evidence — all of it, with ranges	Chapter 10; reproducible generator in Appendix A
failed approaches — curated narrative	Chapter 11
full failure log — all 239 entries	Chapter 12; alphabetical index §12.1
literature	Chapter 14; what each external theorem does and does not give
current frontier	§9.2 first (canonical threshold programme, v4.79–v4.81); then the rest of Chapter 9; the five best next moves are in Chapter 16
exact product locks (T_S , T_R , overlap, orthogonality, full- Q)	§5.3
residue no-gos ($q \geq 5$ saturation, mod 3, topology)	§5.4
FRESH/SMOOTH , smoothness tax, first-front conservation and split	§7.3
lossless prime-power root ledger, collision budget, weighted cut	§7.4
Exact-D pair compensation, primitive normal form, gap filters, branch $k = 1$	§7.5
sparse geometric core lock (FFS layer)	§7.6
surviving lemmas of killed theories (CMFT, CGST, CSD, CAPT, RSD, rough- P_2 , H1-FP)	§11.3
rejected literature — what was tried and why it failed	§14.3; the 52-source audit §14.3.2; the frontier salvage audit §14.3.3; named missing lemmas §14.4

If you need ...	Go to ...
scope classes A / B / C / D — is this really about p_0 ?	§5.1; full classification table there
what to work on next — triage rules and tiers	§13.1; the five best moves in §16.4
Euler defect — and why its normalisation matters	§5.7
shadow / retention arithmetic	§5.5
path invariants (Bézout, winding)	§5.6
$T = UJ$, Euler defect , η , ν	§3.7, §5.7
D_q , M_q , FRESH/SMOOTH	§7.1
B_{abs} , B_{re} , incidence ledger	§7.2
genealogical defect rank κ_{gen}	§6.1
boundary filtration	§6.2
phase diagram $\theta = h/G$, Type I/II/III	§6.3
bidirectional rank ρ_0	§6.4
six special N (128, 1718, 1862, 1928, 2200, 6142)	§6.6
escape margin G_{esc}	§6.7
post-CRD successor theories (CMFT, CGST, CSD, CAPT, RSD)	§11.2
parity barrier / why it is the real wall	§11.4
historical / superseded results	Chapter 15
what not to try again	Chapter 16

Status vocabulary

Status	Meaning
PROVED	A proof is reproduced here or was independently reconstructed from definitions in the latest hostile audit, and no later correction is known.
PROVED WITH REPAIR	Proved after an explicit repair to hypotheses or to a step. The repair is always stated.
AUDITED IDENTITY	An exact identity that survived independent reconstruction on a declared domain, but has not been promoted to canonical PROVED .
VERIFIED COMPUTATION	An exact finite computation with a stated range and provenance. <i>Never</i> evidence for a universal statement.
EMPIRICAL PATTERN	A pattern observed in data, with no causal isolation. Not a theorem, not evidence for one.

Status	Meaning
CONJECTURE	A precisely stated conjecture, usually with computational support.
OPEN	A precisely stated target for which no proof is known.
KILLED	Killed by a proof, a domain correction, or an exact counterexample. Not usable as a lemma.
SUPERSEDED	Retained historically; not a live target. Its correct lemmas may still be used, its programme may not be restarted.
SCOPE-CORRECTED	The statement survives on a narrower domain; what died is a use outside that domain. Used inside the imported failure log, where it is the entry-level analogue of PROVED WITH REPAIR , and for imported results whose domain was narrowed.
<i>Two statuses carried over verbatim from the sources, so that no import is silently promoted:</i>	
PROVED – PENDING RED TEAM	The source states a complete proof but records that the result has not been adversarially audited ([BIB] PROVED-PENDING-REDTEAM). Stronger than CONJECTURE , weaker than PROVED . Do not upgrade it by citing it.
NEEDS VERIFICATION	The source states the result but flags an unverified step or an unverified uniform dependence ([BIB] NEEDS VERIFICATION). Treat as CONJECTURE until the flagged step is discharged.

Status mapping used when importing from the sources. [BIB]/[CORE]
 PROVED → **PROVED**; PROVED-PENDING-REDTEAM → **PROVED – PENDING RED TEAM**;
 PROVED-PENDING-INDEPENDENT-AUDIT → **PROVED – PENDING RED TEAM**; NEEDS
 VERIFICATION → **NEEDS VERIFICATION**; TRANSFERRED-PROVED → **PROVED**; SOURCE
 CLAIM: NEW-PROVED and SOURCE: PROVED-PENDING → **PROVED – PENDING RED TEAM**;
 CONDITIONAL → **CONJECTURE** with the condition stated; COMPUTATIONAL/EXPERIMENT
 → **VERIFIED COMPUTATION**; HEURISTIC → **EMPIRICAL PATTERN**; WORKING → **OPEN**;
 REJECTED/DISPROVED/ABANDONED → **KILLED**; CORRECTED → **SCOPE-CORRECTED**; RETIRED →
SUPERSEDED. The mapping never raises a status.

Chapter 2

Current State of the Project

Three sentences. The project studies binary Goldbach through a factorization graph on primes $p < N$ with edges $p \rightarrow q$ whenever $q \mid N - p$ and $q \nmid N$, and through one distinguished starting vertex, the *canonical selector* $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$. A substantial exact theory exists around p_0 — boundary geometry, a lossless genealogical ledger, path and valuation invariants, a fixed-point characterisation of canonical failure — and essentially all of it is correct. None of it closes Goldbach, and the latest audit [PCRD] establishes that no purely p_0 -local, finite-certificate architecture can.

2.1 Top-level status

Statement	Status	Where
Binary Goldbach: every even $N \geq 4$ is a sum of two primes	OPEN	external; §14.2
Every even $N \geq 6$ with $N/2$ composite has $\text{Good}_N \neq \emptyset$ where the canonical traversal from p_0 terminates in SUCCESS	OPEN	Chapter 9
The p_0 -centred proof programme (finite canonical certificate implying Goldbach)	SUPERSEDED	§8.2, §11.2
Canonical failure has an exact fixed-point characterisation	PROVED	§3.10
Exactly six $N \leq 6 \cdot 10^4$ have $B_N^{\text{low}} \neq \emptyset$	VERIFIED COMPUTATION	§6.6
p_0 is a genuinely distinguished vertex (boundary, genealogy, uniqueness)	PROVED	Chapter 6
p_0 is distinguished <i>dynamically</i> (better escape, smaller world, easier success)	KILLED	§6.5

2.2 What is actually proved

The following are theorems with reproduced or reconstructed proofs. They are safe to use as lemmas; their exact statements and hypotheses are in Chapter 4 and their proofs in Chapters 5–8.

1. **Canonical gap equivalence** (§3.6). For an upper root $r = x + h_r$: $r = p_0 \iff [x, r) \cap \mathbb{P} = \emptyset \iff h_r < G_r$. This is the whole content of “ p_0 is special” at the level of the boundary, and it is categorical, not quantitative.

2. **Strict non-reuse and the reflection gate** (§3.6). Every prime $\ell \geq h$ divides at most one element of the strict reflected block $\mathfrak{B}(p_0)$; every upper mirror c_j is composite. The gate gives compositeness only — *not* activation ($N = 242$).
3. **Genealogical defect rank** (§6.1). $\kappa_{\text{gen}}(p_k) = k$ on the non-diagonal domain, the defect ledger is a lossless code of local prime history, and p_0 is the unique reflection-complete upper root.
4. **Boundary filtration** (§6.2). The reflected blocks of consecutive upper roots nest with a shift, and the gcd-coupling matrix of level k embeds as a submatrix of level $k + 1$.
5. **Phase trichotomy** (§6.3). $m_0 - p_- = G - 2h$ splits the canonical cell into Type I / II / III with exact consequences for immediate success.
6. **Shadow arithmetic** (§5.5). The prime-gap shadow identity $r \mid N - p \iff r \mid \delta - p$, retention–escape $\text{PF}(N - q) \cap Q = \text{PF}(d_q)$, minimum-factor escape, shadow cycle localisation.
7. **Path invariants** (§5.6). The full path Bézout invariant $K_t p_t - C_t x = (-1)^t h$, the winding normal form, the $\leq 2\tau(h)$ reduced-state bound, no-bad-return-to- p_0 , and the global two-sheet collision budget.
8. **Valuation layer** (§5.7). $T_S = U_S J_S$ and the layered Euler defect $\eta(S) + \nu(S) \geq |S| + |\text{Src}(S)|$, canonically $\eta(R) + \nu(R) \geq |R| + 1$.
9. **Post-root factorization ledger** (§7.1, §7.2). $D_q = \gcd(m_0, N - q) = \gcd(B_q, |2h - q|)$, the FRESH/SMOOTH split, and the incidence identities IL1–IL6.
10. **CRD fixed point** (§3.10). $B_N^{\text{low}} = \bigcap_i B_i$ and p_0 fails $\iff N - p_0$ is composite and $\text{PF}(N - p_0) \subseteq B_N^{\text{low}}$.
11. **C0a** (§8.3). The small-gap canonical obligation, via the residual argument.
12. **Exact product locks** (§5.3). The multi-source lock $T_S \equiv (-1)^{|S|} \prod_{\text{Src}} d \pmod{N}$, the rooted full-front lock $T_R \equiv (-1)^{|R|} p_0$, the source-overlap identity, the orthogonality of disjoint child-closed worlds, and the full- Q ledger and production bounds.
13. **Residue no-gos** (§5.4). The residue-only route saturates for every $q \geq 5$; modulus 3 is the sole exception; and topology alone provably cannot supply incremental J .
14. **The rooted first-front layer** (§7.3–§7.5). Conservation and split identities (source status **PROVED – PENDING RED TEAM**), and — fully **PROVED** — the actual prime-power collision budget, the exact weighted cut identity, joint D_q, D_r rigidity, Exact-D pair compensation with its primitive normal form, and the two canonical gap filters.
15. **The canonical threshold engine** (§9.2). The container identity, its slack bound, the charge-free overlap normal form, and the trichotomy of surviving routes.

2.3 What is only computational

Everything in Chapter 10 is **VERIFIED COMPUTATION** or **EMPIRICAL PATTERN** and never more. The load-bearing numbers are: the exhaustive canonical scan to $N \leq 10^7$ with zero canonical failures; the six values of $N \leq 6 \cdot 10^4$ with $B_N^{\text{low}} \neq \emptyset$, namely 128, 1718, 1862, 1928, 2200, 6142; the ten $N \leq 2 \cdot 10^5$ with escape margin one; and the 301 failing non-canonical upper roots for $N \leq 2 \cdot 10^4$ concentrated on exactly those six N . External verification of Goldbach itself reaches $4 \cdot 10^{18}$ [12]; the project has never verified anything at that scale and must not cite it as its own.

2.4 The current frontier in one paragraph

Current (v4.79–v4.81). The live programme targets a *threshold* rather than all N : $\exists N_0^{p_0} < \infty$ such that $p_0(N)$ succeeds for every even $N \geq N_0^{p_0}$. On the primitive branch the selected charge $\mathcal{C}_Q$ is proved to divide the rooted container $U_R J_R D_{\lambda_N}$, and the container is proved to be the row product of the world with slack $> 5N/6$ — which *kills the archimedean size route outright*, together with the reflected/selected overlap route. Exactly three routes remain: (A) population capacity, (B) a residue obstruction inside a bounded container, (C) an explicit GOOD witness. The preferred target is C-P0-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPACITY, an effective upper bound for the

low-core size $|C_\lambda|$. Full details: §9.2.

Historical chain (still correct, no longer where the work is).

$$C0a \Rightarrow C0b \text{ branch obligations (TI, TD, TS, TF)} \Rightarrow P1^{\text{br}} \Rightarrow L1/E1/W1/K1.$$

C0a is **PROVED**. Its smallest missing bridge is the root-anchored TD obligation feeding $P1^{\text{br}}$: a lower bound on realised production measured by the *same* functional as the capacity bound. K1 fails for the identical reason. Every attempt to bridge two different functionals is recorded as dead in Chapter 12. Per-target details: Chapter 9.

2.5 The architecture wall

KILLED — the p_0 engine as a standalone Goldbach proof. [PCRD, final verdict]

Three independent results close the local route.

1. *Every prime can be a canonical root*: for P prime, $N = 2(P - 1)$ gives $p_0(N) = P$. So no property of p_0 can exclude any prime, and no finite affine canonical certificate can be universal (*Finite Affine Canonical Programmability*).
2. *Finite Bad-Tree Universality*: every finite admissible bad tree is realised by some N , so no finite local obstruction can be contradiction-generating.
3. *Finite-local universality wall*: any statement decided by a bounded neighbourhood of p_0 is realised in both truth values.

Consequence: a proof must import global input — a sieve/analytic bound, or a genuinely non-local combinatorial mechanism. The successor theories CMFT, CGST, CSD, CAPT, RSD and the global rough- P_2 matrix all died at this same wall (§11.2), and the H1-FP revival died at Pillai/Bennett $a^\alpha - a = b^\beta - b$ (§11.2).

2.6 What a new model should not do

Short list; the full list is Chapter 16 and the log itself.

1. Do not restart the local certificate programme in a new coordinate system. Five successor theories already did; all five hit the same wall.
2. Do not treat canonical failure as Goldbach failure. Almost every scope error in the log is a version of this.
3. Do not promote a finite scan. The scan to 10^7 says nothing about 10^8 .
4. Do not re-derive proved theorems. Use them by ID.
5. Do not bridge two different functionals with an inequality that holds “morally”. That is precisely what $P1^{\text{br}}$ and K1 need and precisely what has never been produced.

Chapter 3

Canonical Notation and Definitions

One symbol, one meaning, for the whole file. If a future revision needs a new object, give it a new symbol; do not overload $h, m, Q, R, T, U, J, \eta, \nu, \gamma, \kappa$. Several failure-log entries are nothing but collisions of notation (FAIL-NOTATION-KAPPA-001, FAIL-SYMBOL-R-COLLISION-001).

3.1 Symbol registry

Symbol	Meaning (fixed throughout)
N	the even target, $N \geq 6$; always $N = 2x$
x	the midpoint $N/2$
$\mathbb{P}$	the primes
p_0	the canonical selector $\min\{p \in \mathbb{P} : p \geq x\}$
p_k	the k -th upper root: $p_0 < p_1 < p_2 < \dots$, all primes $\geq x$
r	a generic upper root (any prime $\geq x$), <i>not necessarily</i> p_0
h, h_r	the offset $p_0 - x$, resp. $r - x$
m_0, m_r	the complement $N - p_0 = x - h$, resp. $N - r = x - h_r$
δ	$2h$; note $N = 2m_0 + \delta$
p_-	the prime predecessor of p_0
G, G_r	the gap $p_0 - p_-$, resp. $r - r^-$
θ	the phase $h/G \in (0, 1)$
Q	$\text{PF}(m_0)$, the canonical prime set
$\text{PF}(n)$	the set of distinct prime factors of n ; $\text{PF}(1) = \emptyset$
$P^+(n), P^-(n)$	largest, resp. smallest, prime factor of n
$\Omega(n)$	number of prime factors of n with multiplicity
$\tau(n)$	number of divisors of n
$\mathcal{A}_N$	the active primes: $\{p \in \mathbb{P} : p < N, p \nmid N\}$
Good_N	$\{p \in \mathcal{A}_N : N - p \in \mathbb{P}\}$ — the Goldbach witnesses
Bad_N	$\mathcal{A}_N \setminus \text{Good}_N$
$\mathfrak{B}(r)$	the strict reflected block of r
c_j, a_j	upper mirror $r - 2j$, lower mirror $N - c_j = m_r + 2j$
t_r	$\lfloor (h_r - 1)/2 \rfloor$, the block half-length
$\kappa_{\text{gen}}(r)$	genealogical defect rank of the upper root r
ρ_0	bidirectional prime rank, $\#\{s \geq 1 : h < D_s < 2h\}$

Symbol	Meaning (fixed throughout)
$R, R_N(r)$	the stopped failed world of the root r
$\text{Src}(S)$	the sources of S : vertices of S with no parent in S
Reach	reachability closure under $p \rightarrow q$
B_i, B_N^{low}	the CRD filtration and its fixed point
G_{esc}	escape capacity (number of escape edges)
D_q, M_q, B_q	$\gcd(m_0, N - q), (N - q)/D_q, m_0/q$
T_S, U_S, J_S	total valuation mass, unit part, layer part on S
η, ν	Euler defect terms: excess valuation, resp. layer count
γ	the coupling/capacity functional in P1^{br} and K1
K_t, C_t, W_t	path Bézout and winding coefficients

Warning 3.1 (Reserved letters). R is *always* the failed world, never a ring, a remainder, or a generic set. κ with no subscript is never used; only κ_{gen} and $\kappa^{\pm}$ appear. Q is $\text{PF}(m_0)$ and nothing else — the smooth domain in the incidence ledger is written Q_{smooth} .

3.2 The active prime graph

Definition 3.2 (Activity, edges, GOOD/BAD). Fix an even $N \geq 6$.

- p is *active* iff $p \in \mathbb{P}$, $p < N$ and $p \nmid N$. Write $\mathcal{A}_N$ for the set of active primes.
- The *row* of p is the integer $N - p$ together with its factorization.
- There is a directed edge $p \rightarrow q$ iff $q \mid N - p$ and $q \nmid N$, i.e. $q \in \text{PF}(N - p) \cap \mathcal{A}_N$. Write $\Gamma_N(p) = \text{PF}(N - p) \cap \mathcal{A}_N$ for the children of p .
- p is *GOOD* iff $N - p \in \mathbb{P}$ (equivalently $(p, N - p)$ is a Goldbach pair), and *BAD* otherwise.

The condition $q \nmid N$ is not cosmetic: without it the graph acquires absorbing cycles through the divisors of N and every statement about sources fails. [CORE, ARS-ACTIVITY]

Remark 3.3 (Unit complement). $N - p = 1$, i.e. $p = N - 1$, is neither GOOD nor a normal BAD row: its row has empty factorization. It is excluded from every count in this file. Failing to exclude it is FAIL-UNIT-COMPLEMENT-001.

Definition 3.4 (Sources). For $S \subseteq \mathcal{A}_N$, $\text{Src}(S) = \{p \in S : \nexists q \in S, q \rightarrow p\}$. A *closed* set is one with $\Gamma_N(S) \subseteq S$.

3.3 The stopped traversal

Definition 3.5 (Stopped traversal / factor-flow). Given N and an active root r , run breadth-first search from r in the graph of Definition 3.2 with one modification: a popped GOOD vertex is *recorded and not expanded*, and a GOOD child is recorded as an *escape edge* and not enqueued. The traversal returns SUCCESS at depth d if a GOOD vertex is first recorded at depth d , and FAILURE otherwise. The set of BAD vertices actually expanded is the *stopped failed world* $R_N(r)$.

If $2r = N$ (the diagonal case, $x \in \mathbb{P}$) the traversal returns SUCCESS at depth 0 with $R = \emptyset$.

Definition 3.6 (Two modes). Two implementations appear in the corpus and are both correct.

census stops at the first popped GOOD vertex. This reproduces the historical node and depth statistics.

world continues to exhaustion, building the full $R_N(r)$ and the full escape-edge set.

They agree on the verdict and on the success depth; they differ only in how much of the world is materialised. [CORE, ARS-TRAVERSAL-MODES]

Lemma 3.7 (Mode equivalence). **PROVED**. *For every N and every active r , *census* and *world* return the same verdict, and when the verdict is *SUCCESS* the same depth.*

Proof. BFS pops vertices in non-decreasing depth order, and neither mode expands a GOOD vertex, so the multiset of vertices popped strictly before the first GOOD pop is identical in the two modes. The first GOOD pop is therefore the same vertex at the same depth, and the verdict is FAILURE in one mode iff no GOOD vertex is ever popped, iff the same in the other. $\square$

3.4 Reference implementation

The following is the canonical executable semantics of Definitions 3.2–3.6. It is self-contained (standard library only), deterministic (fixed Pollard-rho seed), and is the file `p0_algorithm.py` of the project repository with its comments translated into English; the executable code is unchanged. A future model that is unsure what a definition means should run this rather than guess.

Listing 3.1: `p0_algorithm.py` — canonical selector and stopped traversal, runnable as given.

```

1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  p0_algorithm.py -- minimal, fully self-contained implementation of the
5  canonical selector p_0 and of the historical stopped traversal.
6
7  No imports outside the standard library. Copy and run:
8
9      python3 p0_algorithm.py 1928 3 # N = 1928, roots p_0, p_1, p_2
10     python3 p0_algorithm.py # demonstration on the six EAC
11 """
12 import sys
13 from collections import deque
14 from math import gcd
15 from random import Random
16
17 # --- 1. primality: deterministic Miller-Rabin for n < 3.3 * 10^24 -----
18 BASES = (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37)
19
20
21 def is_prime(n):
22     if n < 2:
23         return False
24     for p in BASES:
25         if n % p == 0:
26             return n == p
27     d, s = n - 1, 0
28     while d % 2 == 0:
29         d //= 2
30         s += 1
31     for a in BASES:
32         x = pow(a, d, n)
33         if x in (1, n - 1):
34             continue
35         for _ in range(s - 1):
36             x = x * x % n
37             if x == n - 1:
38                 break
39     else:
40         return False
41     return True
42
43
44 def next_prime(n):
45     m = n + 1
46     while not is_prime(m):
47         m += 1
48     return m
49
50
51 def prev_prime(n):

```



```

52 m = n - 1
53 while m >= 2 and not is_prime(m):
54     m -= 1
55 return m if m >= 2 else None
56
57
58 # --- 2. factorization: trial division + Pollard rho (deterministic seed) ---
59 def _rho(n, rng):
60     if n % 2 == 0:
61         return 2
62     while True:
63         c, y = rng.randrange(1, n), rng.randrange(0, n)
64         g, r, q, x, ys = 1, 1, 1, 0, 0
65         while g == 1:
66             x = y
67             for _ in range(r):
68                 y = (y * y + c) % n
69             k = 0
70             while k < r and g == 1:
71                 ys = y
72                 for _ in range(min(128, r - k)):
73                     y = (y * y + c) % n
74                     q = q * abs(x - y) % n
75                     g, k = gcd(q, n), k + 128
76             r *= 2
77         if g == n:
78             g = 1
79             while g == 1:
80                 ys = (ys * ys + c) % n
81                 g = gcd(abs(x - ys), n)
82         if g != n:
83             return g
84
85
86 def factorize(n):
87     """Returns {p: v_p(n)}; factorize(1) = {}."""
88     f, rng = {}, Random(20260904)
89     for p in range(2, 100000):
90         if p * p > n:
91             break
92         while n % p == 0:
93             f[p] = f.get(p, 0) + 1
94             n //= p
95     stack = [n] if n > 1 else []
96     while stack:
97         m = stack.pop()
98         if m == 1:
99             continue
100         if is_prime(m):
101             f[m] = f.get(m, 0) + 1
102             continue
103         d = _rho(m, rng)
104         stack += [d, m // d]
105     return dict(sorted(f.items()))
106
107
108 # --- 3. canonical selector and the successive upper roots -----
109 def p0(N):
110     """p_0(N) = min{ p prime : p >= N/2 }. """
111     if N % 2 or N <= 2:
112         raise ValueError("N must be even and greater than 2")
113     x = N // 2
114     return x if is_prime(x) else next_prime(x)
115
116
117 def upper_roots(N, count):
118     """[p_0, p_1, ..., p_{count-1}] -- consecutive primes >= N/2. """
119     out, p = [], p0(N)
120     while len(out) < count and p < N:
121         out.append(p)
122         p = next_prime(p)
123     return out
124
125
126 # --- 4. graph: edge, GOOD vertex, activity -----

```

```

127 def is_active(N, p):
128     """p active for N <=> p prime, p < N, p does not divide N."""
129     return is_prime(p) and p < N and N % p != 0
130
131
132 def is_good(N, p):
133     """GOOD <=> N - p prime, i.e. N = p + (N-p) is a Goldbach pair."""
134     return N - p >= 2 and is_prime(N - p)
135
136
137 def children(N, p):
138     """Active children of the row N - p: {q prime : q | N-p, q does not divide N}."""
139     return sorted(q for q in factorize(N - p) if N % q != 0)
140
141
142 # --- 5. HISTORICAL STOPPED TRAVERSAL / FACTOR-FLOW -----
143 def stopped_traversal(N, root):
144     """Full BFS with stopping at GOOD vertices.
145
146     Returns a dict:
147     verdict 'SUCCESS' or 'FAILURE' of the root,
148     depth depth of the first GOOD (None on failure),
149     witness Goldbach pair (p, N-p) found by the traversal,
150     R the stopped bad world (BAD vertices that were expanded),
151     escapes escape edges (p, q): p in R, q GOOD,
152     rows factorizations of N - p for p in R.
153     """
154     x = N // 2
155     if 2 * root == N: # diagonal case
156         return dict(verdict="SUCCESS", depth=0, witness=(root, root),
157                     R=set(), escapes=set(), rows={}, diagonal=True)
158     if not is_active(N, root):
159         raise ValueError("root must be an active prime")
160
161     queue, queued = deque([(root, 0)]), {root}
162     R, rows, escapes = set(), {}, set()
163     verdict, depth, witness = "FAILURE", None, None
164
165     while queue:
166         p, d = queue.popleft()
167         if p in R:
168             continue
169         if is_good(N, p): # GOOD: record, do NOT expand
170             if witness is None:
171                 verdict, depth, witness = "SUCCESS", d, (p, N - p)
172             continue
173         R.add(p) # BAD: expand the whole row
174         rows[p] = factorize(N - p)
175         for q in sorted(q for q in rows[p] if N % q != 0):
176             if is_good(N, q):
177                 escapes.add((p, q))
178                 if witness is None:
179                     verdict, depth, witness = "SUCCESS", d + 1, (q, N - q)
180                 continue # a GOOD child is not enqueued
181             if q not in R and q not in queued:
182                 queued.add(q)
183                 queue.append((q, d + 1))
184
185     return dict(verdict=verdict, depth=depth, witness=witness, R=R,
186                 escapes=escapes, rows=rows, diagonal=False)
187
188
189 # --- 6. report -----
190 def show(N, k=1):
191     x, r0 = N // 2, p0(N)
192     pm = prev_prime(r0)
193     h, m = r0 - x, N - r0
194     fm = factorize(m)
195     print(f"N = {N}, x = N/2 = {x}")
196     print(f"p_0 = {r0}, h = {h}, m_0 = {m} = "
197           + (" * ".join(f"{p}^{e}" if e > 1 else str(p) for p, e in fm.items())
198             or "1"))
199     print(f"previous prime = {pm}, G = {r0 - pm}, h < G : {h < r0 - pm}")
200     print(f"delta = 2h = {2*h}, Q = PF(m_0) = {sorted(fm)}")
201     for i, r in enumerate(upper_roots(N, k)):

```

```

202     t = stopped_traversal(N, r)
203     hr = r - x
204     tt = (hr - 1) // 2
205     kap = sum(1 for j in range(1, tt + 1) if is_prime(r - 2 * j))
206     print(f" p_{i} = {r:<10d} {t['verdict']:<8s} depth={t['depth']}"
207           f" |R|={len(t['R']):<4d} escapes={len(t['escapes']):<3d}"
208           f" kappa_gen={kap} witness={t['witness']}")
209
210
211 if __name__ == "__main__":
212     if len(sys.argv) > 1:
213         show(int(sys.argv[1]), int(sys.argv[2]) if len(sys.argv) > 2 else 1)
214     else:
215         for N in (128, 1718, 1862, 1928, 2200, 6142, 11822, 14198):
216             show(N, 3)
217             print()

```

Remark 3.8 (Reading the code as a definition). Three lines carry all the semantics that prose gets wrong. `if is_good(N, p): ...continue` is the stopping rule (a GOOD vertex is never expanded). `escapes.add((p, q)); ...continue` is the escape rule (a GOOD child is recorded, not enqueued). `if N % q != 0 in children` is activity. Every historical variant that differs from the corpus differs in exactly one of these three lines.

3.5 Historical variants and how they differ

Variant	Difference	Consequence
Canonical (audited)	As Definition 3.5	The reference. All statistics in this file use it.
<i>Expanding GOOD</i>	Expands GOOD vertices too	Depth statistics change; the world is no longer the failed world; several early tables in the corpus were produced this way and do not match later ones. FAIL-TRAVERSAL-EXPANDGOOD-001
<i>Ignoring activity</i>	Allows $q \mid N$ as a child	Creates absorbing cycles; “source” becomes ill-defined; kills the Euler defect bound.
<i>DFS instead of BFS</i>	Depth-first order	Verdict unchanged, depth statistic meaningless. Depth is only defined relative to BFS.
<i>Unit row admitted</i>	$p = N - 1$ counted	Off-by-one in every census; see Remark 3.3.
<i>Non-canonical root</i>	Starts at p_k , $k \geq 1$, or below x	Legitimate, but its results belong to scope class B (root-generic) and must never be quoted as canonical. FAIL-F1-DOMAIN-MIX-001

3.6 The canonical selector and midpoint geometry

Definition 3.9 (Normal form). Write

$$\begin{aligned}
 N &= 2x, & p_0 &= x + h, & m_0 &:= N - p_0 = x - h, & \delta &:= 2h, \\
 p_- &:= \max\{p \in \mathbb{P} : p < p_0\}, & G &:= p_0 - p_-, & Q &:= \text{PF}(m_0), & \theta &:= \frac{h}{G}.
 \end{aligned}$$

If $h = 0$ then $x \in \mathbb{P}$ and $p_0 = x$: the *diagonal* case. Unless stated otherwise the *non-diagonal* case is assumed: x composite, $h \geq 1$. The exact identity

$$N = 2m_0 + \delta$$

holds always. The same symbols with subscript r denote the analogous quantities for an arbitrary upper root $r = x + h_r \geq x$.

Theorem 3.10 (Canonical gap equivalence). **PROVED**. *Let $r = x + h_r \geq x$ be prime, r^- its predecessor and $G_r = r - r^-$. Then*

$$r = p_0 \iff [x, r) \cap \mathbb{P} = \emptyset \iff h_r < G_r.$$

Proof. If $r = p_0$ then $r^- < x$, so $G_r = r - r^- > r - x = h_r$. Otherwise some prime lies in $[x, r)$, hence $r^- \geq p_0 \geq x$ and $G_r \leq h_r$. [CORE, C-GAP-EQUIVALENCE] $\square$

Warning 3.11 (How *not* to read $h < G$). The inequality $h < G$ looks like a statement about “a large prime gap before p_0 ”. That reading is wrong and was a recurring source of error. Theorem 3.10 is *categorical*, not quantitative: no earlier upper prime lies inside the reflected interval. The correct formulation [CORE, V71.43] is

p_0 is a defect-free boundary condition selected by midpoint minimality.

Equivalently [CORE, V71.42]: p_0 is the unique upper root whose predecessor lies geometrically *outside* its reflected block. Nothing about the *size* of G is asserted, and no bound of Baker–Harman–Pintz type is used or needed here.

Definition 3.12 (Strict reflected block). For an upper root $r = x + h_r$ with $h_r \geq 3$ put

$$t_r = \left\lfloor \frac{h_r - 1}{2} \right\rfloor, \quad \mathcal{J}_{h_r} = \{1, \dots, t_r\}, \quad c_j = r - 2j, \quad a_j = N - c_j = m_r + 2j,$$

$$\mathfrak{B}(r) = \{a_1, \dots, a_{t_r}\} \cup \{c_1, \dots, c_{t_r}\}.$$

Then $m_r < a_j < x < c_j < r$ and $a_j + c_j = N$. The endpoints $u = 0$ and $u = h_r$, i.e. m_r and r themselves, are *not* in $\mathfrak{B}(r)$. Call c_j an *upper mirror* and a_j a *lower mirror*.

Lemma 3.13 (Strict non-reuse of large primes). **PROVED**. *If an odd prime ℓ divides two distinct elements of $\mathfrak{B}(r)$ then $\ell < h_r$. Equivalently, every prime $\ell \geq h_r$ occurs in exactly one element of $\mathfrak{B}(r)$.*

Proof. For $i \neq j$, $|a_i - a_j| = |c_i - c_j| = 2|i - j|$, and for any i, j , $c_j - a_i = 2(h_r - i - j)$. Since $i + j \leq 2t_r < h_r$ the last quantity is positive, so all three differences are nonzero. In each case half the difference is an integer in $\{1, \dots, h_r - 1\}$; an odd prime dividing twice that half divides the half itself, hence is smaller than h_r . [CORE, C-STRICT-NONREUSE] $\square$

Warning 3.14 (No widening of the block). Lemma 3.13 may *not* be applied to the widened block $\{m_r + 2u : 0 \leq u \leq h_r\}$. The endpoint pair $(u, v) = (0, h_r)$ has difference exactly $2h_r$ and is not covered by the proof; it would require a separate canonical input $\gcd(h, m_0) = 1$ and is nowhere used that way below. [FAIL, FAIL-BROADBLOCK-ENDPOINT]

Lemma 3.15 (Reflection gate). **PROVED**. *Let $r = p_0$ and $0 < 2j < h$. Then $c_j \in (x, p_0)$ and c_j is composite. Its full factorization is a graph row with parent a_j if and only if a_j is an active prime; in that case $\Gamma_N(\{a_j\}) = \text{PF}(c_j)$.*

Proof. Compositeness of c_j is immediate from $[x, p_0) \cap \mathbb{P} = \emptyset$; the rest is the definition of an edge. [CORE, ARS-REFLECTION-GATE] $\square$

Warning 3.16 (The gate gives compositeness, not activation). **KILLED** is every strengthening of the gate to activation. The gate gives *only* compositeness of each c_j : it does not give primality of a_j , does not give incidence coupling, does not give basin coupling and does not give capacity. Exact counterexample: $N = 242$, $x = 121$, $p_0 = 127$, $p_- = 113$, so $h = 6 < G = 14$; both reflected values $a_1 = 117 = 3^2 \cdot 13$ and $a_2 = 119 = 7 \cdot 17$ are composite, so *not a single active reflected row exists*. [FAIL, FAIL-REFLECTION-ACTIVATION-001]

This is not an accident: $N = 242$ is in phase Type I, and a guaranteed active lower mirror exists exactly in phase Type III (§6.3).

Scope note 3.17 (Diagonal correction to the genealogical layer). **SCOPE-CORRECTED**. The corpus states the genealogy and reflection-completeness uniqueness theorems for “every prime $r \geq x$ ” [CORE, V71.21, V71.31], while the proof uses the strict inequality $p_i > x$. The difference is real.

If $x = N/2$ is *prime* then $p_0 = x$, $h = 0$, and the hole corresponding to p_0 sits at index $h_k/2 > t_k$, i.e. *outside* the strict reflected block. Then $\kappa_{\text{gen}}(p_k) = k - 1$ for $k \geq 1$, and the root p_1 is also reflection-complete — violating literal uniqueness.

VERIFIED COMPUTATION: the smallest witness is $N = 14$, $x = 7 \in \mathbb{P}$, $p_0 = 7$, $p_1 = 11$, block $\{c_1 = 9 = 3^2\}$, $\kappa_{\text{gen}}(p_1) = 0$ although $p_1 \neq p_0$. For $N \leq 6 \cdot 10^4$, $k \leq 5$, *every* violation has x prime, and for x composite there is not a single violation.

What survives. Theorem 3.10 is unaffected in all cases. Only the equivalence with “every $r - 2j$ is composite” fails, because the strict reflected block covers the *open* interval (x, r) while the canonical condition concerns the *closed* interval $[x, r)$. Correct repair: add the hypothesis “ x composite”, or replace $[x, r)$ by (x, r) .

Why this does not disturb the rest of the file. All failure analysis assumes $h \geq 1$: for x prime the canonical root wins immediately and there is no closed world. The domain of every p_0 -specific theorem is therefore non-diagonal by definition.

3.7 The valuation layer

Definition 3.18 (Valuation mass, unit part, layer part). For a finite set S of active rows put, for each active prime q ,

$$d_S(q) = \sum_{p \in S} v_q(N - p), \quad \kappa_S(q) = \#\{p \in S : q \mid N - p\},$$

and define, the products running over $q \in \Gamma_N(S)$,

$$T_S = \prod_q q^{d_S(q)-1}, \quad U_S = \prod_{p \in S} \frac{N - p}{\text{rad}(N - p)}, \quad J_S = \prod_q q^{\kappa_S(q)-1}.$$

U_S is the *unit part* (excess valuation, i.e. repeated primes inside one row), J_S the *layer part* (support reuse, i.e. the same prime appearing in several rows).

Theorem 3.19 (Exact minimal-collision factorization). **PROVED**. $T_S = U_S J_S$.

Proof. Fix $q \in \Gamma_N(S)$ with positive valuations $e_1, \dots, e_k$ ($k = \kappa_S(q)$, $\sum e_i = d_S(q)$). Its exponent in T_S is $\sum e_i - 1$, in U_S it is $\sum e_i - k$, and in J_S it is $k - 1$; these add up. [CORE, G-TUJ] $\square$

Remark 3.20 (Reading). The radical defect of the world splits *exactly* into excess valuation and support reuse. This is completely generic (scope class B): it uses nothing about p_0 .

The Euler-defect terms η and ν built on top of this, and the inequality $\eta(S) + \nu(S) \geq |S| + |\text{Src}(S)|$ that the corpus records, are discussed with their exact status in §5.7. Read that section before using them: the normalisation matters and is easy to get wrong.

Warning 3.21 (η is not a capacity). η measures valuation leaving the set. It is *not* the functional γ appearing in $P1^{\text{br}}$ and K1, and the repeated attempt to identify them is the single most common dead end in the log (FAIL-ETA-GAMMA-IDENTIFY-001 and its successors).

3.8 Exceptional autonomous components

Definition 3.22 (EAC). An *exceptional autonomous component* for N is a nonempty set $C \subseteq \text{Bad}_N$ that is closed under Γ_N ($\Gamma_N(C) \subseteq C$) and contains no GOOD vertex. Equivalently: a nonempty closed bad set. Božek [8] introduced this notion for the Goldbach factorization graph.

Warning 3.23 (Orientation). Two orientations of the Goldbach graph exist in the literature and in this project. The corpus uses $p \rightarrow q$ iff $q \mid N - p$ (“the row of p points to its prime factors”). Reversing the arrows turns closed sets into reachable sets and *silently changes what “autonomous” means*. Several early comparisons with [8] were wrong for this reason (FAIL-EAC-ORIENTATION-001). Always state the orientation.

Remark 3.24 (EAC and canonical failure). The canonical root p_0 fails exactly when it lies in an EAC. This is the qualitative form of the fixed-point characterisation in §3.10; the quantitative form is the statement that the relevant EAC is B_N^{low} restricted to the reachable set of p_0 .

3.9 The canonical failed world

Definition 3.25 (Stopped failed world). $R_N(r)$ is the set of BAD vertices expanded by the stopped traversal from r (Definition 3.5). An *escape edge* is a pair (v, q) with $v \in R_N(r)$, $q \in \Gamma_N(v)$ and q GOOD. The traversal fails iff there is no escape edge and r itself is BAD.

Lemma 3.26 (Closure and unique source). **PROVED**. *If the traversal from r fails then $R_N(r)$ is closed ($\Gamma_N(R_N(r)) \subseteq R_N(r)$), consists entirely of BAD vertices, and $\text{Src}(R_N(r)) = \{r\}$ provided no $v \in R_N(r)$ has $r \in \Gamma_N(v)$. For $r = p_0$ the proviso holds unconditionally (no-bad-return-to- p_0 , §5.6), so*

$$\text{Src}(R_N(p_0)) = \{p_0\}.$$

Proof. Closure and badness are immediate from the traversal rules (a BAD vertex is expanded and all its active children enqueued; failure means no child is GOOD). r is a source by construction. Any other source would be a vertex enqueued with no parent in the world, impossible since every enqueued vertex other than r was enqueued from its parent. The canonical case uses Theorem 5.33. [CORE, ARS-CLOSED-WORLD] $\square$

Scope note 3.27 (Failure of the canonical traversal is not Goldbach failure). $R_N(p_0) \neq \emptyset$ and no escape means only that *this traversal from this root* found nothing. Good_N may still be large. The six values of N with $B_N^{\text{low}} \neq \emptyset$ all satisfy Goldbach comfortably. Every theorem whose hypothesis is $\text{Good}_N = \emptyset$ is a *class-D* statement (Goldbach-counterexample-conditioned) and may never be invoked under canonical failure alone. FAIL-UPPERHALF-SCOPE-001

3.10 The CRD fixed point B_N^{low}

Definition 3.28 (CRD filtration). Put

$$B_0 = \{p \in \mathcal{A}_N : p < N/3, N - p \text{ composite}\}, \quad B_{i+1} = \{p \in B_i : \text{PF}(N - p) \subseteq B_i\},$$

and $B_N^{\text{low}} = \bigcap_{i \geq 0} B_i$. The sequence is decreasing and finite, so the intersection is reached in finitely many steps.

Theorem 3.29 (Fixed-point characterisation of canonical failure). **PROVED**. B_N^{low} is the largest Γ_N -closed subset of $\{p < N/3 : N - p \text{ composite}\}$, and

$$p_0 \text{ fails} \iff N - p_0 \text{ is composite and } \text{PF}(N - p_0) \subseteq B_N^{\text{low}}.$$

Proof. B_{i+1} removes from B_i exactly the elements with a child outside B_i , so B_N^{low} is closed and contains every closed subset. For the equivalence: if p_0 fails then $R_N(p_0) \setminus \{p_0\}$ is closed, bad, and contained in the $N/3$ range (a child $q \mid N - p$ with $q \geq N/3$ would force $N - p \in \{q, 2q\}$, both excluded for a BAD row with $p > N/3$), hence contained in B_N^{low} ; and $\text{PF}(N - p_0) \subseteq R_N(p_0)$. Conversely if $\text{PF}(N - p_0) \subseteq B_N^{\text{low}}$ then the traversal never leaves B_N^{low} after the first step and B_N^{low} contains no GOOD vertex. [PCRD, CRD fixed point] $\square$

Remark 3.30 (Why the $N/3$ threshold). For $p > N/3$ and $N - p$ composite, every prime factor q of $N - p$ satisfies $q \leq (N - p)/2 < N/3$. So the interesting closed sets live below $N/3$ and the threshold costs nothing. Using $N/2$ instead re-admits p_0 itself and makes the filtration non-monotone in N ; this is FAIL-BLOW-THRESHOLD-001.

Scope note 3.31 (B_N^{low} is a computation, not a mechanism). Theorem 3.29 converts “does the canonical traversal fail” into “is a certain explicitly computable set nonempty”. That is an exact reformulation and it made the $6 \cdot 10^4$ census possible. It is *not* progress towards a proof: deciding $B_N^{\text{low}} = \emptyset$ for all N is at least as hard as the original question, and the finite-local universality wall (§2.5) says no bounded certificate decides it.

Chapter 4

Theorem Registry

How to use this chapter. Every named result of the project has one row. Before using a result anywhere, read its row: the *status* tells you whether it is a theorem, the *dependencies* tell you what breaks if a dependency is later corrected, and the *location* points at the proof or at the failure-log entry that killed it.

Rules. (i) A row is never edited to a higher status. If you prove something that was **CONJECTURE**, add a new row and mark the old one **SUPERSEDED**. (ii) A **VERIFIED COMPUTATION** row always carries its range. A range is part of the statement. (iii) A **KILLED** row is not a lemma. It may not be used “in the direction that still works” without a new row stating precisely that direction. (iv) Dependencies are transitive; the registry lists only immediate ones.

4.1 Foundational and boundary layer

ID	Statement / short name	Status	Dependencies	Location
C-GAP-EQ	Canonical gap equivalence: $r = p_0 \iff [x, r) \cap \mathbb{P} = \emptyset \iff h_r < G_r$	PROVED	—	Thm. 3.10
C-BOUNDARY-READING	$\mathbb{P}$ is a defect-free boundary condition selected by midpoint minimality; <i>not</i> a large-gap statement	PROVED	C-GAP-EQ	Warn. 3.11
C-STRICT-NONREUSE	Every prime $\ell \geq h_r$ divides at most one element of $\mathfrak{B}(r)$	PROVED	Def. 3.12	Lem. 3.13
C-BROADBLOCK-FAIL	The widened block $\{m_r + 2u : 0 \leq u \leq h_r\}$ does <i>not</i> satisfy strict non-reuse (endpoint pair)	KILLED	C-STRICT-NONREUSE	Warn. 3.14
ARS-REFLECTION-GATE	Every upper mirror c_j of p_0 is composite; its row has parent a_j iff a_j is an active prime	PROVED	C-GAP-EQ	Lem. 3.15
C-GATE-ACTIVATION	The gate also gives activation of some a_j	KILLED	—	Warn. 3.16; $N = 242$
C-DIAGONAL-SCOPE	Genealogy and uniqueness hold on the non-diagonal domain (x composite); the diagonal case is a genuine exception	SCOPE-CORRECTED	V71.21, V71.31	Scope 3.17; $N = 14$

ID	Statement / short name	Status	Dependencies	Location
C-NORMALFORM	$N = 2m_0 + \delta$, $\delta = 2h$, $\theta = h/G$	AUDITED IDENTITY		Def. 3.9

4.2 Graph, traversal and world layer

ID	Statement / short name	Status	Dependencies	Location
ARS-ACTIVITY	Activity ($q \nmid N$) is required for the source theory; without it the graph has absorbing cycles	PROVED	—	Def. 3.2
ARS-TRAVERSAL-MONOTONUS	Monotonus and world agree on verdict and success depth	PROVED	Def. 3.6	Lem. 3.7
ARS-CLOSED-WORLD	On failure, $R_N(r)$ is closed and all-BAD; $\text{Src}(R_N(p_0)) = \{p_0\}$	PROVED	ARS-NORETURN	Lem. 3.26
ARS-NORETURN	No-bad-return-to- p_0 : no $v \in R_N(p_0)$ has $p_0 \in \Gamma_N(v)$	PROVED	C-GAP-EQ	Thm. 5.33
CRD-FIXPOINT	B_N^{low} is the largest closed bad set below $N/3$; p_0 fails iff $N - p_0$ composite and $\text{PF}(N - p_0) \subseteq B_N^{\text{low}}$	PROVED	ARS-CLOSED-WORLD	Thm. 3.29
CRD-THRESHOLD	The $N/3$ threshold is the correct one; $N/2$ breaks monotonicity	PROVED	CRD-FIXPOINT	Rem. 3.30
EAC-DEF	EAC = nonempty closed bad set; p_0 fails iff p_0 lies in an EAC	AUDITED IDENTITY	CRD-FIXPOINT	§3.8
EAC-ORIENT	The orientation convention is $p \rightarrow q \iff q \mid N - p$; reversing it changes “autonomous”	PROVED	—	Warn. 3.23
C-SCOPE-D	Canonical failure does <i>not</i> imply $\text{Good}_N = \emptyset$	PROVED	—	Scope 3.27

4.3 Genealogy, filtration and phase

ID	Statement / short name	Status	Dependencies	Location
V71.21	Genealogical defect rank: $\kappa_{\text{gen}}(p_k) = k$	PROVED	C-DIAGONAL-SCOPE	Thm. 6.2
V71.31	p_0 is the unique reflection-complete upper root ($\kappa_{\text{gen}} = 0$)	PROVED	V71.21	Cor. 6.5
C-LEDGER-LOSSLESS	The defect ledger is a lossless code of local prime history: hole positions recover the prime gaps	PROVED	V71.21	Thm. 6.3
C-LEDGER-EXPLAINS	The lossless ledger <i>explains</i> why p_0 succeeds	KILLED	—	Warn. 6.7
C0b-RD	The canonical ledger partitions as $\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$ with no fifth class	PROVED	V71.21	Thm. 8.6

ID	Statement / short name	Status	Dependencies	Location
C-FILTRATION	Boundary filtration: $c_{i+s}^{(k+1)} = c_i^{(k)}$, $a_{j+s}^{(k+1)} = a_j^{(k)}$, $t_{k+1} = t_k + s$, $s = (p_{k+1} - p_k)/2$	PROVED	Def. 3.12	Thm. 6.8
C-GCD-EMBED	The gcd-coupling matrix $M_{ij}^{(k)} = \gcd(c_i, a_j)$ embeds as a submatrix of level $k + 1$	PROVED	C-FILTRATION	Cor. 6.9
C-PHASE	Phase trichotomy from $m_0 - p_- = G - 2h$: Type I ($2h < G$), Type II ($2h = G$), Type III ($2h > G$)	PROVED	C-GAP-EQ	Thm. 6.12
C-PHASE-I	Type I $\Rightarrow m_0$ composite $\Rightarrow$ immediate success impossible	PROVED	C-PHASE	Thm. 6.12
C-PHASE-II	Type II $\Rightarrow m_0 = p_- \Rightarrow$ immediate success automatic	PROVED	C-PHASE	Thm. 6.12
C-PHASE-III	Type III $\Rightarrow p_- = a_{j^*}$ with $j^* = h - G/2$	PROVED	C-PHASE	Thm. 6.12
C-PHASE-CELL	Cell decomposition $1 + (G/2 - 1) + 1 + (G/2 - 1) = G$; θ symmetric under $\theta \mapsto 1 - \theta$	AUDITED IDENTITY	C-PHASE	§6.3
C-RHO	$\rho_0 = \#\{s \geq 1 : h < D_s < 2h\}$, $D_s = p_0 - p_{-s}$; and $\rho_0 \geq 1 \iff$ Type III	PROVED	C-PHASE	Thm. 6.17
C-RHO-USERFORM	The originally proposed form of ρ_0 (without the lower bound $D_s > h$)	KILLED	C-RHO	Warn. 6.18
C-SIGNATURE	Bidirectional signature $(\kappa^+, \kappa^-) = (\kappa_{\text{gen}}, \rho_0)$	AUDITED IDENTITY	7.21, C-RHO	§6.4

4.4 Shadow, path and valuation layers

ID	Statement / short name	Status	Dependencies	Location
S-SHADOW-ID	Prime-gap shadow identity: $r \mid N - p \iff r \mid \delta - p$ for $r \mid m_0$	PROVED	C-NORMALFORM	Thm. 5.22
S-RETENTION	Retention-escape: $\text{PF}(N - q) \cap Q = \text{PF}(d_q)$ with $d_q = \gcd(m_0, \delta - q)$	PROVED	S-SHADOW-ID	Thm. 5.24
S-MINFACTOR	Minimum-factor escape: if $P^-(N - q) \notin Q$ the traversal escapes Q at q	PROVED	S-RETENTION	Cor. 5.25
S-CYCLE-LOC	Shadow cycle localisation: a cycle inside Q forces all its vertices into an explicit residue window	PROVED	S-SHADOW-ID	Thm. 5.27
S-TWOCYCLE	Shadow two-cycle equation $\delta = q + s + tqs$	PROVED	S-CYCLE-LOC	Prop. 5.28
S-PPWITNESS	Prime-power witness: a prime power row forces an explicit escape or an explicit divisibility	PROVED	S-RETENTION	Prop. 5.29

ID	Statement / short name	Status	Dependencies	Location
P-BEZOUT	Full path Bézout invariant $K_t p_t - C_t x = (-1)^t h$	PROVED	—	Thm. 5.30
P-WINDING	Winding normal form $K_t p_t = W_t N + (-1)^t p_0$	PROVED	P-BEZOUT	Cor. 5.31
P-STATES	At most $2\tau(h)$ reduced states per vertex	PROVED	P-WINDING	Thm. 5.32
P-COLLISION	Global two-sheet collision budget: at most two collision endpoints	PROVED	P-STATES	Thm. 5.34
P-COLLISION-CONTRA	The collision budget yields a contradiction from canonical failure	KILLED	P-COLLISION	§5.6
V-TUJ	$T_S = U_S J_S$ (excess valuation $\times$ support reuse)	PROVED	Def. 3.18	Thm. 3.19
V-EULER	Layered Euler defect $\eta(S) + \nu(S) \geq S + \text{Src}(S) $; canonically $\eta(R) + \nu(R) \geq R + 1$ — <i>normalisation of η, ν not reconstructed here</i>	CONJECTURE	V-TUJ, ARS-CLOSED-WORLD	§5.7
V-EULER-A	Active-mass form $\sum_q d_R(q) \geq R + 1$	VERIFIED COMPUTATION	V-TUJ	§5.7
V-EULER-B	Trivial form $\sum_{p \in R} (\Omega(N-p) - 1) + \nu(R) \geq R + 1$	PROVED	—	§5.7
V-ETA-GAMMA	η is the capacity functional γ	KILLED	—	FAIL-ETA-GAMMA-IDENTIFY

4.5 Post-root factorization and incidence

ID	Statement / short name	Status	Dependencies	Location
PR-DQ	$D_q = \gcd(m_0, N - q) = \gcd(B_q, 2h - q)$ for $q \mid m_0$	PROVED	S-SHADOW-ID	Thm. 7.1
PR-MQ	$M_q = (N - q)/D_q$ is coprime to Q	PROVED	PR-DQ	Cor. 7.3
PR-FRESH	FRESH/SMOOTH split: q is FRESH iff $P^+(N - q) \geq B_q$	AUDITED IDENTITY	PR-MQ	Def. 7.4
IL1-IL6	Incidence ledger $\prod_q M_q = U_F(J_F B_{\text{re}}) B_{\text{abs}}$ on $S = Q_{\text{smooth}}$	AUDITED IDENTITY	PR-MQ	Thm. 7.6
IL-DOMAIN	IL1–IL6 hold on Q_{smooth} only; the FRESH part is excluded	SCOPE-CORRECTED	IL1–IL6	Warn. 7.7
PR-BABS	$B_{\text{abs}}, B_{\text{re}}, B_{\text{abs,int}}$: definitions and the exact relation between them	AUDITED IDENTITY	IL1–IL6	§7.2
PR-HIGHGAP	High-gap amplification $h^2 > 4m_0$	SCOPE-CORRECTED	PR-DQ	Warn. 7.5

4.6 CRD obligations and open targets

ID	Statement / short name	Status	Dependencies	Location
C0a	Small-gap canonical obligation (residual argument)	PROVED	C-GAP-EQ, C0b-RD	§8.3
TI	Injective branch obligation of C0b	OPEN	C0b-RD	§8.5
TD	Root-anchored dominance obligation of C0b	OPEN	C0b-RD	§8.5
TS	Stability branch obligation of C0b	OPEN	C0b-RD	§8.5
TF	Fresh-layer branch obligation of C0b	OPEN	C0b-RD	§8.5
TD-DENSITY	Density form of TD	SUPERSEDED	TD	§8.5
P1BR	Branchwise production upgrade: realised production bounded below by the <i>same</i> functional as capacity	OPEN	TD	§9.4
L1	Actual layer realisation	OPEN	V-EULER	§9.5
E1	Escape or coupling	OPEN	S-RETENTION	§9.6
W1	Simultaneous clean witnesses	OPEN	L1	§9.7
K1	Source-sensitive capacity $K_{\times}(R, \mathcal{P}) + K(\mathcal{P}) \leq \mathcal{C}_N(R)$	OPEN	P1BR	§9.8
X1	Canonical failure exclusion	OPEN	K1	§9.9
RPD	Repeated prime-power redundancy: $D_{\text{high}}(H_Q) \leq O(\log \log N) \eta_{\text{high}} + o(M)$	OPEN	V-EULER	§9.10
C-SAMEFUNC-BRIDGE	Global same-functional margin bridge	OPEN	P1BR, K1	§9.11
C-GENEALFREE-DYN	Genealogy-free boundary-to-dynamic escape for p_0	OPEN	V71.31	§9.11
C-POP1-SEP	One-defect dynamic separation between p_0 and p_1	OPEN	V71.21	§9.11

4.7 Computational rows

ID	Statement / short name	Status	Dependencies	Location
EXP1-SCAN	No canonical failure for even $6 \leq N \leq 10^7$	VERIFIED COMPUTATION		§10.2
EXP7-BLOW	$B_N^{\text{low}} \neq \emptyset$ for exactly six $N \leq 6 \cdot 10^4$: 128, 1718, 1862, 1928, 2200, 6142	VERIFIED COMPUTATION	CRD-FOXP	§6.6
EXP3-CENSUS	301 failing non-canonical upper roots for $N \leq 2 \cdot 10^4$, all at those six N ; 2261 unit rows excluded	VERIFIED COMPUTATION	Rem 103	§10.3
EXP4-MARGIN	Ten $N \leq 2 \cdot 10^5$ with escape margin $G_{\text{esc}} = 1$; per-dyadic-block minimum rises $1 \rightarrow 2 \rightarrow 3$	VERIFIED COMPUTATION	Def 105	§6.7
EXP5-PHASE	Phase-type frequencies for $N \leq 2 \cdot 10^5$	VERIFIED COMPUTATION	CF-PHASE	§6.3

ID	Statement / short name	Status	Dependencies	Location
EXP6-FILT	Filtration embedding verified for $N \leq 2 \cdot 10^4$, $k \leq 5$	VERIFIED COMPUTATION	C-FILTRATION	§6.2
CERT-34	All 34 hostile certificates of the corpus reproduced (regressions 11822, 3542948; counterexamples 242, 128, 322, 858, 4736, 6142, 14198)	VERIFIED COMPUTATION		§10.4
EMP-DEPTH	Success depth of the canonical traversal is almost always ≤ 2	EMPIRICAL PATTERN		§10.5
EMP-POFIRST	p_0 is not measurably better than p_1 at escaping	EMPIRICAL PATTERN	EXP3-CENSUS	§6.5

4.8 Killed and superseded programmes

ID	Statement / short name	Status	Dependencies	Location
P0-ENGINE	p_0 as a standalone Goldbach proof engine	KILLED	FACP, FBTU	§2.5
FACP	Finite Affine Canonical Programmability: every prime is $p_0(2(P-1))$, so no finite affine canonical certificate is universal	PROVED	—	§8.2
FBTU	Finite Bad-Tree Universality: every finite admissible bad tree is realised	PROVED	—	§8.2
FLUW	Finite-local universality wall	PROVED	FACP, FBTU	§8.2
CPCT	Canonical Prime Cell Theorem	PROVED	FACP	§8.2
CMFT	Canonical Multiplicative Flow Theory	KILLED	FLUW	§11.2
CGST	Canonical Graph Spectral Theory	KILLED	FLUW	§11.2
CSD	Canonical Shadow Descent	KILLED	FLUW	§11.2
CAPT	Canonical Affine Programmability Transfer	KILLED	FACP	§11.2
RSD	Root Structure Descent	KILLED	FLUW	§11.2
ROUGH2	Global rough- P_2 matrix	KILLED	parity	§11.4
H1-FP	H1 root-factor descent with shift-2 orthogonality	KILLED	Pillai/Bennett	§11.2
TWOCORE	Two-core wall $a^\alpha - a = b^\beta - b$	PROVED	[79]	§11.2

4.9 Imported results: exact product locks and no-gos (CRD Core)

ID	Statement / short name	Status	Dependencies	Location
G-MULTISOURCE-LOCK	$T_S \equiv (-1)^{ S } \prod_{\text{Src}} d \pmod{N}$ for child-closed S	PROVED	V-TUJ	Thm. 5.7
G-ROOT-LOCK	$T_R \equiv (-1)^{ R } p_0 \pmod{N}$; $T_R = (2a + 1)x + (-1)^{ R }h$, $a \geq 3$; $\max(U_R, J_R) > \sqrt{N}$	PROVED	G-MULTISOURCE-LOCK, ARS-CLOSED-WORLD	Thm. 5.8
G-CHILD-CLOSED-OVERLAP-IDENTITY	Overlap identity (OV1), (OV2)	PROVED	V-TUJ	Thm. 5.9
G-DISJOINT-CHILD-SETS-ORTHOGONALITY	Disjoint child-closed worlds are arithmetically orthogonal (OB1)–(OB3)	PROVED	ARS-ACTIVITY, G-COLLISION-CUT	Thm. 5.10
G-FULLQ-LEDGER	$\prod_{q \in Q_r} M_q = U_F(J_F B_{\text{re}}) B_{\text{abs}}$ on the whole Q_r under failure	PROVED	ARS-CLOSED-WORLD, V-TUJ	Thm. 5.11
G-FULLQ-PRODUCTION	$ Q_r > (5/3)^{ Q_r } \text{rad}(m_r) > 1$	PROVED	—	Thm. 5.13
G-COLLISION-CUT	$K(S) = \sum \log \gcd$; $C_{\text{cut}} = (S - 1)L - 2K$; every $\gcd \mid s - s' $	PROVED	—	Thm. 5.14
G-CLEAN-POLYLOG-PARENT-SELECTION	Clean selection of arithmetic parent selection; support load $\leq \rho(\mathcal{L})$	PROVED	Siegel–Walfisz, lower sieve, C-GAP-EQ	Thm. 5.15
W1-MISATTRIBUTION	Earlier listing of G-CLEAN-POLYLOG-PARENT-SELECTION as W1’s open target	SCOPE-CORRECTED	G-CLEAN-POLYLOG-PARENT-SELECTION	Parent 5.15
C-RESIDUE-ONLY-RETURN-MOD3	$\text{TURN}(\{m\})^3 = G$ for $q \geq 5$: no residue-only forced return	PROVED	—	Thm. 5.17
C-MOD3-TAGGED-FORCED-RETURN	$\text{MOD} = \text{RETURN}(\text{MOD } 3)$ forces a sibling $r \equiv 2 \pmod{3}$ with $r \rightarrow 3$	PROVED	—	Thm. 5.18
C-TOPOLOGY-NOGO	Topology alone cannot force $\Delta \log J \gg W(E)$	PROVED	—	Thm. 5.20

4.10 Imported results: the rooted first-front layer (Biblia v7.3–v7.5)

ID	Statement / short name	Status	Dependencies	Location
B-FRESH-PACKING	(FP): $\prod_{q \in E} s_q \geq m_0^{ E } / \prod q$	PROVED – PENDING RED TEAM	B-COLLISION	Prop. 7.10
B-SMOOTH-TAX	(ST): $B_q^{e_q} > q$, $e_q = j_q + u_q$	PROVED – PENDING RED TEAM	B-FRESH	Prop. 7.11
B-SMOOTH-EXCESS	$\Omega(U_R) + \Omega(J_R) - 1 \geq \sum e_q$	PROVED – PENDING RED TEAM	B-SMOOTH-TAX	Cor. 7.12
B-FFC	(FFC), (FFC- Ω): root first-front conservation	PROVED – PENDING RED TEAM	V-TUJ	Thm. 7.14
B-FFRE	(FFRE): $\prod \text{rad}(N - q) = J_{S_1} R_{\partial}(S_1)$	PROVED – PENDING RED TEAM	B-FFC	Cor. 7.15
B-UJD-SPLIT	(U-split), (JR-split): exact first-front $U/J/\partial$ split	PROVED – PENDING RED TEAM	B-FFRE	Thm. 7.16
B-GCD-PACK	$\nu_\ell \leq \min(Q , 1 + \lfloor m_0/\ell \rfloor)$	PROVED – PENDING RED TEAM	—	Lem. 7.18
B-POWERFUL-TAIL	Powerful-integer density is too weak for the adaptive first front	KILLED	—	Scope 7.19

ID	Statement / short name	Status	Dependencies	Location
B-RSCP	Root Smoothness Conservation Problem	OPEN	B-UJD-SPLIT, B-SMOOTH-TAX	Target 7.20
B-PP-LEDGER	(PP-I), (PP-D), (PP): lossless prime-power root ledger	PROVED – PENDING RED TEAM	PR-DQ	Thm. 7.25
B-POSTROOT-DEMAND	$D_q = \gcd(B_q, 2h - q)$; (Demand); (Post-root tax)	PROVED – PENDING RED TEAM	B-SHADOW TEAM	Thm. 7.26
B-ROOT-VAL-RIGID	$v_r(N - q) > a \Rightarrow v_r(2h - q) = a$	PROVED – PENDING RED TEAM	B-SHADOW TEAM	Lem. 7.27
B-SHARP-PP-PACK	$\ell^k \mid n_q, n_r \Rightarrow \ell^k \mid q - r$; the diameter bound on $\nu_{\ell,k}$	PROVED – PENDING RED TEAM		Lem. 7.28
B-VPC	(VPC): actual prime-power collision budget	PROVED	PR-MQ	Thm. 7.30
B-SHARED-MASS	$L_{\text{shared}} \leq 2 \log \Delta_{\text{odd}}(S)$	PROVED	B-VPC	Cor. 7.31
B-PAIR-FLOOR	$L_{\text{pair}}(q, r) \geq \log 5$	PROVED	PR-FRESH	Lem. 7.32
B-WEIGHTED-CUT	(Cut): $\sum L_{\text{pair}} = \sum_{\lambda} \mu_{\lambda}(s - \mu_{\lambda})w_{\lambda}$; $C_{\text{cut}} \geq \binom{s}{2} \log 5$	PROVED	B-PAIR-FLOOR	Thm. 7.33
B-LOW-MULT	Low-multiplicity layer lower bound $W_{\leq u}, W_1$	PROVED	B-WEIGHTED-CUT, B-VPC	Cor. 7.34
B-EXACTD-OUTSIDE	$t < r \Rightarrow M_q M_r > \delta^2$; failure regime (ED-res)	PROVED	PR-DQ	Thm. 7.36
B-GAP-FILTER-1	$9p_0 \geq 18G + 5G^2 \Rightarrow M_q M_r > \delta^2$	PROVED	B-EXACTD-OUTSIDE, C-GAP-EQ	Cor. 7.37
B-JOINT-D	(Joint-D): $D_q D_r \mid m_0 \gcd(c, r - q)$	PROVED	PR-DQ	Thm. 7.38
B-PNF	(PNF): primitive residual normal form	PROVED	B-EXACTD-OUTSIDE	Thm. 7.39
B-HIGH-GAP	(HG-r), (HG): $h^2 > 4m_0$ under pair failure	PROVED	B-PNF	Thm. 7.40
B-GAP-FILTER-2	$p_0 < 2G + G^2/4$	PROVED	B-HIGH-GAP	Cor. 7.42
B-K1-GAP	$r > 4^{d+1} q u^2$; $p_0 < 2G + G^2/12$	PROVED	B-HIGH-GAP	Thm. 7.43
B-K1-QUOTIENT	$k = 1$ quotient normal form in (A, B)	PROVED	B-PNF	Thm. 7.44
B-K1-COVERAGE	Structured composite coverage; $\gcd(B, qu - 1) = 1$	PROVED	B-K1-QUOTIENT	Prop. 7.46
B-K1-BARRIER	Canonical $k = 1$ gap/covering barrier	OPEN	B-K1-COVERAGE	Target 7.47
B-MULTIPLEX-RIGID	$D \geq \delta m^{\kappa_{\text{col}}}$	PROVED – PENDING RED TEAM	B-CUT	Thm. 7.49
B-COMP-BARRIER	$m < (N/6)^{\chi/\eta}$; $\eta \geq t\chi \Rightarrow m < (N/6)^{1/t}$	PROVED – PENDING RED TEAM	B-MULTIPLEX-RIGID	Cor. 7.50
B-SPARSE-CORE-LOCK	$X - 1 \leq U_C J_C \leq U_C (D/2)^{\chi_{\tau}}$	PROVED – PENDING RED TEAM	B-MULTIPLEX-RIGID	Cor. 7.51
B-SUPPORT-EXPLOSION	$\log S \geq \lceil \log A_S / \log B_S \rceil$	PROVED – PENDING RED TEAM		Thm. 7.53

4.11 Imported results: the canonical threshold programme (v4.79–v4.81)

ID	Statement / short name	Status	Dependencies	Location
V79-NAGURA	$p_0 < 3N/5$, $m > 2N/5$ for $N \geq 50$	PROVED	[1], C-GAP-EQ	Prop. 9.1
V79-FLOOR	(V79.4): $U_R^2 J_R B_{\text{abs}} > m(5/3)^{ Q } > 2N/3$	PROVED	G-FULLQ-PRODUCT	Thm. 9.2
V79-COEFF-DICH	$F_Q^{\text{coeff}} = 1 \iff \text{primitive};$ $F_Q^{\text{coeff}} > 1 \Rightarrow T_R > \frac{5}{2}N$	PROVED	G-ROOT-LOCK	Thm. 9.3
V80-LOADONE	Primitive $\Rightarrow D_q = 1; \prod_{q \in Q} (N - q) > \frac{25}{36}N^2$	PROVED	V79-COEFF-DICH	Thm. 9.5
C-PO-PRIMITIVE-OWERPRESERVING-QUADRATIC-UJD-CHAF	OWERPRESERVING-QUADRATIC-UJD-CHAF	PROVED	V80-LOADONE	Thm. 9.6
C-PO-PRIMITIVE-FIRSTFRONT-BLITMARCON-PRODUCTION	FIRSTFRONT-BLITMARCON-PRODUCTION	PROVED	V79-FLOOR	§9.2.1
C-PO-PRIMITIVE-BASES-ESCAPE-OR-REUSE-CHARGE	BASES-ESCAPE-OR-REUSE-CHARGE	SUPERSEDED	C-PO-PRIMITIVE-OWERPRESERVING-QUADRATIC-UJD-CHAF	§9.2.1
V81-CONTAINER	(V81.1): $U_R J_R D_{\lambda_N}$ is the row product of the world	PROVED	V80-LOADONE	Thm. 9.7
V81-SLACK	(V81.2): container slack $> 5N/6$ for $N \geq 500$	PROVED	V81-CONTAINER	Thm. 9.8
V81-SIZE-DEATH	The archimedean size route is closed	PROVED	V81-SLACK	Cor. 9.9
V81-OVERLAP	(V81.3): reflected/selected overlaps are charge-free	PROVED	ARS-REFLECTION-CHAF	Thm. 9.10
C-PO-PRIMITIVE-FIRSTFRONT-REFLECTION-DISJOINT-CHAF	FIRSTFRONT-REFLECTION-DISJOINT-CHAF	KILLED	V81-SLACK, V81-OVERLAP	§9.2.2
V81-C1-SHADOW	(V81.11): primitive load-one $\iff \gcd(m, 2h - u) = 1$ on $R \setminus \{p_0\}$	PROVED	S-SHADOW-ID	Thm. 9.12
V81-C2-RESIDUAL	R' child-closed, nonempty, $\min R' < \sqrt{N}$	PROVED	ARS-CLOSED-WORLD	Thm. 9.14
V81-C3-FACTOR	$T_R \Sigma' = (\prod_{q \in Q} (N - q)) T_{R'}$	PROVED	V81-C2-RESIDUAL	Thm. 9.15
V81-C4-RESLOCK	(V81.13): selected residue lock	PROVED	G-MULTISOURCE-LOCK	Thm. 9.16
V81-C5-ALPHABET	Adaptive fringe cofactor alphabet $\mathcal{L}_N \subseteq [\lambda, 2\lambda)$	PROVED	V80-LOADONE	Thm. 9.17
V81-IMMUNITY	Product-comparison immunity of the minimal survivor	PROVED	V81-SLACK	Thm. 9.19
V81-TRICHOTOMY	Only routes (A) population, (B) residue, (C) GOOD remain	PROVED	V81-IMMUNITY	Thm. 9.20
C-PO-PRIMITIVE-LOWGROW-POPULATION-CAPACITY	LOWGROW-POPULATION-CAPACITY	OPEN	V81-TRICHOTOMY, V81-C5-ALPHABET	§9.2.4
C-PO-PRIMITIVE-SELECTED-RESIDUE-LOCK-CAPACITY	SELECTED-RESIDUE-LOCK-CAPACITY	OPEN	V81-C4-RESLOCK, G-ROOT-LOCK	§9.2.4
C-PO-PRIMITIVE-SELECTED-ULN-CAPACITY	SELECTED-ULN-CAPACITY	OPEN	C-PO-PRIMITIVE-OWERPRESERVING-QUADRATIC-UJD-CHAF	§9.2.1
G-CONDITIONAL-EFFECTIVE-POPULATION	Conditional effective $N_0^{p_0}$	PROVED (conditional)	V79-FLOOR	§9.2.1
FAIL-SUNIT-COUNT-BOUNDS-THETALOWCORREDOG	BOUNDS-THETALOWCORREDOG bounds $ C_\lambda $	KILLED	[18]	§9.2.4

4.12 Imported results: surviving lemmas of killed programmes

ID	Statement / short name	Status	Parent (killed)	Location
PCRD-CMFT-1	Lossless anchor filtration; $\mathcal{H}_{j+1} = \mathcal{H}_j \sqcup \mathcal{A}_j$	PROVED	CMFT	Thm. 11.2
PCRD-CMFT-2	Tail hard-edge universality: local tail geometry does not distinguish p_0	PROVED	CMFT	Thm. 11.3
PCRD-CMFT-3	Canonical least-factor split	PROVED	CMFT	Thm. 11.4
PCRD-CMFT-4	All-root collision identity	PROVED	CMFT	Thm. 11.5
PCRD-CMFT-5	Prime-pressure failure ceiling: $\mathcal{P}_N > \sqrt{x} \log x \Rightarrow \text{Goldbach}$	PROVED	CMFT	Thm. 11.6
PCRD-CGST-1	Canonical straddle theorem	PROVED	CGST	Thm. 11.7
PCRD-CGST-2	Arbitrarily deep finite reflection obstruction	PROVED	CGST	Thm. 11.8
PCRD-CGST-3	Diagonal collapse: the cell dimension is false	PROVED	CGST	Thm. 11.9
PCRD-CSD-1	Canonical conductor-gap compatibility	PROVED	CSD	Thm. 11.11
PCRD-CAPT-1	Fixed- (r, s) compression to two linear prime forms	PROVED	CAPT	Thm. 11.12
PCRD-RSD-1	Exact rough signed identity	PROVED	RSD	Thm. 11.14
PCRD-RSD-2	Distribution-roughness phase law $\theta > 2/(k+1)$	PROVED	RSD	Thm. 11.15
PCRD-M-1	$K_{3,3}$ explosion at the natural scales	CONJECTURE (conditional)	ROUGH2	Thm. 11.17
PCRD-M-2	Boundary blindness of generic operator norms	PROVED	ROUGH2	Thm. 11.18
PCRD-M-3	$h = 1$ matrix invisibility	PROVED	ROUGH2	Thm. 11.19
PCRD-M-4	Fixed-modulus spectrum recovers only the singular series	PROVED	ROUGH2	Thm. 11.20
PCRD-PO-3	Gap scalar freedom: $h, G, h/G$ have no closing regime	PROVED	PO-ENGINE	Thm. 11.22
PCRD-H1-1	H1 root-factor descent; $\text{PF}(N - q_*) \cap \text{PF}(m_0) = \emptyset$	PROVED	H1-FP	Thm. 11.24
PCRD-H1-2	H1 shift-2 orthogonality $\gcd(N - u, u + 2) = 1$	PROVED	H1-FP	Thm. 11.25
PCRD-H1-3	3-core normal form; no 3-core found in finite boxes	VERIFIED COMPUTATION	UTAF	Rem. 11.26
MATCHED-EUCLID-GM	Matched Euclidean gluing lemma (named missing theorem)	OPEN	—	§14.4

Completeness statement. This registry is *not* an exhaustive index of the sources. The CRD Theory Core alone contains 801 live titled results and the Biblia 673; what is imported here is the load-bearing subset — exact identities, no-go theorems, frontier reductions, and the lemmas that survive their killed parent programmes. If a needed result is not here, it may still exist

in the sources; search [CORE] by THEOREM-ID and [BIB] by theorem title before concluding that something is unproved.

Chapter 5

Core Results (the Biblia layer)

This chapter holds the exact machinery that the project actually owns: shadow arithmetic, path invariants, and the valuation ledger. Almost all of it is *root-generic* — it works from any upper root and never uses minimality. Reading it with that in mind is the fastest cure for the most common error in the project’s history, which is to quote a generic lemma as a canonical one.

5.1 Scope classes A / B / C / D

Definition 5.1 (The four scope classes). A result T is

- class A — genuine p_0 -specific if there is an *explicit proof step* that fails when p_0 is replaced by another upper root, *and* the content of the conclusion actually changes;
- class B — root-generic if every step goes through verbatim for $r = x + h_r$ with h_r, m_r in place of h, m_0 ;
- class C — failure-conditioned if its hypothesis is that the traversal from the root fails;
- class D — Goldbach-counterexample-conditioned if its hypothesis is $\text{Good}_N = \emptyset$.

Separately there is the *vacuity-canonical* case: the construction goes through for p_k on those coordinates where it is defined at all, and canonicity only guarantees that it is defined everywhere. This is **not** class A. The corpus repeatedly confused the two.

Object / result	Class	What happens under $p_0 \rightarrow p_k$
Relation $p \rightarrow q$, $\mathcal{A}_N$, valuations, components	B	nothing; this is prior art §14.2
$T_S = U_S J_S$	B	nothing
High-root compression, unique source	B + C	nothing; $r > N/3$ and $N - r$ composite suffice
$q \leq m_r/3 < N/6$	B + C	nothing
$\gcd(h_r, m_r) = 1$, q^e congruences	B	nothing; root algebra
Shadow identity, retention, D_q , M_q	B	nothing FAIL-MR2-CANONICAL-ONLY-001
Bézout / winding / state count	B	nothing; $h \mapsto h_r$
Shadow cycles, prime-power witness	B + C	nothing
Strict non-reuse of the block	B	nothing; the proof uses index differences only
No-bad-return, two-sheet budget	C	nothing beyond the failure hypothesis

Object / result	Class	What happens under $p_0 \rightarrow p_k$
Fixed point B_N^{low} and the failure characterisation	C	nothing; the object depends on N alone
Reflection gate (<i>activation</i>)	vacuity	for p_k exactly k coordinates are prime and there is nothing there for the gate to open
Exact bundle lift	vacuity	the clauses go through verbatim on every row with composite c'_j ; 25.2% of rows are lost in the recorded scan [CORE, P0-CANONICITY-ABLATION]
Zero hole count as <i>mass</i>	KILLED	loss is $O_k(\log N)$; not a mechanism
$h < G$ / empty interval	A	everything; this is the definition and the only canonical input
$\mathcal{P}_{\text{def}} = \emptyset$	A	exactly k prime atoms appear
$\kappa_{\text{gen}} = 0$	A	the rank rises to k ; the equivalence is exact in both directions
Reflection-completeness	A	uniqueness is lost
Ledger completeness $\mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$	A	a fifth class $\mathcal{P}_{\text{def}}$ appears
Lossless code of prime history	A	p_k encodes k earlier roots; p_0 encodes none
Phase trichotomy and $j^* = h - G/2$	A	the notion “predecessor outside the block” exists only for the canonical root
$\rho_0 \geq 1 \iff \theta > 1/2$	A	for p_k the predecessor is p_{k-1} , so $D_1 \leq h_k$ and the whole characterisation collapses
Exact-D filter $p_0 < 2G + G^2/4$	A + C	without $h < G$ the filter does not exist; non-canonical countermodels exist
Exact-D model elimination	A + C	the models return FAIL-EXACTD-NONCANONICAL-001
Upper-half compression	D	requires $\text{Good}_N = \emptyset$ FAIL-UPPERHALF-SCOPE-001
Symmetric fracture, forbidden off-set band	D	require the minimal-counterexample hypothesis

The separation result in three sentences.

1. Almost the whole local arithmetic of p_0 — shadow, D_q , M_q , Bézout, winding, cycles, the U/J ledger — is *generic upper-root algebra*. Minimality is used nowhere in it.
2. The only genuine canonical input is $h < G$, equivalently emptiness of $[x, p_0) \cap \mathbb{P}$. It enters proofs in exactly one place: it makes *every* c_j composite.
3. The consequences of that single input — $\kappa_{\text{gen}} = 0$, reflection-completeness, ledger completeness, the phase trichotomy, the ρ_0 formula — are the only known properties of p_0 that have an *exact converse*. Every other candidate either has no converse (least-upper-parent) or was refuted (mass, density, local algebra, statistics).

5.2 The generic layer

Theorem 5.2 (High-root compression and unique source). **PROVED**, class $B + C$. Let $r > N/3$ be an active odd prime, $N - r$ composite, and suppose the traversal from r fails. Then every non-root state lies below $N/3$ and

$$\Gamma_N(R_N(r)) = R_N(r) \setminus \{r\}.$$

Proof. For a bad row, $N - p$ is odd composite, so every prime factor q has an odd cofactor ≥ 3 ,

whence $q \leq (N - p)/3 < N/3$. If r had a parent $p \in R$ then $N - p = kr$ with k odd and $k < 3$, so $k = 1$ and $p = N - r$ would be prime, contradicting compositeness of the complement. [CORE, G-HIGH-ROOT-SOURCE; canonically ARS-UPPER-EXCLUSION] $\square$

Reading. The bad world has one source and the whole upper half apart from the root is external to it. **This is not a property of p_0** — it is a property of being a high root with composite complement. **VERIFIED COMPUTATION** on 301 real closed worlds ($N \leq 2 \cdot 10^4$), zero violations.

Theorem 5.3 (Canonical prime-free geometry). **PROVED**. Hypotheses: *canonical failure, non-diagonal case*. Conclusion:

$$[x, p_0) \cap \mathbb{P} = \emptyset; \quad m_0 \text{ odd composite, } m_0 \geq 9; \quad \gcd(m_0, N) = \gcd(h, m_0) = 1;$$

$$q \in Q \Rightarrow q \leq \frac{m_0}{3} < \frac{N}{6}; \quad q^e \parallel m_0 \Rightarrow x \equiv h, \quad p_0 \equiv 2h = \delta \pmod{q^e}.$$

Classes: *emptiness of the interval* — A ; $\gcd(h, m_0) = 1$ and the q^e congruences — B (identical for any upper root with h_r, m_r); the inequality $q \leq m_r/3$ — $B + C$.

Proof. Emptiness is minimality. m_0 is odd as the complement of an odd prime, and composite, else the root is GOOD. Next $\gcd(m_0, N) = \gcd(p_0, N)$, and $p_0 \mid N$ with $p_0 \geq N/2$ would force $N = 2p_0$, the diagonal case. If $d \mid h$ and $d \mid m_0 = x - h$ then $d \mid x$, hence $d \mid x + h = p_0$; from $0 < d \leq h < p_0$ and primality of p_0 , $d = 1$. The inequality: the cofactor m_0/q is odd and ≥ 3 . The congruences are $m_0 = x - h$ and $p_0 = x + h$. [BIB, normal form; CORE, ARS-CANON-GEOMETRY] $\square$

VERIFIED COMPUTATION: 4915 canonical bad roots for $N \leq 1.5 \cdot 10^4$, zero violations.

Scope note 5.4 (Inequality repair). **SCOPE-CORRECTED**. Some earlier notes wrote $q < m_0/3$. The correct inequality is $q \leq m_0/3$; equality occurs e.g. for $N = 32$, $p_0 = 17$, $m_0 = 15$, $q = 5$. The consequence $q < N/6$ remains strict. [BIB, minor inequality repair]

Theorem 5.5 (Least upper parent). **PROVED**, class A . If $r = p_0$ then for every prime power $q^e \mid N - r$

$$r = \min(\mathcal{P}_N(q^e) \cap [x, N)), \quad \mathcal{P}_N(q^e) = \{p \in \mathcal{A}_N : q^e \mid N - p\}.$$

[CORE, ARS-LEAST-UPPER-PARENT]

Warning 5.6 (No converse). **KILLED** as a selector. In the recorded scan of 301 failing roots, eleven non-canonical roots satisfy the same first-front minimality, among them (128, 73), (128, 79), (1718, 877), (1862, 941), (1928, 971), (6142, 3089); exactly one has the stronger world-level property, (1928, 971). So *first-front parent minimality does not by itself explain p_0* . [CORE, V71.41]

5.3 Exact product locks and world identities

These are the exact multiplicative identities of the CRD Core. All are **PROVED** with short proofs, all are safe to use as lemmas, and none of them was in earlier compressions of this project. They are the algebraic backbone that every capacity argument silently uses.

Theorem 5.7 (Multi-source product lock, G-MULTISOURCE-LOCK). **PROVED**, class B . If a finite $S \subseteq \mathcal{A}_N$ is child-closed and $D_S^{\text{src}} = S \setminus \Gamma_N(S)$, then

$$T_S \equiv (-1)^{|S|} \prod_{d \in D_S^{\text{src}}} d \pmod{N}.$$

Proof. $T_S = \prod_{s \in S} (N - s) / \prod_{q \in \Gamma_N(S)} q$. Modulo N replace each $N - s$ by $-s$ and cancel the invertible active labels of $\Gamma_N(S)$; exactly the sources survive. [CORE, G-MULTISOURCE-LOCK] $\square$

Limitation. A rooted one-source lock cannot upper-bound an enlarged ambient multi-source system; adjoining external parents changes the source term.

Theorem 5.8 (Rooted full-front lock, G-ROOT-LOCK). **PROVED**, class C. If canonical p_0 fails, $R = R_N(p_0)$, $r = |R|$, then

$$T_R = U_R J_R \equiv (-1)^r p_0 \pmod{N}, \quad \text{and for an integer } a \geq 3, \quad \boxed{T_R = (2a + 1)x + (-1)^r h.}$$

In particular $\max(U_R, J_R) > \sqrt{N}$.

Proof sketch. Unique-source (Lemma 3.26) gives $T_R = m_0 \prod_{p \in R \setminus \{p_0\}} (N - p)/p$, and the congruence follows from Theorem 5.7. Every non-root row lies below x , so each ratio exceeds one. Writing $q = P^-(m_0)$, $B = m_0/q$: if $m_0 \neq 9$ then $B \geq 5$, so $T_R \geq B(N - q) \geq 5N - m_0$, exceeding every affine value with $a \leq 2$. If $m_0 = 9$ then $3 \nmid N$, row 3 is bad, and Theorem 5.2 places its non-root child s below $N/3$, giving $T_R \geq 9 \cdot \frac{N-3}{3} \cdot \frac{N-s}{s} > 3N$ for $N \geq 6$, while every affine value with $a \leq 2$ is at most $5x + h < 3N$. [CORE, G-ROOT-LOCK] $\square$

Reading. This is the sharpest *unconditional* statement the project has about a canonical failed world: its total valuation mass is pinned to p_0 modulo N and forced into an explicit affine form. It is the container of route (B) in the surviving-route trichotomy (§9.2).

Theorem 5.9 (Exact source-overlap identity, G-CHILD-CLOSED-OVERLAP-IDENTITY). **PROVED**, class B. Let $A, B \subseteq \mathcal{A}_N$ be finite child-closed sets with $C = A \cap B \neq \emptyset$, $U = A \cup B$, and write $\Sigma(S) = \prod_{d \in D_S^{\text{src}}} d$. Then

$$\boxed{T_A T_B \Sigma(U) \Sigma(C) = T_U T_C \Sigma(A) \Sigma(B).} \quad (\text{OV1})$$

If moreover $D_U^{\text{src}} = D_A^{\text{src}} \uplus D_B^{\text{src}}$, then

$$\boxed{T_A T_B \Sigma(C) = T_U T_C.} \quad (\text{OV2})$$

Proof. For finite child-closed S , $\Gamma_N(S) = S \setminus D_S^{\text{src}}$, so $T_S = \Sigma(S) \prod_{s \in S} (N - s)/s$. The multiplicative row ratio satisfies exact inclusion–exclusion over A, B, U, C ; substituting gives (OV1), and the disjoint-source hypothesis cancels $\Sigma(U) = \Sigma(A) \Sigma(B)$. [CORE, G-CHILD-CLOSED-OVERLAP-IDENTITY] $\square$

Theorem 5.10 (Disjoint child-closed worlds are arithmetically orthogonal, G-DISJOINT-CHILD-CLOSED-ORTHOGONALITY). **PROVED**, class B. Let $S_1, \dots, S_b \subseteq \mathcal{A}_N$ be finite, pairwise disjoint and child-closed, $U = \bigcup_i S_i$. Then for $i \neq j$,

$$\boxed{\gcd(N - u, N - v) = 1 \quad (u \in S_i, v \in S_j),} \quad (\text{OB1})$$

the supports $\Gamma_N(S_i)$ are pairwise disjoint, and

$$\boxed{T_U = \prod_i T_{S_i}, \quad K(U) = \sum_i K(S_i), \quad \text{Src}(U) = \bigsqcup_i \text{Src}(S_i),} \quad (\text{OB2})$$

$$\boxed{\eta(U) = \sum_i \eta(S_i), \quad \nu(U) = \sum_i \nu(S_i).} \quad (\text{OB3})$$

Proof. A prime q dividing both $N - u$ and $N - v$ is active (Definition 3.2), so child-closure puts it in $S_i \cap S_j$ — impossible. Hence no prime-power layer occurs in two sets, and the primewise definitions of T , K , η , ν are additive; absence of cross edges splits the source sets. [CORE, G-DISJOINT-CHILD-CLOSED-ORTHOGONALITY] $\square$

Reading. Disjoint failed packets carry *no* shared arithmetic whatsoever. This is why “combine several failed worlds and count” arguments gain nothing: the combination is a product, not an interaction.

Theorem 5.11 (Full- Q failure ledger, G-FULLQ-LEDGER). **PROVED**, class C. If the stopped traversal from $r > N/2$ fails and $N - r$ is composite, then with $S = Q_r$

$$\prod_{q \in Q_r} M_q = U_F(J_F B_{\text{re}}) B_{\text{abs}}, \quad U_F \mid U_R, \quad J_F B_{\text{re}} \mid J_R.$$

Every residual prime lies in R , and the same formulas hold for any subfamily of Q_r .

Proof. Failure puts every first-front row q in R , so $\kappa_R(\ell) \geq m_\ell$, and closure puts every residual factor in R . For $\ell \notin Q_r$ split $E_\ell = (E_\ell - m_\ell) + (m_\ell - 1) + 1$: the final copy is reused when $\kappa_R > m_\ell$ and absorbed otherwise. For $\ell \in Q_r$ the root row already supplies the base copy and all residual copies enter U_F . The same primewise comparison gives the two divisibilities. [CORE, G-FULLQ-LEDGER] $\square$

Warning 5.12 (This is the correct domain of the incidence ledger). Theorem 5.11 *supersedes* the reading of IL1–IL6 as a Q_{smooth} -only statement: on the failed branch the identity holds on the whole of Q_r . The Q_{smooth} version (Theorem 7.6) is the restriction. The unconditioned extension to arbitrary Q_0 still fails at $N = 20$ (FAIL-Q0-LEDGER-001), because a successful first-front row can give $\kappa_R < m_\ell$.

Theorem 5.13 (Full- Q production, G-FULLQ-PRODUCTION). **PROVED**, class B. For an active prime $r > N/2$ with odd composite $m_r = N - r$,

$$\mathfrak{F}_{Q_r} := \prod_{q \in Q_r} M_q > \left(\frac{5}{3}\right)^{|Q_r|} \text{rad}(m_r) > 1.$$

Proof. For $q \mid m_r$, activity gives $q \nmid N$, hence $q \nmid D_q$ and $D_q \leq m_r/q$. An odd composite has $q \leq m_r/3$, and $m_r < N/2$, so $M_q \geq q(N - q)/m_r > \frac{5}{3}q$. Multiply over distinct $q \in Q_r$. [CORE, G-FULLQ-PRODUCTION] $\square$

Limitation. The unit complement has $Q = \emptyset$ and is excluded (Remark 3.3, FAIL-UNIT-COMPLEMENT-001).

Theorem 5.14 (Exact collision and cut identities, G-COLLISION-CUT). **PROVED**, class B. For odd row values Y_s ($s \in S$), with $L(S) = \sum_{s \in S} \log Y_s$ the total row mass, $K(S)$ the collision mass and $C_{\text{cut}}(S)$ the cut functional,

$$K(S) = \sum_{s < s'} \log \gcd(Y_s, Y_{s'}), \quad C_{\text{cut}}(S) = (|S| - 1)L(S) - 2K(S).$$

If $Y_s = N - s$ then every such gcd divides $|s - s'|$.

Proof. Use $v_\ell(\gcd(Y_s, Y_{s'})) = \sum_{k \geq 1} \mathbf{1}_{\ell^k \mid Y_s} \mathbf{1}_{\ell^k \mid Y_{s'}}$ and sum; the cut identity is $(n-1)\mu - 2\binom{\mu}{2} = \mu(n-\mu)$. A common divisor of $N - s$ and $N - s'$ divides $s - s'$. [CORE, G-COLLISION-CUT] $\square$

Limitation. A singleton layer contributes nothing ($\binom{1}{2} = 0$); the identity is generic and cannot by itself supply a canonical absorption trichotomy (FAIL-GENERIC-ABSORPTION-001).

Theorem 5.15 (Clean polylogarithmic parent selection, G-CLEAN-POLYLOG-PARENT-SELECTION). **PROVED**. Fix $A > 0$. For all sufficiently large even N , let canonical p_0 fail, $R = R_N(p_0)$, and let $\mathcal{L} \subseteq \{(\ell, k) : \ell \in R, \ell^k \leq (\log N)^A\}$ be finite and nonempty, with $\rho_\ell = \#\{k : (\ell, k) \in \mathcal{L}\}$ and $\rho(\mathcal{L}) = \max_\ell \rho_\ell$. Then there exist distinct upper-half external active primes $p_{\ell,k} > p_0$, one per $(\ell, k) \in \mathcal{L}$, with

$v_\ell(N - p_{\ell,k}) = k$ and the clean-support property: if a prime r divides $N - p_{\ell,k}$ and $N - p_{\ell',k'}$ for two distinct selected parents, then

$$r = \ell = \ell'.$$

Consequently $\max_r \#\{p \in \mathcal{P} : r \mid N - p\} \leq \rho(\mathcal{L})$, and if the layers have pairwise distinct base primes the selected complements are pairwise coprime and the global support load is one.

Hypotheses beyond canonical failure: Siegel–Walfisz, a standard lower-bound sieve, Mertens, Theorem 5.2 and Theorem 3.10. [CORE, G-CLEAN-POLYLOG-PARENT-SELECTION]

Warning 5.16 (Correction to an earlier reading of W1). **SCOPE-CORRECTED**. Earlier compressions of this project listed G-CLEAN-POLYLOG-PARENT-SELECTION as the *open target* of W1. It is not: it is a **PROVED** theorem of the CRD Core. What remains open in W1 is the passage from the *polylogarithmic* layer family $\ell^k \leq (\log N)^A$ to the *actual* layers of the world — i.e. W1 is blocked on L1 (§9.5), not on the selection lemma. See §9.7 as corrected.

5.4 Residue no-go theorems

Theorem 5.17 (Residue-only return no-go for $q \geq 5$, C-RESIDUE-ONLY-RETURN-NOGO). **PROVED**. Let $q \geq 5$ be prime, $G = (\mathbb{Z}/q\mathbb{Z})^\times$, $a = N \bmod q$. Then

$$(G \setminus \{a\})^2 = G.$$

Hence every possible tagged quotient residue is a product of two sibling residues neither congruent to $N \bmod q$. Consequently the single residue $b \bmod q$, even together with $\Omega(b) \leq 3$, cannot force a direct sibling return to q . [CORE, C-RESIDUE-ONLY-RETURN-NOGO]

Theorem 5.18 (Exceptional forced return at modulus 3, C-MOD3-TAGGED-FORCED-RETURN). **PROVED**. Assume $N \equiv 2 \pmod{3}$ and $N - p = 3^e b$ with $3 \nmid b$, $b \equiv 2 \pmod{3}$. Then some sibling prime $r \mid b$ satisfies $r \equiv 2 \pmod{3}$, hence $3 \mid N - r$ and $r \rightarrow 3$. [CORE, C-MOD3-TAGGED-FORCED-RETURN]

Warning 5.19 (Modulus 3 is an exception, not a mechanism). Theorems 5.17 and 5.18 together say exactly this: the pure local-residue route *saturates* for every $q \geq 5$, and 3 is the single exception. Any programme that hopes to iterate the mod-3 forced return into a general mechanism is dead before it starts; this is one of the convergent no-gos that killed CRD1 (§8.1).

Theorem 5.20 (Topology alone cannot convert external seed weight into ambient incremental J , C-TOPOLOGY-NOGO). **PROVED**. No theorem using only finite child closure, source counts, support-graph topology, edge multiplicities and the reverse-extension ledger can force $\Delta \log J \gg W(E)$ for arbitrarily large external source weights $W(E)$.

Proof. A finite realisation permits a family with one small receiving target q and arbitrarily many external components feeding it. Topology and attachment count grow linearly, while the horizontal weight at the receiving label grows only like $e \log q$; external source labels may carry arbitrarily larger evaluation weights without altering the recorded topology. The missing ambient factor must therefore come from arithmetic/order/root information, not from the listed ledgers. [CORE, C-TOPOLOGY-NOGO] $\square$

Reading. This is the precise reason the “ η is the capacity” family of attempts is hopeless (Warning 3.21): the quantity that would have to do the work is provably not a function of the topology.

Remark 5.21 (First-occurrence subtlety). Fresh support is *not* incremental J : if $\kappa_R(\ell) = 0$ then the first occurrence of ℓ belongs to the fresh-support factor $F_R(A)$, and only a *later* occurrence contributes a full $\log \ell$ to J_S/J_R . Confusing the two is a recurring source of phantom capacity.

5.5 Shadow and retention arithmetic

Theorem 5.22 (Prime-gap shadow identity). **PROVED**, class B. For $r \in Q$ and any prime p ,

$$r \mid (N - p) \iff r \mid (\delta - p),$$

and the same equivalence holds modulo r^e when $r^e \mid m_0$.

Proof. $N - p = 2m_0 + \delta - p \equiv \delta - p \pmod{r^e}$. [BIB, thm:shadow; CORE, ARS-SHADOW] $\square$

Reading. The whole part of the support that returns to the first alphabet Q is governed by the single number $\delta = 2h$. This is the first point at which the gap parameter enters the multiplicative dynamics. *Limitation:* the theorem does not replace the full factorization of $N - p$; a factor outside Q may lead to success, to a new transient component, or to another sink.

Warning 5.23 (This is *not* a p_0 selector). **KILLED** as a canonical statement. For *any* upper root $r = x + h_r$ with $m_r = x - h_r$ we have $r = m_r + 2h_r$, so for every odd $q \mid m_r$, $q \mid (r - 2j) \iff q \mid (h_r - j)$. Minimality is not used. [FAIL, FAIL-MR2-CANONICAL-ONLY-001]

Theorem 5.24 (Retention-escape). **PROVED**, class B. For $q \in Q$ and $d_q = \gcd(m_0, |\delta - q|)$,

$$\text{PF}(N - q) \cap Q = \text{PF}(d_q).$$

In particular $d_q = 1$ means every active child of the row $N - q$ is new relative to Q .

Proof. For $r \in Q$: $r \mid N - q \iff r \mid \delta - q \iff r \mid \gcd(m_0, |\delta - q|)$; collecting over all r gives the claim. [BIB, thm:retention] $\square$

VERIFIED COMPUTATION, $N \leq 10^4$, zero violations.

Corollary 5.25 (Minimum-factor escape). **PROVED**, class B. Let $a = P^-(m_0)$. If $a > h$ then $d_a = 1$.

Proof. Every prime divisor of d_a lies in Q , hence is $\geq a$; but $a > h$ gives $0 < |2h - a| < a$, and $2h = a$ is impossible by parity. [BIB, cor:min-factor-escape] $\square$

Warning 5.26. A child outside Q does *not* mean success; it means only that the first alphabet has been left.

Theorem 5.27 (Shadow cycle localisation and normal form). **PROVED**, class B + C. On Q consider the auxiliary edge $q \rightsquigarrow s$ whenever $s \mid \delta - q$. Then every directed cycle lies in $(0, \delta)$; a simple cycle of length ≥ 3 has all vertices $< \delta/3 = 2h/3$; and a two-cycle $\{q, s\}$ satisfies

$$\delta = q + s + tqs, \quad t \geq 0 \text{ even},$$

with $t \geq 2$ inside a larger terminal core; hence a terminal component $C \subseteq Q$ with $|C| \geq 3$ satisfies $C \subset (0, 2h/3)$. [BIB, lem:qcycle-below-delta, lem:shadow-two-cycle, cor:qcore-compression]

Reading. The scale $\delta = 2h$ controls the *size* of the whole recursive part of the first front — the cleanest example of a mechanism in which the gap steers multiplicatively. But δ is a function of the root, not of its minimality.

Proposition 5.28 (Shadow two-cycle). **PROVED**, class B. Contained in Theorem 5.27: a two-cycle $\{q, s\}$ in the shadow graph is exactly a solution of $\delta = q + s + tqs$ with $t \geq 0$ even.

Proposition 5.29 (Prime-power witness). **PROVED**, class C. Let $q \in Q$ with $\text{PF}(N - q) \subseteq Q$. Then there is $r \in Q$ with

$$v_r(m_0) = v_r(\delta - q) > 0.$$

Proof. Put $s = \delta - q$, so $N - q = 2m_0 + s$. If for every $r \mid N - q$ the valuations $v_r(m_0)$ and $v_r(s)$ differed, then for odd r we would have $v_r(N - q) = \min\{v_r(m_0), v_r(s)\} \leq v_r(s)$; since all factors of $N - q$ lie in Q , we would get $N - q \mid |s|$, impossible because $N - q > |\delta - q|$. [BIB, thm:witness] $\square$

5.6 Path invariants and winding

Theorem 5.30 (Full path Bézout invariant). **PROVED**, class B. For a path $p_0 \rightarrow p_1 \rightarrow \dots \rightarrow p_t$ with $N - p_i = k_i p_{i+1}$, and $K_0 = C_0 = 1$, $K_{t+1} = K_t k_t$, $C_{t+1} = 2K_t - C_t$:

$$K_t p_t - C_t x = (-1)^t h, \quad \gcd(K_t, C_t) = \gcd(K_t, h). \quad (\text{PBI})$$

Proof. For $t = 0$ the left side is $p_0 - x = h$. Induction: from $p_t = 2x - k_t p_{t+1}$ we get $K_t k_t p_{t+1} - (2K_t - C_t)x = (-1)^{t+1} h$. Each k_i divides $N - p_i$ and is coprime to N , so $\gcd(K_t, x) = 1$; the second identity follows. [BIB, thm:bezout-path; CORE, ARS-BEZOUT] $\square$

Reading. Every path from the root remembers h as an exact Bézout invariant. This is the strongest sense in which “gap geometry” propagates dynamically — but h is replaced by h_r for any root, so the class is B. **VERIFIED COMPUTATION** on real paths, $N \leq 5 \cdot 10^3$, zero violations.

Corollary 5.31 (Irreducible winding normal form). **PROVED**, class B. With $W_t = (C_t - (-1)^t)/2$,

$$K_t p_t = W_t N + (-1)^t p_0, \quad \gcd(W_t, K_t) = 1, \quad (\text{WN})$$

where $W_0 = 0$, $W_{t+1} = K_t - W_t$. On a bad path $W_t > 0$, $W_t \equiv t \pmod{2}$, and for $t \geq 2$, $K_t p_t > 2N$. [BIB, thm:winding-normal-form]

Reading. The root enters the equation multiplicatively and modulo N , not merely as a starting point.

Theorem 5.32 (Finitely many states per vertex). **PROVED**, class B. After reduction by $d = \gcd(K_t, h)$, with $B = K_t/d$, $A = C_t/d$, $H = h/d$, the state has primitive form $Bp - Ax = \pm H$ with $\gcd(A, B) = \gcd(B, H) = 1$; for fixed p , fixed divisor $H \mid h$, fixed sign and $0 < B < x$ the number B is determined uniquely modulo x , so a single vertex carries at most $2\tau(h)$ reduced states. [BIB, cor:path-state-count]

Reading. The divisor count of the gap bounds the richness of the path state — the only place where the size of h (rather than its relation to G) enters.

Theorem 5.33 (No-bad-return-to- p_0). **PROVED**, class C. Under failure no bad path contains an edge $u \rightarrow p_0$. Consequently a non-root endpoint of a bad path satisfies $r < N/3$.

Proof. Under failure $p_0 > N/2$; if $p_0 \mid N - u$ then from $0 < N - u < N < 2p_0$ we get $N - u = p_0$, so $N - u$ is prime and u is GOOD. [BIB, lem:v71-no-return-p0; I160] $\square$

VERIFIED COMPUTATION: all canonical worlds $N \leq 10^4$, zero violations.

Theorem 5.34 (Global two-sheet collision budget). **PROVED**, class C. In the whole bad world reachable from the canonical p_0 there are at most two distinct endpoints r reached by two distinct bad paths whose label products $K_1, K_2 < N$ have opposite winding parities.

Sketch. The two-sheets base gives $K_1 + K_2 = N$ and $m_0 = q_1 q_2$; comparing two collisions gives $m_0 \mid K - K'$, and oddness of the products gives $2m_0 \mid K - K'$. After cutting off small N (Nagura) we have $m_0 > 2N/5$, so $2m_0 > 4N/5$: three distinct numbers in one class modulo $2m_0$ cannot lie in $(0, N)$. For fixed K the winding form gives $Kr \equiv \pm p_0 \pmod{N}$, and the endpoint satisfies $r < N/3$ (Theorem 5.33), so each K encodes at most one endpoint. [BIB, thm:v71-global-two-sheet; l160] $\square$

Reading. The first *global* result over the whole p_0 world, independent of component size and topology.
Limitation: it is not an escape theorem and does not exclude a closed bad world. The attempt to turn it into a contradiction is **KILLED**.

5.7 The Euler-defect layer

Read this box before using η or ν . The corpus records a “layered Euler defect” in the form

$$\eta(S) + \nu(S) \geq |S| + |\text{Src}(S)|, \quad \text{canonically} \quad \eta(R) + \nu(R) \geq |R| + 1,$$

but the material available to this file does not fix the normalisation of η and ν unambiguously. Two natural readings were tested here on all canonical failed worlds with $50 \leq N \leq 6 \cdot 10^4$ (21 093 worlds), and one obvious reading is *false*. Do not use the inequality without saying which normalisation you mean.

Definition 5.35 (Candidate normalisations). With $d_S(q)$, $\kappa_S(q)$ as in Definition 3.18, and $\nu(S)$ the number of BFS layers of S :

(N1) row-excess form $\eta_1(S) = \sum_{p \in S} (\Omega(N - p) - |\text{PF}(N - p) \cap S|)$.

(N2) valuation/reuse split $\eta_2(S) = \sum_q (d_S(q) - \kappa_S(q))$, $\nu_2(S) = \sum_q (\kappa_S(q) - 1)$, over $q \in \Gamma_N(S)$; these are the logarithmic masses of U_S and J_S counted in prime factors.

(N3) active-mass form $\sum_q d_S(q)$, the total active valuation mass of the world.

(N4) trivial form $\eta_4(S) = \sum_{p \in S} (\Omega(N - p) - 1)$ with ν the layer count.

VERIFIED COMPUTATION — canonical failed worlds, $50 \leq N \leq 6 \cdot 10^4$, 21 093 worlds, all with $|\text{Src}| = 1$:

- | | | | |
|------|-------------------------------------|--------------------------|---|
| (N1) | $\eta_1(R) + \nu(R) \geq R + 1$ | 6 044 violations; | smallest $N = 556$ ($ R = 17$, $\eta_1 = 10$, $\nu = 6$) |
| (N2) | $\eta_2(R) + \nu_2(R) \geq R + 1$ | 3 804 violations; | smallest $N = 50$ |
| (N3) | $\sum_q d_R(q) \geq R + 1$ | 0 violations | |
| (N4) | $\eta_4(R) + \nu(R) \geq R + 1$ | 0 violations | |

Proposition 5.36 (The trivial form is a theorem). **PROVED**. For a canonical failed world R , $\eta_4(R) + \nu(R) \geq |R| + 1$.

Proof. Every $p \in R$ is BAD, so $N - p$ is composite and $\Omega(N - p) \geq 2$, whence $\eta_4(R) \geq |R|$; and $\nu(R) \geq 1$. $\square$

Remark 5.37 (What this means for the programme). (N4) is a theorem but carries no information — it is $\Omega \geq 2$ in disguise. (N3) is the only nontrivial normalisation that survived; it says the world carries at least $|R| + 1$ units of active valuation mass, i.e. strictly more mass than vertices. It is here **VERIFIED COMPUTATION** only, and a proof is a reasonable small target (see §13.1). (N1) and (N2) are **KILLED** as stated. Any argument in the corpus that used the Euler defect in the form (N1) or (N2) must be re-derived.

Warning 5.38 (Do not conflate the layers). $T_S = U_S J_S$ (Theorem 3.19) is exact and safe. The Euler defect built on it is not. Warning 3.21 remains in force: none of these quantities is the capacity functional γ of $P1^{\text{br}}/K1$.

Chapter 6

The Canonical p_0 Programme

This chapter is where p_0 genuinely differs from every other upper root, and also where the project has most often overclaimed. Sections 6.1–6.4 are the positive results; §6.5 is the mandatory negative section and is, for a future model, the most valuable part of the chapter.

6.1 Genealogical defect rank

Definition 6.1 (Prime-defect set and genealogical rank). For the upper root $p_k = x + h_k$ (the k -th prime $\geq x$) put

$$\mathcal{P}_{\text{def}}(p_k) = \{j \in [1, t_k] : c_j^{(k)} = p_k - 2j \in \mathbb{P}\}, \quad \kappa_{\text{gen}}(p_k) = |\mathcal{P}_{\text{def}}(p_k)|.$$

Theorem 6.2 (Genealogy of prime holes). **PROVED**, class A. Domain hypothesis: *non-diagonal case* (x composite; see Scope note 3.17). For every $k \geq 0$,

$$\{c_j^{(k)} : j \in \mathcal{P}_{\text{def}}(p_k)\} = \{p_0, \dots, p_{k-1}\}, \quad \kappa_{\text{gen}}(p_k) = k,$$

the unique hole carrying p_i sits at index

$$j_i^{(k)} = \frac{p_k - p_i}{2}$$

and its lower mirror is

$$a_{j_i^{(k)}}^{(k)} = N - c_{j_i^{(k)}}^{(k)} = N - p_i.$$

Proof. Every prime reflected value lies in (x, p_k) , and the primes in that interval are exactly $p_0, \dots, p_{k-1}$. Conversely, for $i < k$ the number $p_k - p_i$ is positive and even, and $p_i > x$ gives $0 < p_k - p_i < h_k$, so $1 \leq (p_k - p_i)/2 \leq \lfloor (h_k - 1)/2 \rfloor = t_k$; at that index $c_j^{(k)} = p_k - (p_k - p_i) = p_i$. Uniqueness follows from injectivity of $j \mapsto p_k - 2j$. The mirror is $a_j = N - c_j$. [CORE, G-UPPERROOT-PRIMEHOLE-GENEALOGY, V71.21–V71.24] $\square$

VERIFIED COMPUTATION: all $N \leq 10^3$ with composite x and $k \leq 5$, zero violations; the diagonal case is isolated separately (Scope note 3.17).

Theorem 6.3 (The defect ledger is a lossless code of prime history). **PROVED**, class A. Knowing p_k and the index set $\mathcal{P}_{\text{def}}(p_k)$ recovers all earlier upper primes,

$$p_i = p_k - 2j_i^{(k)},$$

and the differences of consecutive hole positions recover the consecutive prime gaps:

$$j_i^{(k)} - j_{i+1}^{(k)} = \frac{p_{i+1} - p_i}{2} = \frac{g_{i+1}}{2}, \quad 0 \leq i < k-1.$$

Proof. The first identity is a rewriting of $j_i^{(k)} = (p_k - p_i)/2$; the second is its difference. $\square$

Reading. $\mathcal{P}_{\text{def}}(p_k)$ is not a “counter of missing rows”. It is a *complete, lossless record of the local history of the primes above the midpoint*, gaps included. In exactly that sense p_k “remembers” all its predecessors, and p_0 remembers nobody.

Theorem 6.4 (Defect-rank-zero characterisation). **PROVED**, class A (non-diagonal). For a prime $r = x + h_r \geq x$ the following are equivalent:

- (i) $r = p_0$;
- (ii) $[x, r) \cap \mathbb{P} = \emptyset$;
- (iii) $h_r < G_r$;
- (iv) $\mathcal{P}_{\text{def}}(r) = \emptyset$;
- (v) $\kappa_{\text{gen}}(r) = 0$;
- (vi) every strictly reflected value $r - 2j$, $1 \leq j \leq \lfloor (h_r - 1)/2 \rfloor$, is composite.

[CORE, C-P0-DEFECTRANK-ZERO-EQUIVALENCE]

Proof. (i)–(iii) is Theorem 3.10. Items (ii), (iv), (vi) are equivalent because the strictly reflected values enumerate exactly the numbers of the same parity as r lying *strictly* between x and r — this is where non-diagonality enters. (v) is the cardinality form of (iv). $\square$

Corollary 6.5 (Uniqueness of the reflection-complete root). **PROVED**, class A (non-diagonal). Call an upper root r reflection-complete if every strictly reflected value $r - 2j$ is composite. Then

$$r \text{ is reflection-complete} \iff r = p_0.$$

[CORE, C-P0-UNIQUE-REFLECTION-COMPLETE-ROOT]

VERIFIED COMPUTATION: $N \leq 10^3$, zero violations for composite x ; the diagonal exception is located exactly ($N = 14$, $r = 11$).

Theorem 6.6 (Reflection ledger completeness). **PROVED**, class A. Among all upper prime roots, p_0 is the only one whose strictly reflected ledger is exhausted by four multiplicative-dynamical classes with no prime-defect class:

$$\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F} \quad \text{versus} \quad \mathcal{J}_{h_k} = \mathcal{P}_{\text{def}} \uplus \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}, \quad |\mathcal{P}_{\text{def}}| = k.$$

[CORE, C-P0-REFLECTION-LEDGER-COMPLETENESS, V71.32–V71.34]

Warning 6.7 (What these theorems do *not* say). **Do not sell this as an automatic explanation of the success/failure anomaly.** Theorems 6.2–6.6 are an *exact structural exceptionality*, but:

- there is no proved implication $\kappa_{\text{gen}} = 0 \Rightarrow$ traversal success; the target C-P0-GENEALOGYFREE-BOUNDARY-TO-DYNAMIC is **OPEN**;
- the property may be *partly equivalent to the ordering of the primes itself*: it says that p_0 is the first element of the sequence $p_0 < p_1 < \dots$, translated into the language of the reflected block. Theorem 6.3 cuts both ways here: since the defect ledger is a *lossless code* of prime history, the statement “ $\kappa_{\text{gen}}(r) = 0$ ” carries exactly as much information as the statement “ r is the first element of that sequence” — no more, no less;
- the logarithmic mass of the holes is negligible: passing from p_0 to p_k the loss is $\sum_{i < k} \log p_i = O_k(\log N)$, which formally *kills* the mass interpretation (§6.5, FAIL-PO-ZEROHOLE-ESSENTIAL-001).

Honest formulation: *the type of relation available on the boundary coordinate changes, not the amount of information.* A prime hole $c_j = p_i$ has no nontrivial factorization $c_j = s\ell$ with $s, \ell > 1$, so on that coordinate there is no multiplicative carrier of the kind that modern factor-flow machinery requires [CORE, V71.44]. That is *eligibility*, not automatic activation — Warning 3.16 remains binding.

6.2 Boundary filtration

Theorem 6.8 (Embedding of reflected blocks). **PROVED**, class B (the filtration) + A (its initial object). Let $p_k < p_{k+1}$ be consecutive upper primes with $p_{k+1} - p_k = 2s$. Then

$$t_{k+1} = t_k + s, \quad c_{i+s}^{(k+1)} = c_i^{(k)}, \quad a_{j+s}^{(k+1)} = a_j^{(k)} \quad (1 \leq i, j \leq t_k),$$

so $\mathfrak{B}(p_k) \subseteq \mathfrak{B}(p_{k+1})$ and the new coordinates are exactly the indices $1, \dots, s$ of the block of p_{k+1} . In particular there is a filtration

$$\mathfrak{B}_0 \subset \mathfrak{B}_1 \subset \mathfrak{B}_2 \subset \dots, \quad \mathfrak{B}_k := \mathfrak{B}(p_k),$$

whose unique initial object is p_0 .

Proof. Since $h_{k+1} = h_k + 2s$, $t_{k+1} = \lfloor (h_k + 2s - 1)/2 \rfloor = \lfloor (h_k - 1)/2 \rfloor + s = t_k + s$. Further $c_{i+s}^{(k+1)} = p_{k+1} - 2(i + s) = p_k + 2s - 2i - 2s = p_k - 2i = c_i^{(k)}$, and since $a_j = N - c_j$ in both blocks the same index shift gives $a_{j+s}^{(k+1)} = a_j^{(k)}$. Index ranges agree: $i \leq t_k$ gives $i + s \leq t_{k+1}$. $\square$

VERIFIED COMPUTATION: 731 446 checked indices ($N \leq 2 \cdot 10^4$, $k \leq 6$), zero violations; EXP6.

Corollary 6.9 (Embedding of the coupling matrix). **PROVED**, class B. With $M_{ij}^{(k)} = \gcd(c_i^{(k)}, a_j^{(k)})$, $1 \leq i, j \leq t_k$,

$$M_{ij}^{(k)} = M_{i+s, j+s}^{(k+1)},$$

i.e. $M^{(k)}$ is exactly the submatrix of $M^{(k+1)}$ cut out by shifting both indices by s . The entire coupling geometry of the earlier root is inherited unchanged.

Proof. Immediate from Theorem 6.8: both gcd arguments are identical. $\square$

Corollary 6.10 (New strip and new hole). **PROVED**, class A. The new coordinates of the block of p_{k+1} are exactly $j = 1, \dots, s$ with $s = (p_{k+1} - p_k)/2$; among them exactly one, $j = s$, is a prime hole and carries the root pair $(p_k, N - p_k)$. The other $s - 1$ new coordinates are composite.

Proof. New indices are $1 \leq j \leq s$, since indices $> s$ come from the embedding. For $j < s$ the value $c_j^{(k+1)} = p_{k+1} - 2j$ lies strictly between p_k and p_{k+1} , hence is composite; for $j = s$ it equals p_k . The lower mirror is $N - p_k$. $\square$

Reading the filtration.

- p_0 is the rank-zero initial object: $\mathfrak{B}_0$ has no hole.
- p_1 contains the *whole* boundary of p_0 plus one new strip of width $s = (p_1 - p_0)/2$ containing exactly one new hole.
- Every further root inherits the entire earlier geometry (coupling matrix included) and adds exactly one new genealogical hole.

This is interesting independently of Goldbach: the reflected boundaries of the upper primes form a natural ascending chain in which the defect rank is the height.

6.3 The canonical phase diagram $\theta = h/G$

Theorem 6.11 (Canonical prime cell). **PROVED**, class A. Let P be an odd prime, P_- its predecessor and $G = P - P_-$. Then for every $0 \leq h < G$,

$$p_0(N_h) = P, \quad N_h := 2(P - h).$$

In particular every odd prime is a canonical root: for $N = 2(P - 1)$ we get $p_0(N) = P$ and $h = 1$.

Proof. $x = P - h > P - G = P_-$, so $[x, P)$ contains no prime and P is the least prime $\geq x$. [PCRD, Canonical Prime Cell Theorem] $\square$

VERIFIED COMPUTATION: full cells for $P \in \{67, 127, 1009, 104729\}$, agreement at every h . Reading. The label “ p_0 ” does *not* single out any class of primes. All canonical information sits in the pair (P, h) , and the natural parameter is $\theta = h/G \in [0, 1)$. This theorem is also the seed of the architecture wall (§8.2).

Theorem 6.12 (Gap-position identity and trichotomy). **PROVED**, class A. For canonical $p_0 = x + h$ with predecessor p_- and $G = p_0 - p_-$,

$$m_0 - p_- = G - 2h,$$

and in the non-diagonal case ($h \geq 1$, so G is even) exactly one of the following holds.

Type I ($2h < G$, i.e. $\theta < 1/2$). Then $p_- < m_0 < x < p_0$, and since p_- and p_0 are consecutive primes and $m_0 \neq p_-$, the number m_0 is composite. Hence the canonical root is BAD and immediate success is impossible.

Type II ($2h = G$, i.e. $\theta = 1/2$). Then $m_0 = p_-$, so m_0 is prime and $N = p_0 + p_-$ is a Goldbach representation: immediate success is automatic.

Type III ($2h > G$, i.e. $\theta > 1/2$). Then $m_0 < p_- < x < p_0$ and the previous prime appears as a lower mirror of the reflected block at index

$$j^* = h - \frac{G}{2} \in \{1, \dots, t\}, \quad a_{j^*} = p_-.$$

Proof. $m_0 - p_- = (x - h) - (x + h - G) = G - 2h$; its sign gives the trichotomy. Type I: $G - 2h > 0$ gives $m_0 > p_-$; also $m_0 < x < p_0$, and (p_-, p_0) contains no prime, so m_0 is composite (for N large enough that $m_0 > 1$). Type II: $G = 2h$ gives $m_0 = p_-$ directly. Type III: setting $a_j = m_0 + 2j = p_-$ gives $2j = p_- - m_0 = 2h - G$, i.e. $j^* = h - G/2$, an integer because G is even. Range: $j^* \geq 1 \iff 2h \geq G + 2$, which is Type III; and $j^* \leq t = \lfloor (h - 1)/2 \rfloor$ follows from $2j^* = 2h - G \leq h - 1$, i.e. $h \leq G - 1$, which always holds since $h < G$ (Theorem 3.10). $\square$

VERIFIED COMPUTATION: all even $N \leq 2 \cdot 10^5$: zero Type I cases with m_0 prime; zero Type II cases with $m_0 \neq p_-$; zero index violations in Type III.

Corollary 6.13 (Cell decomposition). **PROVED**, class A. The cell of a prime P with even gap G splits into exactly

$$\underbrace{1}_{\text{diagonal } (h=0)} + \underbrace{\left(\frac{G}{2} - 1\right)}_{\text{Type I}} + \underbrace{1}_{\text{Type II } (h=G/2)} + \underbrace{\left(\frac{G}{2} - 1\right)}_{\text{Type III}} = G$$

cases. In particular the number of diagonal cases equals the number of Type II cases, and the distribution of θ is symmetric under $\theta \mapsto 1 - \theta$.

Proof. h runs over $0, \dots, G - 1$; $h = 0$ is diagonal, $h = G/2$ is Type II, and the rest split symmetrically about $G/2$ via the bijection $h \mapsto G - h$. $\square$

EXP5 — phase census. **VERIFIED COMPUTATION**, all even $N \leq 2 \cdot 10^5$:

	diagonal	Type I	Type II	Type III
count	9 591	40 407	9 591	40 409
with immediate success	9 591	0	9 591	7 422

The second row is exactly the content of Theorem 6.12: in Type I immediate success is *impossible*, in Type II and the diagonal case it is *automatic*, and in Type III it happens (here in 18.4% of cases). The difference of 2 between Type I and Type III counts comes from the incomplete cell at the end of the range.

Corollary 6.14 (Reflection-gate phase transition). **PROVED**, class A. A *guaranteed* active prime lower-mirror coordinate in the strictly reflected block exists exactly in phase Type III:

$$\theta > \frac{1}{2} \implies a_{j^*} = p_- \in \mathbb{P}, \quad j^* = h - \frac{G}{2}.$$

If in addition $p_- \nmid N$ (**VERIFIED COMPUTATION**: true in all 40 409 Type III cases with $N \leq 2 \cdot 10^5$) then a_{j^*} is an active row and the reflection gate (Lemma 3.15) gives the edge $a_{j^*} \rightarrow \text{PF}(c_{j^*})$. In Types I and II no such guarantee exists.

Remark 6.15 (This explains the counterexample $N = 242$). Warning 3.16 cites $N = 242$ as proof that an empty interval does *not* activate any reflected row. The phase diagram shows this is not an accident: $N = 242$ has $h = 6$, $G = 14$, so $\theta = 3/7 < 1/2$ — Type I, where the guaranteed-mirror theorem does not apply. By contrast $N = 240$ (same $p_0 = 127$, $h = 7$) is Type II and the canonical root wins immediately, and $N = 124$ ($h = 5$, $G = 6$, $\theta = 5/6$) is Type III with two prime lower mirrors, 59 and $61 = p_-$.

6.4 Bidirectional prime rank ρ_0

Definition 6.16 (Lower rank). For canonical $p_0 = P$ let $P_{-1} > P_{-2} > \dots$ be the consecutive earlier primes and $D_s = P - P_{-s}$ (so $D_1 = G$). Put

$$\rho_0 := \#\{j \in [1, t] : a_j = m_0 + 2j \in \mathbb{P}\},$$

the number of *prime lower mirrors* in the strictly reflected block.

Theorem 6.17 (Lower-rank formula). **PROVED**, class A. For $h \geq 3$,

$$\rho_0 = \#\{s \geq 1 : h < D_s < 2h\}.$$

Proof. A lower mirror $a_j = m_0 + 2j = P - 2h + 2j$ equals $P_{-s} = P - D_s$ iff $2j = 2h - D_s$, i.e. $j = h - D_s/2$ (D_s is even for $P > 3$). The condition $j \geq 1$ gives $D_s \leq 2h - 2$, and $j \leq t = \lfloor (h-1)/2 \rfloor$ gives $2h - D_s \leq h - 1$, i.e. $D_s \geq h + 1$. Since D_s is even, $D_s \leq 2h - 2 \iff D_s < 2h$. The map $s \mapsto j$ is injective because D_s is strictly increasing. $\square$

Warning 6.18 (Correction to the originally proposed formula). **KILLED** as stated. The version proposed in the input material had only the upper bound, $\rho_0 = \#\{s \geq 1 : D_s < 2h\}$. It is missing the lower condition $D_s > h$, which is exactly what confines the mirror to the *strictly* reflected block (index $j \leq t$, not $j \leq h - 1$). Without it the formula also counts mirrors lying outside the strict block. **VERIFIED COMPUTATION**: all even $N \leq 2 \cdot 10^5$ with $h \geq 3$ (72 451 cases): the corrected formula agrees with direct counting in *every* case, zero violations.

Corollary 6.19 (The lower rank detects the phase exactly). **PROVED**, class A.

$$\rho_0 \geq 1 \iff G < 2h \iff \theta > \frac{1}{2} \iff \text{Type III}.$$

Proof. ($\Leftarrow$) In Type III, $h < G = D_1 < 2h$, so $s = 1$ satisfies the condition of Theorem 6.17. ($\Rightarrow$) D_s is strictly increasing, so if $D_1 = G \geq 2h$ then $D_s \geq 2h$ for all s and no s qualifies. $\square$

VERIFIED COMPUTATION: $N \leq 2 \cdot 10^5$: the number of cases with $\rho_0 \geq 1$ is 40 409 — *exactly* the number of Type III cases. Distribution of ρ_0 over the 72 451 cases with $h \geq 3$:

ρ_0	0	1	2	3	4	5	6	7	8	9
$\#N$	32 042	24 288	10 200	3 936	1 331	404	170	57	16	7

Definition 6.20 (Bidirectional signature).

$$(\kappa^+, \kappa^-) := (\kappa_{\text{gen}}, \rho_0).$$

The upper coordinate counts primes *above* the midpoint recorded in the reflected blocks, the lower one counts primes *below*.

Why the signature is a better descriptor than κ_{gen} alone. For the canonical root $\kappa^+ = 0$ always, so κ^+ separates no two canonical worlds — it is a constant. But $\kappa^- = \rho_0$ is non-constant and, by Corollary 6.19, encodes exactly the phase:

$$\kappa^- = 0 \iff \theta \leq \frac{1}{2}, \quad \kappa^- \geq 1 \iff \theta > \frac{1}{2}.$$

The signature $(0, \rho_0)$ is therefore the first quantity in this file that is simultaneously (i) an invariant of the boundary geometry, (ii) p_0 -specific in its first coordinate, and (iii) *nontrivially varying* between canonical worlds.

6.5 What does *not* make p_0 exceptional

Mandatory section. For a future model this is the most valuable part of the chapter: “this does not work, and we know why” is worth as much here as a theorem.

6.5.1 Refutations bearing directly on p_0 -exceptionality

Mass of the reflected boundary. **KILLED**. Hypothesis: the absence of prime holes in the canonical reflected block is itself a significant mass mechanism. Counterexample: for p_1 the block has exactly *one* hole, for p_k exactly k ; iterating the pointwise short-interval bound gives $p_k - x = O_k(x^{0.525})$, so removing the holes costs only $\sum_{i < k} \log p_i = O_k(\log N)$ against a full block mass of order $t_k \log(x/h_k)$. Hence hole mass/block mass $\rightarrow 0$. [FAIL, FAIL-P0-ZEROHOLE-ESSENTIAL-001; CORE, V71.35]

p_0 as an information selector. **SUPERSEDED** (not refuted, but **not established**). The ablation P0-CANONICITY-ABLATION classifies every clause of the bundle-lift construction as “fixed N ” or “arbitrary reflected pair $a + c = N$ ”. Replacing p_0 by p_1 leaves the whole construction *word for word* on every row with composite c'_j . Corpus verdict: p_0 is *quantitatively useful and structurally inessential* for that construction; a non-vacuity guarantee is not a principle of information selection. [FAIL, FAIL-P0-INFORMATION-SELECTOR-001]

Shadow / MR2 congruences. **KILLED** as a canonical statement (Warning 5.23).

First-front parent minimality. **KILLED** as a selector (Warning 5.6).

Activation of reflection by the prime-free boundary. **KILLED** (Warning 3.16, $N = 242$). Structurally explained by the phase diagram (Remark 6.15).

Statistical exceptionalism after controlling for row size. **KILLED**. The median rank percentile of p_0 within fixed N , over all reported metrics, is 0.20982; row-size-aware ranks do not support a single universal claim that p_0 dominates on every axis. Comparing canonical and matched non-canonical B_{abs} on 119 reproducible pairs gives a median difference of $4.6127 \cdot 10^{-5}$ — no clean causal separation. [D4, questions 5, 12] — *reported computational result, not reproduced in this document*.

Minimality does not determine local geometry. **KILLED**. At fixed $P = 927\,961$ ($N = 1\,855\,760$) a single change of h gives $\Delta \log B_{\text{abs}} = 14.434$, $\Delta G_{\text{esc}} = -230$, $\Delta |W| = -240$; full sweeps at fixed p_0 find 23 canonical transitions $B_{\text{abs}} \rightarrow B_{\text{re}}$ and 15 in the reverse direction. [D4, questions 6, 17] — *reported, not reproduced here*.

Largest first-front factor as a heuristic. **KILLED**. The variant “follow the largest factor of m_0 ” fails in two steps where the smallest factor wins (Example 6.21). [PCRD, FAIL-P0-LARGEST-FACTOR-TWOSTEP]

Example 6.21 ($N = 124$). **VERIFIED COMPUTATION**. $x = 62$, $p_0 = 67$, $h = 5$, $G = 6$ (Type III), $m_0 = 57 = 3 \cdot 19$. First-front rows:

$$124 - 3 = 121 = 11^2, \quad 124 - 19 = 105 = 3 \cdot 5 \cdot 7.$$

Following the *largest* factor 19, the children are $\{3, 5, 7\}$ and none is GOOD ($124 - 5 = 119 = 7 \cdot 17$, $124 - 7 = 117 = 3^2 \cdot 13$). Following the *smallest* factor 3, the only child is 11, and $124 - 11 = 113$ is prime. The full traversal succeeds at depth 2 with two escape edges, $3 \rightarrow 11$ and $5 \rightarrow 17$.

6.5.2 The ablation $p_0 \rightarrow p_1$

The ablation removes *only* minimality and keeps everything else. This table is the central empirical result about p_0 .

EXP8 — observables at p_0 and at p_1 . **VERIFIED COMPUTATION**. All even $50 \leq N \leq 2 \cdot 10^5$; 99 976 cases per root; world mode.

observable	p_0	p_1	change
root FAILURES	0	2	$0 \rightarrow 2$
success depth, mean	1.3501	1.5234	+12.8%
world size $ R $, mean	52.71	59.38	+12.7%
margin G_{esc} , mean	24.51	26.96	+10.0%
$\Omega(m)$, mean	2.373	2.521	+6.2%
shadow retention, mean	0.5545	0.5128	−7.5%
$P^+(m)/m$, mean	0.31455	0.23966	−23.8%
κ_{gen} , mean	0.0000	0.9041	$0 \rightarrow \approx 1$
κ_{gen} , maximum	0	1	categorical

Definitions. $\Omega(m)$ — number of prime factors of the complement with multiplicity; shadow retention — $\sum_{q \in Q} |\text{PF}(N - q) \cap Q|$ normalised; $P^+(m)/m$ — normalised largest prime factor; κ_{gen} — genealogical rank (Definition 6.1).

Remark 6.22 (Where 0.9041 comes from, rather than 1). The mean $\kappa_{\text{gen}}(p_1)$ falls short of 1 by exactly $9\,591/99\,976 = 9.59\%$ — exactly the fraction occupied by the diagonal cases (Scope note 3.17 and Corollary 6.13). This is an independent confirmation of the domain repair: for prime x the hole of p_0 falls outside the strict block.

Distribution of Ω (a negative result). [VERIFIED COMPUTATION](#), even $N \leq 2 \cdot 10^5$:

Ω	1	2	3	4	5	6	7	8
$m_0 = N - p_0$	26 585	32 687	24 592	10 967	3 699	1 085	272	71
$m_1 = N - p_1$	18 534	36 385	27 198	12 249	4 048	1 161	307	78

Outside the bucket $\Omega = 1$ (which corresponds to immediate success, hence to a *definition*, not to structure) the two distributions are nearly identical. The local factorization of the canonical complement does *not* look unusual. This is one of the main negative results of the project.

Conclusion of the ablation. All aggregate observables — depth, world size, margin, Ω , retention, normalised largest factor — are *nearly interchangeable* between p_0 and p_1 . The only quantity that changes categorically is the defect rank. This is exactly the content of [CORE, V71.37]: *p_0 -specificity is invisible to many aggregate observables and visible in the exact structure of relations.*

6.5.3 Is the exceptionality local, multiplicative, dynamic, or just the empty interval?

1. **Not local-arithmetic.** Exact-D countermodels satisfy every local condition — normal form, smoothness, pair failure, endpoint primality — and die *only* on canonicity. Smallest recorded model: $q = 3$, $r = 5\,413\,103$, $m_0 = 81\,196\,545$, $N = 297\,720\,668$, endpoint $p = 216\,524\,123$ prime, but $x + 9$ is prime and lies much earlier. [FAIL, FAIL-EXACTD-NONCANONICAL-001]
2. **Not purely multiplicative.** D_q , M_q and the whole prime-power ledger are generic; p_0 enters them only through the value $\delta = 2h$.
3. **Not (yet) dynamic.** There is no proved passage from boundary geometry to traversal behaviour; the target C-P0-GENEALOGYFREE-BOUNDARY-TO-DYNAMIC-ESCAPE is **OPEN**.
4. **But not “just” an empty interval either.** The empty interval is *equivalent* to something substantially richer: the boundary is multiplicatively complete, the ledger has no fifth class, the predecessor lies outside the block, and ρ_0 detects the phase. It is a property of the *type of relation* available on each coordinate, not of the length of an interval.
5. **The reservation in the other direction.** Theorem 6.3 shows the defect ledger is a lossless code of prime history. So “ $\kappa_{\text{gen}}(r) = 0$ ” may be *informationally equivalent* to “ r is the first element of the sequence of upper primes”. We do not claim that the exceptionality of p_0 goes beyond the ordering of the primes; we claim it is an *exact* translation of that ordering into the multiplicative structure of the boundary.

The honest one-sentence summary.

The exceptionality of p_0 is an *exact boundary-closure property*, not a density effect; it is invisible to aggregate observables and visible in the structure of relations; and no theorem is known leading from it to dynamic behaviour.

6.6 The six special N

EXP7 — nonempty fixed points. **VERIFIED COMPUTATION**. For all even $N \leq 6 \cdot 10^4$ the set B_N^{low} is nonempty for *exactly* six values:

N	$ B_N^{\text{low}} $	B_N^{low} (initial segment)
128	10	$\{3, 5, 7, 11, 13, 17, 23, 29, 37, 41\}$
1718	55	$\{3, 5, 7, 11, 13, 17, 29, 31, 37, 41, 43, \dots, 571\}$
1862	55	$\{3, 5, 11, 13, 17, 29, 41, 43, 47, 53, 59, \dots, 619\}$
1928	41	$\{3, 5, 7, 11, 13, 17, 29, 43, 53, 71, 73, \dots, 641\}$
2200	2	$\{3, 13\}$
6142	66	$\{3, 5, 7, 13, 17, 19, 31, 43, 67, 71, 73, \dots, 2011\}$

This replicates exactly the list reported in [PCRD] for the same range.

EXP3 — full census of failing upper roots. **VERIFIED COMPUTATION**. Searching *all* upper roots $p \in (N/2, N)$ with composite complement, for all even $N \leq 2 \cdot 10^4$:

- exactly **301** roots whose traversal fails;
- they sit at exactly *six* values of N , distributed as 128:8, 1718:68, 1862:63, 1928:57, 2200:4, 6142:101;
- **not one** of those 301 roots is canonical;
- unit rows $N - p = 1$ excluded separately: 2261.

The number 301 is the one the corpus carries as “301 composite-complement failed upper roots” FAIL-CENSUS-DOMAIN-001; the computation above fixes its exact domain and structure, which the corpus did not record. The canonical root wins in *all six* cases: $p_0 = 67, 859, 937, 967, 1103, 3079$ at depths 0, 0, 2, 2, 0, 2. These six N are exactly the six numbers with an exceptional autonomous component reported up to 10^8 by Božek [8].

Remark 6.23 (The two censuses agree although they measure different things). EXP3 asks whether some failing upper root exists; EXP7 asks whether a closed bad set exists. The lists coincide on $N \leq 2 \cdot 10^4$. That is expected in one direction (a failing root forces a closed bad set) but the converse is not a theorem, and the agreement is data, not proof.

Example 6.24 (Failure structure at $N = 128$). **VERIFIED COMPUTATION**. For $N = 128$ ($x = 64$, $p_0 = 67$):

k	p_k	$N - p_k$	verdict
0	67	61 (prime)	SUCCESS, $d = 0$
1	71	$57 = 3 \cdot 19$	SUCCESS, $d = 1$ ($19 \rightarrow 109 \in \mathbb{P}$)
2	73	$55 = 5 \cdot 11$	FAILURE
3	79	$49 = 7^2$	FAILURE
4	83	$45 = 3^2 \cdot 5$	FAILURE
5	89	$39 = 3 \cdot 13$	FAILURE
6	97	31 (prime)	SUCCESS, $d = 0$
7	101	$27 = 3^3$	FAILURE
8	103	$25 = 5^2$	FAILURE
9	107	$21 = 3 \cdot 7$	FAILURE
10	109	19 (prime)	SUCCESS, $d = 0$
11	113	$15 = 3 \cdot 5$	FAILURE

All eight failures share the *same* bad world $\{3, 5, 7, 11, 13, 23, 29, 41\}$. The set of failing roots is neither an initial nor a final segment: $p_1 = 71$ wins, $p_2 = 73$ loses. *Minimality alone does not order*

the outcome — the first signal that “ p_0 is simply the smallest” is not an explanation.

Remark 6.25 (The witness $N = 2200$ and the two-core wall). **VERIFIED COMPUTATION**. The core $B_N^{\text{low}} = \{3, 13\}$ at $N = 2200$ is an exact witness that a source-free set is *not* by itself contradictory:

$$2200 - 3 = 2197 = 13^3, \quad 2200 - 13 = 2187 = 3^7, \quad 3^7 - 3 = 13^3 - 13 = 2184.$$

At the same time $p_0(2200) = 1103$ and $m_0 = 1097$ is prime, so the canonical root wins immediately and never enters that core. Excluding such cores in general leads to a Pillai/Bennett-type equation $a^\alpha - a = b^\beta - b$, i.e. to an external Diophantine wall. [PCRD, 2-core wall]

6.7 Escape margin

Success depth turned out to be a *bad* measure of difficulty. The right measure is the margin: how many ways out exist at all.

Definition 6.26 (Escape capacity). For even N with BAD p_0 , let $W = R_N(p_0)$ and

$$G_{\text{esc}}(N) = \#\{(v, q) : v \in W, q \mid N - v, q \in \mathcal{A}_N, N - q \in \mathbb{P}\}.$$

Then Goldbach failure $\Rightarrow G_{\text{esc}}(N) = 0$, and $G_{\text{esc}}(N) > 0 \Rightarrow$ Goldbach for N . The converse implication is *not* proved. [BIB, escape capacity; repair relative to v5.8]

EXP4 — escape margin. **VERIFIED COMPUTATION**. All even $N \leq 2 \cdot 10^5$:

- 73 394 numbers have BAD p_0 ; $G_{\text{esc}} = 0$ *never* occurs;
- $G_{\text{esc}} = 1$ for 52 numbers, of which exactly **10** have $|W| \geq 3$:

$$368, 632, 1258, 1598, 1862, 1928, 3692, 4582, 6142, 14198,$$

with world sizes 12, 14, 25, 22, 23, 16, 21, 22, 12, 64 — *exactly* the list and sizes of [BIB, ten numbers with margin one], obtained there by a scan to $3 \cdot 10^6$;

- distribution of small margins at $|W| \geq 3$: $G_{\text{esc}} = 1$: 10, 2:38, 3:122, 4:477, 5:571, 6:656;
- largest canonical bad world: $|W| = 204$ at $N = 152\,288$.

The minimum margin grows — answer to an open point of the corpus.

VERIFIED COMPUTATION. The corpus left as a task the question “does the $G_{\text{esc}} = 1$ list really end at 14198” [BIB, next move]. Minimum G_{esc} over dyadic blocks $[2^{k-1}, 2^k)$, with $|W| \geq 3$ and $N \leq 2 \cdot 10^5$:

k	7	8	9	10	11	12	13	14	15	16	17	18
$\min G_{\text{esc}}$	2	2	1	1	1	1	1	2	2	3	3	3

In this range the $G_{\text{esc}} = 1$ list at $|W| \geq 3$ does end at $14198 \in [2^{13}, 2^{14})$, and the minimum margin in higher blocks rises to 2 and 3. This is a *signal*, not a proof: the range is finite and an order of magnitude smaller than the scan in [BIB].

Channel substitution. **EMPIRICAL PATTERN**, not a theorem. For $N = 14198$ a world of 64 bad vertices hangs on a *single* edge $71 \rightarrow 277$. Preserving the congruence behind that edge at $N \pm 554$ loses the canonical p_0 ; preserving p_0 in the nearest tested perturbations destroys that edge but creates 10–26 replacement edges. Across the whole campaign 459 compensation candidates were recorded with *zero* true orphans [D4, q. 18; ORPH, q. 9–10]. Reading: closing one channel opens another. Neighbouring sweeps do not isolate causality, so this remains a pattern, not a law.

Chapter 7

Factorization and Reachability Machinery

7.1 Post-root factorization

Theorem 7.1 (The post-root gcd). **PROVED**, class B. Let $q \in Q$ (so q is an odd prime dividing m_0) and put $B_q = m_0/q$. Then

$$D_q := \gcd(m_0, N - q) = \gcd(m_0, |\delta - q|) = \gcd(B_q, |2h - q|), \quad q \nmid D_q.$$

Proof. The first equality is the shadow identity (Theorem 5.22) applied to $N - q = 2m_0 + (\delta - q)$. For the rest: $\gcd(h, m_0) = 1$ (Theorem 5.3) and $q \mid m_0$ give $q \nmid h$, and q is odd, so $q \nmid \delta = 2h$; hence $q \nmid (\delta - q)$ and $q \nmid D_q$. Writing $m_0 = q^a m'$ with $q \nmid m'$ we get $\gcd(m_0, \delta - q) = \gcd(m', \delta - q) = \gcd(q^{a-1} m', \delta - q) = \gcd(B_q, \delta - q)$, and $\gcd(\cdot, y) = \gcd(\cdot, |y|)$. [LOG; FND] $\square$

Remark 7.2 (The identity is root-generic). Every step uses only $m_r = x - h_r$ and $\gcd(h_r, m_r) = 1$, so Theorem 7.1 holds verbatim for any upper root. It is class B and must not be quoted as canonical.

Corollary 7.3 (The cofactor is Q -free). **PROVED**, class B. With $M_q = (N - q)/D_q$ we have $\text{PF}(M_q) \cap Q = \emptyset$: all of the first alphabet that survives in the row $N - q$ is concentrated in D_q .

Proof. By Theorem 5.24, $\text{PF}(N - q) \cap Q = \text{PF}(d_q)$ and $d_q \mid D_q$; dividing out D_q removes every prime of Q from the row. $\square$

Definition 7.4 (FRESH / SMOOTH split). $q \in Q$ is *FRESH* if $P^+(N - q) \geq B_q$, and *SMOOTH* otherwise. Write Q_{smooth} for the smooth part. The split is the domain boundary of the incidence ledger below.

Warning 7.5 (High-gap amplification is conditional). **SCOPE-CORRECTED**. The inequality $h^2 > 4m_0$ (“high-gap amplification”) is *not* a consequence of CRD failure. Its actual hypothesis is failure of the Exact-D pair, and the corpus states it under that hypothesis [CORE, ARS-EXACTD]. Using it under canonical failure alone is an error; this is the **SCOPE-CORRECTED** noted against [BIB].

7.2 The incidence ledger

Theorem 7.6 (Incidence ledger IL1–IL6). **AUDITED IDENTITY** on the domain $S = Q_{\text{smooth}}$. With U_F , J_F the unit and layer parts of the ledger restricted to S , and B_{re} , B_{abs} the reused and absorbed base parts,

$$\prod_{q \in S} M_q = U_F (J_F B_{\text{re}}) B_{\text{abs}}.$$

Two independent reconstruction routes, 1 000 worlds, 0 errors [ILA].

Warning 7.7 (Domain). **SCOPE-CORRECTED**. IL1–IL6 hold on $S = Q_{\text{smooth}}$ only. The FRESH part is excluded. Quoting the identity on all of Q is a recorded error.

Definition 7.8 (Base atoms). B_{abs} is the absorbed base (atoms consumed inside the world), B_{re} the reused base (atoms appearing in more than one row), and $B_{\text{abs,int}}$ the internal absorbed base. The corpus proves $B_{\text{abs,int}} = B_{\text{abs}}$ without vacuity [CORE, ARS-BABS-INTERNAL].

Killed here. *Unconditional CAB*: refuted by $N = 322$ ($U_R = J_R = 1$, $B_{\text{abs}} = 11 \cdot 29$) and, in the internal version, by $N = 858$ FAIL-CAB-001, FAIL-CAB-INTERNAL-001. $\mathcal{C}_S < \mathcal{D}_S$ from the first front: refuted by $N = 4736$ ($\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$) FAIL-POTENTIAL-CAPACITY-001. *Failure always produces nontrivial B_{abs}* : refuted FAIL-BABS-PRODUCTION-001. *Four-channel ledger on all of Q_0* : refuted by $N = 20$ (a GOOD row outside R gives $\kappa_R < m_\ell$) [ILA]. *Constant $C = 1$ in charging*: 53 violations, maximum 9.42243 FAIL-CHILD-C1-001. *Naive support expansion of all sources*: Hall compression $456 \rightarrow 17$ at $N = 51\,032$ FAIL-HALL-SUPPORT-001.

Escape-or-reuse (**OPEN** as a theorem, **VERIFIED COMPUTATION** as data). Every internal base atom $\ell \in B_{\text{abs,int}}$ has either a GOOD edge reachable inside one of its first-front branches, or an active parent outside R . Both alternatives are necessary: $(N, \ell, p) = (6142, 877, 3511)$ shows the second is needed. Bounded charging multiplicity: observed maximum 1, boundedness unproved. Data: 61 452 instances, 0 orphans; independent red team 60 413 atoms, 0 orphans [ORPH, RT]. Known scope corrections around it: external reuse is *not* local ($(N, \ell) = (1\,825\,372, 52\,153)$, distance 53 750, FAIL-LOCAL-REUSE-001); the external parent is *not* necessarily a success root ($(6142, 877, 3511)$: 3511 is active, above p_0 , and its own traversal fails, FAIL-EXTERNAL-SUCCESS-001); escape need not be directly adjacent ($N = 14198$: atoms 19, 167 at distances 8 and 7, FAIL-DIRECT-ADJACENCY-001).

7.3 The rooted first-front layer

This section and the two that follow are the Biblia's v7.3–v7.5 rooted first-front theory. It is the most quantitatively developed part of the whole corpus and it is where the exact ledgers live. Read the statuses carefully: the *conservation and split identities* carry the source's **PROVED – PENDING RED TEAM** label, while the *collision budget and Exact-D geometry* of §7.4–§7.5 are fully **PROVED**.

Throughout, $q \in Q = \text{PF}(m_0)$, and

$$n_q := N - q, \quad B_q := \frac{m_0}{q}, \quad D_q := \gcd(m_0, n_q), \quad I_q := \text{rad } D_q, \quad M_q := \frac{n_q}{D_q},$$

$$U_q^{\text{row}} := \frac{n_q}{\text{rad}(n_q)}, \quad E_q := \frac{\text{rad}(n_q)}{I_q}, \quad j_q := \omega(n_q) - 1, \quad u_q := \Omega(n_q) - \omega(n_q),$$

so that $e_q := \Omega(n_q) - 1 = j_q + u_q$ splits the row cost exactly into a *collision/branching channel* j_q and a *squarefull channel* u_q .

7.3.1 FRESH / SMOOTH dichotomy

Definition 7.9 (FRESH and SMOOTH, restated). For $q \in Q$: the row is **FRESH** if $P^+(n_q) \geq B_q$, in which case it has a subthreshold prime child $s_q \geq B_q$; it is **SMOOTH** if $P^+(n_q) < B_q$, in which case every child of the row exceeds the threshold already at the second step.

Proposition 7.10 (Fresh prime packing, B-FRESH-PACKING). **PROVED – PENDING RED TEAM**. For distinct first-front primes q_i choose FRESH children s_i with two-step cost $< N$. The paths have the same parity, so the two-sheet budget (Theorem 5.34) forbids their merging; hence the s_i are pairwise distinct and for every $E \subseteq Q_{\text{fresh}}$

$$\prod_{q \in E} s_q \geq \frac{m_0^{|E|}}{\prod_{q \in E} q}. \quad (\text{FP})$$

[BIB, prop:v73-fresh-packing]

Proposition 7.11 (Smoothness Complexity Tax, B-SMOOTH-TAX). **PROVED – PENDING RED TEAM**. On the SMOOTH branch, with $e_q = \Omega(n_q) - 1$,

$$B_q^{e_q} > q. \quad (\text{ST})$$

Moreover $e_q = (\omega(n_q) - 1) + (\Omega(n_q) - \omega(n_q)) = j_q + u_q$, so the cost splits exactly into the collision/branching channel J and the squarefull channel U . [BIB, prop:v73-smooth-tax]

Corollary 7.12 (Root smoothness excess tax). **PROVED – PENDING RED TEAM**. For the full closed world,

$$\Omega(U_R) + \Omega(J_R) - 1 \geq \sum_{q \in Q_{\text{smooth}}} e_q > \sum_{q \in Q_{\text{smooth}}} \frac{\log q}{\log(m_0/q)}.$$

Warning 7.13 (The two-channel wall, and what it forbids). The external argument “many SMOOTH rows $\Rightarrow$ many powerful integers” is **KILLED**. A SMOOTH row can pay almost its whole cost through support *diversity* — the J channel — with no high prime multiplicity at all. This is exactly why Proposition 7.11 must keep both j_q and u_q . The correct picture is two-channel:

$$\text{SMOOTH} \longrightarrow \begin{cases} j_q \text{ large} & \longrightarrow J/R_\partial \text{ load} \\ u_q \text{ large} & \longrightarrow U \text{ load} \end{cases}$$

and what is needed is a theorem bounding the *joint* capacity of both channels while keeping the root geometry of p_0 . That is the *Root Smoothness Conservation Problem* (§7.3.4).

7.3.2 Root First-Front Conservation

Put $S_1 = \{p_0\} \cup Q$ and $b_1 := |\Gamma_N(S_1) \setminus S_1|$. The root row contains every $q \in Q$, so p_0 is the unique source of S_1 .

Theorem 7.14 (Root First-Front Conservation, B-FFC). **PROVED – PENDING RED TEAM**. The fundamental factor-flow decomposition gives

$$m_0 \frac{\prod_{q \in Q} (N - q)}{\prod_{q \in Q} q} = U_{S_1} J_{S_1} R_\partial(S_1). \quad (\text{FFC})$$

Applying Ω ,

$$\Omega(U_{S_1}) + \Omega(J_{S_1}) + |\Gamma(S_1) \setminus S_1| = \Omega(m_0) + \sum_{q \in Q} (\Omega(N - q) - 1). \quad (\text{FFC-}\Omega)$$

[BIB, thm:v73-ff-conservation]

Corollary 7.15 (Exact recycle–escape identity, B-FFRE). **PROVED – PENDING RED TEAM**. Since $U_{S_1} = \frac{m_0}{\text{rad}(m_0)} \prod_{q \in Q} \frac{N - q}{\text{rad}(N - q)}$ and $\prod_{q \in Q} q = \text{rad}(m_0)$, the squarefull factors cancel and

$$\prod_{q \in Q} \text{rad}(N - q) = J_{S_1} R_\partial(S_1). \quad (\text{FFRE})$$

The first front therefore has an exact decomposition of its support into recycle/collision and true boundary escape. [BIB, cor:v73-recycle-escape]

7.3.3 The exact first-front split ledger

Theorem 7.16 (Exact First-Front $U/J/\partial$ Split, B-UJD-SPLIT). **PROVED – PENDING RED TEAM**.
 With $S_1 = \{p_0\} \cup Q$ and $b_1 = |\Gamma(S_1) \setminus S_1|$,

$$\boxed{\Omega(U_{S_1}) = \Omega(m_0) - |Q| + \sum_{q \in Q} u_q,} \quad (\text{U-split})$$

$$\boxed{\Omega(J_{S_1}) + b_1 = |Q| + \sum_{q \in Q} j_q.} \quad (\text{JR-split})$$

Summing recovers Root First-Front Conservation: $\Omega(U_{S_1}) + \Omega(J_{S_1}) + b_1 = \Omega(m_0) + \sum_{q \in Q} e_q$.

Proof. By definition $U_{S_1} = \frac{m_0}{\text{rad}(m_0)} \prod_{q \in Q} \frac{N-q}{\text{rad}(N-q)}$; applying Ω and $\omega(m_0) = |Q|$ gives (U-split). From Corollary 7.15, $\prod_{q \in Q} \text{rad}(N-q) = J_{S_1} R_\partial(S_1)$, and the boundary radical is squarefree with b_1 prime atoms, so $\sum_{q \in Q} \omega(N-q) = \Omega(J_{S_1}) + b_1$; since $\omega(N-q) = 1 + j_q$ this is (JR-split). [BIB, thm:v74-exact-split] $\square$

Corollary 7.17 (Smoothness tax in the separated channels). **PROVED – PENDING RED TEAM**. For $q \in Q_{\text{smooth}}$ one has canonically $B_q^{e_q} > q$ with $e_q = j_q + u_q$. Hence every SMOOTH row must pay its required cost through exactly one of three routes: larger u_q feeding U , larger j_q feeding J or the boundary support, or a mixture splitting the cost between the channels. **There is no “free” SMOOTH variant** — but at the present stage there is still no global upper capacity bound producing a contradiction.

Lemma 7.18 (First-front GCD packing, B-GCD-PACK). **PROVED – PENDING RED TEAM**. For a prime ℓ put $\nu_\ell := \#\{q \in Q : \ell \mid N-q\}$. All q counted by ν_ℓ lie in one class modulo ℓ , so

$$\boxed{\nu_\ell \leq \min\left(|Q|, 1 + \left\lfloor \frac{m_0}{\ell} \right\rfloor\right).}$$

[BIB, lem:v74-gcd-pack]

Limitation. The lemma is correct but is *not* a knockout: it bounds incidence multiplicity, not total capacity.

Scope note 7.19 (Why the classical powerful-tail bound is too weak). **KILLED** as a route. The global sparsity of powerful integers gives $O(N^{(1+\alpha)/2})$ exceptions up to N , whereas the first front requires control of only $|Q| = \omega(m_0) \leq \log_2 m_0$ special points. Global density of powerful integers is therefore far too weak to force even one non-powerful row in the *adaptive* set $\{N-q : q \mid m_0\}$. [BIB, v7.4 powerful-tail remark]

7.3.4 The Root Smoothness Conservation Problem

Open target 7.20 (Root Smoothness Conservation Problem, B-RSCP). **OPEN**. For a hypothetical failure, find a uniform, deterministic capacity theorem linking

$$B_q^{j_q+u_q} > q \quad (q \in Q_{\text{smooth}})$$

to the exact ledger

$$\Omega(U_{S_1}) = \Omega(m_0) - |Q| + \sum u_q, \quad \Omega(J_{S_1}) + b_1 = |Q| + \sum j_q,$$

together with source attenuation and terminal absorption taxes, and leading to a contradiction or a forced escape. [BIB, def:v74-rscp]

Warning 7.21. This target is the Biblia’s own statement of the *same* obstruction that Chapter 9 records as the same-functional bridge (§9.11). It is not an independent problem: solving one solves the other. What the Biblia adds is the exact two-channel bookkeeping the bridge has to respect.

7.4 The lossless prime-power root ledger

Proposition 7.22 (Root-recycle-free row identity). **PROVED – PENDING RED TEAM**. For every $q \in Q$,

$$I_q = \text{rad gcd}(B_q, |2h - q|), \quad U_q^{\text{row}} E_q = \frac{N - q}{I_q}.$$

In particular the first copy of every prime recycled from the root alphabet is removed exactly from the weighted tax. [BIB, prop:v75-root-recycle-free]

Proposition 7.23 (Full first-front recycle split). **PROVED – PENDING RED TEAM**. For the full first front $S_1 = \{p_0\} \cup Q$ the recycling factor splits as

$$J_{S_1} = J^{\text{rec}} J_Q^{\text{ext}}, \quad J^{\text{rec}} = \prod_{q \in Q} I_q,$$

and after removing the root recycle the remaining lower demand is

$$U_{S_1} J_Q^{\text{ext}} R_{\partial}(S_1) > \left(\frac{9}{5}\right)^{|Q|} m_0.$$

This is a lower-capacity statement; it does not yet give an upper bound closing SMOOTH.

Warning 7.24 (Two sharp interpretive countermodels). SMOOTH does *not* force $E_q > 1$: one can have $E_q = 1$ with the whole cost staying in the valuations. And the same external prime may occur in several rows. Externalisation must therefore never be identified with pairwise-disjoint support, nor with automatic escape.

Let $\emptyset \neq S \subseteq Q_{\text{smooth}}$, $\nu_{\ell,k}(S) := \#\{q \in S : \ell^k \mid n_q\}$ and $a_r := v_r(m_0)$ for $r \in Q$.

Theorem 7.25 (Lossless Prime-Power Root Ledger, B-PP-LEDGER). **PROVED – PENDING RED TEAM**. Three exact decompositions hold:

$$\log \prod_{q \in S} \frac{n_q}{I_q} = \sum_{\ell \notin Q} \sum_{k \geq 1} \nu_{\ell,k} \log \ell + \sum_{r \in Q} \sum_{k \geq 2} \nu_{r,k} \log r, \quad (\text{PP-I})$$

$$\log \prod_{q \in S} \frac{D_q}{I_q} = \sum_{r \in Q} \sum_{k=2}^{a_r} \nu_{r,k} \log r, \quad (\text{PP-D})$$

$$\log \prod_{q \in S} \frac{n_q}{D_q} = \sum_{\ell \notin Q} \sum_{k \geq 1} \nu_{\ell,k} \log \ell + \sum_{r \in Q} \sum_{k \geq a_r + 1} \nu_{r,k} \log r. \quad (\text{PP})$$

Proof. For each prime ℓ the layer-cake identity $\sum_{q \in S} \nu_{\ell}(n_q) = \sum_{k \geq 1} \nu_{\ell,k}(S)$ holds. Dividing by I_q subtracts one copy of every root prime present in the row, so the root sum starts at $k = 2$. Dividing by the full D_q subtracts $\min\{a_r, \nu_r(n_q)\}$, leaving exactly the layers $k \geq a_r + 1$. External primes do not divide m_0 , so their layers start at $k = 1$. [BIB, thm:v75-pp-root-ledger] $\square$

Reading. This is the exact sense in which the root-supported valuation mass can be subtracted *without losing demand*: (PP-D) isolates precisely what the root alphabet already paid for.

Theorem 7.26 (Exact post-root demand). **PROVED – PENDING RED TEAM**. For $q \in Q$,

$$D_q = \text{gcd}(B_q, |2h - q|) \quad \text{and} \quad \frac{n_q}{D_q} \geq \frac{N - q}{\min\{B_q, |2h - q|\}}. \quad (\text{Demand})$$

Setting $w_q := \sum_{\ell \notin Q} \nu_{\ell}(n_q) + \sum_{r \in Q} (v_r(n_q) - a_r)_+$, every SMOOTH row satisfies

$$B_q^{w_q} > \frac{n_q}{D_q} \geq \frac{N - q}{\min\{B_q, |2h - q|\}}. \quad (\text{Post-root tax})$$

Proof. Since $q \mid m_0$ and $\gcd(h, m_0) = 1$ we get $q \nmid 2h - q$, so $\gcd(m_0, n_q) = \gcd(m_0, 2h - q) = \gcd(B_q, |2h - q|)$. Hence $D_q \leq B_q$ and $D_q \leq |2h - q|$, which is (Demand). After dividing by D_q all remaining prime factors of a SMOOTH row are still smaller than B_q , and their number with multiplicity is exactly w_q . [BIB, thm:v75-post-root-demand] $\square$

Lemma 7.27 (Root valuation rigidity). **PROVED – PENDING RED TEAM**. Let $r^a \parallel m_0$. Then

$$\boxed{v_r(N - q) > a \implies v_r(|2h - q|) = a,} \quad \text{and hence} \quad \boxed{r^a > |2h - q| \implies v_r(N - q) \leq a.}$$

Proof. Write $m_0 = r^a M$ with $r \nmid M$ and $b = v_r(|2h - q|)$. From $n_q = 2m_0 + (2h - q)$,

$$v_r(n_q) = \begin{cases} b, & b < a, \\ a, & b > a, \\ a + v_r(2M + \frac{2h-q}{r^a}), & b = a. \end{cases}$$

Excess above a can occur only in the case $b = a$; and if $r^a > |2h - q|$ then $v_r(|2h - q|) = a$ is impossible. [BIB, lem:v75-root-valuation-rigidity] $\square$

Lemma 7.28 (Sharp Prime-Power Packing, B-SHARP-PP-PACK). **PROVED – PENDING RED TEAM**. If $\ell^k \mid n_q$ and $\ell^k \mid n_r$ then

$$\boxed{\ell^k \mid q - r.}$$

All incident roots therefore lie in one class modulo ℓ^k , and for any eligible set $\mathcal{E}_{\ell,k}$ containing all actually incident roots,

$$\boxed{\nu_{\ell,k} \leq \min(|\mathcal{E}_{\ell,k}|, 1 + \left\lfloor \frac{\text{diam } \mathcal{E}_{\ell,k}}{\ell^k} \right\rfloor).}$$

In particular $\ell^k > \text{diam } \mathcal{E}_{\ell,k}$ forces $\nu_{\ell,k} \leq 1$. [BIB, lem:v75-pp-pack]

Warning 7.29 (Potential capacity is not actual capacity). **KILLED**. One may *not* assume $\mathcal{C}_S < \mathcal{D}_S$ uniformly on the basis of the first front alone. The canonical model

$$N = 4736, \quad p_0 = 2371, \quad h = 3, \quad m_0 = 2365 = 5 \cdot 11 \cdot 43, \quad Q = S = \{5, 11, 43\}$$

has all three rows SMOOTH,

$$N - 5 = 3 \cdot 19 \cdot 83, \quad N - 11 = 3^3 \cdot 5^2 \cdot 7, \quad N - 43 = 13 \cdot 19^2,$$

and for the potential-capacity definition used, $\mathcal{C}_S \approx 22.60 > 20.16 \approx \mathcal{D}_S$. This is not a closed bad world, so it does *not* refute a future theorem with a closure hypothesis; it does refute deriving the needed inequality from local eligibility geometry alone. **FAIL-POTENTIAL-CAPACITY-001**

7.4.1 Actual prime-power collision budget

For residual rows $M_q = (N - q)/D_q$ set $\mu_{\ell,k}(S) := \#\{q \in S : \ell^k \mid M_q\}$ and $\Delta_{\text{odd}}(S) := \prod_{q < r, q, r \in S} |q - r|_{\text{odd}}$.

Theorem 7.30 (Actual Prime-Power Collision Budget, B-VPC). **PROVED**. Exactly,

$$\boxed{\sum_{\ell} \sum_{k \geq 1} \binom{\mu_{\ell,k}}{2} \log \ell = \sum_{q < r} \log \gcd(M_q, M_r) \leq \log \Delta_{\text{odd}}(S).} \quad (\text{VPC})$$

Proof. For a fixed pair $q < r$, $\sum_{\ell,k} \mathbf{1}_{\ell^k \mid M_q, M_r} \log \ell = \log \gcd(M_q, M_r)$. Summing over pairs, the number of pairs sharing layer (ℓ, k) is $\binom{\mu_{\ell,k}}{2}$. Moreover $\gcd(M_q, M_r) \mid \gcd(N - q, N - r) \mid r - q$, and the M_q are odd, so the common divisor divides the odd part $|r - q|_{\text{odd}}$. Multiplying over pairs gives the last inequality. [BIB, thm:v75-actual-pp-collision] $\square$

Corollary 7.31 (Actual shared-mass bound). **PROVED**. With $L_{\text{shared}} := \sum_{\mu_{\ell,k} \geq 2} \mu_{\ell,k} \log \ell$,

$$L_{\text{shared}} \leq 2 \log \Delta_{\text{odd}}(S).$$

Proof. For $\mu \geq 2$ one has $\mu \leq 2 \binom{\mu}{2}$; sum over the layers that actually occur and apply Theorem 7.30. $\square$

For a pair $q < r$ put $g_{q,r} := \gcd(M_q, M_r)$ and $L_{\text{pair}}(q, r) := \log \frac{M_q M_r}{g_{q,r}^2}$.

Lemma 7.32 (Pair isolation floor). **PROVED**. For two distinct SMOOTH roots, $L_{\text{pair}}(q, r) \geq \log 5$.

Proof. $g_{q,r}$ is odd and divides $|r - q|_{\text{odd}} < r/2$. For the larger SMOOTH root r , $M_r > \frac{9}{5}r$, so $M_r/g_{q,r} > 18/5$. That quotient is a positive odd integer, hence at least 5; and $M_q/g_{q,r} \geq 1$. [BIB, lem:v75-pair-log5] $\square$

Theorem 7.33 (Exact weighted cut identity, B-WEIGHTED-CUT). **PROVED**. Let $s = |S|$ and let the layer $\lambda = (\ell, k)$ carry weight $w_\lambda = \log \ell$ and multiplicity μ_λ . Then

$$\sum_{q < r} L_{\text{pair}}(q, r) = \sum_{\lambda} \mu_\lambda (s - \mu_\lambda) w_\lambda, \quad (\text{Cut})$$

and consequently $C_{\text{cut}} := \sum_{\lambda} \mu_\lambda (s - \mu_\lambda) w_\lambda \geq \binom{s}{2} \log 5$.

Proof. Layer λ contributes w_λ to $L_{\text{pair}}(q, r)$ exactly when it belongs to precisely one of the two rows; the number of such unordered pairs is $\mu_\lambda (s - \mu_\lambda)$. Summing over layers gives the identity, and Lemma 7.32 gives the lower bound. [BIB, thm:v75-weighted-cut] $\square$

Corollary 7.34 (Low-multiplicity layer lower bound). **PROVED**. For $1 \leq u \leq s - 1$ put $W_{\leq u} := \sum_{1 \leq \mu_\lambda \leq u} w_\lambda$, $R_u := \frac{2(s-u-1)}{u}$ and $H_u := \frac{s-1}{u} \lfloor \frac{(u+1)^2}{4} \rfloor$. Then

$$W_{\leq u} \geq \frac{[\binom{s}{2} \log 5 - R_u \log \Delta_{\text{odd}}(S)]_+}{H_u},$$

and in particular

$$W_1 \geq \frac{[\binom{s}{2} \log 5 - 2(s-2) \log \Delta_{\text{odd}}(S)]_+}{s-1}.$$

Proof. Subtract a multiple of the collision budget from (Cut). For $\mu \geq u + 1$, $\frac{\mu(s-\mu)}{\binom{\mu}{2}} = \frac{2(s-\mu)}{\mu-1} \leq R_u$; for $1 \leq \mu \leq u$ the coefficient $\mu(s-\mu) - R_u \binom{\mu}{2} = \frac{s-1}{u} \mu(u+1-\mu)$ does not exceed H_u . Apply $K \leq \log \Delta_{\text{odd}}(S)$ and take positive parts. $\square$

Warning 7.35 (The sharp abstract obstruction). Cut geometry alone does *not* force singletons once the right-hand side of Corollary 7.34 vanishes. Abstract systems in which every layer has multiplicity exactly 2 (or generally $u + 1$) realise the obstruction. Globalisation needs congruences, source structure or absorption data in addition.

7.5 Exact-D pair compensation and canonical pair geometry

This section reduces a hypothetical Exact-D failure to an explicit rigid regime, then to an explicit normal form, and finally to a canonical gap filter. Everything here is **PROVED**. The

branch $k = 1$ ends at a stated **OPEN** barrier. Notation: $t := 2h$, $m_0 = qrc$ with $q < r$ distinct in Q_{smooth} , and $\delta := |r - q|_{\text{odd}}$.

Theorem 7.36 (Exact-D pair compensation outside the regime $r < t$, B-EXACTD-OUTSIDE). **PROVED**.

If $t < r$ then $M_q M_r > \delta^2$; more precisely $\frac{M_q M_r}{\delta^2} > \frac{36q}{5}$. Consequently every potential Exact-D failure satisfies

$$\boxed{q < r < t, \quad r \mid D_q, \quad t \equiv q \pmod{r}.} \quad (\text{ED-res})$$

Proof. Suppose $M_q M_r \leq \delta^2$. Since r is SMOOTH, $M_r > \frac{9}{5}r$, so

$$D_q = \frac{N - q}{M_q} > \frac{9r(N - q)}{5\delta^2} > \frac{9qr^2c}{5\delta^2} > \frac{36}{5}qc, \quad (\text{ED1})$$

using $\delta < r/2$. If $t < r$ then $|t - q| < r$; from $D_q \mid rc$, $D_q \mid |t - q|$ and $\gcd(t, m_0) = 1$, D_q cannot contain r , so $D_q \mid c$, contradicting (ED1). Hence failure forces $r < t$ (equality is excluded by parity). By (ED1), $D_q > c$, so every divisor $D_q \mid rc$ must contain r , i.e. $r \mid D_q \mid (t - q)$. [BIB, thm:v75-exactd-pair-outside] $\square$

Corollary 7.37 (First canonical gap filter). **PROVED**. Writing $G = p_0 - p_-$ (so canonicity gives $h < G$, hence $t < 2G$), every Exact-D pair failure satisfies

$$p_0 < 2G + \frac{5(2G - q)\delta^2}{9r} < 2G + \frac{5}{9}G^2,$$

and in particular

$$\boxed{9p_0 \geq 18G + 5G^2 \implies M_q M_r > \delta^2 \text{ for all SMOOTH pairs.}}$$

Theorem 7.38 (Joint D_q, D_r rigidity, B-JOINT-D). **PROVED**. If $s^a \parallel m_0$ and $\alpha = v_s(D_q)$, $\beta = v_s(D_r)$, then

$$\boxed{\alpha + \beta \leq a + \min\{a, v_s(r - q)\},}$$

and globally

$$\boxed{D_q D_r \mid m_0 \gcd(m_0, r - q) = m_0 \gcd(c, r - q).} \quad (\text{Joint-D})$$

Proof. $\max(\alpha, \beta) \leq a$, while $s^{\min(\alpha, \beta)} \mid \gcd(D_q, D_r) \mid \gcd(N - q, N - r) \mid r - q$. Adding exponents gives the first bound; the primewise product gives (Joint-D). Since neither q nor r divides $r - q$, the common part $\gcd(m_0, r - q)$ comes from c . [BIB, thm:v75-joint-D] $\square$

Theorem 7.39 (Primitive residual normal form, B-PNF). **PROVED**. Assume the regime (ED-res) and write $t - q = rk$, $m_0 = qrc$ with k positive and odd; put $g := \gcd(c, k)$, $c = gu$, $k = gv$, $\gcd(u, v) = 1$. Then exactly

$$\boxed{\begin{array}{lll} t = q + rgv, & m_0 = qrgu, & D_q = rg, \\ N - q = rg(2qu + v), & M_q = 2qu + v, & p_0 = q + rg(qu + v), \\ h = \frac{q+rgv}{2}, & D_r = \gcd(qgu, q + (gv - 1)r), & M_r = \frac{2qrgu + q + (gv - 1)r}{D_r}. \end{array}} \quad (\text{PNF})$$

The formulas remain correct when q or r occur to higher powers in m_0 ; the extra copies are contained in $c = gu$.

Proof. The key identity is $D_q = \gcd(rc, rk) = r \gcd(c, k) = rg$; the rest follows by substituting $m_0 = qrgu$ and $t = q + rgv$ into $N = 2m_0 + t$ and $p_0 = m_0 + t$. For D_r use $D_r = \gcd(B_r, t - r) = \gcd(qgu, q + (gv - 1)r)$. Since $\gcd(t, m_0) = 1$ we have $q \nmid gv$. [BIB, thm:v75-primitive-normal] $\square$

Theorem 7.40 (High-gap amplification, B-HIGH-GAP). **PROVED** under the *Exact-D pair-failure hypothesis*. With $d := v_2(r - q) \geq 1$, every *Exact-D pair failure* in primitive normal form satisfies

$$r > 2^{2d+1}(2qu + v) > 2^{2d+2}qu \quad (\text{HG-r})$$

and

$$\frac{h^2}{m_0} > 4^d gv^2 + 2^{2d-1} \frac{gv^3}{qu} + \frac{v}{2u} > 4^d gv^2 \geq 4, \quad (\text{HG})$$

so in particular $h^2 > 4m_0$.

Proof. Put $A = 2qu + v$. From $D_r \leq B_r = qgu$ we get $M_r \geq (2 + \frac{qv-1}{qgu})r + \frac{1}{gu} > 2r$. Failure $M_q M_r \leq (r - q)^2 / 4^d$ then gives $2 \cdot 4^d Ar < r^2$, which is (HG-r). Moreover $\frac{h^2}{m_0} = \frac{rgv^2}{4qu} + \frac{v}{2u} + \frac{q}{4rgu}$; substituting $r > 2^{2d+1}(2qu + v)$ into the first term gives (HG). [BIB, thm:v75-high-gap] $\square$

Warning 7.41 (Hypothesis, restated). **Warning 7.5** applies verbatim: the hypothesis is *Exact-D pair failure*, not CRD failure. Using $h^2 > 4m_0$ under canonical failure alone is an error.

Corollary 7.42 (Quadratic canonical gap filter). **PROVED**. With $\Lambda := 4^d gv^2 + 2^{2d-1} \frac{gv^3}{qu} + \frac{v}{2u}$ and $h < G$,

$$p_0 < 2G + \frac{G^2}{\Lambda} < 2G + \frac{G^2}{4^d gv^2} \leq 2G + \frac{G^2}{4}.$$

Finite check of the quadratic filter. **VERIFIED COMPUTATION** as reported by the source. An exact check to $p_0 \leq 10^9$ found no canonical failure satisfying the filter $p_0 < 2G + G^2/4$; of the candidates only $p_0 = 11$ passed, and it admits no permissible pair of distinct roots. This is a finite verification, not a global proof. — *reported computational result, not reproduced in this document.*

7.5.1 The branch $k = 1$

For $k = 1$ one has $g = v = 1$, so $t = q + r$, $m_0 = qru$ with $u \geq 3$ odd, and exactly $D_q = r$, $M_q = 2qu + 1$, $D_r = q$, $M_r = 2ru + 1$.

Theorem 7.43 ($k = 1$ high-gap strengthening). **PROVED**. For $k = 1$ every pair failure satisfies

$$r > 4^{d+1}qu^2, \quad \frac{h^2}{m_0} > 4^d u, \quad \text{hence} \quad p_0 < 2G + \frac{G^2}{4^d u} \leq 2G + \frac{G^2}{12}.$$

The only non-automatic SMOOTH condition on the larger row is $P^+(2ru + 1) < qu$.

Proof. Failure gives $(2qu + 1)(2ru + 1) \leq (r - q)^2 / 4^d < r^2 / 4^d$; the lower bounds on both factors give $r > 4^{d+1}qu^2$. Then $\frac{h^2}{m_0} = \frac{(q+r)^2}{4qru} > \frac{r}{4qu} > 4^d u$, and the filter for p_0 follows from $h < G$. The condition $u \geq 3$ is necessary: at $u = 1$ the factor $r = D_q = B_q$ breaks the strict SMOOTH threshold. [BIB, thm:v75-k1-gap] $\square$

Theorem 7.44 ($k = 1$ quotient normal form). **PROVED**. With $A := 2qu + 1$, $B := 2ru + 1$ one has exactly

$$q = \frac{A - 1}{2u}, \quad r = \frac{B - 1}{2u}, \quad A \equiv B \equiv 3 \pmod{4},$$

$$4um_0 = (A - 1)(B - 1), \quad 4uh = A + B - 2, \quad 4ux = AB - 1, \quad 4up_0 = AB + A + B - 3,$$

$$up_0 + 1 = \frac{(A+1)(B+1)}{4} = (qu+1)(ru+1).$$

Moreover $d = v_2(B-A) - 1$ and $\delta = \frac{B-A}{2^{d+1}u}$; pair failure is equivalent to

$$4^{d+1}u^2AB \leq (B-A)^2,$$

SMOOTH of the larger row reads $P^+(B) < \frac{A-1}{2}$, and canonicity reads

$$\left[\frac{AB-1}{4u}, \frac{AB+A+B-3}{4u} \right) \cap \mathbb{P} = \emptyset$$

with prime endpoint $p_0 = \frac{AB+A+B-3}{4u}$. [BIB, thm:v75-k1-quotient]

Remark 7.45 (The continued-fraction relation, and why it is not a descent). With $a := qu+1 = \frac{A+1}{2}$ and $b := ru+1 = \frac{B+1}{2}$ one has $ab - up_0 = 1$, and the Euclidean algorithm gives explicitly

$$\frac{a}{p_0} = [0; r, u, q], \quad \frac{u}{b} = [0; r, u].$$

The modular-inverse relation is therefore an exact relation between consecutive convergents — but by itself it supplies no descent. Compare the *Matched Euclidean Gluing Lemma* recorded as missing in §14.4.

Proposition 7.46 (Structured composite coverage in the prime-free interval). **PROVED**. For odd s with $\frac{r-q}{2q} < s < \frac{r}{q}$, the number $z = r - sq$ satisfies $0 < z < h$ and

$$p_0 - z = q(ru + s + 1).$$

Moreover for every prime $\ell \mid B$, $2up_0 \equiv qu - 1 \pmod{\ell}$, hence $\gcd(B, qu - 1) = 1$. Each such ℓ covers exactly one residue class among the shifts $p_0 - 2j$, and under pair failure the number of its hits in $0 < 2j < h$ is at least $4^d u$. [BIB, prop:v75-k1-coverage]

Open target 7.47 (Canonical $k = 1$ gap/covering barrier). **OPEN**. The canonical branch $k = 1$ has *not* been eliminated. The present apparatus produces a gap filter, a smooth-shift condition and a covering requirement, but no contradiction. This is the sharpest surviving remnant of the Exact-D programme and is a self-contained problem: exhibit a canonical $k = 1$ Exact-D pair failure, or prove none exists. [BIB, Canonical $k = 1$ Gap/Covering Barrier]

7.6 Sparse geometric core locks (FFS layer)

The factor-flow-system (FFS) layer of the Biblia proves that *collision compression must be paid for with geometry*. These results are stated for a finite factor-closed core C of a prime affine FFS; the Goldbach specialisations take $\delta = 2$ because all active primes are odd. Statuses are the source's own: **tagN** = source claim “new, proved” and **tagG** = Goldbach specialisation, both imported as **PROVED – PENDING RED TEAM**.

Definition 7.48 (Collision multiplexing ratio). For a finite factor-closed core C let $\eta(C) = \Omega(J_C) > 0$, let $\chi(C)$ be the minimal number of distinct parent-pairs in a collision witness forest, and put

$$\kappa_{\text{col}}(C) := \frac{\eta(C)}{\chi(C)} \geq 1.$$

$\kappa_{\text{col}} = 1$ means no multiplexing: each unit of collision excess needs its own pair. A value above 1 means some witness pairs carry several distinct prime labels at once.

Theorem 7.49 (Multiplex collision rigidity, B-MULTIPLEX-RIGID). **PROVED – PENDING RED TEAM**.
 With $m = \min C$, $D = \max C - \min C$, $\eta = \Omega(J_C) > 0$ and $\chi = \chi(C)$,

$$D \geq \delta m^{\eta/\chi} = \delta m^{\kappa_{\text{col}}(C)}.$$

Proof. Closure puts every child in C , so every prime label occurring in J_C is at least m and $J_C \geq m^\eta$. Choose a collision witness forest realising χ ; lattice collision divisibility bounds each of its χ aggregated labels by D/δ , and the product of the aggregated labels is exactly J_C , so $J_C \leq (D/\delta)^\chi$. Combine and take χ -th roots. [BIB, Multiplex Collision Rigidity] $\square$

Reading. If many independent collision-excess labels are packed into few parent-pairs, those pairs must be extremely far apart: at $\kappa_{\text{col}} = 2$ the diameter grows at least quadratically in the minimal atom, at 3 at least cubically. Topological excess is converted into a power lower bound for the diameter.

Corollary 7.50 (Goldbach collision-compression barrier). **PROVED – PENDING RED TEAM**. Let C be a terminal bad SCC of the graph for even N that is not a pure support cycle, $m = \min C$, $D = \max C - \min C$. Then $D \geq 2m^{\eta/\chi}$, and since $\max C < N/3$,

$$m < \left(\frac{N}{6}\right)^{\chi/\eta}.$$

In particular for every integer $t \geq 2$: $\eta \geq t\chi \Rightarrow m < (N/6)^{1/t}$; equivalently a hypothetical core with $m > N^{1/t}$ must satisfy $\eta < t\chi$.

Proof. All active primes are odd, so $\delta = 2$; apply Theorem 7.49 and $\max C < N/3$ (Theorem 5.2). $\square$

Reading. This converts a structural question into a very concrete one: instead of killing all large cores directly, it suffices to exhibit *triple* collision multiplexing in the relevant class. It does not solve Goldbach — χ may be close to η , recovering only a linear diameter barrier.

Corollary 7.51 (Sparse Geometric Core Lock, B-SPARSE-CORE-LOCK). **PROVED – PENDING RED TEAM**.
 Under the same hypotheses, for every collision witness forest with χ_τ distinct parent-pairs,

$$N - 1 \leq U_C J_C \leq U_C \left(\frac{D}{2}\right)^{\chi_\tau}, \quad \text{hence} \quad D \geq 2 \left(\frac{N - 1}{U_C}\right)^{1/\chi_\tau}.$$

Proof. For a terminal bad SCC the unit-slope product lock gives $T_C = U_C J_C \geq N - 1$, and lattice collision divisibility with $\delta = 2$ gives $J_C \leq (D/2)^{\chi_\tau}$. [BIB, Sparse Geometric Core Lock] $\square$

Remark 7.52. Unlike the older bound $U_C G_C \geq N - 1$, the right-hand side here uses only the *sparse* family of parent-pairs needed to realise the collision fibres. It is a geometric version of exact- J , not a product over all pairwise gcds of rows.

Theorem 7.53 (Quantitative support explosion). **PROVED – PENDING RED TEAM**. Let S be a finite subsystem of an exact prime affine FFS and $K_\tau(S)$ any collision witness product. Put

$$\mathcal{M}(S) = P_{\text{src}}(S) \frac{\prod_{p \in S} R(p)}{\prod_{p \in S} p}, \quad B_S = \max_{p \in S} R(p).$$

If $A_S := \mathcal{M}(S)/(U_S K_\tau(S)) > 1$ then the number of distinct boundary atoms satisfies

$$|\partial S| \geq \left\lceil \frac{\log A_S}{\log B_S} \right\rceil,$$

and more generally $\mathcal{M}(S) > U_S K_\tau(S) B_S^r \Rightarrow |\partial S| \geq r + 1$.

Proof. The geometric escape criterion gives $R_\partial(S) \geq A_S$. Every boundary atom divides some $R(p)$, hence is at most B_S , and R_∂ is a product of $|\partial S|$ distinct primes, so $R_\partial(S) \leq B_S^{|\partial S|}$. Take logarithms and use integrality. [BIB, Quantitative Support Explosion] $\square$

Reading. The escape criterion is not merely “ $R_\partial > 1$ ”; the same computation can force *many* new boundary atoms. Together with Corollary 7.51 the FFS layer therefore has two independent elimination mechanisms, one for existence of escape and one for its multiplicity.

Chapter 8

Canonical Root Dynamics (CRD)

8.1 Epistemic convention

CRD is a *reformulation*, not a mechanism. Its content is:

1. the failed world is a closed bad set with a unique source (Lemma 3.26);
 2. the union of all closed bad sets below $N/3$ is computable by a decreasing filtration (Definition 3.28);
 3. canonical failure is exactly membership of $\text{PF}(N - p_0)$ in that fixed point (Theorem 3.29).
- Anything beyond that — capacity arguments, container arguments, spectral arguments — belongs to the successor theories of §11.2 and is **KILLED**. When reading older CRD material, keep this boundary in mind: the three items above survived every audit; the rest did not.

8.2 The architecture wall

Theorem 8.1 (Finite Affine Canonical Programmability). **PROVED** in [PCRD] (proof not reproduced here), class A as a statement about canonicity. For any finite family of affine forms $L_j(P) = a_jP + b_j$ and any prescribed finite valuation and compositeness conditions, under the standard coprimality conditions there exist — by the Chinese remainder theorem and Dirichlet’s theorem — infinitely many primes P realising the whole pattern. Setting $N = 2(P - 1)$ gives canonical $p_0 = P$, $h = 1$.

Theorem 8.2 (Finite Bad-Tree Universality). **PROVED** in [PCRD] (proof not reproduced here). Any finite rooted BAD tree under a canonical root with $h = 1$ can be programmed, with prescribed edge valuations.

Corollary 8.3 (Finite-local universality wall). **PROVED** as a no-go; **SUPERSEDED** for the whole family of arguments it covers. No argument whose only canonical input is a finite number of shifts, congruences, valuations, divisibility or compositeness certificates, bounded-depth rows, or neighbouring roots can derive a contradiction from canonicity alone — because every such finite transcript is realisable in infinitely many canonical worlds with $h = 1$. [PCRD, P0 finite-local universality wall]

Remark 8.4 (Link to the phase diagram). All $h = 1$ worlds with $G \geq 4$ lie in phase Type I (Theorem 6.12), hence have m_0 composite and $\rho_0 = 0$. The whole $h = 1$ programme of [PCRD] therefore lives *inside a single phase* — the one in which the reflection gate gives no guarantee. Two reported witnesses, verified here:

N	p_0	G	m_0	type, depth
103 344	51 673	14	$51\,671 = 163 \cdot 317$	Type I, $d = 4$
364 844	182 423	6	$182\,421 = 3^2 \cdot 20\,269$	Type I, $d = 6$

VERIFIED COMPUTATION.

8.3 C0a — the small-gap canonical obligation

Theorem 8.5 (Small-gap residual closure, C0a). **PROVED**, class C . Hypotheses: *canonical failure*, $h \leq 2$. Then $h \in \{1, 2\}$, m_0 is odd composite, $Q \neq \emptyset$, and

$$\max\{U_R, J_R, B_{\text{abs}}\} > \left[\left(\frac{5}{3} \right)^{|Q|} \text{rad}(m_0) \right]^{1/3} > 1.$$

[CORE, C0a-RESIDUAL-CLOSURE]

Reading. For $h \leq 2$ the reflected block is empty, so boundary geometry gives nothing; the theorem replaces it by a nonzero obligation on the row side. *Limitation:* it yields neither a path to GOOD nor a capacity contradiction.

8.4 C0b — the exact canonical reflection-defect ledger

Theorem 8.6 (C0b-RD). **PROVED**, class $A + C$. Hypotheses: *canonical failure*, $N \geq 50$, $h \geq 3$. With $t = \lfloor (h-1)/2 \rfloor$ and

$$\begin{aligned} \mathcal{I} &= \{j : \text{PF}(c_j) \cap R \neq \emptyset, a_j \in \mathcal{A}_N\}, & \mathcal{D} &= \{j : \text{PF}(c_j) \cap R \neq \emptyset, a_j \notin \mathcal{A}_N\}, \\ \mathcal{S} &= \{j : \text{PF}(c_j) \cap R = \emptyset, P^+(c_j) < h\}, & \mathcal{F} &= \{j : \text{PF}(c_j) \cap R = \emptyset, P^+(c_j) \geq h\}, \end{aligned}$$

one has

$$\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$$

and for $j \in \mathcal{F}$ the number $\ell_j = P^+(c_j)$ is active, lies outside R , and occurs in exactly one element of $\mathfrak{B}(p_0)$. Moreover, if $\mathcal{I} = \emptyset$ and $k = \lceil t/3 \rceil$, then at least one of

$$\sum_{j \in \mathcal{D}} \Omega(a_j) \geq 2k, \quad \sum_{j \in \mathcal{S}} \Omega(c_j) > k \frac{\log x}{\log h}, \quad \prod_{\ell \in \Lambda_{\mathcal{F}}} \ell \geq h^k$$

holds. With $Q_{\leq} = \{q \in Q : q \leq t+1\} \neq \emptyset$ and $j(q) \equiv h \pmod{q}$, $1 \leq j(q) \leq q-1$, one has $q \mid c_{j(q)}$, $j(q) \in \mathcal{I} \cup \mathcal{D}$ and $B_{\leq} < h^{\lceil \mathcal{J}_{\leq} \rceil}$; and if $Q_{\leq} = \emptyset$ then $\mathfrak{F}_Q > (5(t+1)/3)^{|Q|}$. [CORE, C0b-RD]

Reading. This is the strongest *quantitative* obligation that canonical boundary geometry issues at all: every coordinate of the reflected block must land in one of the four classes. *Known limitation:* C0b-RD supplies neither a path to GOOD nor a capacity bound.

Scope note 8.7 (Why the hypothesis $Q_{\leq} \neq \emptyset$ is needed). **SCOPE-CORRECTED**. In version v1 this condition was omitted; with $Q_{\leq} = \emptyset$ the inequality would read $1 < 1$. The configuration is not rare: among canonical instances with $h \geq 3$ and $N < 2 \cdot 10^5$, 47 551 of 72 446 have $Q_{\leq} = \emptyset$; the smallest case is $N = 50$, $p_0 = 29$, $h = 4$, $t = 1$, $m_0 = 21$, $Q = \{3, 7\}$. [CORE, C0b-RD audit repair] — *reported computational result, not reproduced in this document.*

8.5 The C0b branch targets: TI, TD, TS, TF

C0b-RD says one of four classes must carry the obligation. The four branch targets are the statements “if the obligation lands in class X , then the dynamics must realise it”. All four are **OPEN**. They are the entry point to the frontier; full frontier entries with obstructions and success criteria are in Chapter 9.

ID	Branch	Obligation to be discharged
TI	$\mathcal{I}$ (incident, active mirror)	Turn an active reflected row into realised production: the mirror a_j is an active parent of c_j , so it <i>could</i> feed the world; TI asks that it must feed it <i>measurably</i> .
TD	$\mathcal{D}$ (incident, inactive mirror)	The root-anchored dominance obligation. The mirror is not active, so the mass must be recovered from $\Omega(a_j)$; TD asks for a lower bound on realised production anchored at the root. This is the smallest missing bridge (§9.11).
TS	$\mathcal{S}$ (h -smooth mirror)	The coordinate is smooth, so its factors are small and lie in the low range; TS asks that smooth coordinates cannot all be absorbed.
TF	$\mathcal{F}$ (fresh large factor)	$\ell_j = P^+(c_j) \geq h$ is active and outside R ; TF asks that a fresh external atom forces an escape or a coupling.
TD-DENSITY	density form of TD	SUPERSEDED : replaced by the root-anchored form after the density version was found to require a bound of the same strength as its conclusion.

Scope note 8.8 (TF is local, and locality is not enough). **SCOPE-CORRECTED**. “A local $\mathcal{F}$ certificate together with $h < G$ forces routing” is false: at $N = 128$, $\ell = 13$ the certificate holds, but p_0 wins and $R = \emptyset$. FAIL-TF-LOCAL-SCOPE-001

Chapter 9

Current Proof Frontier

This is the only chapter where new work should start. Everything before it is either settled or dead. Each target below carries seven fields: exact statement, why it matters, what is already proved, the exact remaining obligation, known obstructions, what would count as success, and the failure-log entries that constrain it.

Before reading further, internalise the shape of the problem. The chain is

$$\underbrace{\text{C0a}}_{\text{PROVED}} \Rightarrow \underbrace{\text{C0b-RD}}_{\text{PROVED}} \Rightarrow \underbrace{\{\text{TI, TD, TS, TF}\}}_{\text{OPEN}} \Rightarrow \underbrace{\text{P1}^{\text{br}}}_{\text{OPEN}} \Rightarrow \underbrace{\{\text{L1, E1, W1, K1}\}}_{\text{OPEN}} \Rightarrow \underbrace{\text{X1}}_{\text{OPEN}} .$$

The chain does *not* end at Goldbach even if every link is proved: it ends at “the canonical traversal succeeds”, which is strictly stronger than Goldbach for each N but is not implied by it. The architecture wall (§8.2) says the chain cannot be closed by finite local data alone, so at least one link must import global input.

Read §9.2 first. That chain is the historical frontier. The project’s *current* frontier is the canonical threshold programme of v4.79–v4.81 (§9.2), whose target is deliberately weaker than X1 — an asymptotic $N_0^{p_0}$ rather than all N — and which has already converted two of its own demanded outputs into proved impossibilities. The two live targets are named there. The older chain is retained because its statements are correct and its obstructions are real, and because C0b-RD still feeds it.

9.1 The exact Goldbach boundary

What is proved. For every even N in the non-diagonal case: the canonical root is well defined and unique; if it fails, the failed world is closed, all-BAD, has p_0 as its unique source, and lies below $N/3$; the reflected boundary is defect-free and its ledger has exactly four classes; and the row side satisfies C0a when $h \leq 2$ and C0b-RD when $h \geq 3$.

What would imply Goldbach. If for every N the canonical traversal succeeded, then in particular $\text{Good}_N \neq \emptyset$ for every N , which is binary Goldbach. So a proof of universal canonical success is *sufficient*.

What does not follow. Goldbach does *not* imply universal canonical success; the six N of §6.6 have $B_N^{\text{low}} \neq \emptyset$ while satisfying Goldbach comfortably, and 301 non-canonical upper roots fail while their N satisfy Goldbach. Every statement conditioned on $\text{Good}_N = \emptyset$ is class D and is unusable under canonical failure alone. **FAIL-UPPERHALF-SCOPE-001**

The current smallest missing bridge. The root-anchored TD obligation feeding P1^{br} . Everything upstream is proved; everything downstream is blocked on it or on the same defect in

another guise.

9.2 The canonical threshold programme (post-v4.79/4.80/4.81)

This is the project's actual current frontier, and it is newer than the TI/TD/TS/T-F/P1BR/K1/X1 chain of §9.3–§9.9. The older chain is not wrong; it is simply not where the work is. A model starting fresh should read this section first.

The target here is deliberately weaker than X1 and therefore more attackable:

$$\exists N_0^{p_0} < \infty \quad \forall \text{ even } N \geq N_0^{p_0} : \quad p_0(N) \text{ succeeds.}$$

The verdict after v4.81 is: *yes in architecture, not in closure*. Two of the four demanded outputs of the previous frontier were *proved impossible*; exactly three routes remain, and the preferred one is named.

9.2.1 The v4.79/v4.80 engine

Throughout: $N = 2x$, $p_0 = x + h$, $m = N - p_0$, $Q = \text{PF}(m)$, $R = R_N(p_0)$, canonical failure, $N \geq 50$.

Proposition 9.1 (Nagura window). **PROVED**. For $N \geq 50$, Nagura's explicit interval theorem together with canonical minimality gives

$$p_0 < \frac{3N}{5}, \quad m > \frac{2N}{5}.$$

[CORE, v4.79 (V79.2); Nagura [1]]

Reading. This is why the two-sheet argument (Theorem 5.34) may use $m_0 > 2N/5$; it is the only place an external prime-counting input enters the canonical layer, and it is completely explicit.

Theorem 9.2 (Root-row vertical compensation and weighted residual floor). **PROVED**. The root row itself contributes $V_0 := m/\text{rad}(m)$ to the canonical vertical defect U_R . Combined with full- Q residual production (Theorem 5.13) and the rooted residual-absorption split,

$$U_R^2 J_R B_{\text{abs}} > m \left(\frac{5}{3}\right)^{|Q|} > \frac{2N}{3}. \quad (\text{V79.4})$$

[CORE, vNEXT4.79]

Theorem 9.3 (First-front coefficient dichotomy). **PROVED**. With the exact first-front coefficient defect

$$F_Q^{\text{coeff}} = \frac{m}{\text{rad}(m)} \prod_{\substack{q|m \\ \kappa_R(q) \geq 2}} q,$$

one has the sharp dichotomy

$$F_Q^{\text{coeff}} = 1 \iff m \text{ squarefree and } \kappa_R(q) = 1 \ (q \mid m),$$

while

$$F_Q^{\text{coeff}} > 1 \implies T_R > \frac{5}{2}N,$$

which forces a half-power U/J channel. Nonprimitive first-front failure is therefore already coefficient-heavy. [CORE, vNEXT4.79–4.80]

Definition 9.4 (Primitive branch). The canonical failed world is *primitive* when $F_Q^{\text{coeff}} = 1$, i.e. m is squarefree and every first-front label has load one. Otherwise it is *nonprimitive*.

Theorem 9.5 (Primitive load-one and selected first-front production). **PROVED**. *On the primitive branch the load-one condition is exactly*

$$D_q = \gcd(m, N - q) = 1 \quad (q \in Q),$$

so the selected first-front product is exactly

$$U_F(J_F B_{\text{re}}) B_{\text{abs}} = \prod_{q \in Q} (N - q).$$

With the adaptive least non-dividing prime $\lambda_N = \min\{\ell \in \mathbb{P} : \ell \geq 3, \ell \nmid N\}$ one has $q < N/(2\lambda_N)$ for every $q \in Q$, hence

$$\prod_{q \in Q} (N - q) > \left(1 - \frac{1}{2\lambda_N}\right)^{|Q|} N^{|Q|} \geq \frac{25}{36} N^2.$$

The primitive selected production therefore has leading logarithmic coefficient $|Q| \geq 2$, not merely one. [CORE, vNEXT4.80]

Theorem 9.6 (Owner-preserving quadratic UJD charge, C-PO-PRIMITIVE-OWNERPRESERVING-QUADRATIC-UJD-CHARGE). **PROVED**. *The adaptive high range forms a unique-parent acyclic forest. Each absorbed high atom is replaced by a selected terminal descendant whose support lies in $C_{\lambda_N} \setminus Q$; low absorbed support and selected terminal support are disjoint, so vertical, horizontal and denominator support can each be single-charged. The resulting owner-preserving selected divisor $\mathcal{C}_Q$ satisfies*

$$\mathcal{C}_Q \mid U_R J_R D_{\lambda_N}.$$

[CORE, vNEXT4.80]

Consequences recorded by v4.80. C-PO-PRIMITIVE-BASEABS-ESCAPE-OR-REUSE-CHARGE is **SUPERSEDED**: no external-parent escape/reuse theorem is needed to charge primitive base absorption. C-PO-PRIMITIVE-FIRSTFRONT-UNITMARGIN-PRODUCTION ($U_F(J_F B_{\text{re}}) B_{\text{abs}} > \frac{10N}{9}$, **PROVED** in v4.79) is *retained* but superseded as the binding primitive floor. C-PO-PRIMITIVE-SELECTEDFAMILY-CAPACITY is refined to C-PO-PRIMITIVE-SELECTED-UJD-CAPACITY, which remains **OPEN**. G-CONDITIONAL-EFFECTIVE-PO-NO is **PROVED** only as a conditional theorem: no numerical $N_0^{p_0}$, no universal p_0 -success theorem and no Goldbach proof is claimed.

9.2.2 The v4.81 container theorem and what it kills

The v4.80 preferred frontier C-PO-PRIMITIVE-FIRSTFRONT-REFLECTION-DISJOINT-CAPACITY asked for one of four outputs: (1) reflected production disjoint from $\mathcal{C}_Q$ at fixed positive logarithmic scale; (2) same-label overlap forcing an additional exact J/U layer; (3) a GOOD event; (4) strict scale descent. v4.81 proves that **(1) and (2) cannot exist**, by exact identity rather than heuristic.

Theorem 9.7 (Exact rooted container identity). **PROVED**. *On the primitive canonical failed branch,*

$$U_R J_R D_{\lambda_N} = \frac{\prod_{u \in R} (N - u)}{m \prod_{r \in H_\lambda} r}. \quad (\text{V81.1})$$

So the “container” of the v4.80 selected charge is, up to an explicit denominator, the row product of the world itself. [CRD v4.81, Thm. container]

Theorem 9.8 (Container slack for $\mathcal{C}_Q$). **PROVED**. For every $N \geq 500$,

$$\frac{U_R J_R D_{\lambda_N}}{\mathcal{C}_Q} = \frac{\prod_{r \in C_{\lambda} \setminus Q} (N-r) \prod_{r \in H_{\lambda} \setminus Z} (N-r)}{\prod_{r \in H_{\lambda} \setminus A_H} r} > \frac{5N}{6}. \quad (\text{V81.2})$$

The selected charge therefore never approaches its container: the gap is at least one full logarithmic unit, and grows by one further unit for every low non-front state. [CRD v4.81, Thm. slackCQ]

Corollary 9.9 (Death of the size route). **PROVED**. Two consequences close the archimedean size route completely.

1. Trivially: a divisor never exceeds what it divides. Once $\mathcal{C}_Q \mid U_R J_R D_{\lambda_N}$ is a theorem (Theorem 9.6), no lower bound for $\mathcal{C}_Q$ can contradict any true upper capacity for the container.
2. Quantitatively: any adjoined production P with $\mathcal{C}_Q P \mid U_R J_R D_{\lambda_N}$ is priced exactly by (V81.2) — it must supply one factor $\geq \frac{5}{6}N$ for every low non-front state before it can begin to bite.

Hence “new selected storage disjoint from $\mathcal{C}_Q$ at fixed positive logarithmic scale” — the frontier’s own demanded output (1) — is insufficient by construction, not merely unproved.

Theorem 9.10 (Exact overlap normal form: overlaps are charge-free). **PROVED**. If a prime ℓ divides a selected row $N - u$ ($u \in Q \cup Z$) and a strict reflected carrier $c_j = p_0 - 2j$, then the entire content of that coincidence is

$$\ell \mid a_j - u, \quad 0 < a_j - u < x, \quad (\text{V81.3})$$

which is equivalent to the already-counted divisibility $\ell \mid N - u$ together with the residue condition $2j \equiv p_0 \pmod{\ell}$. Because $a_j \notin R$, the coincidence creates no rooted parent, no κ_R increment, no J -layer and no U -layer.

Corollary 9.11. Alternative (2) of the v4.80 frontier is empty.

Reading. This is the sharpened, final form of the reflection-gate warning: not only does the prime-free boundary fail to *activate* a reflected row (Warning 3.16, $N = 242$), an overlap between the reflected block and the selected family carries *no charge at all*.

9.2.3 New positive structure surviving the attack

All of the following is owner-preserving and reconstructible from the rooted CRD definitions.

Theorem 9.12 (C1: gap-scale shadow normal form). **PROVED**. $D_q = \gcd(m, 2h - q)$, and primitive load-one is equivalent to

$$\gcd(m, 2h - u) = 1 \quad (u \in R \setminus \{p_0\}). \quad (\text{V81.11})$$

This puts the first front and the reflected block on a single scale $\delta = 2h$.

Remark 9.13. This is the Prime-Gap Shadow Identity (Theorem 5.22) re-proved from the v4.80 definitions and used for the first time in the v4.79/v4.80 chain. The v4.81 salvage audit records it as the *one genuinely missing tool* the older corpus contributed to the current frontier.

Theorem 9.14 (C2: primitive residual world). **PROVED**. $R' := R \setminus (\{p_0\} \cup Q)$ is child-closed, nonempty, satisfies $\text{Src}(R') \subseteq \Gamma_N(Q)$ and $\min R' < \sqrt{N}$.

Theorem 9.15 (C3: owner-preserving capacity factorisation). **PROVED**. With $\Sigma' = \prod_{d \in \text{Src}(R')} d$,

$$T_R \Sigma' = \left(\prod_{q \in Q} (N - q) \right) T_{R'}.$$

Theorem 9.16 (C4: selected residue lock). **PROVED**.

$$\mathcal{C}_Q B_H \equiv (-1)^{|Q|+|Z|} m \prod_{z \in Z} z \pmod{N}. \quad (\text{V81.13})$$

Theorem 9.17 (C5: adaptive fringe cofactor alphabet). **PROVED**. *Every edge into H_λ has cofactor a prime in $[\lambda, 2\lambda]$ coprime to N , and that prime is itself a low state. The alphabet is the full prime range $\mathcal{L}_N \subseteq [\lambda, 2\lambda]$.*

Warning 9.18. This is *not* the rejected two-letter claim $\{\lambda, 2\lambda-1\}$, which is **KILLED** as FAIL-FRIDGE-ALPHABET-TRANSI. Theorem 9.17 is precisely the “what survives” clause of that failure entry.

9.2.4 The minimal hostile survivor and the surviving routes

Theorem 9.19 (Product-comparison immunity). **PROVED**. *A primitive canonical failed world survives every archimedean product comparison between a single-charged selected family and the rooted $U/J/D$ container. The reason is now exactly quantified: a selected family can beat its container only if it leaves fewer than $|H_\lambda|$ rows unselected, whereas $\mathcal{C}_Q$ leaves at least $|C_\lambda \setminus Q| + |H_\lambda| - k$.*

Theorem 9.20 (Trichotomy of surviving routes). **PROVED**. *Any proof that the minimal hostile survivor class is empty must proceed through at least one of*

(A) population capacity $\vee$ (B) residue/congruence obstruction $\vee$ (C) a GOOD witness.

- (A)** an upper bound for $|C_\lambda|$ (or $|R|$) contradicting a lower bound for the same quantity. This is the only archimedean route left, because by (V81.1) every product container scales with $|R|$.
- (B)** an obstruction inside a container of bounded size. The only such containers in the corpus are the residue class modulo N (Theorem 9.16, Theorem 5.8) and the gap-scale divisors $h-j < h$ of $\text{COB-RD}(3)$.
- (C)** an actual reachable $g \in R$ with $N-g$ prime.

Proof. Immediate from Theorem 9.19 together with the observation that every capacity object in the corpus (T_R , U_R , J_R , D_λ , the FC and LCAP envelopes) is a product over rows or labels of R , hence has logarithm $\Theta(|R| \log N)$, whereas any selected subfamily has logarithm at most $|S| \log N$ with $S \subseteq R$. $\square$

C-P0-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPACITY — the current preferred target

Status. **OPEN**.

Statement. Primitive canonical failed world, $N \geq 500$, with the partition $R = \{p_0\} \uplus C_\lambda \uplus H_\lambda$. By Theorem 9.17 exactly $|H_\lambda|$ rows carry a high factor, each of the form $L\ell$ with $L \in \mathcal{L}_N$. Hence

exactly $|R| - |H_\lambda|$ distinct integers $N - u$ ($u \in R$) are C_λ -smooth and lie in $[m, N)$,

all of them $> (1 - \frac{1}{2\lambda})N$ except the root row $m > 2N/5$. Wanted: an effective upper bound

$$|C_\lambda| \leq \Psi^*(N, \lambda_N)$$

for the number of integers in the window $((1 - \frac{1}{2\lambda})N, N)$ all of whose prime factors lie in a prescribed set of $|C_\lambda|$ primes below $N/(2\lambda)$, contradicting the matching lower bound $|C_\lambda| \geq |R| - |H_\lambda| - 1$ forced by child-closure.

Why it matters. By Theorem 9.20 route (A) is the only archimedean route left; by Theorem 9.17 the high band is completely described, so the whole unknown is the low core; and by (V81.1) a bound on $|C_\lambda|$ is exactly what converts every existing product capacity into a finite quantity.

Already proved. The container identity (V81.1), the slack bound (V81.2), the fringe alphabet, the shadow normal form (V81.11), the residual world R' , and the capacity factorisation.

Honest difficulty statement. This is an analytic question about S -smooth integers in a short interval with prescribed *sparse* support. $\Psi(x, y)$ technology does not apply directly, because the support set is not an initial segment of the primes.

Known obstruction. Evertse's S -unit bound is *vacuous* here: the equation $u + w = N$ with $u \in S$, w S -smooth, $S = R \setminus \{p_0\}$ has $|R| - 1$ solutions, while the bound is $3 \cdot 7^{2|S'|+3}$ with $|S'| \leq |S| + \omega(N)$ — exponential in the very quantity to be bounded. Recorded as FAIL-SUNIT-COUNT-BOUNDS-THE-LOWCORE-001.

Success criterion. The effective bound Ψ^* above, or a proof that no such bound can exist at this scale.

C-P0-PRIMITIVE-SELECTED-RESIDUE-LOCK-CAPACITY — the secondary target

Status. **OPEN**.

Statement. Produce a second, *independent* determination of $\mathcal{C}_Q \bmod N$ (or of $T_{R'} \bmod N$) from the reflection-complete boundary, incompatible with (V81.13) above a computable threshold.

Input. (V81.13), G-ROOT-LOCK (Theorem 5.8), the capacity factorisation (Theorem 9.15), and the gap-scale normal form (V81.11). Reflected carriers entering the exact lock are the natural source of a second determination.

Guard — read before attempting. A congruence that is a formal consequence of the *same* factorisation is *vacuous*. The check performed in Theorem 9.16 shows that combining (V81.12), (V81.13) and G-ROOT-LOCK reproduces only $m \equiv -p_0$. Any candidate must import an *independent* evaluation.

Why it is route (B). It lives inside a container of bounded size (the residue class modulo N), which is the only kind of container Theorem 9.20 leaves for a non-archimedean argument.

The nonprimitive branch. Separately alive but without a consumer. The coefficient lock gives $T_R > \frac{5}{2}N$; a large F_Q^{coeff} yields a superlinear amplifier, while cheap nonprimitive defect bases localise below a small power and into a Baker-accessible reduction. *These are reductions, not a capacity theorem*, and the same container degeneracy (Corollary 9.9) forbids the scalar route there too. The exact Baker requirement recorded in v4.81 is *not* met by the available effective machinery.

Status line, verbatim from the source. Binary Goldbach remains **OPEN**; universal canonical p_0 -success remains **OPEN**; $N_0^{p_0}$ is **not proved**, neither effectively nor ineffectively; no status of G-CONDITIONAL-EFFECTIVE-P0-NO is changed.

9.2.5 Updated proof DAG

$$\begin{array}{l}
p_0(N) \text{ fails, } \quad N \geq 500 \\
\Downarrow \\
F_Q^{\text{coeff}} > 1 \vee F_Q^{\text{coeff}} = 1 \\
\Downarrow \\
\left\{ \begin{array}{l} \textbf{nonprimitive: } T_R > \frac{5}{2}N, \text{ split at } N^{3/40} \quad (\text{no consumer; the scalar route is forbidden}) \\ \textbf{primitive: } m \text{ squarefree, } \kappa_R(q) = 1 \end{array} \right. \\
\Downarrow \quad (\text{primitive}) \\
D_q = 1 \iff \gcd(m, 2h - q) = 1 \iff \gcd(m, 2h - u) = 1 \quad \forall u \in R \setminus \{p_0\} \quad (\text{V81.11}) \\
\Downarrow \\
U_F(J_F B_{\text{re}}) B_{\text{abs}} = \prod_{q \in Q} (N - q) > \left(\frac{5N}{6}\right)^{|Q|} \\
\Downarrow \\
\mathcal{C}_Q \mid U_R J_R D_{\lambda_N} \quad \text{and} \quad \frac{U_R J_R D_{\lambda_N}}{\mathcal{C}_Q} > \frac{5N}{6} \quad (\text{V81.1, V81.2}) \\
\Downarrow \\
\text{archimedean product comparison is immune} \implies \text{routes (A), (B), (C) only}
\end{array}$$

9.3 The C0b branch targets

TI — injective branch obligation

Statement. If the C0b-RD obligation is carried by the class $\mathcal{I}$ (incident coordinates with an *active* lower mirror), then the map $j \mapsto (a_j, \text{PF}(c_j))$ realises production in the world: the active mirrors inject into distinct parent–child incidences, and the number of such incidences is bounded below by $|\mathcal{I}|$ up to an explicit constant.

Why it matters. $\mathcal{I}$ is the only class whose coordinates are already *active graph rows*. If the obligation lands there and TI holds, the boundary obligation converts directly into world structure with no further input. It is the cheapest of the four branches.

Already proved. The reflection gate (Lemma 3.15): for $j \in \mathcal{I}$ the edge $a_j \rightarrow \text{PF}(c_j)$ exists and $\Gamma_N(\{a_j\}) = \text{PF}(c_j)$. Strict non-reuse (Lemma 3.13): every prime $\ell \geq h$ occurs in at most one block element, so distinct $j \in \mathcal{I}$ with a large factor give distinct incidences.

Remaining obligation. Injectivity for the *small* factors, i.e. for primes $\ell < h$, where strict non-reuse gives nothing. Equivalently: bound $\#\{j \in \mathcal{I} : \text{PF}(c_j) \subseteq [2, h)\}$ from above by a fraction of $|\mathcal{I}|$.

Obstructions. (i) The class $\mathcal{I}$ can be empty — $N = 242$ has no active reflected row at all (Warning 3.16); TI is then vacuous and the obligation moves to $\mathcal{D}, \mathcal{S}, \mathcal{F}$. (ii) The guaranteed active mirror exists only in phase Type III (Corollary 6.14), so TI can only be the operative branch on $\theta > 1/2$. (iii) Small-factor collisions are exactly what the widened block would have controlled, and the widened block is refuted (Warning 3.14).

Success criterion. An unconditional inequality of the form $\#\{\text{distinct incidences from } \mathcal{I}\} \geq c|\mathcal{I}|$ with $c > 0$ absolute, valid in all phases where $\mathcal{I} \neq \emptyset$, and *stated in the same functional* that P1^{br} consumes.

Constraining log entries. FAIL-REFLECTION-ACTIVATION-001, FAIL-BROADBLOCK-ENDPOINT, FAIL-TF-LOCAL-SCOPE-001.

TD — root-anchored dominance obligation

Statement. If the obligation is carried by $\mathcal{D}$ (incident coordinates whose lower mirror is *inactive*), then the recovered mass $\sum_{j \in \mathcal{D}} \Omega(a_j) \geq 2\lceil t/3 \rceil$ of C0b-RD translates into a lower bound on *realised production in the world*, anchored at the root, measured by the same functional that bounds capacity.

Why it matters. This is **the smallest missing bridge in the whole project**. C0b-RD is proved and hands over a genuine quantitative obligation; TD is the first step that turns a boundary count into a dynamical count. Every downstream target (P1^{br}, L1, W1, K1, X1) either depends on TD or fails for the same reason TD fails.

Already proved. C0b-RD (Theorem 8.6) including the $\mathcal{D}$ -branch inequality $\sum_{j \in \mathcal{D}} \Omega(a_j) \geq 2k$ under $\mathcal{I} = \emptyset$; the closed-world and unique-source structure (Lemma 3.26); no-bad-return (Theorem 5.33); the exact factorization $T_S = U_S J_S$ (Theorem 3.19).

Remaining obligation. Produce a lower bound

$$(\text{realised production of the world}) \geq \gamma(\text{obligation mass of } \mathcal{D})$$

in which γ is *the same* functional as the one appearing on the capacity side of P1^{br} and K1. Not a comparable functional, not a functional that dominates it on average: the same one.

Obstructions. (i) a_j inactive means a_j is composite or divides N ; the mass $\Omega(a_j)$ is therefore not attached to any graph vertex, and there is no proved route from it to an incidence. (ii) The density formulation TD-DENSITY is **SUPERSEDED** precisely because it needed a bound of the same strength as its own conclusion. (iii) The finite-local universality wall (Corollary 8.3) forbids recovering the bound from any bounded canonical transcript. (iv) Every recorded attempt to substitute a different but “morally equivalent” functional is in the log as a scope error.

Success criterion. A proof of the displayed inequality with an explicit γ , plus a verification that the γ used is literally the one in §9.4. A proof with a different functional is *not* partial progress; it is a new open problem.

Constraining log entries. FAIL-ETA-GAMMA-IDENTIFY-001, FAIL-POTENTIAL-CAPACITY-001, FAIL-CAB-001, FAIL-CAB-INTERNAL-001, FAIL-BABS-PRODUCTION-001.

TS — smooth branch obligation

Statement. If the obligation is carried by $\mathcal{S}$ (coordinates with $P^+(c_j) < h$ and $\text{PF}(c_j) \cap R = \emptyset$), then the smooth mass $\sum_{j \in \mathcal{S}} \Omega(c_j) > k \log x / \log h$ cannot be entirely absorbed inside the world: some smooth coordinate must contribute an escape or a coupling.

Why it matters. $\mathcal{S}$ is the branch where the mass bound is largest (it grows like $\log x / \log h$ per unit), so if the obligation lands there the quantitative margin is greatest. It is the branch most likely to be provable by elementary means.

Already proved. The $\mathcal{S}$ -branch inequality of C0b-RD; minimum-factor escape (Corollary 5.25); shadow cycle localisation (Theorem 5.27), which confines a terminal core inside Q to $(0, 2h/3)$ — exactly the range in which smooth coordinates live.

Remaining obligation. Convert “ c_j is h -smooth and its factors avoid R ” into a statement about the world. The factors of c_j are active primes outside R ; they are candidate escape targets, but their rows have not been shown to reach GOOD.

Obstructions. (i) A factor outside Q or outside R does not mean success — only that the current alphabet was left (warning after Corollary 5.25). (ii) Smoothness bounds *numbers*, not *vertices*; the passage from $\Omega(c_j)$ to distinct active vertices needs the same injectivity that TI lacks.

Success criterion. An unconditional statement: if $\mathcal{I} = \emptyset$ and the $\mathcal{S}$ inequality is the operative one, then $G_{\text{esc}}(N) > 0$. That alone would be a real theorem, since it would give Goldbach for every N in that regime.

Constraining log entries. FAIL-HALL-SUPPORT-001, FAIL-DIRECT-ADJACENCY-001.

TF — fresh branch obligation

Statement. If the obligation is carried by $\mathcal{F}$ (coordinates with $\ell_j = P^+(c_j) \geq h$, active and outside R), then the product bound $\prod_{\ell \in \Lambda_{\mathcal{F}}} \ell \geq h^k$ forces an escape edge or an external coupling.

Why it matters. $\mathcal{F}$ is the only branch that hands over *named active primes outside the world*. Those are exactly the objects an escape argument needs.

Already proved. The $\mathcal{F}$ -branch product inequality; that each ℓ_j is active, outside R , and occurs in exactly one block element (Theorem 8.6 combined with Lemma 3.13).

Remaining obligation. Show that an active prime outside R occurring as a large factor of a reflected coordinate must be, or must lead to, a GOOD vertex.

Obstructions. (i) **Locality is not enough:** at $N = 128$, $\ell = 13$, the local $\mathcal{F}$ certificate holds but the canonical root wins outright and $R = \emptyset$, so the certificate constrains nothing. FAIL-TF-LOCAL-SCOPE-001 (ii) An external parent need not be a success root: (6142, 877, 3511) has 3511 active and above p_0 , yet its own traversal fails. FAIL-EXTERNAL-SUCCESS-001 (iii) External reuse is not local: $(N, \ell) = (1\,825\,372, 52\,153)$ at distance 53 750. FAIL-LOCAL-REUSE-001

Success criterion. A theorem of the form: under canonical failure with $\mathcal{I} = \mathcal{D} = \mathcal{S} = \emptyset$, some $\ell \in \Lambda_{\mathcal{F}}$ has $N - \ell$ prime, or has an active parent outside R whose row meets Good_N . The second alternative must be included — it is provably necessary.

Constraining log entries. FAIL-TF-LOCAL-SCOPE-001, FAIL-EXTERNAL-SUCCESS-001, FAIL-LOCAL-REUSE-001.

9.4 P1^{br} — branchwise production upgrade

P1BR

Statement. Let γ be the capacity functional of the canonical world. P1^{br} asserts that the production realised branchwise — summed over the branches of the first front, each measured on its own subtree — admits a lower bound in γ that exceeds the capacity bound in the same γ :

$$\sum_{\text{branches } \beta} \gamma(\text{production}(\beta)) > \gamma(\mathcal{C}_N(R)).$$

Why it matters. P1^{br} is the node where the boundary obligations become a contradiction. Everything downstream (L1, E1, W1, K1, X1) consumes its conclusion.

Already proved. The branchwise decomposition itself is legitimate: the world is closed with a unique source, so the first front partitions it into branches. The exact ledger $T_S = U_S J_S$ holds on every branch separately. The capacity side $\mathcal{C}_N(R)$ is well defined.

Remaining obligation. A *positive external packet* feeding the *same* P1 capacity — the corpus target C-POSITIVE-EXTERNAL-PACKET-TO-SAME-P1-CAPACITY. Concretely: exhibit a quantity that (a) is provably positive under canonical failure, (b) is external to the world (so it is not already counted in $\mathcal{C}_N(R)$), and (c) is measured in γ rather than in a functional that merely correlates with γ .

Obstructions. (i) The status of P1^{br} was overridden several times in the corpus ([CORE, vNEXT4.25–4.33]) as successive candidate packets were found to be either non-positive or already internal. (ii) $\mathcal{C}_S < \mathcal{D}_S$ from the first front is **KILLED** by $N = 4736$ ($\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$). FAIL-POTENTIAL-CAPACITY-001 (iii) Unconditional CAB is **KILLED** by $N = 322$ and $N = 858$. (iv) The naive support expansion of all sources is **KILLED** by Hall compression $456 \rightarrow 17$ at $N = 51\,032$. FAIL-HALL-SUPPORT-001

Success criterion. A packet satisfying (a), (b), (c) above, with the positivity proved and the functional identity checked symbol by symbol. Any proof that silently changes γ between the production and capacity sides is invalid; that substitution is the single most repeated error in the log.

Constraining log entries. FAIL-POTENTIAL-CAPACITY-001, FAIL-CAB-001,
 FAIL-CAB-INTERNAL-001, FAIL-HALL-SUPPORT-001, FAIL-ETA-GAMMA-IDENTIFY-001,
 FAIL-CHILD-C1-001.

9.5 L1 — actual layer realisation

L1

Statement. The layer part J_R of the canonical world is realised by *actual* distinct layers of the internal filtration, i.e. the support reuse counted by J_R corresponds to genuinely distinct BFS levels rather than to repeated incidences inside one level.

Why it matters. Every capacity argument that uses J_R implicitly assumes this. Without L1, J_R is a bookkeeping quantity with no dynamical meaning, and the Euler-defect route collapses — as the computation in §5.7 shows it does under the naive normalisations.

Already proved. $T_S = U_S J_S$ exactly (Theorem 3.19); the world is closed with a unique source, so the BFS filtration is well defined and canonical.

Remaining obligation. A lower bound on the number of layers in terms of $\log J_R$, or a counterexample. Note that the two false normalisations (N1) and (N2) of §5.7 are exactly failed attempts at L1 in disguise.

Obstructions. The measured data go the wrong way: 3 804 of 21 093 canonical worlds violate the (N2) inequality, so support reuse is systematically *larger* than the layer count can absorb.

Success criterion. Either a proof of $\nu(R) \geq c \log J_R$ for some $c > 0$, or an explicit family showing $\nu(R) = O(1)$ with J_R unbounded (which would kill L1 and, with it, every capacity route that passes through J). The second is probably easier and is listed as an experiment in Chapter 16.

Constraining log entries. FAIL-ETA-GAMMA-IDENTIFY-001 and the computational refutations of (N1), (N2) recorded in §5.7.

9.6 E1 — escape or coupling

E1

Statement. Under canonical failure, every internal base atom $\ell \in B_{\text{abs,int}}$ either has a GOOD edge reachable inside one of its first-front branches, or has an active parent outside R . (This is the *escape-or-reuse* alternative promoted to a theorem.)

Why it matters. E1 is the only frontier target with overwhelming computational support and a completely explicit statement: 61 452 instances with 0 orphans, plus an independent red team over 60 413 atoms with 0 orphans. If any target is true, this one is.

Already proved. Retention-escape (Theorem 5.24); minimum-factor escape (Corollary 5.25); the prime-power witness (Proposition 5.29); $B_{\text{abs,int}} = B_{\text{abs}}$ without vacuity [CORE, ARS-BABS-INTERNAL].

Remaining obligation. The disjunction itself, unconditionally. Both alternatives are provably necessary: (6142, 877, 3511) shows the second cannot be dropped.

Obstructions. (i) External reuse is not local (FAIL-LOCAL-REUSE-001), so the “active parent outside R ” cannot be found by bounded search. (ii) The external parent need not itself be a success root (FAIL-EXTERNAL-SUCCESS-001). (iii) Escape need not be directly adjacent (FAIL-DIRECT-ADJACENCY-001: at $N = 14198$ the atoms 19 and 167 escape at distances 8 and 7). (iv) Charging multiplicity is observed to be 1 but is *not* proved bounded, and the constant- $C = 1$ charging scheme is **KILLED** (53 violations, maximum 9.42243, FAIL-CHILD-C1-001).

Success criterion. A proof of the disjunction, or a bounded-charging lemma (multiplicity $\leq C$ for an absolute C) which the corpus identifies as the missing ingredient.

Constraining log entries. FAIL-LOCAL-REUSE-001, FAIL-EXTERNAL-SUCCESS-001, FAIL-DIRECT-ADJACENCY-001, FAIL-CHILD-C1-001.

9.7 W1 — simultaneous clean witnesses

W1 — corrected

Statement. Several clean witnesses can be selected simultaneously at the *actual* layer scale of the world: a selection of parents, one per branch, simultaneously clean (no shared support, no double charging), covering the layers that really occur in R rather than only the polylogarithmic ones.

Correction to an earlier reading. **SCOPE-CORRECTED**. Earlier compressions listed G-CLEAN-POLYLOG-PARENT-SELECTION as W1's open target. It is *not* open: it is a **PROVED** theorem of the CRD Core (Theorem 5.15), giving exactly such a clean selection for every finite layer family with $\ell^k \leq (\log N)^A$, with global support load at most $\rho(\mathcal{L})$ — and load one when the base primes are distinct. See Warning 5.16.

Why it matters. Capacity arguments need many witnesses at once, and the selection problem is *solved* in the polylogarithmic range. What is missing is the range, not the selection.

Already proved. Theorem 5.15 (the selection itself); strict non-reuse (Lemma 3.13) for primes $\ell \geq h$; the global two-sheet collision budget (Theorem 5.34).

Remaining obligation. Extend the selection from $\ell^k \leq (\log N)^A$ to the actual layers of R . That is precisely the content of L1 (§9.5): **W1 is blocked on L1, not on a selection lemma**. If L1 is refuted — and §5.7 suggests it may be — then W1 is not merely open but misconceived, and the correct object is the polylogarithmic family alone.

Obstructions. Hall-type compression destroys naive selections outside the proved range: at $N = 51\,032$ the naive support expansion of all sources compresses $456 \rightarrow 17$ (FAIL-HALL-SUPPORT-001). This is not a technical nuisance — it says the bipartite incidence structure has small vertex covers, which is exactly what a simultaneous selection must avoid, and it is why Theorem 5.15 needs Siegel–Walfisz and a lower-bound sieve rather than a combinatorial argument.

Success criterion. Either a clean selection at the actual layer scale, or a proof that the polylogarithmic range is the maximum possible — which would close W1 negatively and remove it from the frontier.

Constraining log entries. FAIL-HALL-SUPPORT-001, FAIL-CHILD-C1-001.

9.8 K1 — source-sensitive capacity

K1

Statement.

$$K_{\times}(R, \mathcal{P}) + K(\mathcal{P}) \leq \mathcal{C}_N(R)$$

where $K_{\times}$ is the cross term between the world R and the selected parent set $\mathcal{P}$, $K(\mathcal{P})$ the internal term of $\mathcal{P}$, and $\mathcal{C}_N(R)$ the capacity of the world — all three measured in the *same* functional as $P1^{\text{br}}$.

Why it matters. K1 is the quantitative closure: with $P1^{\text{br}}$ above and K1 below, the two bounds cross and canonical failure is excluded.

Already proved. The capacity side $\mathcal{C}_N(R)$ is well defined and computable; the world's ledger identities (IL1–IL6) hold on Q_{smooth} .

Remaining obligation. *Exactly the same defect as TD.* K1 needs the cross term and the capacity to be expressed in one functional. Every recorded attempt produced two functionals with a comparison that holds on average and fails pointwise. The corpus’s vertical-origin overrides ([CORE, vNEXT4.8–4.13]) end at the target **C-NATURAL-SCALE-WEIGHTED-CONSUMER**: a consumer of the packet weighted by a natural scale, which has never been produced.

Obstructions. (i) The same-functional obstruction, shared with TD and P1^{br} — these three are not independent problems. (ii) $N = 4736$ kills the first-front form of the capacity inequality. (iii) $N = 322$ and $N = 858$ kill the unconditional CAB form.

Success criterion. The displayed inequality with all three terms written in one explicit functional and proved unconditionally under canonical failure. Equivalently: solve TD and K1 together, since a solution to one in the correct functional very likely gives the other.

Constraining log entries. FAIL-POTENTIAL-CAPACITY-001, FAIL-CAB-001, FAIL-CAB-INTERNAL-001, FAIL-ETA-GAMMA-IDENTIFY-001.

9.9 X1 — canonical failure exclusion

X1

Statement. For every even N in the non-diagonal case, the canonical traversal from p_0 terminates in SUCCESS. Equivalently: $\text{PF}(N - p_0) \not\subseteq B_N^{\text{low}}$ whenever $N - p_0$ is composite (Theorem 3.29).

Why it matters. X1 is the terminal node. It implies binary Goldbach. It is the central hypothesis of the whole project, written here as (**H**- p_0).

Already proved. Nothing towards X1 itself. The fixed-point reformulation (Theorem 3.29) is exact but not progress.

Remaining obligation. Everything above, plus the passage from “the capacity inequality fails” to “the traversal succeeds”.

Obstructions. The architecture wall (§8.2) — no finite local certificate can do it; the parity barrier (§11.4) — no purely sieve-theoretic route reaches a prime complement; and the fact that X1 is *strictly stronger* than Goldbach for each N , so any proof strategy that would also prove Goldbach must be at least as hard as Goldbach.

Success criterion. A proof. Nothing weaker counts; in particular no finite scan, however large, changes the status of X1.

Computational status. **VERIFIED COMPUTATION**: no canonical failure for even $N \leq 10^7$ (this project, exhaustive). Independent Goldbach verification reaches $4 \cdot 10^{18}$ [12] but says nothing about X1.

9.10 RPD — repeated prime-power redundancy

RPD

Statement. The repeated external prime-power layer redundancy is bounded:

$$D_{\text{high}}(H_Q) \leq C \log \log N \cdot \sum_{\text{layers}} (\kappa - 1)_+ + o(M),$$

equivalently $D_{\text{high}}(H_Q) \leq O(\log \log N) \cdot \eta_{\text{high}} + o(M)$, where M is the total mass and η_{high} the high-range excess valuation.

Why it matters. If RPD is true then $\Delta_J \gg M$: the layer defect dominates the mass, which is exactly the quantitative separation that the capacity route needs. RPD is the one open target that would supply the *margin* rather than merely the structure.

Already proved. $T_S = U_S J_S$; the high/low decomposition of the valuation mass; the prime-power witness (Proposition 5.29).

Remaining obligation. The $\log \log N$ bound. The natural approach is a divisor-type bound on how often a fixed prime power can recur across layers, which is where $\log \log N$ would come from.

Obstructions. (i) The bound is asserted for the *external* redundancy; the internal part is already controlled and is not the difficulty. (ii) The $o(M)$ term hides the interaction between layers, and no proof technique in the corpus separates layers without losing a factor of the size that RPD is supposed to save.

Success criterion. A proof of the displayed inequality with an explicit C , or a family of N showing D_{high} can exceed $\eta_{\text{high}} \log \log N$ by an unbounded factor.

Provenance note. RPD is recorded in the corpus as a v4.45 open target inside a commented archive block of the CRD failure log; it is reproduced here because it is the only surviving open target that carries an explicit quantitative margin. Its exact original phrasing should be checked against the source before it is used as a hypothesis.

9.11 Open targets specific to p_0 alone

These are stated *without* reference to a Goldbach proof and each is a self-contained problem.

C-SAMEFUNC-BRIDGE — global same-functional margin bridge

Statement. There is a single functional γ in which both the production lower bound (TD, $P1^{\text{br}}$) and the capacity upper bound (K1) can be stated, with a positive margin between them.

Why it matters. This is the abstract form of the obstruction shared by TD, $P1^{\text{br}}$ and K1. Solving it solves all three; failing to solve it blocks all three. Anyone attacking one of them individually should first decide whether they are really attacking this.

Already proved. Nothing. Several *pairs* of functionals are known in which one side is provable; no pair is known in which both are.

Remaining obligation / obstruction / success criterion. As for TD; see §9.3.

Constraining log entries. FAIL-ETA-GAMMA-IDENTIFY-001 and every entry filed under capacity.

C-GENEALFREE-DYN — boundary to dynamic escape

Statement. Does the absence of any genealogical boundary defect ($\kappa_{\text{gen}} = 0$) force the whole C0b-RD obligation into *actual* dynamical channels?

Why it matters. This is the single missing implication between Chapter 6 (exact boundary structure) and any dynamical statement. The entire “why p_0 works” question reduces to it.

Already proved. The boundary side completely (Theorems 6.2–6.6); the obligation side completely (Theorem 8.6). The implication between them: nothing.

Minimum acceptable form. A theorem whose conclusion *actually fails*, or acquires an explicit leakage term, when the defect rank is one. Anything that also holds at rank one is not progress — it is a class-B result in disguise.

Obstruction. Warning 6.7(2): the statement $\kappa_{\text{gen}}(r) = 0$ may be informationally equivalent to “ r is the first upper prime”. If so, no theorem can extract more from it than from the ordering of the primes, and the target should be restated.

Constraining log entries.

FAIL-P0-ZEROHOLE-ESSENTIAL-001,

FAIL-P0-INFORMATION-SELECTOR-001.

C-POP1-SEP — one-defect dynamic separation

Statement. Prove a dynamical difference between p_0 and p_1 that is caused by the single embedded channel of the earlier root, not by aggregate size.

Why it matters. This is the sharpest available test of C-GENEALFREE-DYN and the one with the best data.

Already proved. The categorical separation on the boundary (κ_{gen} : 0 versus 0.9041 mean, maximum 1) and the aggregate near-identity of every other observable (§6.5.2).

Remaining obligation. Isolate causality. The aggregate observables differ by 6–24%, which is consistent with pure size effects; the defect rank differs categorically but has negligible logarithmic mass.

Obstruction. FAIL-P0-ZEROHOLE-ESSENTIAL-001: the mass of the hole is $O(\log N)$, so any argument that routes the separation through mass is dead before it starts.

Success criterion. A statement true for p_0 and false for p_1 whose proof uses the hole *structurally* (as absence of a multiplicative carrier, [CORE, V71.44]) and not quantitatively.

9.12 Smaller open questions about p_0

Each of the following is a self-contained problem that does not mention Goldbach.

Q1. Does p_0 have dynamical consequences? **OPEN**. The signature $(\kappa^+, \kappa^-) = (0, \rho_0)$ is the first boundary invariant that varies nontrivially between canonical worlds. Does $\rho_0 \geq 1$ (Type III, where a guaranteed active reflected row exists) correlate with, or imply, anything about the traversal? Preliminary data: immediate success occurs in 18.4% of Type III cases and is *impossible* in Type I — but that is a consequence of the definition, not of dynamics.

Q2. Does the escape margin grow? **OPEN**. Is $G_{\text{esc}}(N) \rightarrow \infty$? Data (§6.7): the per-dyadic-block minimum runs 1, 1, 1, 1, 1, 1, 2, 2, 3, 3, 3 up to $2 \cdot 10^5$. A theorem “ $G_{\text{esc}}(N) \geq f(N)$ for increasing f ” would be strictly stronger than the existence of a Goldbach pair and is, as far as is known, untouched.

Q3. Is there a missing state variable? **OPEN**. Four pairs of structural twins survived enrichment by all global-source information; they are separated only by the *internal witness signature* (localisation, nearest escape distance in the same branch, external-reuse flag) [F3]. Is there a scalar invariant, or is the system state intrinsically a profile?

Q4. A second canonical anchor? **OPEN**. Let $p^* = \max\{p \in \mathbb{P} : p < N\}$ and $g = N - p^*$. Then $g \leq N - p \leq m_0$ for upper sources, $p^* - p_0 = m_0 - g$, and $\gcd(m_0, g) = \gcd(m_0, p^* - p_0) = \gcd(g, p^* - p_0)$ **PROVED** [BIB]. Does p^* have its own rigidity? Evidence against: p^* finances on average only 25.07% of the B_{abs} atoms against 81.44% for the whole ensemble of upper sources; priority assessed as low [D4, q. 11, 21].

Q5. Is p_0 exceptional outside this one graph? **OPEN**. An analogous selector appears in the Erdős–Straus model and in the Schinzel model [BIB]. Does “the least element above the midpoint, chosen unconditionally” have the same reflection-completeness property there? The definition of the reflected block transfers verbatim, so the question is well posed.

Q6. Diagonal closure. **OPEN** (Scope note 3.17). What is the right formulation of the genealogical layer covering $x \in \mathbb{P}$? Replacing $[x, r)$ by (x, r) makes the theorem unconditionally true, but then p_0 is no longer uniquely characterised. Is there a simultaneous formulation?

Q7. Invariants of the boundary filtration. **OPEN**. The filtration $\mathfrak{B}_0 \subset \mathfrak{B}_1 \subset \dots$ (Theorem 6.8) with the coupling-matrix embedding (Corollary 6.9) is a purely combinatorial structure independent of Goldbach. Does the sequence $(M^{(k)})_k$ have invariants — rank, spectrum, zero pattern — that do not reduce to the gap sequence $g_1, g_2, \dots$?

Q8. Prove the active-mass Euler bound. **OPEN**. Normalisation (N3) of §5.7, $\sum_q d_R(q) \geq |R| + 1$, held in all 21 093 canonical worlds tested. It is the only nontrivial surviving form; a proof would be small but real, and would repair the part of the corpus that used the false normalisations.

Chapter 10

Computational Evidence

Everything in this chapter is **VERIFIED COMPUTATION** or **EMPIRICAL PATTERN**. **Nothing here is evidence for a universal statement.** A range is part of a claim: quoting a number without its range is an error, and quoting a number without its *domain* (which roots, which complements, whether unit rows are included) is the error that produced the “301” confusion (FAIL-CENSUS-DOMAIN-001).

Every number below is regenerated by the single script in Appendix A, which has no external dependencies. The block IDs EXP1–EXP8 are the script’s own.

10.1 Provenance and reproduction

ID	What it computes	Range	Key result
EXP1	exhaustive canonical scan, census mode	even $N \leq 10^7$	0 failures; mean 7.0527 nodes
EXP2	verdict of roots $p_0, \dots, p_6$	even $N \leq 2 \cdot 10^5$	0 for $k = 0$; 2, 4, 5, 3, 2, 1 for $k \geq 1$
EXP3	census of all failing upper roots	even $N \leq 2 \cdot 10^4$	301 roots, six N , 0 canonical
EXP4	escape margin G_{esc}	even $N \leq 2 \cdot 10^5$	52 numbers with margin 1; 10 with $ W \geq 3$
EXP5	phase trichotomy and ρ_0	even $N \leq 2 \cdot 10^5$	0 violations; $\rho_0 \geq 1 \iff \text{Type III}$
EXP6	embedding of reflected blocks	$N \leq 2 \cdot 10^4, k \leq 6$	731 446 indices, 0 violations
EXP7	fixed point B_N^{low}	even $N \leq 6 \cdot 10^4$	nonempty for exactly six N
EXP8	observables p_0 vs p_1	even $N \leq 2 \cdot 10^5$	aggregates $\pm 6\text{--}24\%$; $\kappa_{\text{gen}}: 0 \rightarrow 0.9041$
EULER	Euler-defect normalisations (N1)–(N4)	even $50 \leq N \leq 6 \cdot 10^4$	(N1) 6 044 and (N2) 3 804 violations; (N3), (N4) none
CERT	the 34 hostile certificates of the corpus	fixed witnesses	34/34 pass

Exact invocation reproducing the numbers in this file:

```
python3 p0_experiments.py -scan 10000000 -ablation 200000 -census 20000
                        -margin 200000 -phase 200000
                        -filtration 20000 -fixedpoint 60000
                        -observables 200000 -json data_p0.json
```

EXP1 at `-scan 10000000` takes about 75 s and about 50 MB; EXP7 at `-fixedpoint 60000` about 70 s. The defaults (`-scan 1000000 -fixedpoint 20000`) finish in a few minutes and reproduce everything except the 10^7 scan records.

10.2 The canonical scan

EXP1. **VERIFIED COMPUTATION**. All even $4 \leq N \leq 10^7$, canonical root, **census** mode:

cases	4 999 999
failures of p_0	0
mean number of processed vertices	7.0527
record vertex count	151 ($N = 3\,542\,948$)
record success depth	10 ($N = 11\,822$)

The same scan restricted to $N \leq 10^6$ gives 499 999 cases, 0 failures, mean 5.8044, record 133 ($N = 976\,904$), record depth 10 ($N = 11\,822$) — reproducing the table of [BIB] to the last digit. The depth record 10 is attained only by $N = 11\,822$ and remains stable through the whole 10^7 range.

Distribution of the success depth $d(N)$ over all 4 999 999 cases:

d	0	1	2	3	4	5	6	7	8	9	10
$\#N$	983 722	1 426 153	1 500 435	802 099	240 796	42 059	4 436	276	18	4	1

$d = 0$ means immediate success ($N - p_0$ prime, or the diagonal case).

Warning 10.1 (What the scan does not say). 0 failures up to 10^7 is consistent with a failure at $10^7 + 2$. The scan is **VERIFIED COMPUTATION** and stays **VERIFIED COMPUTATION**. In particular it is *not* evidence for X1 (§9.9); it is the statement that X1 has no small counterexample.

10.3 Root ablation and the census

EXP2 — verdict of the root p_k . **VERIFIED COMPUTATION**. All even $N \leq 2 \cdot 10^5$, roots $p_0, \dots, p_6$; failures on a *composite* row and degenerate failures on the unit row $N - p_k = 1$ counted separately:

k	0	1	2	3	4	5	6
failures, composite row	0	2	4	5	3	2	1
failures, unit row	0	4	2	3	3	2	2

Smallest failures on a composite row: $k = 1$: $(N, p_1) = (1862, 941)$; $k = 2, \dots, 5$: $(128, 73), (128, 79), (128, 83), (128, 89)$; $k = 6$: $(1862, 977)$.

The full census EXP3 and the six special N are in §6.6; the escape margin EXP4 is in §6.7; the

phase census EXP5 in §6.3; the filtration check EXP6 in §6.2; the fixed point EXP7 in §6.6; the ablation EXP8 in §6.5.2. They are not repeated here.

10.4 Hostile certificates

CERT — 34/34 pass. **VERIFIED COMPUTATION**. Every finite certificate cited anywhere in the corpus was re-run against the reference implementation of §3.4. Groups:

A1 regression on the record cases $N = 11\,822$ (depth 10) and $N = 3\,542\,948$ (151 vertices).

B1 gap equivalence (Theorem 3.10) on all even $N \leq 2 \cdot 10^5$.

C1–C5 genealogy, reflection completeness, filtration embedding, and the diagonal calibration ($N = 14$, $r = 11$).

D1–D5 canonical geometry, shadow identity, Bézout invariant, no-bad-return, unique source.

E1–E14 the hostile counterexamples: $N = 242$ (reflection activation), $N = 128$ (TF local scope), $N = 322$ and $N = 858$ (CAB), $N = 4736$ (potential capacity), $N = 6142$ (external success), $N = 14198$ (direct adjacency), $N = 20$ (four-channel ledger), $N = 1718$ (generic absorption), $N = 51\,032$ (Hall compression), Exact-D countermodels, the $K_{3,3}$ grid at $N = 10^6$, the $N = 2200$ two-core, and the $h = 1$ witnesses 103 344 and 364 844.

F1–F5 ablation and the 301 census with its exact domain.

All 34 pass. Where a corpus claim and a certificate disagreed, the disagreement is recorded as a scope correction in §11.1, not silently resolved.

10.5 Other observables

Depth. **EMPIRICAL PATTERN**. The success depth of the canonical traversal is ≤ 2 for 3 910 310 of 4 999 999 cases (78.21%) and ≤ 4 for 4 953 205 cases (99.06%). The tail is extremely thin: a single case reaches $d = 10$. No mechanism is known that would force a bounded depth, and the depth is *not* the right measure of difficulty — see §6.7.

p_0 is not measurably better at escaping. **EMPIRICAL PATTERN**. Beyond the aggregate table of §6.5.2: within the 301 failing upper roots of EXP3, the canonical root wins at all six N , but the set of failing roots is neither an initial nor a final segment of $p_1, p_2, \dots$ (Example 6.24: p_1 wins, p_2 fails). Ordering by height does not order the outcome.

Two censuses agree. **EMPIRICAL PATTERN**. The six N with $B_N^{\text{low}} \neq \emptyset$ (EXP7, $N \leq 6 \cdot 10^4$) coincide with the six N carrying failing upper roots (EXP3, $N \leq 2 \cdot 10^4$) and with the six numbers with an exceptional autonomous component reported to 10^8 by Božek [8]. The three lists measure different things; their coincidence is data, not a theorem, and it has not been checked beyond $6 \cdot 10^4$ in this project.

Chapter 11

Failed and Killed Approaches

This chapter is the curated narrative; Chapter 12 is the complete raw record. Read this one first: it groups 239 individual entries into the dozen mistakes that actually recur. A model that internalises this chapter will not need the raw log except to check a specific ID.

The five recurring shapes of error.

1. **Domain mixing** — quoting a result outside the class (A/B/C/D) on which it was proved. Most common single failure.
2. **Functional substitution** — proving a bound in one functional and consuming it in another. Kills TD, P1^{br}, K1.
3. **Vacuity mistaken for canonicity** — a construction that “only works for p_0 ” because for p_k it is undefined on k coordinates.
4. **Finite certificate** — deriving a contradiction from a bounded canonical transcript. Provably impossible (§8.2).
5. **Promotion** — turning a scan with zero counterexamples into a theorem, or an exact identity into a mechanism.

11.1 Curated ledger of negative results

The table is complete with respect to all sources. Rows are grouped by theme; each entry carries an exact counterexample or an exact reason.

Claim / route	Status	Exact reason
Exceptionality of p_0		
Zero hole count as a mass mechanism	KILLED	loss $O_k(\log N)$ against block mass $t_k \log(x/h_k)$ FAIL-P0-ZEROHOLE-ESSENTIAL-001
p_0 as information selector	SUPERSEDED	every clause survives ablation; not refuted, but not established FAIL-P0-INFORMATION-SELECTOR-001
MR2 as a canonical identity	KILLED	$q \mid (r - 2j) \iff q \mid (h_r - j)$ for any upper root FAIL-MR2-CANONICAL-ONLY-001
Least-upper-parent as selector	KILLED	eleven non-canonical roots share the property [CORE, V71.41]
Prime-free boundary activates a reflected row	KILLED	$N = 242$: $a_1 = 117$, $a_2 = 119$ both composite FAIL-REFLECTION-ACTIVATION-001
Statistical dominance of p_0	KILLED	median rank percentile 0.20982 [D4, q. 12]

Claim / route	Status	Exact reason
Minimality determines local geometry	KILLED	23+15 transitions $B_{\text{abs}} \leftrightarrow B_{\text{re}}$ at fixed p_0 [D4, q. 17]
Largest-factor heuristic	KILLED	$N = 124$ [PCRD, FAIL-P0-LARGEST-FACTOR-TWOSTEP]
Finite canonical data give a contradiction	KILLED	affine programmability and finite bad-tree universality [PCRD]
Mechanisms around p_0		
Literal label conservation along a path	KILLED	$v_\ell(N - \ell) = v_\ell(N) = 0$ [LOG]
Absorption trichotomy from factor-flow alone	KILLED	$N = 1718$, $\ell = 59$: both payments have zero valuation FAIL-GENERIC-ABSORPTION-001
Four-channel ledger on all of Q_0	KILLED	$N = 20$: a GOOD row outside R gives $\kappa_R < m_\ell$ [ILA]
Unconditional CAB	KILLED	$N = 322$: $U_R = J_R = 1$, $B_{\text{abs}} = 11 \cdot 29$; internal version $N = 858$ FAIL-CAB-001, FAIL-CAB-INTERNAL-001
$\mathcal{C}_S < \mathcal{D}_S$ from the first front	KILLED	$N = 4736$: $\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$ FAIL-POTENTIAL-CAPACITY-001
Escape must be directly adjacent	KILLED	$N = 14198$: atoms 19,167 at distances 8 and 7 FAIL-DIRECT-ADJACENCY-001
External reuse is local	KILLED	$(N, \ell) = (1825372, 52153)$, distance 53750 FAIL-LOCAL-REUSE-001
External parent is a success root	KILLED	(6142, 877, 3511): 3511 active, above p_0 , own traversal fails FAIL-EXTERNAL-SUCCESS-001
Constant $C = 1$ in charging	KILLED	53 violations, maximum 9.42243 FAIL-CHILD-C1-001
Naive support expansion of all sources	KILLED	Hall compression $456 \rightarrow 17$ at $N = 51032$ FAIL-HALL-SUPPORT-001
Exact-D local algebra replaces minimality	KILLED	non-canonical models die only on $h < G$ FAIL-EXACTD-NONCANONICAL-001
Failure always produces nontrivial B_{abs}	KILLED	FAIL-BABS-PRODUCTION-001
Widened block in the non-reuse lemma	KILLED	the endpoint pair has difference $2h$ FAIL-BROADBLOCK-ENDPOINT
Euler defect in normalisation (N1) or (N2)	KILLED	6044 resp. 3804 violations for $N \leq 6 \cdot 10^4$ (§5.7)
Scope corrections		
Upper-half compression under p_0 failure	SCOPE-CORRECTED	the theorem assumes $\text{Good}_N = \emptyset$ FAIL-UPPERHALF-SCOPE-001
Local $\mathcal{F}$ certificate + $h < G$ forces routing	SCOPE-CORRECTED	$N = 128$, $\ell = 13$: certificate holds, p_0 wins, $R = \emptyset$ FAIL-TF-LOCAL-SCOPE-001
Source-free kernel = EAC = terminal component	SCOPE-CORRECTED	three different constructions FAIL-KERNEL-EAC-001
Parent lattice over all primes	SCOPE-CORRECTED	the correct object is the intersection with $\mathcal{A}_N$ FAIL-PARENT-LATTICE-DOMAIN-001

Claim / route	Status	Exact reason
“301” counts all failing roots	SCOPE-CORRECTED	the census covers composite complements only FAIL-CENSUS-DOMAIN-001
Genealogy and reflection-completeness without hypothesis	SCOPE-CORRECTED	diagonal case, $N = 14$ (Scope note 3.17)
Formula $\rho_0 = \#\{D_s < 2h\}$	SCOPE-CORRECTED	missing lower condition $D_s > h$ (Warning 6.18)
High-gap amplification as unconditional	SCOPE-CORRECTED	hypothesis is <i>pair failure</i> , not CRD failure (Warning 7.5)
$q < m_0/3$	SCOPE-CORRECTED	conject is $q \leq m_0/3$; $N = 32$ attains equality (Scope note 5.4)
IL1–IL6 on all of Q	SCOPE-CORRECTED	domain is Q_{smooth} (Warning 7.7)

11.1.1 What survived hostile testing

- Incidence ledger IL1–IL6 on the domain $S = Q_{\text{smooth}}$: two independent reconstruction routes, 1 000 worlds, 0 errors [ILA] — AUDITED IDENTITY.
- The escape-or-reuse alternative: 61 452 instances, 0 orphans; independent red team 60 413 atoms, 0 orphans [ORPH, RT] — VERIFIED COMPUTATION, as a theorem OPEN.
- The whole genealogical layer (§6.1) and the filtration (§6.2) — PROVED.
- The phase diagram and the bidirectional rank (§6.3, §6.4) — PROVED.
- Global two-sheet collision budget, no-bad-return-to- p_0 , the B_N^{low} characterisation — PROVED.
- All 34 hostile certificates of the corpus were reproduced here and pass (§10.4).

11.2 Post-CRD successor theories

[PCRD] documents a sequence of theories built after CRD1, each intended to preserve information lost by its predecessor. All reached a wall. The table summarises them; their proofs are *not* reproduced here.

Theory	Fundamental object		What it was to recover	What stopped it
CRD1	support/valuation dynamics from p_0		reachability, shadow, capacity	the “same container” argument, generic smooth counting, residue saturation, exact self-consistency of the fixed point
CMFT	lossless factorization (δ, q, a, b)	mirror-tensor	parent-valuation-cofactor correlation	the least-factor tree <i>is</i> the Buchstab/sieve decomposition; the cofactor star at fixed q is an affine bijection
CGST	canonical cell N_h and the convolution $a_P * a_P$		exact additive pairing + phase	the factorization layer collapses diagonally in (h, t) ; exception propagation reduces to prime-pair geometry
CSD	exceptional spectral configurations + the gap		a pointwise inverse anomaly	finite conductor patterns are consistent with an arbitrarily long <i>fixed</i> canonical gap

Theory	Fundamental object	What it was to recover	What stopped it
CAPT	Li–Liu almost-prime witnesses	a massive real family of sources	at fixed (r, s) the system reduces to two linear prime forms; $(3, 3)$ is the natural degeneration
RSD	rough reservoir + Liouville sign + P_3 correction	explicit exposure of parity	either level of distribution $> 2/3$, or a pointwise binary prime–Liouville correlation
rough- P_2 matrix	$A_N(q, r) = \mathbf{1}_{\mathbb{P}}(N - qr)$	global higher-order correlation	boundary blindness, local spectrum = singular series, co-degrees = prime-tuple problem, $K_{3,3}$ grids exist arithmetically
H1-FP	root-factor descent at $h = 1$ with shift-2 orthogonality	elementary closure of the 2-core	reduces to a Pillai/Bennett equation $a^\alpha - a = b^\beta - b$

Example 11.1 (A $K_{3,3}$ grid at $N = 10^6$). **VERIFIED COMPUTATION**. For $q \in \{61, 67, 73\}$ and $r \in \{113, 179, 599\}$ all nine numbers $N - qr$ are prime, and all six factor vertices are BAD. A full rank-two bipartite rough- P_2 grid is therefore arithmetically real and is *not* a contradiction. [PCRD, hostile exact survivor]

Post-CRD verdict on the p_0 programme. **SUPERSEDED**. [PCRD] closes the proof programme based on p_0 -exceptionality: “ p_0 as a standalone Goldbach proof engine — killed / superseded”. The scope of that verdict is precise and is *not* an impossibility theorem: it means that every mechanism identified so far in which p_0 -exceptionality was to generate closure has been either killed, or reduced to a known open barrier, or left without an independent consumer.
What remains alive: p_0 as a selector, a coordinate, a source of exact lemmas, and a laboratory for failure — which is exactly the subject of this file.

11.3 Surviving lemmas of the killed programmes

A killed theory is not a theory whose theorems are false. Each successor programme of §11.2 died at an *architecture* wall, and each left behind correct, reusable lemmas. Compression is exactly where those get thrown out with the dead programme, so they are collected here with their statuses intact. *Every item below is **PROVED** in [PCRD] unless marked otherwise; the proofs are not reproduced here.* Using one of them is legitimate; restarting its parent programme is not.

11.3.1 From CMFT (Canonical Mirror Factorization Theory)

CMFT’s objects: for $N = 2x$, $\mathcal{U}_N = \{p \in \mathbb{P} : x \leq p < N\}$, $\mathcal{M}_N = \{m \leq x : N - m \in \mathbb{P}\}$, so that Goldbach is exactly $\mathcal{M}_N \cap \mathbb{P} \neq \emptyset$; the canonical root has mirror anchor $m_0 = N - p_0 = \max \mathcal{M}_N$; and the CMFT atom is $\mathfrak{e} = (\delta, q, a, b)$ with $m_0 - \delta = q^a b$ and $p_0 + \delta + q^a b = N$.

Theorem 11.2 (Lossless anchor filtration, PCRD-CMFT-1). **PROVED**. For upper primes $p_0 < p_1 < \dots$ with $m_j = N - p_j$, the full left history of the anchor p_j grows by adjoining the entire hyperedge row m_j :

$$\mathcal{H}_{j+1} = \mathcal{H}_j \sqcup \mathcal{A}_j, \quad L_{j+1}(\ell, k) - L_j(\ell, k) = \mathbf{1}_{\ell^k | m_j}.$$

The ablation $p_0 \rightarrow p_1$ loses exactly the full factorization of m_0 .

Why it is worth keeping. This is the precise *information-theoretic* form of the ablation result of §6.5.2: it says what is lost, not merely that something is.

Theorem 11.3 (Tail hard-edge universality, PCRD-CMFT-2). **PROVED** / *no-go*. For any later anchor p_j , after truncating the earlier prefix its complement m_j becomes the maximal element of the tail mirror field and satisfies identical local gcd, offset and bilinear identities. Hence

the local geometry of the tail does not distinguish p_0 .

The genuinely unique information carried by p_0 is only the zero lost prefix.

Reading. This is the sharpest available formulation of what §6.5 establishes empirically, and it is a theorem. Combined with Warning 6.7(2) it is the strongest reason to doubt that any purely local property of p_0 can do dynamical work.

Theorem 11.4 (Canonical least-factor split, PCRD-CMFT-3). **PROVED**. For a composite mirror row m choose uniquely $q = P^-(m)$, $b = m/q$. If $q > x^{1/3}$ then b must be prime, hence

$$m \text{ composite} \implies P^-(m) \leq x^{1/3} \quad \vee \quad m = qr \text{ a balanced semiprime.}$$

Why it is worth keeping. It repairs the divisor-multiplicity bias of any raw bilinear form over rows: without it, divisor-rich rows are counted $\tau(m)$ times.

Theorem 11.5 (All-root collision identity, PCRD-CMFT-4). **PROVED**. For a composite row family S ,

$$\sum_{\ell} \sum_{k \geq 1} \binom{\nu_{\ell,k}(S)}{2} \log \ell = \sum_{i < j} \log \gcd(m_i, m_j).$$

A pigeonhole over least factors yields a positive collision floor.

Limitation, recorded by the source. The floor has *no consumer*: nothing in the corpus converts it into a contradiction. Compare Theorem 7.30, which is the residual-row analogue and does have one.

Theorem 11.6 (Prime-pressure failure ceiling, PCRD-CMFT-5). **PROVED**. With $\mathcal{P}_N = \sum_{m \in \mathcal{M}_N} \Lambda(m)$, a hypothetical Goldbach failure eliminates every $k = 1$ contribution, so

$$\mathcal{P}_N \leq E(x) := \sum_{r^k \leq x, k \geq 2} \log r \leq \sqrt{x} \log x, \quad \text{whence} \quad \boxed{\mathcal{P}_N > \sqrt{x} \log x \implies \text{Goldbach.}}$$

Reading. A clean quantitative reformulation of Goldbach for a single N , entirely elementary. It did not supply the matching lower bound, which is why CMFT died — but the reformulation itself is correct and reusable, and it is a class-D-free statement (it does not assume $\text{Good}_N = \emptyset$; it concludes Goldbach).

The CMFT wall. Raw bilinear mass counted divisor-rich rows repeatedly. After the least-factor repair the machinery turned out to be *exactly* the Buchstab/sieve decomposition, and the fixed- q cofactor star $p \leftrightarrow b = (N - p)/q$ is an affine bijection, not an independent Type-II correlation. Giant stars have the natural scale of primes in arithmetic progressions and can be far larger than the “forced” $x^{2/3}$ with no anomaly at all.

11.3.2 From CGST (Canonical Gap-Spectral Theory)

CGST's reformulation: with $A_P = \{u \in \mathbb{Z} : P + 2u \in \mathbb{P}\}$ and $N_h = 2(P - h)$,

$$N_h \text{ is Goldbach} \iff -h \in A_P + A_P.$$

Theorem 11.7 (Canonical straddle theorem, PCRD-CGST-1). **PROVED**. *If no direct pair through P exists, every Goldbach representation has the form*

$$P + 2t + (P - 2(h + t)) = 2(P - h), \quad t \geq 1,$$

and the lower prime must lie below the previous prime P_- .

Reading. This is the additive counterpart of the phase trichotomy (Theorem 6.12) and explains why Type I forces the search strictly past p_- .

Theorem 11.8 (Arbitrarily deep finite reflection obstruction, PCRD-CGST-2). **PROVED**. *For any fixed T , CRT and Dirichlet construct infinitely many canonical $h = 1$ worlds in which $P - 2, P - 4, \dots, P - 2(T + 1)$ are all composite. Hence no bounded-depth reflection search can be a universal closer.*

Reading. The reflection-specific instance of the architecture wall (§8.2). If a proposed argument inspects the reflected block to any fixed depth, this theorem kills it immediately.

Theorem 11.9 (Diagonal collapse, PCRD-CGST-3). **PROVED**. *In cell coordinates the reflection row is $C_P(h, t) = P - 2(h + t)$, so the entire valuation/factor tensor on the (h, t) plane depends only on $h + t$. The extra “cell dimension” is a false degree of freedom for the factorization layer.*

Warning 11.10. Any construction that treats (h, t) as two independent parameters at the factorization layer is already dead. The exception-amplification wall follows: two cells h and $h + s$ share exact rows only when the upper-prime offsets form a prime pair of difference $2s$, and same-row factor certificates satisfy $\gcd(C_P(h, t), C_P(h + s, t)) = \gcd(C_P(h, t), s)$, which is 1 for $s = 1$. Factorization-driven propagation of one Goldbach exception to neighbouring N is therefore far too weak.

11.3.3 From CSD (Canonical Spectral Defect)

Theorem 11.11 (Canonical conductor-gap compatibility, PCRD-CSD-1). **PROVED**. *For any fixed modulus pattern M and any fixed length L , CRT and Dirichlet construct infinitely many canonical $h = 1$ worlds with $M \mid N$ and a prime-free interval of length at least L immediately before p_0 . Hence any finite conductor pattern is compatible with an arbitrarily long fixed canonical gap.*

Reading. The bridge “exceptional characters $\Rightarrow$ canonical gap contradiction” does not exist, and this theorem says why in one line. Keep it: it is the cheapest available refutation of an entire family of circle-method hybrids.

11.3.4 From CAPT (Canonical Almost-Prime Transfer)

Theorem 11.12 (Fixed- (r, s) compression, PCRD-CAPT-1). **PROVED**. *After freezing (r, s) in the Li-Liu representation $N = p + rq$ with $N - q = sb$, one has $q = N - sb$ and*

$$p = rsb - (r - 1)N,$$

so the system becomes two linear prime forms in the single variable b .

Remark 11.13 (The $(3, 3)$ degeneration). For $(r, s) = (3, 3)$: $q = N - 3b$, $p = 9b - 2N$ — an ordinary two-linear-prime pattern that occurs abundantly in real canonical worlds. Second-layer compression is therefore not a contradiction. Large-factor consequences of Li-Liu that remain correct: under Goldbach failure $r \neq 1$, and from $rq < N$ with $r \leq q^{9/10}$ one gets $r < N^{9/19}$ and $q > N^{10/19 - o(1)}$ for almost all of the relevant mass.

11.3.5 From RSD (Rough Signed Decomposition)

Theorem 11.14 (Exact rough signed identity, PCRD-RSD-1). **PROVED**. *On the rough support $(\rho_z(n) = \mathbf{1}_{(n, P(z))=1}$, $z > X^{1/4}$, so $\Omega(n) \leq 3$),*

$$\mathbf{1}_{\mathbb{P}}(n)\rho_z(n) = \rho_z(n) \left[\frac{1 - \lambda(n)}{2} - \mathbf{1}_{\Omega(n)=3} \right],$$

and for shifted-prime weights

$$G_z(N) = \frac{1}{2}R_z(N) - \frac{1}{2}L_z(N) - T_{3,z}(N),$$

so that under Goldbach failure $L_z(N) = R_z(N) - 2T_{3,z}(N)$.

Reading. This is the cleanest *exact* statement in the corpus of where the parity obstruction sits for this problem. It is worth keeping precisely because it is an identity, not an estimate.

Theorem 11.15 (Distribution-roughness phase law, PCRD-RSD-2). **PROVED**. *If the usable distribution level is $D = X^\theta$ and one sieves to $z = X^\beta$, the lower linear sieve needs $\beta < \theta/2$, while forcing $\Omega(n) \leq k$ by roughness needs $\beta > 1/(k+1)$. The two are compatible iff*

$$\theta > \frac{2}{k+1}.$$

For $k = 2$ the natural threshold is $\theta > 2/3$.

Reading. A quantitative triage rule: any proposal in this family can be checked against it in one line. Pascadi's $5/8 - o(1)$ [93] for special factorable weights is below $2/3$, so branch A is closed at current technology.

Remark 11.16 (The parity endpoint, exactly). At $z = X^{5/16-\varepsilon}$ the surviving P_3 sits in a narrow balanced box that switching/modern Chen sieves control well, but after reduction to P_1/P_2 the exact identity

$$P_1(N) = \frac{D_{1,2}(N) - \Delta_2(N)}{2}, \quad \Delta_2 = P_2 - P_1$$

remains, and Goldbach failure means full saturation $\Delta_2 = D_{1,2}$. Any theorem giving “even a little desaturation” is *already equivalent* to finding a prime complement on that family. This is the precise sense in which the parity barrier is the endpoint (§11.4).

11.3.6 From the global rough- P_2 matrix route

With $A_N(q, r) = \mathbf{1}_{\mathbb{P}}(N - qr)$:

Theorem 11.17 ($K_{3,3}$ explosion, PCRD-M-1). **CONJECTURE** (conditional **PROVED**). *At rough- P_2 mass of order $X/\log^2 X$, Kővári-Sós-Turán forces a $K_{3,3}$ at the natural scales $|Q| \sim X^{1/2}$, $|R| \sim X^{11/16}$. $K_{4,4}$ is not forced by that density scale.*

Theorem 11.18 (Boundary blindness, PCRD-M-2). **PROVED**. *Extending the bulk matrix by the boundary row/column $r = 1$ encodes the Goldbach entries. Zeroing the whole boundary axis is a perturbation of rank at most 2, whose operator norm is asymptotically small against the bulk scale. A generic operator norm or energy can therefore almost fail to notice the loss of every Goldbach entry.*

Theorem 11.19 ($h = 1$ matrix invisibility, PCRD-M-3). **PROVED**. *For $N = 2(P - 1)$ we have $m_0 = P - 2 = X - 1$, so every odd product $qr \leq X$ automatically satisfies $qr \leq X - 1 = m_0$. The canonical hard edge removes no rough- P_2 matrix cell: p_0 disappears from the domain geometry entirely.*

Theorem 11.20 (Fixed-modulus spectrum, PCRD-M-4). **PROVED**. *The local admissibility matrix modulo a prime $\ell \nmid N$ has the form $J - P_N$; its singular values are explicit and recover only local density / singular-series data. The fixed-modulus spectrum does not see Goldbach failure.*

Remark 11.21 (The codegree wall). The pointwise codegree $C_N(q_1, q_2) = \sum_r A_N(q_1, r) A_N(q_2, r)$ counts simultaneous primality of the three linear forms $r, N - q_1 r, N - q_2 r$, i.e. it enters the prime-tuple problem; average codegrees are boundary-blind by Theorem 11.18. Only a specially constructed bulk-to-boundary inverse theorem would remain, and no candidate is known that does not reduce to an earlier binary-correlation wall.

11.3.7 From the p_0 kill programme and the H1-FP revival

Theorem 11.22 (Gap scalar freedom, PCRD-P0-3). **PROVED** (from known facts). *For a fixed prime P with gap $G = P - P_-$, all $N_h = 2(P - h)$, $0 \leq h < G$, have canonical root P . With unbounded prime gaps the ratio h/G can approximate all of $[0, 1]$; with bounded-gap results there are infinitely many small G . Hence the scalars $h, G, h/G$ have no universal closing regime.*

Warning 11.23. This is the exact statement that kills every attempt to close the problem by a size condition on $\theta = h/G$ (§6.3). The phase diagram is a correct classification; it is not a filter.

Theorem 11.24 (H1 root-factor descent, PCRD-H1-1). **PROVED**. *If $q, r \mid m_0 = P - 2$ then*

$$r \mid N - q \iff r \mid q - 2.$$

The root-factor subsystem is therefore a strictly decreasing DAG, and the least root factor q_ satisfies*

$$\text{PF}(N - q_*) \cap \text{PF}(m_0) = \emptyset,$$

so failure must immediately produce external factors.

Theorem 11.25 (H1 shift-2 orthogonality, PCRD-H1-2). **PROVED**. *For $N = 2(P - 1)$ with P prime and $u < N/3$ prime,*

$$\gcd(N - u, u + 2) = 1.$$

Every terminal fixed core must therefore satisfy full shift-2 coprimality on all of its hyperedges.

Remark 11.26 (The 3-core normal form). The two smallest rows of a terminal core have disjoint factor supports. For $K = \{a < b < c\}$ this leads to one of two coupled forms, e.g.

$$N - b = a^\alpha, \quad N - a = b^\beta c^\gamma, \quad N - c = a^u b^v.$$

A hostile reverse search over large finite boxes found no 3-core; the status is **VERIFIED COMPUTATION**, not a theorem.

Why the H1-FP revival was not enough. Closing $N = 2(P - 1) \Rightarrow B_N^{\text{low}} = \emptyset$ requires classifying complete self-consistent shifted S -unit systems and the primality of the linear form $N/2 + 1$. Already the minimal case $|K| = 2$ touches an unresolved Pillai/Bennett-type uniformity ($a^\alpha - a = b^\beta - b$; Remark 6.25). p_0 supplies no mechanism of its own for that problem.

11.3.8 Live CRD results retained through every pivot

[PCRD] records the following as explicitly retained when CRD1 was frozen. They are the intersection of everything that survived; all appear elsewhere in this file and are listed together here so a future model can see the retained core at a glance.

- the fixed-point characterisation B_N^{low} (Theorem 3.29);
- the C0b decomposition and its correct gap-scale congruence consequences (Theorem 8.6);
- T2 first-front residue closure;
- T3 mod-3 exclusion/rigidity (Theorems 5.17, 5.18);
- the source-free core K with its shifted S -unit equations (Remark 6.25, Remark 11.26);
- the exact container/product identities (§5.3);
- the correct prime-power collision geometry (§7.4);
- the empirical six-EAC corpus (§6.6).

What killed CRD1 — the convergence, not a single false lemma.

1. Selected-divisor capacity routes were *same-container* arguments: one cannot beat a container with one of its own divisors. (This is exactly what Corollary 9.9 later made quantitative.)
2. Generic smooth-population bounds lost self-consistency: the factor alphabet is simultaneously the set of state labels.
3. The pure local-residue route saturates for $q \geq 5$; modulus 3 is an exception, not a mechanism (Theorem 5.17).
4. Reflection is not a universal closer: $h = 1$ gives an *empty* strict reflected block.
5. Nonprimitive large product/reuse is compatible with a failed world and gives no capacity contradiction.
6. A source-free core is not itself contradictory: $N = 2200$, $K = \{3, 13\}$ is an exact witness.
7. After removing the intermediate ledgers the frontier returned to the exact self-consistency $\text{PF}(m_0) \subseteq B_N^{\text{low}}$ with no independent intermediate parameter.

11.4 The parity barrier and why it is the real wall

Two distinct walls stand in front of this project and they are often confused.

The architecture wall (§8.2) is internal and *proved*: no finite local canonical certificate can decide the question, because every finite transcript is realisable in infinitely many canonical worlds with $h = 1$.

The parity barrier is external and classical: sieve methods cannot distinguish numbers with an even number of prime factors from those with an odd number, so no purely sieve-theoretic argument produces a *prime* complement.

The architecture wall says “not locally”. The parity barrier says “not by sieving”. Together they say a proof needs a genuinely non-local, non-sieve input — which is why the successor theories all failed at the same point regardless of their coordinates.

RSD and parity. RSD attempted to expose parity explicitly through a rough reservoir with a Liouville sign and a P_3 correction. It requires either level of distribution $> 2/3$ — beyond Pascadi’s $5/8$ [93] — or a pointwise binary correlation between primes and the Liouville function, which is at least as hard as the problems calibrated by Mangerel [95]. **KILLED**.

The weaker P_2 desaturation target. Also **KILLED**: at $\Omega \leq 2$ a sign- (-1) witness *is* a prime complement, so the “weaker” target is the original problem.

Chen-type input. R. Li’s bound $D_{1,2}(N) \geq 1.9728 C(N)N/\log^2 N$ [94] and Li–Liu’s $(1 + 1.9)$ [15] supply almost-prime representations in quantity, but not the last bit distinguishing P_1 from P_2 . That last bit is the parity barrier.

Friedlander–Iwaniec. The asymptotic sieve for primes [38] is the template for what the missing input looks like: a bilinear form estimate. No bilinear input of the required shape has been produced in this project, and the rough- P_2 matrix was the attempt to manufacture one.

Warning 11.27 (Do not try to route around parity by enlarging the graph). Several killed routes are variations on “add more structure to the graph until the parity obstruction disappears”. The obstruction is not a property of the graph; it is a property of the sieve information being used.

Adding vertices, weights or higher-order correlations to a construction whose only input is sieve-type data cannot remove it.

Chapter 12

Full Failure Log

This chapter is reproduced in full and unabridged. It contains all 239 live entries of the project failure log, with every field preserved. No entry was removed for appearing unimportant; the record’s value lies precisely in its completeness.

How to use it.

- Entries are ordered alphabetically by FAIL-ID, so an ID cited anywhere in this file can be found directly.
- The index in §12.1 gives ID, status and page.
- Every entry is labelled `fail:FAIL-ID`, so `\ref{fail:FAIL-XXX-001}` resolves.
- Cross-references that pointed into the original document and cannot be resolved here appear as `[target]` rather than being silently dropped.
- Legacy status macros of the source were mapped onto the status vocabulary of §1; the mapping is stated below.

Legacy status mapping. `R, RI, RM` → `KILLED`; `SC` → `SCOPE-CORRECTED`; `SU, RET` → `SUPERSEDED`; `SR, C` → `VERIFIED COMPUTATION`; `P` → `PROVED`; `PA` → `PROVED WITH REPAIR`; `H` → `EMPIRICAL PATTERN`; `O` → `OPEN`. The mapping is status-preserving in the sense that no entry was moved to a stronger status.

12.1 Index of failure-log entries

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FAIL-ABOVE-BV-MASS-INVISIBLE-001	SCOPE-CORR.	103
FAIL-ABSORBED-FRIDGE-RESIDUAL-HAS-ARBITRARY-001	KILLED	103
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FAIL-BABS-PRODUCTION-001	SUPERSEDED.	104
FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-SCOPE-CORR-001	SCOPE-CORR.	105
FAIL-BOUNDED-ARITY-WEIGHT-001	KILLED (infer.).	105
FAIL-BUNDLE-ENLARGES-CAPACITY-001	KILLED.	106
FAIL-BV-BOUNDARY-DESTROYS-MASS-001	KILLED.	106
FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001	SUPERSEDED.	107
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FAIL-CAB-001	KILLED without a failure condition.	108

ID	Status	Str.
FAIL-CAB-INTERNAL-001	KILLED without a failure condition.	108
FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTENSION-CORR	SCOPE-CORR (domain failure).	109
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FAIL-CASCADE-DIRECTION-001	KILLED as stated and as proved.	110
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FAIL-CHILD-C1-001	KILLED by source-recorded computation.	111
FAIL-CLEAN-ROOTED-CAP-P1-001	KILLED (infer.).	112
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FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHKILL-DEAD-001	KILLED.	116
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FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-KILLED	KILLED for the capacity inference.	118
FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPED	KILLED as a proof-design necessity.	118
FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONEKILLED	KILLED as a proof mechanism from one r	119
FAIL-DEEP-ONE-AP-PACKING-CLOSES-001	KILLED as a sufficient closing mechani	119
FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001	KILLED.	119
FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001	KILLED as a proof-design description.	119
FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-SUPERSEDED	SUPERSEDED as a primary frontier.	120
FAIL-DESCENT-ONE-STATE-001	SUPERSEDED.	120
FAIL-DIRECT-ADJACENCY-001	KILLED for direct adjacency.	120
FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001	KILLED (infer.).	121
FAIL-EXACTD-NONCANONICAL-001	KILLED as a substitute for minimality;	121

ID	Status	Str.
FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001	KILLED.	122
FAIL-EXTERNAL-ETA-IDENTIFICATION-001	KILLED.	122
FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING	CLOSED PENDING INDE- PENDENT AUDIT .	123
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FAIL-FIRSTFRONT-SCALAR-ROUTE-001	KILLED (mech.) with the scope below.	128
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FAIL-FRIDGE-CONGRUENCE-CONTRADICTION-001	SCOPE-CORR.	131
FAIL-GAMMA-BOUND-ROUTES-001	KILLED for all four, as routes; the un	131
FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001	KILLED (infer.).	132
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FAIL-GENERIC-PRIME-ROW-RPD-001	KILLED as a generic theo- rem.	133
FAIL-GREEDY-AVOIDANCE-NAIVE-001	KILLED as an argument. The conclusion	134
FAIL-HALL-SUPPORT-001	KILLED (mech.).	135
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FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP	KILLED.	135
FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001	KILLED.	136
FAIL-HIGHFLOOR-DOMAIN-VACUITY-001	SCOPE-CORR. The theo- rem is PROVED; its	136
FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWN-001	KILLED.	137
FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001	KILLED.	137

ID	Status	Str.
FAIL-INDIVIDUAL-MESOSCALE-BV-001	KILLED (infer.).	137
FAIL-INVERSE-SIBLING-INJECTIVE-001	KILLED (infer.).	138
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FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-KILLED as supported only on the recorded	KILLED as supported only on the recorded	144
FAIL-LITERAL-LINEAGE-001	KILLED.	145
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FAIL-LOSSLESS-INTERFACE-001	KILLED for lossless transfer, under ev	146
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FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOUT-001	KILLED for $h > N^{1/3}$.	149
FAIL-MATCHING-CAPACITY-001	KILLED (infer.) — the matching theor	149
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FAIL-MERGER-WEIGHT-TO-J-TOPOLOGY-001	KILLED (infer.).	150
FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001	SCOPE-CORR.	150
FAIL-MIRROR-MASS-ONE-SCALE-001	KILLED.	150
FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001	KILLED (infer.).	151
FAIL-MONOMIAL-MOD3-NOVELTY- $\forall$ NEXT4.11	SCOPE-CORR (provenance correction).	151
FAIL-MONOMIAL-NEEDS-SPARSITY-001	KILLED (mech.).	151
FAIL-MONOMIAL-NODESCENT-CHEAP-001	KILLED.	152
FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001	KILLED.	152
FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001	KILLED.	152
FAIL-MR2-CANONICAL-ONLY-001	KILLED as a scope claim.	153
FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001	KILLED.	153
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FAIL-NAIVE-MU-LE-3ETA-001	KILLED.	154
FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001	KILLED.	154
FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-001	KILLED (mech.).	154
FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001	KILLED as a description.	155
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FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001	SCOPE-CORR.	156
FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-001	KILLED in the v18 owner-completed doma	157
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FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001	KILLED.	159
FAIL-PARENT-LATTICE-DOMAIN-001	SCOPE-CORR.	160
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FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001	KILLED for ledger-only derivations. No	163
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FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-CHARACTERISTICS-001	KILLED (mech.).	165
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FAIL-Q0-LEDGER-001	KILLED.	167
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FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001	KILLED.	168
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FAIL-RDS-GEOMETRY-DENSITY-001	KILLED (infer.).	169
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FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-AC-KILLED-001	KILLED	171
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FAIL-RETURN-COFACTOR-IS-FREE-001	KILLED as a description.	171
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FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COST-FREE-001	SCOPE-CORR.	172
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FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001	KILLED (infer.).	194
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FAIL-VERTICAL-TO-ETA-LOSES-LOG-001	KILLED (mech.).	196
FAIL-VERTICAL-UNLOCALISED-001	KILLED as a description.	197
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FAIL-WORLD-SIZE-NAIVE-001	KILLED as stated.	198
FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-0	SUPERSEDED.	199
FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001	SCOPE-CORR.	199
FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONST	KILLED	199

ID	Status	Str.
FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PERKILLED.	KILLED.	199
FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-KILLED-SINK-001	KILLED-SINK-001 counting diagnosis.	200
FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-0KILLED.	KILLED.	200
FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-KILLED-A-CONTRADICTION-001	KILLED-A-CONTRADICTION-001	200
FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-KILLING-SPACING-001	KILLING-SPACING-001	200
FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001	MONOTONICITY-001	201
FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001	COEFFICIENT-SINK-001	201
FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-KILLED.	KILLED.	201
FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-KILLED-001	CLOSES-DEEP-001	201
FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-SCHEME-CORR.	SCHEME-CORR.	201
FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001KILLED.	KILLED.	202
FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001	KILLED.	202
FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTARTED-as a packet-design inference.	RESTARTED-as a packet-design inference.	202
FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-KILLED.	KILLED.	203

12.2 Entries

FAIL-ABOVE-BV-MASS-INVISIBLE-001

FAIL-ID:

FAIL-ABOVE-BV-MASS-INVISIBLE-001.

Date / iteration:

vNEXT4.19.

Old description:

Mirror mass above one scalar BV boundary is unavailable to clean selection.

Status:

SCOPE-CORRECTED.

Replacement:

Track both designated-base scale Q and roughness scale z . With polylogarithmic z , the clean designated-base range reaches $Q_* = N^{1/2}/\text{polylog } N$. Significant mass beyond Q_* forces a polynomial actual-state packet rather than disappearing.

FAIL-ABSORBED-FRIDGE-RESIDUAL-HAS-ARBITRARY-COFACTOR-001

FAIL-ID:

FAIL-ABSORBED-FRIDGE-RESIDUAL-HAS-ARBITRARY-COFACTOR-001.

Status:

KILLED.

Rejected statement:

A first-front residual prime $\ell \in (N/6, N/3)$ may occur with an arbitrary cofactor in $N - q$.

Certificate:

since $q < N/6$, $(N - q)/\ell$ is an odd integer strictly between 1 and 6, hence $N - q = 3\ell$ or $N - q = 5\ell$. The root-supported gcd then divides 3 or 5.

FAIL-AP2-HIDDEN-OMEGA-DEPENDENCY-001

FAIL-ID:

FAIL-AP2-HIDDEN-OMEGA-DEPENDENCY-001.

Date / iteration:

vNEXT4.5 audit of vNEXT4.2.

Claim:

The recorded AP1/AP2 consequence follows merely from “standard Mertens/PNT estimates”.

Status:

SCOPE-CORRECTED.

Exact repair:

The support-capacity inequality contains $\omega_h(x) \log c_{\max}/t$, so one also needs

$$\omega_h(x) \leq \omega(x) = O\left(\frac{\log x}{\log \log x}\right).$$

With that explicit dependency, the quantified population conclusion survives.

Do not repeat:

Do not suppress the $\omega(x)$ dependency when invoking this population lemma.

FAIL-BABS-PRODUCTION-001**FAIL-ID:**

FAIL-BABS-PRODUCTION-001.

Date / iteration:

Canonical bridge audit §4, 31 August 2026.

Claim:

Every failed world produces a nontrivial absorbed base product $B_{\text{abs}} > 1$, which is therefore the universal carrier of failure mass.

Original motivation:

B_{abs} is the natural object of ambient promotion and was observed nontrivial in most instances.

Status:

SUPERSEDED.

Minimal / strong counterexample:

$(N, r) = (128, 79)$.

Exact certificate:

Full- Q production is nontrivial for this root while the recorded B_{abs} equals 1. Production and absorption are different channels, and only the former is always nonvacuous on the composite-complement domain.

Failed hypothesis or mechanism:

One channel of a four-channel ledger was promoted to the whole ledger.

Why the argument failed:

B_{abs} collects exactly the residual primes with $\kappa_R(\ell) = m_\ell$; nothing forces that set to be nonempty.

What survives:

ARS-BABS-INTERNAL, that $\mathcal{B}_{\text{abs}, \text{int}} = \mathcal{B}_{\text{abs}}$ under failure — which explicitly yields *no* nonvacuity conclusion — and B_{abs} as one channel and the proper object of ambient promotion.

Safe weaker version:

Carry failure mass on $\mathfrak{F}_Q = \prod_{q \in Q_r} M_q$, not on B_{abs} .

Current replacement:

Theory Core, G-FULLQ-PRODUCTION, which gives $\mathfrak{F}_{Q_r} > (5/3)^{|Q_r|} \text{rad}(m_r) > 1$ unconditionally on the composite-complement domain.

Do not repeat:

Do not make one ledger channel universal.

Related current target:

$P1^{\text{br}}$, K1.

FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-RPD-WITNESS-001

FAIL-ID:

FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-RPD-WITNESS-001.

Status:

SCOPE-CORRECTED.

Correction:

G-BAKER-SPARSE-EXACT-LAYER-FAN packages Baker's analytic theorem together with extra CRD domain assumptions and an upper-half conclusion. The v4.46 witness uses only the analytic sparse-moduli input and repeats the endpoint subtraction on a different fixed proportional interval.

What survives:

the Level-C prime-row no-go is a legitimate new arithmetic consequence, but its provenance must be stated correctly.

FAIL-BOUNDED-ARITY-WEIGHT-001

FAIL-ID:

FAIL-BOUNDED-ARITY-WEIGHT-001.

Date / iteration:

Bombieri-Vinogradov four-ary continuation, 1 September 2026.

Claim:

Once every selected parent has at most four unintended children, the large-state packet E is automatically a lower-order cost, or the clean parent family closes P1BR by cardinality alone.

Status:

KILLED.

Minimal / strong counterexample:

Structural; one unintended prime factor is already enough.

Exact certificate:

A squarefree selected row may have

$$N - p = \ell r, \quad r > N^{1/4-\varepsilon},$$

with $r \in R$. The four-ary conclusion permits this configuration, and the only universal weight estimate it supplies is

$$\log r = \log(N - p) - \log \ell.$$

Thus even $\omega(E_\ell) = 1$ permits the unintended label to carry the entire non-base cofactor mass. More generally

$$\log E \leq L(\mathcal{P}) - \log \prod \ell,$$

with equality when every unintended child lies in R .

Failed hypothesis or mechanism:

An unweighted degree bound was used as a weighted logarithmic capacity bound.

Why the argument failed:

The number of factors and the sum of their logarithms are different functionals. Roughness makes each surviving factor larger, not cheaper.

What survives:

The four-ary theorem turns E into a bounded-degree dynamic packet. The exact clean source-surplus identity then cancels the base copy of every internal unintended factor and exposes its true cost as $(\kappa_R(r) - 1) \log r$.

Safe weaker version:

Use bounded arity only for dynamic branching and population statements. For P1BR prove an external-mass-versus-rooted-reuse inequality with logarithmic weights.

Current replacement:

Theory Core v1.6, G-CLEAN-MULTIPARENT-SURPLUS-IDENTITY and
G-BV-FOUR-ARY-DISPOSITION.

Do not repeat:

$\omega(E) = O(|A_0|)$ does not imply $\log E = o(L(\mathcal{P}))$.

Related current target:

P1BR, K1, large-layer disposition.

FAIL-BUNDLE-ENLARGES-CAPACITY-001**FAIL-ID:**

FAIL-BUNDLE-ENLARGES-CAPACITY-001.

Date / iteration:

vNEXT4.1, Phase 7.

Claim:

That preserving a canonical bundle D in the lifted row forces a canonical contribution — to the denominator of (RC4'), or to a state-difference packing budget — and so extracts more from R than an arbitrary external row does.

Original motivation:

The lifted row shares exact prime powers with a canonical reflected row, and empirically bundle rows are much more often nonpositive than single-base rows on the same reflected row (87% against 72% in COMP-VNEXT41-DISPOSITION). Nonpositive rounds are the ones with a recorded consumer, so this looked exploitable.

Status:

KILLED.

Exact certificate:

Theory Core G-EXTERNAL-INTERFACE-CAPACITY. For any globally coprime family $\mathcal{P}$ of external rows and any failed world R , the integers $\gcd(N - p, N - u)$, $p \in \mathcal{P}$, are pairwise coprime divisors of $N - u$, so $\sum_p \sum_u \log \gcd(N - p, N - u) \leq \sum_u \log(N - u) < |R| \log N$. The bound has *no* hypothesis on how $\mathcal{P}$ was chosen. Consequently at most $2|R| < 6.6\gamma \log N$ members with $N - p > \sqrt{N}$ can be nonpositive, whatever the provenance; the empirical bias towards nonpositivity is therefore not exploitable beyond that ceiling.

Failed hypothesis or mechanism:

Confusing a richer *description* of a row with a larger *charge* on R . The charge is a gcd against the fixed integers $N - u$, and those integers have finitely many divisors.

What survives:

The bundle lift, the exact valuations and the mirror relation, all **PROVED**; and the ceiling (IC3), which is new and which also strengthens the vNEXT4 statement (MD1) by dropping the designated-base hypothesis.

Do not repeat:

Before claiming a selection strategy extracts more from R , check it against (IC1), which is provenance-blind.

Related current target:

P1BR-ADDITIVE-COUPLING.

FAIL-BV-BOUNDARY-DESTROYS-MASS-001**FAIL-ID:**

FAIL-BV-BOUNDARY-DESTROYS-MASS-001.

Date / iteration:

vNEXT4.15.

Claim:

Low-child states above the current BV-accessible range lose the natural reflected mass because they cannot be cleanly selected.

Status:

KILLED.

Exact correction:

Above $Y_{BV} = N^{1/(6+2\vartheta)}(\log N)^{-O(1)}$, each distinct state already carries $(1/6 - o(1)) \log N$ weight. A proportional family therefore retains the original $t \log \Lambda$ scale as weighted state mass.

Do not repeat:

The analytic boundary changes the type of object, not its total natural logarithmic weight.

FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001**FAIL-ID:**

FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001.

Date / iteration:

vNEXT4.17; interpretation superseded by vNEXT4.19.

Old description:

Above the then-used same-scale BV threshold the mirror packet becomes analytically invisible.

Status:

SUPERSEDED.

Replacement:

Already vNEXT4.17 converted significant above-threshold mass to distinct active states. vNEXT4.19 further shows that the relevant selector geometry has two independent scales Q, z , so the designated-base range extends almost to $\sqrt{N}$ when z is polylogarithmic.

FAIL-C-SYMBOL-COLLISION-001**FAIL-ID:**

FAIL-C-SYMBOL-COLLISION-001.

Date / iteration:

vNEXT2 independent audit.

Claim:

The symbol c may be read uniformly across the Theory Core.

Original motivation:

Both quantities are small integers attached to a packet and were introduced in different rounds.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

Internal. In (LE1)–(LE6), $c = |D_S^{\text{src}}|$ is the *source count*; in (FC7)–(FC10), $c = |C|$ is the *low-core cardinality*. For a canonical R the first is 1 and the second is typically large, so any statement that combines an LE identity with an FC bound while reading c uniformly is wrong by a factor that grows with the world.

Exact certificate:

(LE4) reads $\eta(S) + \nu(S) \geq |S| + c$ with $c = 1$ for a canonical rooted world, whereas (FC8) reads $J_R < N^{c+\tau}(25/18)^\tau / \prod_{q \in C \setminus Q} q$ with $c = |C|$. Substituting one into the other is a category error.

Failed hypothesis or mechanism:

Symbol reuse across rounds without a concordance entry.

What survives:

Both families of statements, unchanged, on their own readings.

Safe weaker version:

Use $\gamma = |C|$ for the low core and reserve c for the source count, as fixed by Convention [conv:c-collision] in the Theory Core. (FC7)–(FC10) are not restated; their c is γ .

Current replacement:

Theory Core, Convention [conv:c-collision].

Do not repeat:

Do not introduce a symbol already bound in an earlier round without a concordance line.

Related current target:

(FC8), K1.

FAIL-CAB-001**FAIL-ID:**

FAIL-CAB-001.

Date / iteration:

Post-v7.5 CAB audit, 30 August 2026.

Claim:

For every canonical stopped world, $\log B_{\text{abs}} \leq A(\log U_R + \log J_R)$ for a fixed finite A .

Original motivation:

Every absorbed base atom was expected to be financed by power excess or support reuse.

Status:

KILLED without a failure condition.

Minimal / strong counterexample:

$N = 322, p_0 = 163$.

Exact certificate:

$N - p_0 = 159 = 3 \cdot 53$, and $322 - 3 = 319 = 11 \cdot 29$. The audited stopped world has $U_R = J_R = 1$ but $B_{\text{abs}} = 11 \cdot 29 > 1$. The other first-front state 53 is GOOD because $322 - 53 = 269 \in \mathbb{P}$.

Failed hypothesis or mechanism:

A successful-world local ledger was treated as a failure-world capacity law.

Why the argument failed:

Absorption can occur on the bad branch while another first-front branch already succeeds, leaving zero proposed denominator.

What survives:

This witness does not refute a theorem explicitly conditioned on a closed canonical failure.

Safe weaker version:

Any CAB repair must state the failure domain and a nonzero source term; no such repair is currently proved.

Current replacement:

C0b-RD and the branchwise production program.

Do not repeat:

Never cite $N = 322$ as refuting a failure-conditioned theorem, and never cite CAB unqualified.

Related current target:

$P1^{\text{br}}, K1$.

FAIL-CAB-INTERNAL-001**FAIL-ID:**

FAIL-CAB-INTERNAL-001.

Date / iteration:

Post-v7.5 CAB audit, 30 August 2026.

Claim:

CAB becomes valid in every canonical stopped world after discarding boundary atoms and retaining only internal absorbed atoms.

Original motivation:

The $N = 322$ defect looked purely external.

Status:

KILLED without a failure condition.

Minimal / strong counterexample:

$N = 858, p_0 = 431, \ell = 23$.

Exact certificate:

$N - p_0 = 427 = 7 \cdot 61$, $858 - 7 = 851 = 23 \cdot 37$, and the audited stopped world has $U_R = J_R = 1$. The atom 23 is internal and absorbed; 37 is a boundary atom. The state 61 is GOOD because $858 - 61 = 797 \in \mathbb{P}$.

Failed hypothesis or mechanism:

Boundary deletion was assumed to remove the zero-payment phenomenon.

Why the argument failed:

The same phenomenon occurs at an internal atom on a bad branch of a globally successful rooted traversal.

What survives:

Again, this successful-world witness does not refute a statement explicitly conditioned on canonical failure.

Safe weaker version:

None is currently proved; require a genuine failure-specific source mechanism.

Current replacement:

C0b-RD and targets $L1, E1, K1$.

Do not repeat:

“Internal” does not mean “already paid”.

Related current target:

$K1$.

FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTERNAL-HQ-001**FAIL-ID:**

FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTERNAL-HQ-001.

Status:

SCOPE-CORRECTED (domain failure).

Killed transfer:

apply the canonical fringe Brun–Titchmarsh prime-power packing theorem unchanged to all of $H_Q \setminus R$.

Certificate:

its decisive population cap is tied to canonical fringe paths / the γ -controlled canonical domain. External descendants do not inherit that cap.

What survives:

the theorem applies on its canonical domain and the overlap $H_Q \cap R$ is negligible in the v4.45 high-band accounting.

Do not repeat:

replacing γ by $|H_Q|$ is a new crude estimate, not a legal transport of the canonical theorem.

FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-001**FAIL-ID:**

FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-001.

Status:

SCOPE-CORRECTED.

Killed formulation:

the v18–v32 exact-layer/star machinery is external and cannot be applied to an exact layer already contained in the canonical world R .

Certificate:

v4.26 already contains **C-CANONICAL-POLYLOG-STAR-ABSORPTION**; exact-layer parent slots, quotient inflation and cofactor conservation are generic exact-layer results. Moreover v4.35 shows that an already extracted polynomial canonical vertical layer has only the full-scale receiving channels J_R , U_R^{nr} and D_C .

What was actually missing at v4.35:

a termination/payment theorem for repeated q -free $U \rightarrow U$ migration, or a legal generation/owner transfer to the v28–v32 package. v4.36 then supersedes this dynamic issue as the primary vertical frontier by giving a static upper capacity for the whole U -mass.

FAIL-CASCADE-DIRECTION-001**FAIL-ID:**

FAIL-CASCADE-DIRECTION-001.

Date / iteration:

vNEXT3.1 correction round.

Claim:

G-CASCADE-GAMMA-TERMINUS: the (FC10) cascade closes precisely on $\gamma \leq (\log N)^A$, proved by bounding $\sum_i M_i \log z_i \ll (\log N)^A \log N$.

Original motivation:

The selector supplies at most one clean row per designated base and the bases are polylogarithmically bounded, so the total production looked easy to estimate.

Status:

KILLED as stated and as proved.

Minimal / strong counterexample:

The proof itself. (DP5) states that *simultaneous nonpositivity implies* $\sum_i M_i \log z_i < \log J_R$; to conclude that some round is positive one must therefore exhibit a **lower** bound $\sum_i M_i \log z_i \geq \log J_R$. The vNEXT3 proof supplied an **upper** bound $\sum_i M_i \log z_i \ll (\log N)^A \log N$ and used it in that place. An upper bound on a quantity can never certify that the quantity is large; the argument establishes nothing, in either direction.

Failed hypothesis or mechanism:

The two roles of $\sum_i M_i \log z_i$ — what (DP5) concludes about it, and what must be supplied to invoke (DP5) contrapositively — were interchanged.

Why the argument failed:

Counting the *number* of available bases bounds production from above. Forcing a positive round needs the *logarithmic mass* of the bases actually used, bounded from below.

What survives:

The intended conclusion, correctly proved, with the quantifiers made explicit and the domain moved to the global-failure DAG: see G-SELECTABLE-CLEAN-LOG-MASS and G-GAMMA-POLYLOG-FORCES-POSITIVE-ROUND in the Theory Core. The correct relation between the exponents is $A > B + 1$, with B fixed first.

Safe weaker version:

None; the statement must be reproved from a lower bound, which is what the replacement does.

Current replacement:

Theory Core, Theorem [thm:clean-log-mass] and Theorem [thm:gamma-forces-positive].

Do not repeat:

Before invoking a contrapositive, check which side of the inequality must be supplied and which side is concluded.

Related current target:

γ , P1BR-POSITIVE-ROUND-CONSEQUENCE.

FAIL-CENSUS-DOMAIN-001**FAIL-ID:**

FAIL-CENSUS-DOMAIN-001.

Date / iteration:

Canonical bridge audit §12, item 2.

Claim:

The recorded figure “301” counts all foundational failed upper roots in the scanned range.

Original motivation:

The label was abbreviated in later reports.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

The unit-complement row, already registered as FAIL-UNIT-COMPLEMENT-001.

Exact certificate:

The scan enumerated *composite-complement* failed upper roots. A row with $N - r = 1$ also fails, has no children, and is outside that census by construction; $(N, r) = (20, 19)$ is the smallest instance.

Failed hypothesis or mechanism:

A domain-qualified count was quoted without its qualifier.

Why the argument failed:

Every theorem keyed to that census inherits the composite-complement hypothesis.

What survives:

The number itself, with its domain label attached, as a regression fixture. Status **VERIFIED COMPUTATION**.

Safe weaker version:

Always write “301 composite-complement failed upper roots in the recorded scan”.

Current replacement:

Benchmark suite below, where the domain label is preserved.

Do not repeat:

Never quote a census figure without its domain.

Related current target:

C0a and every generic failed-root lemma.

FAIL-CHILD-C1-001**FAIL-ID:**

FAIL-CHILD-C1-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

The selected child-prime normalization has universal charging constant $C = 1$.

Original motivation:

Every selected parent financed at most one audited base atom in the focused incidence table.

Status:

KILLED by source-recorded computation.

Minimal / strong counterexample:

The campaign records 53 violations; the empirical maximum is 9.42243 at $N = 406996$.

Exact certificate:

The numerical ratio and the location of its maximum are preserved from the final report. The individual raw child-mass table was unavailable in this extraction, so this certificate was not independently recomputed.

Failed hypothesis or mechanism:

A one-to-one incidence observation was used as a weighted logarithmic inequality.

Why the argument failed:

Multiplicity of financed atoms and size of their weights are different quantities.

What survives:

A more generous partner-prime normalization had empirical maximum 1, but remains computational only.

Safe weaker version:

Treat every finite observed constant as a conjectural target until an exact source-sensitive bound is proved.

Current replacement:

Theory Core target $K1$.

Do not repeat:

Never promote the best observed normalisation to a universal constant.

Related current target:

$K1$.

FAIL-CLEAN-ROOTED-CAP-P1-001**FAIL-ID:**

FAIL-CLEAN-ROOTED-CAP-P1-001.

Date / iteration:

Clean-support-load continuation, 1 September 2026.

Claim:

Once one selects pairwise-coprime upper-half external parents, the rooted capacity $\mathcal{C}_N(R) = L(R)$ supplied by the clean base-layer theorem can be beaten by the cofactor production of those same parents, closing P1BR.

Status:

KILLED.

Minimal / strong counterexample:

Structural; no finite witness is needed.

Exact certificate:

In a canonical failed world, every non-root state $r \in R \setminus \{p_0\}$ lies below $N/3$, hence $N - r > 2N/3$. Every selected upper-half parent satisfies $p > p_0$, hence $N - p < m = N - p_0 < N/2$. With at most one selected parent per distinct internal base label, $|\mathcal{P}| \leq |R| - 1$, so

$$L(\mathcal{P}) < (|R| - 1) \log m < \log m + (|R| - 1) \log(2N/3) < L(R).$$

Thus the exact rooted upper bound $K_\times + K(\mathcal{P}) \leq L(R)$ is too large for this naive production choice to exceed.

Failed hypothesis or mechanism:

Solving the collision-selection problem was mistaken for solving the production-versus-capacity comparison.

Why the argument failed:

The clean theorem makes external complements small and noncolliding, but the failed-world rows are individually larger: non-root rows exceed $2N/3$, whereas selected upper-half-parent complements are below $m < N/2$. The order geometry points in the wrong direction for $L(\mathcal{P}) > L(R)$.

What survives:

G-CLEAN-POLYLOG-PARENT-SELECTION and G-CLEAN-BASE-ROOTED-CAPACITY remain exact **PROVED** statements. They close the W1/K1 collision side on the fixed-polylog base-layer subdomain.

Safe weaker version:

Use the clean selection to isolate the remaining cost: the base-copy surcharge from unintended labels of R . A future P1BR route must control that smaller support set, or use a different common production/capacity functional; it may not simply take $L(R)$.

Do not repeat:

“Pairwise coprime parents” is a capacity statement, not a production surplus.

Related current target:

P1BR, W1, K1.

FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONE-001

FAIL-ID:

FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONE-001.

Status:

KILLED.

Killed inference:

once cross-row cofactor radical reuse is identified with incremental J , a large load-one packet automatically has ambient $\Omega(M \log N)$ cofactor entropy.

Certificate:

a common tiny factor such as 3 across M rows carries only $(M - 1) \log 3 = O(M)$ entropy while designated high labels have $\Theta(M \log N)$ mass.

What survives:

cofactor entropy is exact horizontal storage, but the correct next reduction is the cube-root row trichotomy / boundary argument.

FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001

FAIL-ID:

FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001.

Date / iteration:

v1.22 cofactor-window continuation, 1 September 2026.

Claim:

For every internal returned fresh seed, the cofactor $r_j = (N - w_j)/\ell_j$ is prime.

Status:

KILLED.

Why the inference fails:

v1.21 proves only that all prime factors of r_j lie in U and are at least $s = \min U$. Primality follows in v1.22 only when $r_j < s^2$, supplied by the bounded window $r_j < 2\Lambda h < h^2 \leq s^2$ under no descent.

What survives:

On $b_j \leq \Lambda h$ with $s \geq h > 2\Lambda$, the return cofactor is prime and yields the sibling packet of TF-COFACTOR-WINDOW-SIBLING-RENEWAL.

Do not repeat:

Do not replace cofactor roughness by primality outside the explicit short-window inequality.

Related current target:

TF, above-BV return transport.

FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001

FAIL-ID:

FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001.

Status:

KILLED as a present engineering requirement.

Killed inference:

first/second occurrence, reflected provenance, parenthood, cofactor data or no-double-charge ownership currently require a new fundamental dynamic state variable.

Certificate:

the v4.39 audit finds all theorem-relevant objects reconstructible from full CRD1 together with derived subsets/matchings. No pair with equal full CRD1 state and different theorem-relevant deterministic outcome has been exhibited.

What survives:

introduce CRD2 only if a future proof exhibits non-reconstructible data, genuine history dependence, same-state/different- outcome behaviour, or irreducible persistent ownership. Until then use CRD1-R / CRD1.5 as a derived relation-preserving interface.

FAIL-CRD2-PREMATURE-001

FAIL-ID:

FAIL-CRD2-PREMATURE-001.

Date / iteration:

vNEXT4.3–vNEXT4.4.

Claim:

Because the $J, U, \eta, \kappa, \Delta$ ledgers forget parent or bundle provenance, CRD1 must be replaced immediately by a richer fundamental CRD2 state space.

Status:

DESIGN PROHIBITION / FRONTIER CERTIFICATE; not a claim that CRD2 will never be needed.

Exact certificate:

The full ambient object

$$\mathfrak{C}_N = (N, \mathcal{A}_N, p_0, \nu_N), \quad \nu_N(p, q) = v_q(N - p),$$

already reconstructs active supports/valuations, parent sets, sibling and factorisation hyperedges, exact cofactors $(N - p)/q^e$, inactive residuals, and reflected provenance (j, a_j, c_j) . The proven vNEXT4 independence certificates concern compressed ledgers, not equality of the full CRD1 state.

What survives:

Use a relation-preserving derived view (CRD1-R/CRD1.5) when a theorem must keep provenance alive.

Migration trigger:

CRD2 becomes justified only after a theorem-relevant non-reconstructible datum, genuine history dependence, a full-state information-loss pair, or an irreducible ownership state is exhibited.

Related current target:

C-VERTICAL-LOWCORE-CAPACITY; CRD2 is dormant.

FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001

FAIL-ID:

FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001.

Date / iteration:

vNEXT4.7.

Claim:

On $h \geq \sqrt{N/3}$, the block-derived vertical family can still absorb a positive proportion of a $t \log N$ -scale obligation.

Status:

KILLED.

Exact certificate:

The monomial collapse puts every vertical base in $(\sqrt{2N/3}, \sqrt{N})$, so standard prime counting gives

$$|\mathcal{V}| \ll \sqrt{N}/\log N = O(t/\log N),$$

and exactly one vertical layer per base gives

$$\log U_{\mathcal{V}} \ll \sqrt{N} = O(t) = o(t \log N).$$

What survives:

The critical square-root branch itself is not closed, because production of enough large block factors degenerates there. What is closed is the possibility that an already-produced critical block family can hide its logarithmic cost in vertical layers.

Related current target:

C-MONOMIAL-REFLECTED-PAIR-SPARSITY.

FAIL-CRITICAL-VERTICAL-GENERIC-001**FAIL-ID:**

FAIL-CRITICAL-VERTICAL-GENERIC-001.

Date / iteration:

vNEXT4.7.

Claim:

Near the critical square-root gap, the surviving vertical branch still has arbitrary prime-power/cofactor structure.

Status:**KILLED** as a description.**Exact certificate:**

On

$$h \geq \sqrt{N/3},$$

every block-derived vertical event is exactly

$$N - w = q^2.$$

What survives:

A specific monomial reflected/quadratic prime-pair correlation, not generic U_R -capacity.

FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001**FAIL-ID:**

FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001.

Status:**KILLED**.**Killed inference:**

a linear aggregate defect on bases $q \leq N^{1/3}$ automatically yields one base of polynomial multiplicity.

Certificate:

there are $\asymp N^{1/3}/\log N$ available prime bases; for a packet of size $M \asymp N^{1/3}$, plain pigeonhole forces only $\gg \log N$ average multiplicity.

Do not repeat:

do not replace the current low-layer consumer by an unweighted single-base pigeonhole.

FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-001**FAIL-ID:**

FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-001.

Status:**SCOPE-CORRECTED**.**Earlier failure:**

FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001 correctly kills raw cube-root pigeonhole as a concentration theorem.

New explanation:

this is exactly the boundary value $m = 2$ of **G-POWER-THRESHOLD-DEFECT-COMPRESSSION**: the threshold $N^{1/(m+1)}$ is then $N^{1/3}$ and the guaranteed star exponent $1/3 - 1/(m+1)$ is zero. Every fixed $m \geq 3$ gives a nontrivial polynomial star.

What survives:

do not infer polynomial concentration from cube-root pigeonhole alone; first split the alphabet at a lower power threshold.

FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEP-CYCLES-001

FAIL-ID:

FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEP-CYCLES-001.

Source:

explicit v4.53 Failure Log addition.

Status:

KILLED as a sufficient uniform closure mechanism.

Reason:

the exact relation

$$Q^L D = mN + (-1)^L$$

is useful, but in the small- Q regime it does not by itself bound the population of non-pure cycles at the required polynomial scale.

FAIL-DC-CAPACITY-COMPRESSION-001

FAIL-ID:

FAIL-DC-CAPACITY-COMPRESSION-001.

Status:

SCOPE-CORRECTED.

Killed form:

within the Q -indexed scalar route of C-Q-INDEXED-SCALAR-PRODUCTION-SUBCRITICALITY, increasing D_C cannot lower FC8 below the hard $|Q|$ -scale floor needed by that same production.

What survives:

once a proposed production violates the one-logarithmic-unit per- Q -index hypothesis, the exact term $-\log D_C$ in A_R matters again. Do not promote this entry to a universal ban on denominator compression.

FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHGAP-PACKET-001

FAIL-ID:

FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHGAP-PACKET-001.

Status:

KILLED.

Killed inference:

the dead-smooth carrier or reflected classes may carry a fixed polynomial fraction of a high-gap reflected packet.

Certificate:

v4.40 counts integers supported on the fixed alphabet $\text{PF}(x)$ by exponent vectors and obtains only $x^{o(1)}$ possibilities below $2x$. Hence $|\text{DS}|, |\text{RDS}| = N^{o(1)} = o(t)$ on $h > N^{1/3}$.

Compatibility:

FAIL-DEADSMOOTH-UNBOUNDED-001 remains valid. It rejected a finite/uniform inference from the old local ceiling. The new statement is asymptotic fixed-support counting and does not assert that two dead-smooth rows are impossible or that the class is absolutely bounded.

FAIL-DEADSMOOTH-PILLAI-001

FAIL-ID:

FAIL-DEADSMOOTH-PILLAI-001.

Date / iteration:

Phase-3 continuation, 31 August 2026.

Claim:

No canonical N carries two simultaneous dead-smooth carriers, since none was found below $3 \cdot 10^5$.

Original motivation:

The search returned nothing at all, and the configuration looks arithmetically absurd.

Status:

SCOPE-CORRECTED. The *search result* stands as COMP-DEADSMOOTH-CENSUS; the *exclusion* does not.

Minimal / strong counterexample:

None is needed: the claim has no proof.

Exact certificate:

By TD-DEADSMOOTH-NORMALFORM(5), two simultaneous dead-smooth carriers with $\omega(a_j) = 1$ are exactly a Pillai coincidence $|q^e - q'^e| < h$ with $q^e, q'^e > 4x/5$, $q, q' < h$ and $qq' \mid x$. Excluding that unconditionally requires effective linear forms in logarithms. The corpus contains no such argument.

Failed hypothesis or mechanism:

An empty finite search was read as an exclusion theorem.

Why the argument failed:

This is PROHIB-STATUS-LADDER: a finite computation, however clean, is not a universal statement.

What survives:

The exact reduction to a Pillai equation, which is a genuine normal form and the right object for a future attack.

Safe weaker version:

“No such pair occurs below $3 \cdot 10^5$ ”, labelled **VERIFIED COMPUTATION**.

Current replacement:

Theory Core, TD-DEADSMOOTH-NORMALFORM(5) and COMP-DEADSMOOTH-CENSUS.

Do not repeat:

Never convert an empty search into an exclusion.

Related current target:

TD-SMALLSMOOTH.

FAIL-DEADSMOOTH-UNBOUNDED-001**FAIL-ID:**

FAIL-DEADSMOOTH-UNBOUNDED-001.

Date / iteration:

Phase-3 continuation, 31 August 2026.

Claim:

The exceptional dead-smooth class is negligible for every N , so the side condition $\lceil t/3 \rceil > \sigma_N^+(h)$ of TD-COB-CLOSURE can be discharged and removed.

Original motivation:

Dead-smooth carriers are extraordinarily rare — 19 among 514 447 tested, none doubled — which invites the conclusion that the class is uniformly negligible.

Status:

KILLED.

Minimal / strong counterexample:

$N = 60\,060$, $p_0 = 30\,047$, $h = 17$, $t = 8$.

Exact certificate:

$x = 30\,030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ is a primorial, so $\mathcal{P}_N(h) = \{3, 5, 7, 11, 13\}$ — every odd prime below h divides x . The ceiling then degenerates: $\sigma_N^+(h) = 6.32$ while $\lceil t/3 \rceil = 3$. The side condition fails.

Failed hypothesis or mechanism:

Rarity of the class in a sample was confused with a bound valid for every N . The ceiling depends on how many small primes divide x , and x may legitimately be a primorial.

Why the argument failed:

σ^{dead} and σ^+ are small *because* x usually has few small prime factors, not because the dead-smooth normal form is intrinsically bounded. No theorem forbids $x = \prod_{\ell \leq h} \ell$.

What survives:

TD-DEADSMOOTH-CEILING and TD-COB-CLOSURE exactly as stated, with the side condition retained as a hypothesis. It is an arithmetic condition on N alone, checkable in $O(\pi(h) \log x)$ operations, and it held in every tested canonical instance with $t \geq 16$.

Safe weaker version:

State the closure with the side condition, and treat the small-smooth- x regime as a separate open case.

Current replacement:

Theory Core, TD-COB-CLOSURE and the retained target TD-DENSITY.

Do not repeat:

Do not discharge a hypothesis because a sample makes it look automatic; check the extremal arithmetic first.

Related current target:

TD-DENSITY, $P1^{\text{br}}$.

FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-001**FAIL-ID:**

FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-001.

Alias:

v4.50 R50-1.

Status:

KILLED for the capacity inference.

Exact theorem retained:

for any actual path $v_0 \rightsquigarrow p$,

$$Q \mid N - p \implies \Xi(\mathcal{P}) \equiv v_0 N^{-1} \pmod{Q}.$$

Why rejected as capacity:

this residue is forced by the endpoint through the exact affine path identity and imposes no independent restriction on intermediate vertices or cofactors.

Replacement theorem:

G-DEEP-PATH-RESIDUE-ENDPOINT-TAUTOLOGY.

FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPE-001**FAIL-ID:**

FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPE-001.

Source:

explicit v4.54 Failure Log addition and correction to v4.53.

Status:

KILLED as a proof-design necessity.

Reason:

every pure path terminates at a non-pure row, its length is $N^{o(1)}$, and the refined fibre theorem already forces polynomial boundary defect/off-layer support without first passing through ancestry predecessors.

Supersession:

C-DEEP-FIRST-ENTRY-OWNER-CAPACITY is **SUPERSEDED** as the binding frontier. The v4.53 first-entry matching remains a valid secondary local diagnostic.

FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONE-001

FAIL-ID:

FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONE-001.

Source:

explicit v4.52 Failure Log addition.

Status:

KILLED as a proof mechanism from one reflected root, common-sink topology, total path cofactor products, and bounded branching alone.

Reason:

the current corpus yields renewal alternatives rather than a subpolynomial fibre bound.

Scope guard:

this does not refute a stronger arithmetic fibre theorem using edge-local valuation or additional selected-family structure.

FAIL-DEEP-ONE-AP-PACKING-CLOSES-001

FAIL-ID:

FAIL-DEEP-ONE-AP-PACKING-CLOSES-001.

Status:

KILLED as a sufficient closing mechanism.

Killed route:

use only the fact that a selected row/index lies in one residue class modulo the deep modulus Q .

Reason:

the resulting ceiling of shape $1 + N/(6Q)$ is too weak when $Q = N^{o(1)}$.

FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001

FAIL-ID:

FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001.

Source:

explicit v4.53 Failure Log addition.

Status:

KILLED.

Killed possibility:

the pure subgraph defined by $N - p = Qq$ can support a nontrivial active cycle.

Reason:

the affine fixed-point identity of v4.53 forbids every nontrivial active cycle in the pure $d = 1$ subgraph.

Scope guard:

non-pure cycles with variable multipliers d_i are not ruled out by this statement; they satisfy the separate multiplier law.

FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001

FAIL-ID:

FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001.

Source:

explicit v4.53 Failure Log addition and v4.53 frontier correction.

Status:

KILLED as a proof-design description.

Killed description:

a cheap polynomial deep fibre may require an independent global theorem controlling exact-layer self-renewal as a terminal phenomenon.

Reason:

the v4.53 stability theorem forces the cheap fibre into short pure affine paths whose starts expose boundary owners / off-layer support.

Supersession:

C-DEEP-SELF-RENEWAL-QCOFACTOR-CAPACITY is **SUPERSEDED** as the standalone binding frontier. This does not say all renewed external population is contradictory.

FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001**FAIL-ID:**

FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001.

Status:

SUPERSEDED as a primary frontier.

Superseded description:

v4.35 temporarily treated repeated migration of deep vertical mass to new q -free bases as the smallest obstruction.

Certificate:

v4.36 proves, pending audit, the static global capacity

$$\log U_R^{\text{nr}} \ll \sqrt{\frac{N\gamma}{\log(2 + \gamma)}},$$

without proving dynamic termination.

What survives:

nested peeling remains useful for structural analysis of near-extremal packets, but it is not necessary merely to prevent unbounded vertical storage.

FAIL-DESCENT-ONE-STATE-001**FAIL-ID:**

FAIL-DESCENT-ONE-STATE-001.

Date / iteration:

vNEXT4.15.

Old description:

Passing immediately to the full descendant closure reduces the descent branch to the existence of one state $r \leq h$.

Status:

SUPERSEDED.

Replacement:

G-FIRST-ROW-HIGHLOW-DISPOSITION retains the selected first parent rows and yields full logarithmic defect, weighted state expansion, explicit horizontal reuse, a fixed-polylog population, or a quantitative reciprocal band.

FAIL-DIRECT-ADJACENCY-001**FAIL-ID:**

FAIL-DIRECT-ADJACENCY-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

Every internally absorbed atom is incident, or uniformly adjacent, to a GOOD escape edge in its first-front branch.

Original motivation:

Most observed witnesses were shallow.

Status:

KILLED for direct adjacency.

Minimal / strong counterexample:

$N = 14198$, internal atoms 19 and 167.

Exact certificate:

The audited sets are $B_{\text{abs,int}} = \{19, 167\}$ and $B_{\text{re}} = \{17, 83, 149\}$. Both internal atoms use the same merged GOOD edge $71 \rightarrow 277$, reached only at branch distances 8 and 7, respectively. Atom 19 has 93 external parents and atom 167 has 12; neither is directly incident to that GOOD edge.

Failed hypothesis or mechanism:

Typical shallow depth was promoted to a local theorem.

Why the argument failed:

Branches can merge long before their eventual escape.

What survives:

A nonlocal branch-escape disjunct remains viable.

Safe weaker version:

Reachability of a GOOD edge, with no fixed adjacency bound.

Current replacement:

Theory Core target $E1$.

Do not repeat:

Do not charge only the nearest one or two edges.

Related current target:

$E1, W1$.

FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001**FAIL-ID:**

FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001.

Status:

KILLED.

Rejected inference:

an unweighted lower bound on $\eta(R) = \sum_q (\kappa_R(q) - 1)$ is equivalent to the corresponding lower bound on $\log J_R$.

Reason:

exactly

$$\log J_R = \sum_q (\kappa_R(q) - 1) \log q,$$

while only $\log J_R \geq \eta(R) \log 3$ follows from the unweighted count.

FAIL-EXACTD-NONCANONICAL-001**FAIL-ID:**

FAIL-EXACTD-NONCANONICAL-001.

Date / iteration:

Exact-D audit; reaffirmed in the canonical bridge audit §11.2.

Claim:

The Exact-D local row algebra — pair compensation, smoothness, primitive normal form, prime endpoint — is enough to isolate the canonical root, so canonicity need not be used as boundary transport.

Original motivation:

The local conditions are sharp and eliminate most models.

Status:

KILLED as a substitute for minimality; **SCOPE-CORRECTED** as machinery.

Minimal / strong counterexample:

The recorded noncanonical normal forms, including $(q, r, m_0) = (3, 5\,413\,103, 81\,196\,545)$ — note $81\,196\,545 = 3 \cdot 5 \cdot 5\,413\,103$ — together with the two recorded $k = 1$ examples.

Exact certificate:

These models satisfy the local D_q, D_r conditions, smoothness, pair failure and a prime endpoint, and are eliminated *only* by $h < G$. Removing $h < G$ restores them.

Failed hypothesis or mechanism:

Row algebra was asked to do the work of the empty prime interval.

Why the argument failed:

Local normal forms constrain a row; they do not constrain the position of the root relative to x .

What survives:

ARS-EXACTD: pair compensation outside $q < r < 2h$, the primitive normal form, and $h^2 > 4m$ in the residual regime — all under an Exact-D *pair failure* hypothesis, not under CRD failure alone.

Safe weaker version:

Use Exact-D only where the pair-failure hypothesis is explicit, and state $h < G$ as a separate input.

Current replacement:

Theory Core, ARS-EXACTD and the ablation discussion of $h < G$.

Do not repeat:

Do not attempt to repair Exact-D by improving a constant; the corpus already localises the missing step as interval-to-ledger transport.

Related current target:

C0a, T_D , $P1^{\text{br}}$.

FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001**FAIL-ID:**

FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001.

Date / iteration:

vNEXT4.20.

Branch:

Retain Exact-D Pair Compensation failure as an obstruction when $h \leq \sqrt{x}$.

Status:

KILLED.

Certificate:

Existing Exact-D high-gap amplification says pair failure forces $h^2 > 4m$. But $h \leq \sqrt{x}$ gives $h^2 \leq x < 4m$ for sufficiently large x . Thus all first-front SMOOTH pairs satisfy Exact-D compensation on this domain.

Scope guard:

This closes only the pair-failure branch; it does not by itself strengthen the global weighted-cut/K1 bound.

FAIL-EXTERNAL-ETA-IDENTIFICATION-001**FAIL-ID:**

FAIL-EXTERNAL-ETA-IDENTIFICATION-001.

Date / iteration:

vNEXT2 independent audit, Phase 3.

Claim:

The high-floor reuse $\eta(U) \geq \lceil n/3 \rceil + c$ of an external orthogonal packet can be charged to the canonical capacity ledger.

Original motivation:

η is the same functional in both places, and a large value looks like a large obligation wherever it occurs.

Status:

KILLED.

Minimal / strong counterexample:

Structural, and it is exactly the orthogonality that was thought to help.

Exact certificate:

The packet is orthogonal, $K_{\times}(R, U) = 0$. That identity says precisely that U contributes nothing to the rooted collision ledger. Any functional built from cross-collision is therefore identically zero on this configuration, so no such functional can carry $\eta(U)$ into $\mathcal{C}_N(R)$.

Failed hypothesis or mechanism:

Orthogonality was read as a licence to add the two ledgers; it is the obstruction to doing so.

What survives:

TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY itself, now **PROVED** with both of its skeleton gaps closed.

Safe weaker version:

Report $\eta(U)$ as a property of U only.

Current replacement:

Theory Core, Proposition [prop:missing-bridge], which states the exact minimal lemma required: a *source-side* statement that re-closing $R \cup U$ changes the multisource product by a factor whose logarithm is bounded below in terms of $\eta(U)$.

Do not repeat:

Never add an orthogonal packet's ledger to the rooted one; the zero cross-term is the reason it cannot be done.

Related current target:

TF, K1, W1.

FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001**FAIL-ID:**

FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001.

Date / iteration:

vNEXT4.15.

Old obstruction:

The fixed-polylog external BAD clean bridge lacked an explicit squarefree-parent theorem.

Status:

CLOSED PENDING INDEPENDENT AUDIT.

Repair:

Large-square thinning applies to the external BAD parent fans as well, yielding squarefree, pairwise-coprime complements with exact designated valuation one and no unintended fixed-polylog factors.

FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001**FAIL-ID:**

FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001.

Date / iteration:

vNEXT4.14.

Strategy:

Partition external seeds into merger/disjoint subfamilies before analysing source retention.

Status:

SUPERSEDED as proof organisation.

Replacement:

Union all failed seed basins first. The union is child-closed below $N/3$, source retention applies to the whole family, and one overlap ledger covers either orthogonality or nonempty intersection with R .

FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-BRIDGE-001**FAIL-ID:**

FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-BRIDGE-001.

Status:

KILLED relative to the current data.

Rejected strategic description:

The next optimal step is to force a contradiction from source-front CRT compatibility of the external v18 star.

Reason:

the source-front lock constrains one affine coefficient but does not upper-bound it, and the non- q_0 moduli are largely pairwise coprime, so CRT compatibility is not itself a capacity obstruction.

Current replacement:

attack the canonical root-tied target **C-FIRSTFRONT-RESIDUAL-TO-LOWCORE-CAPACITY**; retain K1-INC and owner-anchored incremental capacity as secondary global routes.

FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-RADICAL-RESERVOIR-001**FAIL-ID:**

FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-RADICAL-RESERVOIR-001.

Status:

SUPERSEDED as a primary factorisation branch.

Superseded description:

external seed/source weight must remain as a third independent radical-storage reservoir in addition to fresh support and incremental J .

Certificate:

v4.39 proves the exact extension identity

$$\prod_{a \in A} \text{rad}(N - a) = F_R(A) \frac{J_S}{J_R}.$$

All new complement radical mass is therefore stored in one baseline copy of fresh support plus rooted/repeated copies in incremental J .

What survives:

Σ_{ext} remains useful for source-sensitive congruence/product arguments, but it is not needed as an independent primary factorisation leaf.

FAIL-EXTERNAL-SUCCESS-001**FAIL-ID:**

FAIL-EXTERNAL-SUCCESS-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

If an internal atom has an active parent outside the current bad world, then that parent supplies a successful rooted escape.

Original motivation:

Leaving the current closed set was conflated with entering the global success basin.

Status:**KILLED**.**Minimal / strong counterexample:** $(N, \ell, p) = (6142, 877, 3511)$.**Exact certificate:**

$3511 \in \mathcal{A}_{6142}$, it lies outside the audited p_0 -world, and $6142 - 3511 = 2631 = 3 \cdot 877$; hence it is an external parent of 877. The independently checked stopped traversal rooted at 3511 fails.

Failed hypothesis or mechanism:

“External to R ” was replaced by “outside the global bad basin.”

Why the argument failed:

A different failed basin can feed the same atom.

What survives:

The parent is an exact ambient incidence and can be promoted into an enlarged valuation ledger.

Safe weaker version:

External active parent *or* internal branch escape, with no success claim about the parent.

Current replacement:

Theory Core, G-AMBIENT-PROMOTION and targets $E1, W1, K1$.

Do not repeat:

Existence of a parent is not success of that parent.

Related current target:

$E1, W1$.

FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001**FAIL-ID:**

FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001.

Status:**KILLED**.**Killed inference:**

the exact relative identity $T_{S^*}/T_R = (U_{S^*}/U_R)(J_{S^*}/J_R)$ and source lock automatically bound the external vertical factor from above.

Certificate:

increasing an exact valuation from 1 to e enlarges Δ_U without changing the support graph, source set, or source-lock residue class. The known canonical upper capacity is for canonical U_R^{pr} , not arbitrary external incremental U .

What survives:

the identity is a strong exact payment ledger, but a new receiving-capacity theorem would be required to close it.

FAIL-F1-DOMAIN-MIX-001**FAIL-ID:**

FAIL-F1-DOMAIN-MIX-001.

Date / iteration:

Interrupted T_D round, dry run, 31 August 2026.

Claim:

A falsifier run over mixed controls can refute a candidate statement whose domain is “closed failed world”.

Original motivation:

A single sweep over many roots was cheaper than three separate domains.

Status:**SCOPE-CORRECTED**.

Minimal / strong counterexample:

The dry run itself.

Exact certificate:

The recorded log states that the candidate F1 had domain “closed failed world” while the code mixed in successful-envelope controls; the domain filter and the counter for fresh states belonging to R were corrected, and three separate domains were preserved thereafter. This is a **VERIFIED COMPUTATION** record of a tooling defect, not a mathematical failure of F1.

Failed hypothesis or mechanism:

Domain bookkeeping inside the falsifier, not the statement under test.

Why the argument failed:

A counterexample from the wrong domain is not a counterexample.

What survives:

The candidate statement, untouched, still awaiting a correctly scoped test.

Safe weaker version:

Every falsifier must declare and enforce its domain before it may report a kill.

Current replacement:

The benchmark suite below, in which every row carries its domain and evidence class.

Do not repeat:

Never let a falsifier report a kill it obtained outside the declared domain.

Related current target:

TF, T_D .

FAIL-FC8-FLOOR-FOR-RHS-001**FAIL-ID:**

FAIL-FC8-FLOOR-FOR-RHS-001.

Status:

KILLED.

Rejected inference:

proving $\log J_R > (|Q| + \tau) \log N$ automatically contradicts FC8.

Reason:

$(|Q| + \tau) \log N$ is only a lower floor for

$$A_R = (\gamma + \tau) \log N + \tau \log \frac{25}{18} - \sum_{q \in C \setminus Q} \log q.$$

The exact target must use the exact right-hand side, not a lower bound for it.

FAIL-FC8-SELF-CONTRADICTION-001**FAIL-ID:**

FAIL-FC8-SELF-CONTRADICTION-001.

Date / iteration:

vNEXT2 independent audit, Phase 2.

Claim:

The low-core capacity bound (FC8), being an upper bound on J_R under canonical failure, can be contradicted by the forced lower bounds already available, closing the $\mathcal{D}$ /low-core front.

Original motivation:

(FC8) is the first genuine upper capacity for J_R , and upper capacities are what a contradiction architecture wants.

Status:

KILLED. (FC8) itself is **PROVED**; the inference from it is not.

Minimal / strong counterexample:

Structural; no finite witness needed.

Exact certificate:

The only unconditional lower bound on J_R available from canonical failure is $J_R \geq 3^{\eta(R)}$ by (LE3). A contradiction with (FC8) therefore requires $\eta(R) > (\gamma + \tau) \log_3 N + \tau \log_3(25/18) - \sum_{C \setminus Q} \log_3 q$, that is $\eta(R) \gtrsim \gamma \log N$. Nothing in the corpus forces a reuse count carrying a factor $\log N$; the strongest available floor is $\eta(R) \geq |Q| + |H|$ (Theorem [thm:lowcore-floor]), which is bounded by $|R|$.

Failed hypothesis or mechanism:

An upper capacity was treated as self-contradictory without a matching lower bound of the same order.

Why the argument failed:

The two sides differ by a factor $\log N$ per low-core element; that gap is not an accident of effort but of what the available identities can produce.

What survives:

(FC8) as an upper capacity usable through the external comparison (FC10), exactly as its own capacity paragraph states.

Safe weaker version:

Report (FC8) only in the (FC10) form.

Current replacement:

Theory Core, Proposition [prop:fc8-barrier].

Do not repeat:

Do not attempt another self-contained contradiction from (FC8); the barrier is at the level of available bounds.

Related current target:

K1, P1BR.

FAIL-FC8-SLACK-IS-VERTICAL-MASS-001**FAIL-ID:**

FAIL-FC8-SLACK-IS-VERTICAL-MASS-001.

Status:

SCOPE-CORRECTED.

Killed inference:

a large $\Delta_{\text{FC8}} = B_R - \log(D_C J_R)$ by itself forces a comparably large $\log U_R^{\text{nr}}$.

Certificate:

v4.35 gives the exact identity

$$\Delta_{\text{FC8}} = \log U_R^{\text{nr}} + \Delta_{\text{full}}, \quad \Delta_{\text{full}} > 0.$$

The positive full-layer row/path envelope slack is an independent part of the FC8 slack.

What survives:

LCAP1/LCAP4 may still require vertical mass after the size of $D_C J_R$ has been specified.

FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001**FAIL-ID:**

FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001.

Source:

explicit v4.54 Failure Log addition.

Status:

KILLED.

Killed description:

the square first-entry packet can only be spaced using one residue class modulo Q .

Correction:

a pure-path start satisfies

$$p_i \equiv N(1 - Q) \pmod{Q^2},$$

so distinct square first-entry targets have spacing at least $2Q^2$.

FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001**FAIL-ID:**

FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001.

Status:

KILLED.

Rejected description:

$Q \subseteq C$ was missing from the previous Core.

Certificate:

it is explicitly used in the independently audited FC7/FC8 proof and audit. v4.34 only extracted it for a new scale comparison.

FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001**FAIL-ID:**

FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001.

Status:

KILLED on $h \leq \sqrt{x}$.

Rejected statement:

The only general lower scale for $M_q = (N - q)/\gcd(m, N - q)$ is the old $M_q > (5/3)q$ bound.

Certificate:

canonical shadow geometry gives $D_q = \gcd(m, |2h - q|)$ and uniformly $M_q > (5/6)h$; when $q \leq 4h$, $M_q > (5/6)\Lambda h$.

FAIL-FIRSTFRONT-SCALAR-ROUTE-001**FAIL-ID:**

FAIL-FIRSTFRONT-SCALAR-ROUTE-001.

Status:

KILLED with the scope below.

Killed inference:

a production indexed by $S \subseteq Q$, contributing less than one ambient logarithmic unit per index, can close the current FC8 capacity without multiplicity amplification or a new structural upper bound.

Certificate:

$Q \subseteq C$ is already proved and the exact FC8 right-hand side A_R satisfies $A_R > |Q| \log N$, whereas such a production is $< |Q| \log N$.

What survives:

first-front shadow geometry, H_3/H_5 structure, multiplicity-amplified arguments, independent capacity compression, analytic input, and non-scalar contradictions are not killed.

FAIL-FIXED-FINITE-MIRROR-HUB-001**FAIL-ID:**

FAIL-FIXED-FINITE-MIRROR-HUB-001.

Date / iteration:

vNEXT4.16.

Escape:

Store all active smooth-mirror valuation mass on an N -independent finite family of small primes.

Status:**KILLED**.**Exact certificate:**

The mirror support capacity

$$M_{\text{act}}^- \leq t \sum_{r \in P_{\text{mir}}} \frac{\log r}{r-1} + |P_{\text{mir}}| \log N$$

is only $O(t) + O(\log N)$ for fixed finite support, while the natural smooth branch requires $M_{\text{act}}^- \gg t \log \Lambda$ with $\Lambda \rightarrow \infty$.

Scope guard:

This does not say the direct least-prime children of the original parent rows cannot all equal 3; it says the mirror side must then create additional growing active support.

FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001**FAIL-ID:**

FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001.

Date / iteration:

vNEXT4.12.

Claim:A standard fixed- k Selberg sieve immediately gives

$$\#\{q : q, N - kq^2 \text{ prime}\} \ll \sqrt{N/k} / \log^2 N$$

with one absolute constant uniformly over all polylogarithmic odd k .

Status:**KILLED**.**Exact obstruction:**

The fixed- k local root count contains the quadratic character (kN/p) ; uniform singular-series control is an additional theorem obligation.

Replacement:

Sum over k first and use the hybrid dimension-one linear sieve in the coefficient variable.

FAIL-FRESH-BRANCH-PRIMACY-001**FAIL-ID:**

FAIL-FRESH-BRANCH-PRIMACY-001.

Date / iteration:

Phase-3 continuation, 31 August 2026.

Claim:

The external fresh branch $\mathcal{D}_{\text{fr}}^{\text{ext}}$ is where the $\mathcal{D}$ obligation is strongest, so cut statements should be developed there first and restricted to it.

Original motivation:

External carriers are candidate *new sources* and therefore the ones that bear on $P1^{\text{br}}$; that relevance was transferred to the cut itself.

Status:**SCOPE-CORRECTED**.**Minimal / strong counterexample:**

Structural, by TD-LOCALIZED-CUT.

Exact certificate:

For $\tau \geq 2$ distinct active primes bounded by y , $C_{\text{cut}} > \tau(\tau-1) \log((N-y)/y)$. A fresh carrier gives only $y = N/6$ and hence the constant $\log 5$; a live smooth carrier gives $y < h$ and hence $\log((N-h)/h)$, larger by orders of magnitude once $h = o(N)$. The fresh branch is therefore

the *weakest* of the three for cut purposes.

Failed hypothesis or mechanism:

Relevance to $P1^{\text{br}}$ was conflated with strength of the cut.

Why the argument failed:

Two different functionals were being compared. Only external carriers can be new sources; but the cut constant is governed by the *localisation* of the primes, and small primes localise better.

What survives:

Both facts, kept apart. Only the external branch bears on $P1^{\text{br}}$; the live-smooth branch gives the largest cut defect.

Safe weaker version:

State the cut with the explicit bound y and read the constant off it, rather than fixing $\log 5$.

Current replacement:

Theory Core, TD-LOCALIZED-CUT, with TD-FRESH-CUT and TD-FIVEFOLD-CUT as its $y = N/6$ cases.

Do not repeat:

Do not restrict a theorem to the sub-branch that happens to be relevant downstream; state it where its proof lives.

Related current target:

T_D , $P1^{\text{br}}$.

FAIL-FRIDGE-ALPHABET-TRANSFER-001

FAIL-ID:

FAIL-FRIDGE-ALPHABET-TRANSFER-001.

Date / iteration:

vNEXT2 independent audit, Phase 4.

Claim:

For $\lambda_N > 3$ the upper fringe is charged by the alphabet $\{\lambda, 2\lambda - 1\}$, by analogy with $\{3, 5\}$.

Original motivation:

$\{3, 5\} = \{\lambda, 2\lambda - 1\}$ at $\lambda = 3$, and (AN7) features $2\lambda - 1$.

Status:

SCOPE-CORRECTED.

Exact certificate:

The admissible coefficients are determined by (AN5), $N(1 - 1/\lambda)/q < b < N/q$, together with b odd, $(b, N) = 1$, $b \geq \lambda$. For q ranging over $(N/(2\lambda), N/\lambda)$ this interval has length $N/(q\lambda) \in (1, 2)$ only near the top of the band; lower in the band it admits several odd values, so the alphabet is a range, not a pair. The two-letter alphabet at $\lambda = 3$ is an accident of $2 < b < 6$ admitting exactly $\{3, 5\}$.

Failed hypothesis or mechanism:

A coincidence at one value of λ was read as a pattern.

What survives:

(AN5) itself, which is the correct general statement and the right starting point: compute the alphabet from the interval before proposing any potential.

Safe weaker version:

For general λ , state the admissible coefficient set as the odd integers coprime to N in the (AN5) interval.

Current replacement:

Theory Core, Phase-4 note in Section [sec:vnext2].

Do not repeat:

Do not propose a monotone potential of $D(q) = 4q - N$ type for $\lambda > 3$ before the alphabet is computed; D was built from the coefficient 3.

Related current target:

$\lambda_N > 3$ fringe capacity.

FAIL-FRIDGE-BOUNDED-MULTIPLICITY-001**FAIL-ID:**

FAIL-FRIDGE-BOUNDED-MULTIPLICITY-001.

Status:

KILLED.

Certificate:

the first-front-to-fringe map is already injective and the H_3, H_5 bands are disjoint. Thus there is no hidden large fibre to overflow.

What survives:

an independent theorem proving that the relevant incidences cannot exist at all, or cannot coexist with canonical high-gap geometry, remains possible.

FAIL-FRIDGE-BRANCH-DOUBLE-CHARGE-001**FAIL-ID:**

FAIL-FRIDGE-BRANCH-DOUBLE-CHARGE-001.

Status:

SCOPE-CORRECTED.

Killed inference:

count the same fringe entrance/path population τ as a new independent scalar payment after FC8/LCAP4 already price those rows.

What survives:

exact 3/5 arithmetic may still yield a forbidden configuration, a new nonlocal relation, or GOOD. Structural use is not double charging.

FAIL-FRIDGE-CONGRUENCE-CONTRADICTION-001**FAIL-ID:**

FAIL-FRIDGE-CONGRUENCE-CONTRADICTION-001.

Status:

SCOPE-CORRECTED.

Certificate:

the finite exact example $N = 4298$ in the revised report shows that the bare congruence pattern $q \mid m$, $N - q = b\ell$ with $b \in \{3, 5\}$ is arithmetically realisable with several incidences.

Scope:

that example is outside the canonical high-gap/sub-root failed domain. It kills a contradiction from the bare congruences alone, not a future theorem using the missing canonical/high-gap or primality input.

FAIL-GAMMA-BOUND-ROUTES-001**FAIL-ID:**

FAIL-GAMMA-BOUND-ROUTES-001.

Date / iteration:

vNEXT2 independent audit, Phase 2.

Claim:

The adaptive indegree bound, the denominator of (FC8), a distinct resource per low-core element, or the layered Euler defect bounds $\gamma = |C|$.

Original motivation:

Each route looks like it converts local structure into a global count.

Status:

KILLED for all four, as routes; the underlying statements are untouched.

Exact certificate:

(i) Summing (AN2) against Theorem [thm:lowcore-floor] gives $\gamma + |H| \leq \sum_{q \in C} \lceil N/(6q) \rceil$, which every summand already satisfies term by term. (ii) Nothing distinguishes $C \setminus Q$ from an arbitrary set of distinct odd primes, so only the primorial bound applies, and Corollary [cor:primorial-ceiling] then gives $\gamma \log \gamma \lesssim 2\gamma \log N$, i.e. $\gamma \lesssim N^2$. (iii) A single row can host $\log_3 N$ low prime factors, so the parent count bounds γ only by $|R| \log_3 N$. (iv) The layered Euler route produces Theorem [thm:lowcore-floor] itself, which bounds $\eta(R)$ below and the low-core primorial mass above, and γ in neither direction.

What survives:

Theorem [thm:lowcore-floor] and Corollary [cor:primorial-ceiling], both new and both genuine.

Do not repeat:

Do not retry these four; a bound on γ needs a mechanism that is not a summation of an existing local bound.

Related current target:

(FC8), K1.

FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001**FAIL-ID:**

FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001.

Status:

KILLED.

Rejected inference:

$\gamma = |C| = O(\log N / \log \log N)$, and hence $t \gg \gamma$, holds throughout the relevant canonical domain.

Reason:

the standard divisor-count bound concerns $\omega(m)$ and hence $|Q|$, not γ . Existing polylogarithmic statements for γ are hypotheses on subdomains, not universal conclusions.

Do not repeat:

do not compare reflected population t with γ without a proved theorem giving that comparison.

FAIL-GENERIC-ABSORPTION-001**FAIL-ID:**

FAIL-GENERIC-ABSORPTION-001.

Date / iteration:

Post-v7.5 generic absorption audit.

Claim:

Every residual label in a closed world is paid by within-row power, cross-row reuse, or the proposed generic absorption channel.

Original motivation:

$T = UJ$ suggested that no residual singleton could remain unpaid.

Status:

KILLED on the generic/noncanonical domain.

Minimal / strong counterexample:

$N = 1718, r = 1109$, not the canonical root (the canonical selector is the diagonal 859).

Exact certificate:

$N - r = 609 = 3 \cdot 7 \cdot 29$, and

$$1718 - 7 = 1711 = 29 \cdot 59, \quad D_7 = 29, \quad M_7 = 59.$$

The chosen closed recurrent core contains 59, only the row 7 contains the factor 59, and $1718 - 59 = 1659 = 3 \cdot 7 \cdot 79$. Thus the audited 59-valuations of both the within-row and

cross-row payment terms are zero.

Failed hypothesis or mechanism:

A generic identity was asked to produce canonical source information.

Why the argument failed:

A singleton residual can enter the state set without being paid by the proposed local channels.

What survives:

$T = UJ$ itself and any future statement that uses genuine canonical minimality essentially.

Safe weaker version:

Keep the generic identity; add a separately proved canonical transport statement before asserting absorption.

Current replacement:

Theory Core, C0b-RD and branch targets.

Do not repeat:

Do not derive a canonical trichotomy from Factor–Flow alone.

Related current target:

$T_D, T_F, P1^{\text{br}}$.

FAIL-GENERIC-AVERAGE-DEPTH-O-LOGLN-001

FAIL-ID:

FAIL-GENERIC-AVERAGE-DEPTH-O-LOGLN-001.

Status:

SUPERSEDED.

Old diagnostic:

size, exact valuation, parent-slot and scalar Brun–Titchmarsh admit Level-B integer packets with full-scale vertical depth but too little horizontal multiplicity for an $O(\log \log N)$ average-depth bound.

Superseded by:

FAIL-GENERIC-PRIME-ROW-RPD-001, which realises the same obstruction on actual prime rows.

FAIL-GENERIC-PRIME-ROW-RPD-001

FAIL-ID:

FAIL-GENERIC-PRIME-ROW-RPD-001.

Status:

KILLED as a generic theorem.

Killed statement:

for every packet X of genuine prime rows below $N/3$,

$$\sum_{Y < r \leq N^{1/3}} \sum_{k \geq 2} (n_k^X(r) - 1)_+ \ll (\log \log N) \sum_{Y < r \leq N^{1/3}} (\kappa_X(r) - 1)_+ + o(|X|).$$

Certificate:

using the analytic sparse-moduli theorem underlying Baker’s exact-layer fan, choose a mesoscale prime $r > Y$ and an exact depth $e \asymp (\log \log N)^3$. A fixed proportional interval below $N/3$ contains $N^{1-o(1)}$ primes in the required exact residue class. After Brun–Titchmarsh/Mertens control of unintended band supports one can select $h = N^{1/3+o(1)}$ prime rows with

$$R_X \gg h(\log \log N)^3, \quad \eta(X) = O(h \log \log N).$$

Scope guard:

this is Level C only. It does not show that the rows are actual members of $H_Q \setminus R$ or descendants of the designated v4.41 matching.

Do not repeat:

generic primality, spacing, nested packing and standard BT cannot establish the Level-D RPD theorem.

FAIL-GREEDY-AVOIDANCE-NAIVE-001

FAIL-ID:

FAIL-GREEDY-AVOIDANCE-NAIVE-001.

Date / iteration:

vNEXT3 hostile proof of the external selector.

Claim:

In the external selector one may choose complements greedily, avoiding the primes already used, and the surviving proportion of the fan is $\prod_{r \in \mathcal{E}} (1 - \frac{1}{r-1}) \gg 1/\log \log N$.

Original motivation:

$|\mathcal{E}| = O((\log N)^{A+1})$, so the avoided set is small compared with the fan, and the product looks Mertens-like.

Status:

KILLED as an argument. The *conclusion* of the selector is true — see G-BAD-BASE-SQUAREFREE-POLYLOG-EXTENSION-PROVED — but not by this route.

Minimal / strong counterexample:

Structural, two independent failures.

Exact certificate:

(i) Without a prior roughness condition, $\mathcal{E}$ may contain small primes. If $3 \in \mathcal{E}$ then $\sum_{r \in \mathcal{E}} 1/(r-1) \geq \frac{1}{2}$, and the Brun–Titchmarsh union bound over $\mathcal{E}$ already exceeds the whole fan, so the union bound gives nothing. (ii) To use $\prod_{r \in \mathcal{E}} (1 - \frac{1}{r-1})$ as a density one needs a *sieve* over $\mathcal{E}$, not a union bound. The combined modulus is $\prod_{r \in \mathcal{E}} r = \exp(\theta(y))$ with $y \approx (\log N)^{A+2}$, that is $\exp((\log N)^{A+2})$, which vastly exceeds N ; Siegel–Walfisz does not apply there, and a sieve at that level would require a level-of-distribution input such as Bombieri–Vinogradov, which was never declared.

Failed hypothesis or mechanism:

A product of local densities was used where only a union bound was available, and the modulus range needed for the alternative was never checked.

What survives:

Everything, after reordering. Imposing w -roughness *first* — no prime below $w = (\log N)^{A+2}$ divides a complement except its own designated base — forces $\mathcal{E}$ to consist only of primes $\geq w$, whence $\sum_{r \in \mathcal{E}} 1/(r-1) \leq |\mathcal{E}|/(w-1) \ll 1/\log N$ and a plain union bound suffices. The sieve then used has polylogarithmic sifting limit and is truncated at a polylogarithmic level, so every modulus stays inside the Siegel–Walfisz range and no level-of-distribution theorem is imported.

Attribution correction:

the $1/\log \log N$ loss is real, but it belongs to the roughness sieve, where it is $\asymp 1/\log w$. The coprimality step costs only $1 + O(\log w/\log N)$. vNEXT2 attributed the loss to the wrong step.

Safe weaker version:

State the selection as: sieve for roughness, then delete squares, then union-bound the coprimality classes.

Current replacement:

Theory Core, Theorem [thm:selector].

Do not repeat:

Do not multiply local densities across conditions that are only accessible by a union bound, and always check that the combined modulus stays in the range of the distribution theorem being used.

Related current target:

(FC10), K1.

FAIL-HALL-SUPPORT-001

FAIL-ID:

FAIL-HALL-SUPPORT-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

A Hall expansion over the support counts of all sources yields the required capacity bound.

Original motivation:

Every source contributes support, so a global counting argument looked available.

Status:

KILLED.

Minimal / strong counterexample:

The recorded compression figure: 456 sources compress to 17 factor atoms. Status **VERIFIED COMPUTATION**; the raw table was not reloaded in this extraction.

Exact certificate:

With that degree of compression the naive expansion has no room to run: the support side is far smaller than the source side, so no Hall-type surplus exists in the required direction.

Failed hypothesis or mechanism:

Sources and atoms were counted as if they were in near-bijection.

Why the argument failed:

Compression is exactly the regime in which a counting surplus disappears.

What survives:

The precise asymptotic polylog Hall matching theorem is *not* refuted by this: G-POLYLOG-MATCHING runs on a fixed finite family of polylog prime-power layers with an enormous per-layer fan, not on an all-source support count.

Safe weaker version:

Restrict any Hall argument to the polylog layer family where the fan is proved.

Current replacement:

Theory Core, G-POLYLOG-PARENTS, G-POLYLOG-MATCHING, target W1.

Do not repeat:

Do not run a counting argument across a compression.

Related current target:

W1, K1.

FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-ARITY-001

FAIL-ID:

FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-ARITY-001.

Status:

KILLED.

Killed inference:

after selecting a unique-parent target $q > N^{1/3}$, the remaining complement may still contain an uncontrolled high- arity cofactor supported entirely above the cube root.

Certificate:

v4.41 proves exact valuation depth $e \in \{1, 2\}$ and the row trichotomy: a low child $r \leq N^{1/3}$, the square q^2 , or the squarefree semiprime qr with both factors above the cube root.

What survives:

low boundary layers, monomial squares and high semiprime dynamics.

FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001

FAIL-ID:

FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001.

Status:**KILLED**.**Killed inference:**

a linear high-semiprime packet may heavily reuse its second factors while keeping incremental J negligible.

Certificate:

every repeated sibling $r > N^{1/3}$ contributes more than $(\log N)/3$ per excess row occurrence. If that ambient payment is absent, a $> 3/4$ subfamily has distinct siblings and therefore pairwise-coprime squarefree two-prime complements.

What survives:

the clean positive-sign branch itself was not a contradiction; v4.42 later supersedes it as the primary node.

FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001**FAIL-ID:**

FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001.

Status:**KILLED**.**Killed inference:**

above the exact cube-root floor, a large $\nu_{>N^{1/3}}$ term may remain a third independent storage channel alongside horizontal reuse and low boundary descent.

Certificate:

every high squared prime either forms a monomial row or has a cofactor $< N^{1/3}$ and therefore already crosses the low boundary. Together with monomial-component capacity and floor-truncated conservation,

$$\eta_{>N^{1/3}}(H) + 2D_{\leq N^{1/3}}(H) \geq (|H| - 1)/2 + c_{N^{1/3}}.$$

What survives:

a genuine aggregate low-layer defect on bases $q \leq N^{1/3}$.

FAIL-HIGHFLOOR-DOMAIN-VACUITY-001**FAIL-ID:**

FAIL-HIGHFLOOR-DOMAIN-VACUITY-001.

Date / iteration:

vNEXT2 independent audit, Phase 1E.

Claim:

TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY applies to packets that actually occur.

Original motivation:

The theorem is stated unconditionally on its hypothesis and its proof is complete.

Status:

SCOPE-CORRECTED. The theorem is **PROVED**; its *antecedent* is unverified.

Exact certificate:

An exhaustive search over even $N < 3 \cdot 10^4$ and all admissible seeds found *no* bad child-closed packet with $\min U > N^{1/3}$ and $U \subset (0, N/3)$. Status of that search: **VERIFIED COMPUTATION**.

Failed hypothesis or mechanism:

None mathematically; the risk is rhetorical — quoting $\eta(U) \geq \lceil n/3 \rceil + c$ as though it were about an exhibited object.

What survives:

The theorem in full, with both skeleton gaps now proved: σ has no cycles (a 1-cycle forces $N = u(u + 1)$, hence $u \mid N$ and u inactive; a 2-cycle forces $(v - u)(v + u + 1) = 0$; longer cycles are excluded because $f(x) = \sqrt{N - x}$ is strictly decreasing), and there are no three

consecutive square rows (the contraction constant of f puts u_2, u_3 within 0.111 of the fixed point for $N \geq 50$, forcing $u_2 = u_3$ and hence a 1-cycle).

Safe weaker version:

Always state it as a conditional.

Do not repeat:

An empty finite search is not evidence that the hypothesis is unsatisfiable, and a complete proof is not an exhibited instance.

Related current target:

TF, K1.

FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWNER-001

FAIL-ID:

FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWNER-001.

Source:

v4.53 scope correction relative to v4.32.

Status:

KILLED.

Killed import:

transfer the old lower bound

$$k_i > \frac{N}{6s_i}$$

from the upper-half owner geometry directly to the present low descendant basin.

Reason:

the algebraic owner-return normal form transfers, but the old sign and positive lower bound used an upper-half owner which is absent here.

What survives:

the signed deep owner-return normal form and the same-layer second-parent inverse bound remain valid in their stated domains.

FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001

FAIL-ID:

FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001.

Status:

KILLED.

Rejected statement:

$J_{R \cup P} / J_R \leq J_R$ for arbitrary external extensions.

Certificate:

vNEXT4.31 gives the exact primewise increment and permits new support multiplicity to create genuinely new J storage.

Do not repeat:

K1-INC itself is an open upper-capacity theorem, not a consequence of the definition of J .

FAIL-INDIVIDUAL-MESOSCALE-BV-001

FAIL-ID:

FAIL-INDIVIDUAL-MESOSCALE-BV-001.

Date / iteration:

v1.10 scale-wall continuation, 1 September 2026.

Claim:

The v1.6 “sub-sum of ordinary Bombieri–Vinogradov” argument can be transferred unchanged from a polylogarithmic base modulus to one fixed base $q = N^\beta$, $\beta > 0$, and still prove a positive lower- sieve fan.

Original motivation:

Extend the clean-parent theorem directly from polylogarithmic labels to polynomially sized states.

Status:

KILLED.

Exact certificate:

For one polynomial base the expected prime/sieve main term is of size at most

$$\asymp \frac{N}{q \log N \log z} = N^{1-\beta} \text{polylog}(N)^{-1},$$

whereas the ordinary Bombieri–Vinogradov all-moduli error available from simply taking a sub-sum is

$$O_B\left(\frac{N}{(\log N)^B}\right).$$

For every fixed B and every fixed $\beta > 0$, the latter is larger by a factor $N^\beta / \text{polylog}(N)$. Therefore the v1.6 error comparison does not prove positivity for an individual mesoscopic base.

Failed hypothesis or mechanism:

A global mean error was treated as though it scaled with the individual base modulus.

What survives:

Aggregate use over many bases does work at the unsieved parent-incidence level. **G-BV-HARMONIC-PARENT-MASS** proves that fixed positive reciprocal mass below the BV level yields $\gg N / \log N$ parent incidences and $\gg N / \log^2 N$ distinct external parents. **TD-HARMONIC-SUPPORT-DISPOSITION** supplies the reciprocal-mass input on the D branch.

Safe weaker version:

Use the existing individual selector only while the prescribed base modulus is polylogarithmic, or prove a genuinely aggregate many-base theorem with its own remainder accounting.

Current replacement:

v1.10 adaptive lift remains the individual polylog tool. v1.11 adds genuinely aggregate clean extraction: **G-BV-HARMONIC-CLEAN-EXTRACTION** selects a clean load-one subfamily from mesoscopic reciprocal mass, and **G-BV-POLYNOMIAL-HARMONIC-CLEAN-EXTRACTION** gives a polynomially large clean family on fixed power bands. Neither theorem supplies a fan for every prescribed mesoscopic base.

Do not repeat:

Do not cite ordinary BV's total error as an individual polynomial-modulus lower bound.

Related current target:

TD scale bridge, W1, P1BR/K1.

FAIL-INVERSE-SIBLING-INJECTIVE-001**FAIL-ID:**

FAIL-INVERSE-SIBLING-INJECTIVE-001.

Date / iteration:

v1.23 inverse-sibling continuation, 1 September 2026.

Claim:

Distinct returned fresh seeds ℓ necessarily produce distinct prime siblings s satisfying $\ell s < N$.

Status:

KILLED.

Why the inference fails:

A single packet state can divide the return cofactors of several distinct parent rows. The inverse product inequality contains no injectivity information.

What survives:

After first passing to distinct return parents, repeated choice of the same sibling s gives genuine rooted reuse inside U and is charged by

$$\eta(U) \geq |P| - |S|.$$

Thus collisions are an explicit alternative rather than an ignored loss.

Current replacement:

Theory Core, TF-HIGH-RETURN-INVERSE-PACKET.

Do not repeat:

Never convert the inverse-scale map directly into a cardinality lower bound without the $\eta(U)$ collision term.

Related current target:

TF inverse-scale analytic re-entry.

FAIL-KERNEL-EAC-001**FAIL-ID:**

FAIL-KERNEL-EAC-001.

Date / iteration:

v0.1 audit, notation and object review.

Claim:

The iterated source-free core $C_{\text{sf}} = \bigcap_{j \geq 0} \Gamma_N^j(R)$ is an EAC, equivalently a terminal strongly connected component.

Original motivation:

Both objects are “what is left at the bottom”.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

Definitional; the two constructions are not the same operation.

Exact certificate:

C_{sf} is defined by iterated image and need not be strongly connected, nor a single component. A terminal SCC is defined by the condensation having no outgoing edge. G-FAIL-CLOSURE produces a terminal SCC; it does not produce C_{sf} , and neither statement implies the other.

Failed hypothesis or mechanism:

Two distinct objects shared an informal name.

Why the argument failed:

A theorem about one was used as a theorem about the other.

What survives:

Both objects, kept separate, as in the Theory Core notation registry.

Safe weaker version:

Name the construction whenever either object is used.

Current replacement:

Theory Core, G-FAIL-CLOSURE and the notation entry for C_{sf} .

Do not repeat:

Existence of a terminal bad SCC somewhere does not show that p_0 reaches it.

Related current target:

X1.

FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001**FAIL-ID:**

FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001.

Status:**SUPERSEDED .****Superseded description:**

the coefficient $(k\alpha - 1)/(k - 1)^2$ from v4.37 is the primary canonical D/J lower bound.

Certificate:

v4.38 constructs exactly γ fully- C -supported rows and uses the uniform radical-fibre count to obtain

$$\log(D_C J_R) \geq \gamma \log \gamma - o(\gamma \log N).$$

For $\gamma = N^{\alpha+o(1)}$ the coefficient is $\alpha - o(1)$, and at the cube-root splice it is $1/3 - o(1)$ instead of $1/24 - o(1)$.

What survives:

the k -power theorem remains valid and useful for explicit powerful-divisor structure.

FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPEATED-001**FAIL-ID:**

FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPEATED-001.

Status:**KILLED .****Killed inference:**

a large fresh-support factor may consist almost entirely of fresh labels repeated on many extension rows while incremental J stays small.

Certificate:

for fresh labels with $\kappa_A(q) \geq 2$, at least one full copy of $\log q$ already belongs to incremental J . Hence

$$W_1(A) \geq \log F_R(A) - \log(J_S/J_R),$$

and the relative radical floor gives

$$\log(J_S/J_R) \geq \mathcal{R}(A)/3 \quad \vee \quad W_1(A) \geq \mathcal{R}(A)/3.$$

What survives:

the cheap branch is not arbitrary fresh support but the exact unique-parent layer.

FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-MOD3-CLASSES-001**FAIL-ID:**

FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-MOD3-CLASSES-001.

Status:**KILLED .****Killed inference:**

for $3 \nmid N$, a polynomial family $N - w = q^2$ with $q > N^{1/3}$ may occur for either nonzero residue of N modulo 3.

Certificate:

if $N \equiv 1 \pmod{3}$, then $w = N - q^2 \equiv 0 \pmod{3}$; prime w must equal 3, so at most one such edge exists.

What survives:

a polynomial monomial packet can only remain when $N \equiv 2 \pmod{3}$.

FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001**FAIL-ID:**

FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001.

Status:**KILLED** as a live frontier on $q > N^{1/3}$.

Killed description:

genuinely independent cubic-or-deeper vertical storage may persist on bases above $N^{1/3}$.

Certificate:

$q^3 < N$ is necessary for depth at least three. Above the cube-root scale vertical depth is at most two, and the one depth-two copy is termwise accompanied by the same-base radical support already counted by FC7.

What survives:

cubic-and-deeper layers on $q < N^{1/3}$.

FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001**FAIL-ID:**

FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001.

Date / iteration:

vNEXT4.5.

Claim:

Once $\log D + \log J + \log U = \Omega(\gamma \log N)$, FC8 or Goldbach follows automatically.

Status:

KILLED.

Exact certificate:

The same functional has the natural sandwich

$$\log W_C^{\text{nr}} = \gamma \log N + O(\gamma + |H|).$$

The lower scale and upper capacity match in the leading term.

Failed mechanism:

A conservation identity was mistaken for an independent overload inequality.

What survives:

An additive/canonical theorem may still force an incompatible $D/J/U$ partition.

Do not repeat:

Do not count the exact row mass twice as two independent obligations.

FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001**FAIL-ID:**

FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001.

Date / iteration:

vNEXT4.1, Phase 0B.

Claim:

vNEXT4 introduced (LI5) under the heading “that trivial bound is the only one available”, suggesting η_{supp} is the unique ledger-derivable lower bound on $\eta(S)$.

Original motivation:

The witness family drives $\eta(R_n)$ to infinity at constant round datum, which does refute every *growing* bound.

Status:

SCOPE-CORRECTED. G-POSITIVE-ROUND-LEDGER-INDEPENDENCE is preserved in full.

Exact certificate:

The $n = 1$ member has $\eta(R_1) = 2$. Hence any F with $F \leq \eta$ satisfies only $F(\mathfrak{d}) \leq 2$ at that datum; constants up to 2 are consistent with the family. So the family excludes unbounded growth in Δ , not the existence of some bounded lower bound.

What survives:

(LI5'): no universally valid lower bound on $\eta(S)$ expressible solely as a function of the round datum can grow unboundedly with Δ ; in particular none supplies the $\gamma \log N$ -scale bound the (LE3)+(FC8) route consumes. This is the only form used downstream, and it is what makes

the route unavailable.

Do not repeat:

Distinguish “no bound of the required size” from “no bound”. Only the former was proved.

Related current target:

P1BR-ADDITIVE-COUPPLING.

FAIL-LEGACY-CONTINUUM-001

FAIL-ID:

FAIL-LEGACY-CONTINUUM-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

As N grows the discrete CRD structure becomes effectively continuous, and that transition is itself usable.

Original motivation:

Asymptotic densities do appear.

Status:

KILLED.

Exact certificate:

No density or covering statement is attached to the phrase. It must be replaced by a precise density or covering theorem before anything can be deduced.

Failed hypothesis or mechanism:

An informal limit replaced a quantified statement.

What survives:

Genuine asymptotics, such as G-POLYLOG-PARENTS, stated with explicit uniformity and an explicitly INEFFECTIVE threshold.

Current replacement:

Theory Core, the polylog theorems.

Do not repeat:

Do not use a limit as a lemma.

Related current target:

W1.

FAIL-LEGACY-ENCODING-001

FAIL-ID:

FAIL-LEGACY-ENCODING-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

A binary or positional encoding of the states supplies an established CRD invariant.

Original motivation:

Encodings are cheap to define.

Status:

KILLED.

Exact certificate:

No retained theorem uses such an encoding, and none of the exact identities — $T = UJ$, the source lock, the collision and cut identities — is expressible through one.

Failed hypothesis or mechanism:

A representation was mistaken for a structure.

What survives:

Nothing in this direction.

Current replacement:

Theory Core, GEN-SBR and the specialisation dictionaries, which is where representation-level statements belong.

Do not repeat:

A change of notation is not a theorem.

Related current target:

none.

FAIL-LEGACY-PIGEONHOLE-001**FAIL-ID:**

FAIL-LEGACY-PIGEONHOLE-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

A parametric symmetric pigeonhole argument supplies the weighted CRD capacity bound.

Original motivation:

The finite lemma is genuinely true.

Status:

KILLED as a mechanism.

Minimal / strong counterexample:

Structural.

Exact certificate:

The bare finite lemma is true and supplies no weighted logarithmic inequality; capacity in CRD is a statement about $\sum \mu_{\ell,k} \log \ell$, which the lemma never reaches.

Failed hypothesis or mechanism:

A cardinality lemma was used where a weighted one is required.

Why the argument failed:

Counting objects is not bounding their logarithmic mass.

What survives:

The finite lemma itself, as an unrelated true statement.

Safe weaker version:

None in this direction.

Current replacement:

Theory Core, targets $P1^{\text{br}}$ and K1.

Do not repeat:

Do not offer an unweighted lemma as a weighted bound.

Related current target:

$P1^{\text{br}}$, K1.

FAIL-LEGACY-WAVES-001**FAIL-ID:**

FAIL-LEGACY-WAVES-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

Describing residue behaviour as interfering sinusoidal waves supplies a rigorous CRD invariant.

Original motivation:

The picture is suggestive.

Status:

KILLED.

Exact certificate:

Only the underlying residue lattice has proved content; no retained theorem uses the wave

description, and no invariant is derived from it anywhere in the corpus.

Failed hypothesis or mechanism:

Imagery in place of a statement.

What survives:

The residue lattice, used directly.

Current replacement:

Theory Core, prime-power layer formalism.

Do not repeat:

Do not carry a metaphor into a proof.

Related current target:

none.

FAIL-LF2-AS-UPPER-BOUND-001

FAIL-ID:

FAIL-LF2-AS-UPPER-BOUND-001.

Date / iteration:

vNEXT3.1 correction round.

Claim:

In the (FC8) frontier certificate, “(LF2) bounds $\eta(R)$ above by $|R|$ ”.

Original motivation:

$|Q| + |H| \leq |R|$ is true, and the certificate needed $\eta(R)$ to be small.

Status:

KILLED.

Exact certificate:

(LF2) reads $\eta(R) \geq |Q| + |H|$. It is a *lower* bound. The inequality $|Q| + |H| \leq |R|$ says only that this recorded lower bound is itself at most $|R|$; it places no ceiling on $\eta(R)$, which by (LE2) can exceed $|R|$ whenever rows carry repeated support.

Failed hypothesis or mechanism:

A bound on the right-hand side of a lower-bound inequality was read as a bound on the left-hand side.

What survives:

The certificate, correctly stated: to contradict (FC8) through (LE3) one needs a *lower* bound on $\eta(R)$ of order $\gamma \log N$, and the recorded (LF2) does not carry a factor $\log N$.

Current replacement:

Theory Core, Proposition [prop:fc8-corrected], status FRONTIER CERTIFICATE / RELATIVE TO RECORDED BOUNDS.

Do not repeat:

Never infer a ceiling from a floor.

Related current target:

K1, γ .

FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-POLYLOG-SUPPORTED-001

FAIL-ID:

FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-POLYLOG-SUPPORTED-001.

Status:

KILLED asymptotically on the recorded high-gap/BHP domain.

Killed inference:

a linear smooth-mirror family may carry the required active mass on only polylogarithmically many active support primes.

Certificate:

existing mirror active-mass plus support capacity gives, for $|J_{\text{sm}}| \geq \rho t$,

$$|P_{\text{mir}}| \geq N^{0.2375\rho - o(1)}$$

unless there is already a linear active-state population.

What survives:

this is asymptotic and does not replace finite small-gap case analysis.

FAIL-LITERAL-LINEAGE-001**FAIL-ID:**

FAIL-LITERAL-LINEAGE-001.

Date / iteration:

Post-v7.5 layer-lineage audit.

Claim:

Once a prime label ℓ is produced, its valuation can be followed literally through the row indexed by ℓ .

Original motivation:

Residual factors were pictured as conserved tokens moving down a path.

Status:

KILLED.

Minimal / strong counterexample:

Universal for active ℓ .

Exact certificate:

Since $\ell \nmid N$, $N - \ell \equiv N \not\equiv 0 \pmod{\ell}$; hence

$$v_{\ell}(N - \ell) = 0.$$

Failed hypothesis or mechanism:

Row labels and factor labels were identified.

Why the argument failed:

The dynamics changes the row index; it does not conserve a prime token along a path.

What survives:

Global valuation accounting $T = UJ$, collision layers, and residual products.

Safe weaker version:

Track incidence across a finite set of rows, not literal pathwise conservation.

Current replacement:

Theory Core, G-TUJ and the prime-power layer formalism.

Do not repeat:

Never write a lineage law without an exact valuation identity at every transition.

Related current target:

$E1, K1$.

FAIL-LOCAL-REUSE-001**FAIL-ID:**

FAIL-LOCAL-REUSE-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

An upper-source reuse parent occurs within five source indices.

Original motivation:

A local window would make simultaneous witness selection inexpensive.

Status:

KILLED by source-recorded computation.

Minimal / strong counterexample:

The strongest recorded witness is $(N, \ell) = (1825372, 52153)$ with source-index distance 53750.

Exact certificate:

Only 0.0265727 of defined recorded witnesses lay within five indices; the cited distance is the recorded maximum. The raw row list was not reloaded in this extraction.

Failed hypothesis or mechanism:

Source order was treated as a metric controlling arithmetic incidence.

Why the argument failed:

Parent lattices are residue classes, not short intervals in the adaptive source enumeration.

What survives:

Polylogarithmic parent abundance and finite matching for prescribed prime powers.

Safe weaker version:

Use the full active parent lattice or prove a new quantitative selector theorem.

Current replacement:

Theory Core, G-POLYLOG-PARENTS, G-POLYLOG-MATCHING, and W1.

Do not repeat:

Do not impose an unproved absolute locality window.

Related current target:

W1, K1.

FAIL-LOSSLESS-INTERFACE-001**FAIL-ID:**

FAIL-LOSSLESS-INTERFACE-001.

Date / iteration:

vNEXT4.1, Phase 4.

Claim:

That a canonical row c_j can be transferred to a global prime selection with its information preserved — the round's primary success criterion, third arrow.

Original motivation:

The bundle query does preserve strictly more than the old single-base query: exact valuations, the cofactor class and the mirror relation $D \mid p - a_j$. It was natural to hope the preserved part could be made all of A_j .

Status:

KILLED for lossless transfer, under every distribution theorem presently available. The bundle lift itself is **PROVED**.

Exact certificate:

Theory Core P0-QUERY-CAPACITY: if every member of a residue class $n \equiv a \pmod{M}$ satisfies $v_q(N - n) = e$, then $q^{e+1} \mid M$. Taking $n = a$ and $n = a - M$ forces $q^e \mid M$; and if $v_q(M) = e$ exactly then one k in each complete residue system mod q gives $q^{e+1} \mid N - a + kM$. Hence $D \text{ rad } D \mid M$ and $\log D \leq \frac{E}{E+1} \log M$. With Siegel–Walfisz, $\log M \leq K \log \log N$, so the preserved fraction is $< K \log \log N / \log(N/2) \rightarrow 0$. Under GRH the fraction is at most $\frac{1}{2}$ (at most $\frac{1}{4}$ for squarefree bundles), so GRH does not make it lossless either. Losslessness needs a level of distribution $\theta \geq 1$ for an *individually specified* modulus of size $N^{1-o(1)}$ — stronger than GRH, and not the average-level statement of Elliott–Halberstam.

Failed hypothesis or mechanism:

Treating “retain more information” as free. Retention is paid for in the modulus, at a rate fixed by (QC1), and the modulus is capped by the available distribution theorem.

What survives:

The exact capacity: $\log D \leq \frac{E}{E+1} K \log \log N$, and everything the bundle does pre-

serve, which is genuinely more than the old query. See P0-EXACT-BUNDLE-LIFT and P0-ADAPTIVE-BUNDLE-DICHOTOMY.

Do not repeat:

Do not propose an interface that preserves a positive proportion of a canonical row without first naming the level-of-distribution theorem that pays for it, and do not import Bombieri–Vinogradov silently — this corpus does not use it anywhere.

Related current target:

P1BR-ADDITIVE-COUPPLING, P0-BUNDLE-ROUGH-HIGHPOWER-BRANCH.

FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-VERTICAL-WALL-001

FAIL-ID:

FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-VERTICAL-WALL-001.

Status:

KILLED as an independent primary frontier.

Killed description:

a polynomial-size canonical low core may keep an asymptotically negligible $D_C J_R$ while financing essentially all compulsory mass through diffuse low-base powerful rows.

Certificate:

v4.37 proves, pending audit, that for $\gamma = N^{\alpha+o(1)}$ and every fixed $\alpha > 0$,

$$\log(D_C J_R) = \Omega_\alpha(\gamma \log N).$$

Equivalently, near-pure vertical escape forces $\gamma = N^{o(1)}$.

What survives:

a subpolynomial low core or a non-negligible D/J payment; rowwise powerful-divisor structure remains useful downstream.

FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001

FAIL-ID:

FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001.

Date / iteration:

vNEXT4.4.

Claim:

Once the reflected block forces large $\gamma = |C|$, the layered Euler theorem automatically converts this into enough rooted reuse to contradict FC8.

Status:

KILLED.

Exact certificate:

Canonical layered Euler gives only

$$\eta(R) + \nu(R) \geq |R| + 1 \geq \gamma + 2,$$

hence only

$$\max\{\log J_R, \log U_R\} \geq \frac{\gamma + 2}{2} \log 3.$$

This is $O(\gamma)$ guaranteed log mass. The recorded FC8 route through $J_R \geq 3^{\eta(R)}$ needs a lower bound at scale $\gamma \log N$ unless a separate large denominator is supplied. Moreover the population may be paid vertically through prime-power layers U_R ; the generic failed-world prime-power examples show that such vertical storage is a real mechanism.

Failed hypothesis or mechanism:

Cardinality was confused with weighted prime-base mass, and horizontal reuse was treated as if it were the only defect channel.

What survives:

vNEXT4.4 produces a genuine FC8 denominator on the sub-square-root large-factor branch, and the exact horizontal/vertical split remains available. A theorem forcing large bases or upper-bounding cheap vertical storage would reopen the route.

Do not repeat:

A new lower bound on γ , $\eta + \nu$, or an unweighted layer count is not a consumer unless its logarithmic bases are controlled.

Related current target:

C-VERTICAL-LOWCORE-CAPACITY.

FAIL-LOWCORE-CHILDCLOSED-001**FAIL-ID:**

FAIL-LOWCORE-CHILDCLOSED-001.

Date / iteration:

v1.23 low-core continuation, 1 September 2026.

Claim:

Since every reused rooted label is below $N/6$, the set $C = R \cap (0, N/6)$ may be treated as a child-closed failed subsystem.

Status:

KILLED.

Why the inference fails:

The theorem controls the location of *repeated targets*, not all children of low rows. A low row $u < N/6$ has $N - u > 5N/6$, and a prime factor of this complement can lie in $(N/6, N/3)$. Thus an edge may leave C for the upper fringe.

What survives:

Every repeated target and the complete factor J_R are supported in C ; every high state has a unique-parent forest structure and returns to C when parent ancestry is followed backwards.

Current replacement:

Theory Core, G-CANONICAL-REUSE-LOWCORE and G-CANONICAL-UPPER-FRIDGE-FOREST.

Do not repeat:

Do not apply G-MULTISOURCE-LOCK, low-packet product locks, or child-closed component arguments directly to C without separately proving closure for the subset in question.

Related current target:

K1 low-core upper capacity.

FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001**FAIL-ID:**

FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001.

Date / iteration:

vNEXT4.20.

Old description:

The low-polylog $t \log \Lambda$ mirror multiplicity cannot be consumed because there are not enough actual prime parent rows preserving the required layers.

Status:

KILLED as a description of the current obstruction.

Certificate:

After an $o(t \log \Lambda)$ high-power tail is removed, all retained polylog prime-power blocks can be simultaneously matched to distinct external BAD primes satisfying $p_{j,q} \equiv c_j \pmod{q^{e+1}}$.

Replacement:

The live obstruction is joint unintended support load / single charging, not parent abundance.

FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOVE-4H-001

FAIL-ID:

FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOVE-4H-001.

Status:

KILLED for $h > N^{1/3}$.

Rejected statement:

A large fraction of Q may lie above $4h$ without additional restriction.

Certificate:

three such distinct primes would divide $m < N$ but have product $> (4h)^3 > 64N$.

Current safe statement:

$\#\{q \in Q : q > 4h\} \leq 2$.

FAIL-MATCHING-CAPACITY-001

FAIL-ID:

FAIL-MATCHING-CAPACITY-001.

Date / iteration:

Canonical bridge audit §5.3, 31 August 2026.

Claim:

An injective, load-one assignment of external parents to layers bounds the ambient collision cost, closing the capacity target.

Original motivation:

Load one removes repeated use of a single parent, which was mistaken for removing the cost.

Status:

KILLED — the matching theorem itself is **PROVED**; the inference drawn from it is not.

Minimal / strong counterexample:

Structural; no finite witness is needed.

Exact certificate:

Injectivity leaves two costs untouched. One selected row $N - p$ may meet many atoms of R , inflating $K_{\times}(R, \mathcal{P})$; and two distinct selected rows may share unintended new support, inflating $K(\mathcal{P})$. Greedy selection of pairwise-coprime complements would have to avoid, for each previously chosen p' , every class $p \equiv N \pmod{t}$ with $t \mid N - p'$; Siegel–Walfisz for a single polylog modulus gives no uniform positive count under a growing adaptive system of such prohibitions, and some t may be large.

Failed hypothesis or mechanism:

Existence and load were confused with cost.

Why the argument failed:

Abundance is a statement about one modulus at a time; capacity is a statement about a whole selected family at once.

What survives:

G-POLYLOG-PARENTS and G-POLYLOG-MATCHING, now for all polylog prime-power layers, both with an explicitly INEFFECTIVE threshold.

Safe weaker version:

Report load one and the two uncontrolled terms separately; never report their conjunction as capacity.

Current replacement:

Theory Core, target K1, which is exactly the missing bound $K_{\times}(R, \mathcal{P}) + K(\mathcal{P}) \leq \mathcal{C}_N(R)$.

Do not repeat:

This is the precise point at which abundance stops becoming clean capacity; do not cross it silently.

Related current target:

W1, K1.

FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001

FAIL-ID:

FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001.

Status:

KILLED.

Killed inference:

e merger/attachment events automatically force $\Delta \log J = \Omega(e \log N)$.

Certificate:

v4.38 proves a real attachment charge, but if all events reuse a fixed small label q , its weight is only $O(e \log q)$. The ELCFS finite-realisation theorem shows that source counts/topology/reverse-generation bookkeeping alone cannot manufacture the missing ambient $\log N$ factor.

What survives:

attachment forces an unweighted horizontal defect and CRD-specific arithmetic may still retain scale.

FAIL-MERGER-WEIGHT-TO-J-TPOLOGY-001

FAIL-ID:

FAIL-MERGER-WEIGHT-TO-J-TPOLOGY-001.

Status:

KILLED.

Killed sufficient strategy:

prove $\Delta \log J \gg W(E)$ using only child closure, source counts, graph topology, support multiplicities and the v28 reverse-extension ledger.

Certificate:

ELCFS finite realisation permits arbitrarily large source weights attached through a fixed low target. The topology grows while the horizontal receiving weight remains only $e \log q$.

What survives:

CRD arithmetic absent from ELCFS—the map $N - p$, difference divisibility, reflected provenance, exact cofactors and canonical root extremality—may still supply the missing scale.

FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001

FAIL-ID:

FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001.

Date / iteration:

vNEXT4.17, sharpened by vNEXT4.19.

Escape:

Store all natural mirror mass below $(\log N)^D$ for super-polylogarithmic Λ .

Status:

SCOPE-CORRECTED.

Certificate:

For fixed D ,

$$M_{\leq (\log N)^D}^- = O_D(t \log \log N).$$

Thus this reservoir can remain competitive only when $\log \Lambda = O(\log \log N)$. That low-polylog balance is the current open frontier rather than a killed escape.

FAIL-MIRROR-MASS-ONE-SCALE-001

FAIL-ID:

FAIL-MIRROR-MASS-ONE-SCALE-001.

Date / iteration:

vNEXT4.17.

Escape:

Store $\gg t \log \Lambda$ active mirror mass in one dyadic base band.

Status:

KILLED.

Certificate:

Uniformly $M_Y^- \ll t$. Natural mass therefore occupies $\Omega(\log \Lambda)$ scales.

FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001**FAIL-ID:**

FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001.

Date / iteration:

vNEXT4.17.

Claim:

Represent reflected multiplicity m_r^- by the same number of clean selected parent copies carrying designated base r in one family.

Status:

KILLED.

Exact obstruction:

The clean multiparent identity contains

$$-\binom{\rho_r}{2} \log r,$$

a quadratic selected-family self-collision cost.

Safe replacement:

Convert aggregate reflected mass first to distinct-base reciprocal mass/population, or use separate rounds with a proved global single-charge mechanism.

FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11**FAIL-ID:**

FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11.

Date / iteration:

vNEXT4.11.

Claim:

The mod-3 collapse used in the repeated- b classification is new in vNEXT4.11.

Status:

SCOPE-CORRECTED (provenance correction).

Exact certificate:

Its source is the earlier C-MONOMIAL-MOD3-COLLAPSE theorem from vNEXT4.7.

What survives:

The repeated-entry-cofactor classification remains valid subject to its own audit; only the novelty/provenance claim is corrected.

FAIL-MONOMIAL-NEEDS-SPARSITY-001**FAIL-ID:**

FAIL-MONOMIAL-NEEDS-SPARSITY-001.

Date / iteration:

vNEXT4.8.

Claim:

The prime-square branch can only be handled by proving that q and $N - q^2$ are sparse.

Status:

KILLED.

Exact certificate:

Monomial square edges form components with at most two edges. A large monomial family therefore forces many component starts; every nonroot start has a nonmonomial entry edge, and on $h > N^{1/3}$ those starts produce a linear packet of actual entry-parent rows. Existing rooted transport then routes the packet into horizontal reuse, low-scale states, or reciprocal mass.

What survives:

Prime-square sparsity remains an optional analytic strengthening, not the necessary binding step.

FAIL-MONOMIAL-NODESCENT-CHEAP-001**FAIL-ID:**

FAIL-MONOMIAL-NODESCENT-CHEAP-001.

Date / iteration:

vNEXT4.8.

Claim:

Above $N^{1/3}$, a large monomial entry packet whose descendant closure stays above h can remain supported only by cheap vertical layers.

Status:

KILLED.

Exact certificate:

If P is the selected entry-parent packet and its descendant closure U satisfies $\min U > h > N^{1/3}$, then

$$\eta(U) \geq |P|, \quad \log J_R > |P| \log h.$$

What survives:

The downstream question is how to consume this horizontal payment inside K1.

FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001**FAIL-ID:**

FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001.

Status:

KILLED.

Killed inference:

the observed maximum monomial path length two is only a finite-census phenomenon.

Certificate:

the pre-v4.41 Core theorem already proves path length at most two. The interval $(\sqrt{N - \sqrt{N}}, \sqrt{N})$ additionally yields uniqueness of a two-edge component.

FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001**FAIL-ID:**

FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001.

Date / iteration:

vNEXT4.8.

Claim:

A large monomial vertical family may account for the whole canonical layered defect by itself.

Status:

KILLED.

Exact certificate:

$$\eta(R) + \nu_{\neq \square}(R) \geq \left\lceil \frac{M_{\square}}{2} \right\rceil + 1.$$

Every monomial family forces a companion horizontal or nonmonomial vertical defect.

FAIL-MR2-CANONICAL-ONLY-001

FAIL-ID:

FAIL-MR2-CANONICAL-ONLY-001.

Date / iteration:

vNEXT4.2.

Claim:

For $q \mid m$, the equivalence $q \mid c_j \iff q \mid h - j$ is a genuinely canonical p_0 -only relation.

Status:

KILLED as a scope claim.

Exact certificate:

For an arbitrary upper prime root $r = x + h_r$, $m_r = x - h_r$, one has $r = m_r + 2h_r$. Hence for every odd $q \mid m_r$,

$$q \mid (r - 2j) \iff q \mid h_r - j.$$

No minimality of p_0 is used.

What survives:

The relation itself remains valid and useful; only its classification changes from “canonical selector information” to generic upper-root geometry.

Do not repeat:

Reserve “genuinely canonical” for conclusions using minimality/prime-freeness, e.g. $h < G$, not identities valid for every upper root.

FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001

FAIL-ID:

FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001.

Date / iteration:

vNEXT4.20.

Killed description:

$N^{1/4-\varepsilon}$ is an intrinsic ceiling on simultaneous realisation of the low-polylog mirror multiplicity.

Status:

KILLED.

Correction:

That scale belongs to the clean rough-support/lower-sieve architecture. Without demanding global complement coprimality, the full low-polylog block reservoir, of size up to $N^{1/2-o(1)}$, has distinct prime parent representatives.

FAIL-N1/6-DESIGNATED-BASE-BARRIER-001

FAIL-ID:

FAIL-N1/6-DESIGNATED-BASE-BARRIER-001.

Date / iteration:

vNEXT4.19 correction of vNEXT4.15–vNEXT4.17.

Killed claim:

Variable-mass clean selection cannot act on designated bases larger than $N^{1/6-o(1)}$.

Status:

KILLED.

Exact correction:

The selector requires only

$$Qz^{2+\vartheta} \ll N^{1/2}/\text{polylog } N.$$

With $z = (\log N)^D$, designated bases may reach

$$Q = N^{1/2} / \text{polylog } N.$$

Call $N^{1/6}$ only the same-scale specialization $z \asymp Q$.

FAIL-NAIVE-MU-LE-3ETA-001

FAIL-ID:

FAIL-NAIVE-MU-LE-3ETA-001.

Status:

KILLED.

Killed inference:

for $q > N^{1/4}$, valuation depth at most three gives $\mu_H(q) \leq 3(\kappa_H(q) - 1)$.

Certificate:

if $\kappa_H(q) = 1$ and $d_H(q) = 3$, then $\mu_H(q) = 2$ while $\eta_q = 0$.

Safe replacement:

$\mu_H(q) \leq 3\eta_q + 2$; for general depth m , $\mu_H(q) \leq m\eta_q + (m - 1)$.

Do not repeat:

never discard the occupied-base additive term in a power-threshold argument.

FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001

FAIL-ID:

FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001.

Date / iteration:

vNEXT4.14.

Claim:

A high reflected denominator family may carry only its first copies in D_C , with no further canonical consequence.

Status:

KILLED.

Exact certificate:

If its selected parent descendants do not reach $\leq h$, then

$$\log J_R > \frac{\log h}{3 \log N} L(E),$$

so a fixed positive fraction of the same logarithmic family weight is paid horizontally.

FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-SAVING-001

FAIL-ID:

FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-SAVING-001.

Status:

KILLED.

Killed inference:

simultaneous use of the nested spacing bounds for $r^2, r^3, \dots, r^e$ must create an extra cumulative saving beyond the single-layer parent-slot envelope.

Certificate:

if κ rows lie in one exact layer $v_r(N - p) = e$, then $n_k(r) = \kappa$ for every $1 \leq k \leq e$. Consequently all nested layers are realised by the same packet and

$$\sum_{k \geq 2} (n_k(r) - 1)_+ = (e - 1)(\kappa - 1).$$

The support graph sees only $\kappa - 1$ repeated r -edges.

Do not repeat:

nested packing alone cannot turn valuation depth into an $O(1)$ or $O(\log \log N)$ bounded-multiplicity graph charge.

FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001**FAIL-ID:**

FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001.

Date / iteration:

vNEXT4.7.

Claim:

For $h > N^{1/3}$, a nonmonomial event $N - w = kq^2$, $k > 1$, may remain an independent vertical obstruction.

Status:

KILLED as a description.

Exact certificate:

Every prime $r \mid k$ is rooted and satisfies

$$r < N/h^2 < h.$$

For a family of such events, repeated siblings force horizontal/collision mass, while distinct siblings create an explicit lower-scale rooted packet.

What survives:

A downstream consumer/upper-capacity problem for the exported sibling/collision alternatives.

FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICITY-001**FAIL-ID:**

FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICITY-001.

Date / iteration:

vNEXT4.16.

Claim:

The existing clean selector closes the smooth-mirror harmonic branch merely by selecting one representative incidence for each active base.

Status:

KILLED.

Exact obstruction:

Reflected packing permits about t/r incidences at a small base r . Present clean selectors retain one incidence per base, or repeated multiplicity only for a fixed-polylog total multiset.

Replacement target:

Preserve a positive fraction of the weighted multiplicity before applying the signed K1 ledger.

FAIL-ORTHOGONAL-BRIDGE-001**FAIL-ID:**

FAIL-ORTHOGONAL-BRIDGE-001.

Date / iteration:

vNEXT3, Phases 2–3.

Claim:

Although cross-collision is dead under $K_{\times}(R, U) = 0$, a source- or product-side functional can still carry $\eta(U)$ from an external orthogonal packet into a capacity determined by R .

Original motivation:

The source product of G-MULTISOURCE-LOCK does change when a packet is adjoined, so it looked like a live channel.

Status:**KILLED**.**Minimal / strong counterexample:**

Structural, by exact decoupling.

Exact certificate:

For disjoint bad child-closed R, U , **G-ORTHOGONAL-DECOUPLING** gives $\Gamma_N(S) = \Gamma_N(R) \dot{\cup} \Gamma_N(U)$, $\text{Src}(S) = \text{Src}(R) \dot{\cup} \text{Src}(U)$, $T_S = T_R T_U$, $J_S = J_R J_U$, $U_S = U_R U_U$, $\eta(S) = \eta(R) + \eta(U)$, $\nu(S) = \nu(R) + \nu(U)$, and the multisource lock for S factors as the product of the two locks. Every recorded functional is therefore a disjoint-union homomorphism, and the recorded state of R is *identical* for every admissible U . The target inequality $\log(T_{R \cup U}/T_R) \geq c_0 \eta(U)$ does hold, with $c_0 = \log 3$, and is worthless: its left side equals $\log T_U$.

Failed hypothesis or mechanism:

The source product does change, but it changes by exactly U 's own source product; nothing is shared.

Why the argument failed:

$K_{\times}(R, U) = 0$ is not an extra hypothesis to be worked around — it is *automatic* for disjoint child-closed packets, and it is a symptom of total decoupling rather than of one blocked channel.

What survives:

G-BRIDGE-ROUTING-EQUIVALENCE: a bridge can exist only if $U \cap R \neq \emptyset$, which is precisely the TD/TF routing target. The two fronts are one.

Safe weaker version:

Report $\eta(U)$ as a property of U alone.

Current replacement:

Theory Core, Theorem [thm:decoupling], Corollary [cor:worthless-coupling], Theorem [thm:bridge-routing].

Do not repeat:

Do not seek any functional coupling two disjoint child-closed packets; all recorded functionals provably factor.

Related current target:

TD, TF, K1.

FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001**FAIL-ID:**

FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001.

Date / iteration:

vNEXT3, Phase 2.

Claim:

$K_{\times}(R, U) = 0$ is a special favourable hypothesis on the external packet, to be secured by construction.

Original motivation:

It is written as a hypothesis throughout the external-packet material.

Status:**SCOPE-CORRECTED**.**Exact certificate:**

(OD4). If R and U are disjoint and both child-closed, then $\Gamma_N(R) \subseteq R$ and $\Gamma_N(U) \subseteq U$ are disjoint, so no prime divides a complement on both sides and every cross gcd is 1. Orthogonality is a consequence of disjointness plus child-closure, not an independent assumption.

What survives:

Every statement conditioned on orthogonality, now with the hypothesis recognised as redundant given the other two.

Do not repeat:

Do not spend effort “arranging” orthogonality, and do not read it as evidence of a favourable configuration; it signals decoupling.

Related current target:

TF, K1.

FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-LOSS-001**FAIL-ID:**

FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-LOSS-001.

Status:

KILLED on the v18 owner-completed domain.

Rejected statement:

The first later return edge into an external rough seed can always be absorbed solely by destroying that seed as a source.

Certificate:

the owner row p_i already supplies $p_i \rightarrow r$; a later $u \rightarrow r$ is a second occurrence and creates a full $+\log r$ incremental J charge.

Scope guard:

This does not invalidate the generic vNEXT4.28 ledger; it strengthens it on the retained-owner domain.

FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001**FAIL-ID:**

FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001.

Status:

KILLED.

Rejected statement:

A return row using only the owner support $\{q_0\} \cup \text{PF}(b_i)$ may be a cheap squarefree recycle.

Certificate:

its complement is larger than the squarefree owner complement while its radical divides that owner complement; verticality is forced.

Do not repeat:

Owner support preservation is a verticality certificate, not a squarefree escape.

FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDITIVE-STRUCTURE-001**FAIL-ID:**

FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDITIVE-STRUCTURE-001.

Status:

KILLED.

Killed description:

a fresh returned reflected factor with no prior owner is arithmetically anonymous apart from its support incidence.

Certificate:

for $c_j = sq$ and a low active parent $N - w = bq$, v4.39 gives

$$\frac{10}{9}s < b < 2s, \quad a_j - w = (b - s)q,$$

with $b - s$ positive even and

$$a_j - w > N/18.$$

What survives:

this additive relation still needs a capacity / second- occurrence consumer.

FAIL-P0-INFORMATION-SELECTOR-001

FAIL-ID:

FAIL-P0-INFORMATION-SELECTOR-001.

Date / iteration:

vNEXT4.1, Phase 9.

Claim:

The central hypothesis of the round: that p_0 identifies the particular additive–multiplicative correlations a global prime-distribution theorem should preserve.

Original motivation:

The canonical reflected block is the only place in CRD where a *composite* row of size $> N/2$ with a canonical mirror is produced deterministically.

Status:

KILLED. Not refuted; **not established**.

Exact certificate:

Theory Core P0-CANONICITY-ABLATION. Every clause of P0-EXACT-BUNDLE-LIFT classifies as (AB-A) “fixed N only” or (AB-B) “an arbitrary reflected pair $a + c = N$ ”. Replacing p_0 by any upper-half prime p_1 leaves (BL3)–(BL6) and (BL5) word for word intact on every row whose c'_j is composite. Canonicity is used in exactly one place: the prime-free interval $[x, p_0)$ makes *every* c_j composite, so branch (MC1) is available on the whole block. Measured on the 301 recorded noncanonical failed upper roots, 32,766 of 130,090 reflected rows — 25.2% — have c'_j prime and are unbundleable (their single block costs $c_j'^2 > N$).

Failed hypothesis or mechanism:

A non-vacuity guarantee is not an information-selection principle. Canonicity guarantees the construction is always available; it does not make the construction, or its conclusion, different.

What survives:

One genuinely canonical statement, (MR2): for $q^e \parallel m$, $q \mid c_j \iff q \mid h - j$, from ARS-CANON-GEOMETRY. No consequence was built on it in this round.

Do not repeat:

Do not record a P0-INFORMATION-SELECTOR-CERTIFICATE without an ablation in which the conclusion actually fails after the canonical datum is deleted. None was found.

Related current target:

P1BR-ADDITIVE-COUPPLING, P0-BUNDLE-ROUGH-HIGHPOWER-BRANCH.

FAIL-P0-ZEROHOLE-ESSENTIAL-001

FAIL-ID:

FAIL-P0-ZEROHOLE-ESSENTIAL-001.

Date / iteration:

vNEXT4.2.

Claim:

The fact that the canonical reflected block has no prime holes is, by itself, the essential p_0 -specific additive–multiplicative glue.

Status:

KILLED as a mechanism for the active/overflow mass argument.

Exact certificate:

If p_1 is the immediate next prime after canonical $p_0 > x$, the strict p_1 -reflected block has exactly one prime hole, namely p_0 , at index $(p_1 - p_0)/2$. More generally the fixed p_k block has exactly the k earlier primes as holes. Iterating the recorded pointwise short-interval bound a fixed number of times gives $p_k - x = O_k(x^{0.525})$, so deleting the holes costs only $O_k(\log x)$ in the block active-mass identity when the width tends to infinity.

Why the argument failed:

The earlier 25.2% statistic concerned a broad family of noncanonical upper roots; it was not the immediate $p_0 \rightarrow p_1$ ablation. The correct immediate ablation is a one-hole perturbation,

not a positive-density destruction of the block.

What survives:

p_0 still gives the exact zero-hole formulation and may be essential through stronger canonical information such as $h < G$, Exact-D, or another conclusion which actually fails already at p_1 .

Do not repeat:

Do not infer a P0-INFORMATION-SELECTOR-CERTIFICATE from the absence of prime holes alone.

Related current target:

C-CRITICAL-SQRT-GAP-BRANCH.

FAIL-P4-IS-THE-BEST-BV-ARITY-001

FAIL-ID:

FAIL-P4-IS-THE-BEST-BV-ARITY-001.

Date / iteration:

vNEXT4.21.

Killed claim:

Ordinary full-interval Bombieri–Vinogradov can support only the old P_4 rough cofactor output.

Status:

KILLED.

Replacement:

The explicit Richert-weighted dimension-one sieve in vNEXT4.21 gives a tagged squarefree P_3 fan with rough floor $N^{6/25-o(1)}$, pending independent analytic audit of the weighted-sieve interface.

FAIL-PAIR-LOADED-DESCENT-OPEN-001

FAIL-ID:

FAIL-PAIR-LOADED-DESCENT-OPEN-001.

Date / iteration:

vNEXT4.15.

Old obstruction:

Rows $N - w = Bq_1q_2$ with $q_i \geq h$ and $B > 1$ might create an uncontrolled low residual packet.

Status:

CLOSED AS A POLYLOG- Λ SCALE-REENTRY PROBLEM.

Exact certificate:

If $\Lambda \leq (\log N)^A$, then $B < 2\Lambda$; residual prime factors are fixed-polylog BAD states. A large family therefore produces reuse or enters the repaired rooted/external clean ledger.

FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001

FAIL-ID:

FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001.

Status:

KILLED.

Rejected statement:

The final X1 stitching must upper-bound the full pairwise functional $K_{\times}(R, P) + K(P)$.

Certificate:

vNEXT4.31 proves the exact promotion hierarchy

$$L(A) \leq \log \frac{J_{R \cup P}}{J_R} \leq K_{\times}(R, P) + K(P).$$

Thus the weaker target K1-INC, bounding only the actual incremental J by the same P1BR functional, is logically sufficient.

Do not repeat:

Do not identify “K1 is sufficient” with “K1 is logically necessary.”

FAIL-PARENT-LATTICE-DOMAIN-001**FAIL-ID:**

FAIL-PARENT-LATTICE-DOMAIN-001.

Date / iteration:

v0.1 audit, canonical–ambient split correction.

Claim:

The parent lattice of a layer may be treated as the full residue class $p \equiv N \pmod{\ell^k}$ over all primes.

Original motivation:

The analytic count is naturally stated over all primes in an interval.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

Definitional.

Exact certificate:

The correct object is $\mathcal{P}_N(\ell^k) = \{p \in \mathcal{A}_N : \ell^k \mid N - p\}$, the intersection of the congruence class with the *finite active prime universe* $\mathcal{A}_N$. Primes dividing N , and primes at least N , are not states.

Failed hypothesis or mechanism:

An ambient analytic set was used as a set of CRD states.

Why the argument failed:

Counting in the class is legitimate; treating every counted prime as a usable parent is not.

What survives:

G-POLYLOG-PARENTS, which restricts to $p \in (N/2, N)$ and proves activity and externality explicitly.

Safe weaker version:

Intersect with $\mathcal{A}_N$ at the moment the class is introduced.

Current replacement:

Theory Core, notation entry $\mathcal{P}_N(\ell^k)$ and G-POLYLOG-PARENTS.

Do not repeat:

A residue class is not a state set.

Related current target:

W1, K1.

FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001**FAIL-ID:**

FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001.

Source:

explicit proposed Failure Log item in v4.51; also v4.51 R51-2.

Status:

KILLED for the inference

remote reflected root+aggregate affine path product $\implies$ control of the terminal exponent E .

Certificate:

after appending the last edge $p \rightarrow r$, the aggregate affine identity recombines the terminal row into the single integer $N - p$; modulo N , the factor r^{E-1} in the total path product is cancellable because $r \nmid N$.

Scope guard:

this is not a no-go for CRD1-R. The full valuation tensor still knows $v_r(N - p) = E$. The loss

occurs only after compressing the terminal edge into the aggregate path product.

Replacement:

retain the final row/layer datum (p, r, E) edge-locally.

FAIL-PER-ROW-RADC-COMPRESSION-001

FAIL-ID:

FAIL-PER-ROW-RADC-COMPRESSION-001.

Status:

KILLED as a failure certificate.

Reason:

LCAP4 controls the full product $D_C J_R U_R^{\text{nr}}$, not the radical factor $D_C J_R$ alone. Large mass may be stored in U_R^{nr} . The revised report had conflated the three-channel product with one channel.

FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001

FAIL-ID:

FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001.

Status:

SCOPE-CORRECTED.

Killed sufficient strategy:

use Brun–Titchmarsh or Bombieri–Vinogradov pointwise to rule out one dynamically selected canonical exact layer or to infer non-reachability in R .

Safe surviving statement:

ambient AP estimates do not see canonical reachability and need not kill singleton or exceptional layers.

v4.36 correction:

Brun–Titchmarsh becomes effective after summing over *all* repeated prime-power layers together with the canonical γ -row budget and the weighted layer count $A_2(Y) \ll \sqrt{Y}$. Thus the old no-go was correct pointwise and too broad as an aggregate prohibition.

FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-EXACT-STAR-EXPORT-001

FAIL-ID:

FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-EXACT-STAR-EXPORT-001.

Status:

SCOPE-CORRECTED.

Scope correction:

the polylogarithmic restriction belongs to some analytic star-selection / finite-alphabet stages of the old v4.26 apparatus. If a fixed active base already has polynomial row multiplicity, exact-exponent pigeonhole loses only $N^{o(1)}$, and parent-slot / quotient inflation gives a q -free residual export of $\Omega(n \log N)$ for arbitrary base size.

What survives:

the hard part is producing polynomial concentration from the dispersed cube-root defect, not consuming such concentration after it is known.

FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROGRESS-001

FAIL-ID:

FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROGRESS-001.

Date / iteration:

vNEXT4.18.

Claim:

Because a positive signed singleton round is a dead end, the repeated-base fan gives no progress unless all rounds are nonpositive.

Status:**KILLED**.**Correction:**

Independently of signs, every selected row supplies a distinct rough active child above $N^{1/4-\varepsilon}$. The polynomial state/source packet therefore survives the positive-round wall, although it still is not an upper-capacity contradiction.

FAIL-POSITIVE-ROUND-CONSEQUENCE-001**FAIL-ID:**

FAIL-POSITIVE-ROUND-CONSEQUENCE-001.

Date / iteration:

vNEXT3.1, Phase-3 semantic audit.

Claim:

Forcing a positive signed round is progress toward a contradiction, so the (FC10) route closes once γ is controlled.

Original motivation:

The (DP1)–(DP5) depletion apparatus is elaborate and its conclusion is that nonpositive rounds cannot continue; the natural reading is that the alternative must be favourable.

Status:

KILLED. The depletion apparatus is untouched; the inference from its escape branch is not supported.

Exact certificate:

By (EB6), $\Delta > 0$ says exactly $\sum_{r \in B_0} \log r > W_R(B_-)$: the fresh support mass of the selected rows exceeds the rooted reuse mass they touch. A mechanical audit of every DOMAIN line and every proof in the Theory Core found three theorems assuming *nonpositive* rounds — G-ROW-SATURATION-ROOTED-DEFECT, G-EXTERNAL-BAD-SUPPORT-SATURATION-DISPOSITION, G-GLOBALLY-CLEAN-EXTERNAL-SUPPORT-BUDGET — and *none* assuming a positive round. Nothing derives a GOOD state, a capacity violation, or any canonical obligation from $\Delta > 0$.

Failed hypothesis or mechanism:

The escape branch of a dichotomy was treated as a conclusion. Structurally $\Delta > 0$ is *growth* of the enlarged world, not contradiction; and growth cannot be iterated to exhaustion, because designated bases live below $(\log N)^A$ while $|\mathcal{A}_N| \sim N/\log N$.

What survives:

Everything about nonpositive rounds, and G-GAMMA-POLYLOG-FORCES-POSITIVE-ROUND, which correctly delivers the positive round. What is corrected is only what that delivery is worth.

Safe weaker version:

“ γ polylog forces a positive signed round.” Never “... forces a contradiction”.

Current replacement:

Theory Core, Proposition [prop:positive-dead-end] and the new open target P1BR-POSITIVE-ROUND-CONSEQUENCE.

Do not repeat:

Do not announce that a wall has moved because an object is now forced to exist; check first that something consumes that object.

Related current target:

P1BR-POSITIVE-ROUND-CONSEQUENCE, P1BR, K1.

FAIL-POSITIVE-ROUND-EXHAUSTION-001**FAIL-ID:**

FAIL-POSITIVE-ROUND-EXHAUSTION-001.

Date / iteration:

vNEXT4, Phase 9.

Claim:

That repeated positive rounds must eventually run out of fresh primes, giving a contradiction by exhaustion.

Original motivation:

Nonpositive rounds deplete the integer potential $\mathfrak{J}_k$ of (DP4); by symmetry one hopes positive rounds deplete something too.

Status:

KILLED, quantitatively.

Exact certificate:

All fresh support of an external upper-half round lies below $N/6$ by (RC2). The total logarithmic demand of the whole selector output at exponent A is at most $(\log N)^{A+1}$; the supply in that band is $\vartheta(N/6) \geq N/12$. The ratio tends to 0 like $(\log N)^{A+1}/N$. Theory Core, G-AMBIENT-SUPPLY-NONDEPLETION.

Failed hypothesis or mechanism:

$\Delta > 0$ is a statement of growth. Growth has no terminal state when the reservoir is superpolynomially larger than the demand.

What survives:

The depletion statement for *nonpositive* rounds, (DP1)–(DP4), which is unaffected and remains the only monotone integer potential in the DAG.

Do not repeat:

Do not argue by exhaustion without comparing demand to ϑ of the relevant band.

Related current target:

P1BR-ADDITIVE-COUPLING.

FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001**FAIL-ID:**

FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001.

Date / iteration:

vNEXT4, Phase 7.

Claim:

That the failure recorded at FAIL-POSITIVE-ROUND-CONSEQUENCE-001 was a gap in the present list of theorems, so that a consequence of $\Delta > 0$ could be found by writing more theorems in the same language.

Original motivation:

vNEXT3.1 established option F by a mechanical audit of the file's headers. An audit of a text is silent about what could be added to the text.

Status:

KILLED for ledger-only derivations. **Not** a refutation of P1BR-POSITIVE-ROUND-CONSEQUENCE itself.

Exact certificate:

Define the round datum $\mathfrak{d}(S, \mathscr{P})$ as the rows, their supports, the heights on those supports and κ_S restricted to those supports. Every quantity in (EB6)–(EB10), (DP1)–(DP5) and (PD1) is a function of $\mathfrak{d}$. The family $\mathfrak{S}_n$ of Theory Core G-POSITIVE-ROUND-LEDGER-INDEPENDENCE has constant $\mathfrak{d}$, constant $\Delta > 0$, and $\eta(R_n) = n + 1 \rightarrow \infty$; it is child-closed with a unique source, has no unit rows, and is routing-compatible. Its member $n = 1$ has $\eta(R_1) = 2$ while Δ can be made arbitrarily large by raising the two fresh heights. Hence the only lower bound on η expressible in $\mathfrak{d}$ is the trivial $\eta_{\text{supp}} = \sum_{r \in B_-} (\kappa(r) - 1)$, which is 0 whenever $B_- = \emptyset$; in particular no such bound grows with Δ . The witnesses exist by GEN-FINITE-REALISATION.

Failed hypothesis or mechanism:

The (LE3)+(FC8) route consumes a *lower* bound on $\eta(R)$ of order $\gamma \log N$. A nonpositive

round supplies one by (EB8). A positive round supplies an *upper* bound on the rooted mass of its own support and constrains $\Gamma_N(R)$ nowhere else — which is exactly where the witnesses place all their incidence.

What survives:

The asymmetry is proved, not assumed: for a nonpositive round the same datum does bound $\eta(S)$ below. The obstruction is the sign, not the bookkeeping.

Safe weaker version:

A consequence may still be derivable using an axiom that the witness family violates. Three are identified and recorded as live: (L1) $\mathcal{D}(p) = N - p$; (L2) $\gcd(N - p, N - p') \mid p' - p$; (L3) band localisation plus prime counting. All three are additive.

Do not repeat:

Loss-control rule (LR1): a claimed consequence of a positive round must name a quantity that is not a function of the round datum.

Related current target:

P1BR-ADDITIVE-COUPLING.

FAIL-POSITIVE-SEMPIRIME-ROUND-IS-NEXT-PRIMARY-001

FAIL-ID:

FAIL-POSITIVE-SEMPIRIME-ROUND-IS-NEXT-PRIMARY-001.

Status:

SUPERSEDED.

Superseded description:

after v4.41, the binding branch is to convert a positive clean semiprime round directly through P1BR additive coupling.

Certificate:

v4.42 continues through the actual designated BAD states q_i and their stopped descendant closure. It forces ambient incremental J or a linear cube-root boundary load before any positive-sign consumer is needed.

What survives:

positive sign remains a valid certificate where proved, and P1BR-ADDITIVE-COUPLING remains open in its own right.

FAIL-POTENTIAL-CAPACITY-001

FAIL-ID:

FAIL-POTENTIAL-CAPACITY-001.

Date / iteration:

v7.5 first-front audit; retained in v0.1.

Claim:

A first-front-only potential capacity $\mathcal{C}_S < \mathcal{D}_S$ holds canonically and supplies the needed contradiction.

Original motivation:

Pairwise row isolation seemed to make the available factor capacity smaller than root demand.

Status:

KILLED.

Minimal / strong counterexample:

$N = 4736, p_0 = 2371, h = 3, m = 2365 = 5 \cdot 11 \cdot 43$.

Exact certificate:

The three first-front rows are

$$4736 - 5 = 4731 = 3 \cdot 19 \cdot 83, \quad 4736 - 11 = 4725 = 3^3 \cdot 5^2 \cdot 7,$$

$$4736 - 43 = 4693 = 13 \cdot 19^2.$$

With the source definitions, the audited values are $\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$.

Failed hypothesis or mechanism:

Potential parents were counted as if their availability already encoded closure and actual incidence.

Why the argument failed:

Pair isolation controls collisions, not which rows are dynamically realised or simultaneously usable.

What survives:

Exact collision identities for actual rows.

Safe weaker version:

Introduce closure/activation first, then charge only realised rows.

Current replacement:

Theory Core, G-COLLISION-CUT, G-AMBIENT-PROMOTION, and target $P1^{\text{br}}$.

Do not repeat:

Potential incidence is not actual incidence.

Related current target:

$L1, W1, K1$.

FAIL-POWER-THRESHOLD-TO-POLYLOG-001**FAIL-ID:**

FAIL-POWER-THRESHOLD-TO-POLYLOG-001.

Status:

SCOPE-CORRECTED.

Original killed formulation:

power-threshold compression can never force a polylogarithmic / near-logarithmic star, because reaching that scale requires $m \rightarrow \infty$.

Correction:

variable m is legal with its explicit occupied-base error. Taking $m \asymp \log N / \log \log N$ can force $q_0 \leq (\log N)^{1+o(1)}$ with multiplicity $N^{1/3-o(1)}$ on the low branch.

What the no-go really says:

one threshold cannot simultaneously reach that star scale and retain the old $M \log N$ -scale generic high- J lower bound. The high branch falls to $M(\log \log N)^2 / \log N$.

Do not repeat:

do not interpret the v4.43 fixed- m calculation as a ban on growing m .

FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-NEARCRITICAL-STAR**FAIL-ID:**

FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-NEARCRITICAL-STAR.

Status:

KILLED.

Killed mechanism:

preserve $N^{1/3-o(1)}$ rows under repeated exact-layer peeling using only distinct peeled bases, growth of the common modulus, parent-slot, and separate valuation-layer counts.

Sharp cumulative-loss bound:

writing $X = \log N$, $B_i = \log Q_i$ and $\ell_i = \log q_i$,

$$\mathcal{L}_t := \sum_{i < t} \log E_i \leq \sum_{i < t} \log \frac{X - B_i}{\ell_i} \leq \log \frac{Q_t}{Q_{\text{in}}} + (1 + o(1)) \frac{\log N}{(\log \log N)^2}.$$

The coefficient 1 before $\log(Q_t/Q_{\text{in}})$ is asymptotically sharp from the inspected scalar data alone.

Certificate:

a primorial hostile sequence with consecutive small primes up to $a \log N$, $1/3 < a < 2/3$,

satisfies simultaneously

$$Q_t = N^{a+o(1)}, \quad \prod_i E_i = N^{a+o(1)}.$$

Thus the formal cardinality can collapse before the modulus reaches the $N^{2/3}$ parent-slot scale.

Sharp scope guard:

the no-go ignores global compatibility of all valuations in the same integer $N - p$ and therefore does not apply to simultaneous joint-valuation compression.

Current replacement:

G-SUBCRITICAL-ALPHABET-SIMULTANEOUS-VALUATION-FREEZE.

FAIL-PROFILE-CARRIER-001

FAIL-ID:

FAIL-PROFILE-CARRIER-001.

Date / iteration:

Interrupted T_D round, 31 August 2026.

Claim:

The cardinality profile of the four C0b classes separates canonical from noncanonical roots and can therefore carry a theorem.

Original motivation:

Hard canonical controls were reported as D-heavy and $\mathcal{F}$ as associated with easy or direct-GOOD controls.

Status:

SUPERSEDED.

Minimal / strong counterexample:

The recorded profiles themselves.

Exact certificate:

The recovered log states that raw profiles did not separate the relevant outcomes and that the reported associations were explicitly not promoted to theorems. The associated 97.81% D-fresh rate is **VERIFIED COMPUTATION**, with no surviving artifact.

Failed hypothesis or mechanism:

A correlation was treated as a mechanism.

Why the argument failed:

Cardinalities discard the arithmetic that produces the classes.

What survives:

The *negative* conclusion only: cardinality-only theorem design is rejected as a current route.

Safe weaker version:

Select a branch by the theorem it produces, not by the correlation it exhibits.

Current replacement:

Theory Core, TD-CARRIER-SPLIT. T_D is selected because it factorises, not because it correlates.

Do not repeat:

Never promote a computational rate to a mechanism.

Related current target:

T_D .

FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001

FAIL-ID:

FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001.

Source:

explicit v4.54 Failure Log addition.

Status:**KILLED .****Killed description:**

repeated pure edges preserve only the original endpoint congruence modulo Q .

Correction:

the fixed cofactor Q on every pure edge lifts the path-start congruence to modulus Q^{L+1} .

Scope guard:

this does not contradict the earlier aggregate affine-path no-go, which allowed arbitrary path cofactors and therefore could lose the deep exponent.

FAIL-PURE-Q-GRAPH-CAN-BRANCH-001**FAIL-ID:**

FAIL-PURE-Q-GRAPH-CAN-BRANCH-001.

Source:

explicit v4.54 Failure Log addition.

Status:**KILLED .****Killed possibility:**

a pure target q can have two distinct pure parents.

Reason:

a pure parent is uniquely determined by

$$p = N - Qq.$$

Thus the pure graph is injective.

FAIL-Q0-LEDGER-001**FAIL-ID:**

FAIL-Q0-LEDGER-001.

Date / iteration:

Post-v7.5 incidence-ledger audit, 30 August 2026.

Claim:

The four channels proved for the smooth first-front domain cover every $q \in Q_0 = \text{PF}(N - p_0)$ without an under-parented channel.

Original motivation:

The same residual split appeared syntactically available for all root factors.

Status:**KILLED .****Minimal / strong counterexample:**

$N = 20, p_0 = 11, Q_0 = \{3\}$.

Exact certificate:

$20 - p_0 = 9, 20 - 3 = 17 \in \mathbb{P}$, so 3 is GOOD. Under stopped traversal $R = \{11\}$. Moreover $D_3 = \gcd(9, 17) = 1, M_3 = 17$, while the required 17-incidence in R is zero although the root demand is one.

Failed hypothesis or mechanism:

A successful first-front state was treated as though it were an expanded bad-world row.

Why the argument failed:

Stopping at GOOD deletes precisely the row needed by the proposed channel accounting.

What survives:

The exact ledger on its verified smooth/failed domain; the arithmetic identity $D_q M_q = N - q$.

Safe weaker version:

Retain the original domain or add an explicit under-parented/success channel.

Current replacement:

Theory Core, Full-Q Failure Ledger and C0b-RD, both with explicit stopped-world hypotheses.

Do not repeat:

Never enlarge a ledger domain merely because its formulas remain syntactically defined.

Related current target:

T_I, T_D, T_S, T_F .

FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-001**FAIL-ID:**

FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-001.

Status:

KILLED.

Rejected statement:

If owner-sealed verticality occurs only at the small common hub q_0 , the branch is cheap.

Certificate:

polynomial q_0^2 occupancy enters the already retained vNEXT4.26 exact-layer star peeling and exports full q_0 -free logarithmic mass.

FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001**FAIL-ID:**

FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001.

Source:

explicit v4.52 Failure Log addition.

Status:

KILLED.

Killed possibility:

on a polynomial exact fibre with cheap J/U_{res} , all unique external q -free first-copy support can remain at primes $\leq n$.

Reason:

the unique external q -free support has $\Omega(n \log N)$ weight, while all primes at most n carry only $O(n)$ total Chebyshev weight.

Replacement:

a constant-scale portion escapes above n , yielding C-DEEP-CARDINALITY-SCALE-SIBLING-RENEWAL.

FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001**FAIL-ID:**

FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001.

Date / iteration:

vNEXT4.6.

Claim:

The fixed- (s, b) two-linear-form upper sieve can be union-bounded without loss over all $s \leq N^\delta$.

Status:

KILLED.

Exact certificate:

For a given s , there are $O(s)$ possible parent coefficients b , while the reflected progression has only $O(t/s)$ indices. These factors cancel, leaving roughly one $t/\log^2 t$ cost per quotient band. The present union bound therefore gives no polynomial range.

What survives:

Fixed and sufficiently slowly growing S satisfying (QS4).

Do not repeat:

Do not spend another round merely optimising constants in QS3 in hopes of reaching N^δ .

FAIL-RDS-GEOMETRY-DENSITY-001

FAIL-ID:

FAIL-RDS-GEOMETRY-DENSITY-001.

Date / iteration:

v1.13 support-free continuation, 1 September 2026.

Claim killed:

From canonical reflected geometry and $\text{rad}(c_j) \mid d_j$ alone, one may infer a universal bound $|\text{RDS}| \leq t/8$.

Status:

KILLED.

Exact certificate:

Take

$$N = 4368, \quad x = 2184, \quad p_0 = 2203, \quad h = 19, \quad \text{prevprime}(p_0) = 2179, \quad G = 24, \quad t = 9.$$

Thus $h < G$, so the root is canonical. The strict reflected block contains

$$j = 3 : d_j = 13, \quad c_j = 2197 = 13^3,$$

and

$$j = 8 : d_j = 3, \quad c_j = 2187 = 3^7.$$

Hence both indices satisfy $\text{rad}(c_j) \mid d_j$, so $|\text{RDS}| \geq 2 > 9/8 = t/8$.

Scope warning:

This is a successful-world certificate: $m = N - p_0 = 2165 = 5 \cdot 433$ and the first-front state 5 is GOOD because $N - 5 = 4363$ is prime. Therefore the witness kills only a *geometry-only* density inference; it does not refute a stronger theorem conditioned on a genuinely stopped canonical failed world.

What survives:

TS-REFLECTED-DEADSMOOTH-CEILING is exact and remains PROVED. Any stronger failure-conditioned density bound requires additional dynamic input, not the reflected geometry alone.

Do not repeat:

Do not infer a small universal RDS density from powerful-number rarity or from finite empirical scarcity without using a proved failure-conditioned mechanism.

FAIL-REFLECTION-ACTIVATION-001

FAIL-ID:

FAIL-REFLECTION-ACTIVATION-001.

Date / iteration:

Canonical bridge audit, 31 August 2026.

Claim:

Canonical $h < G$ alone forces at least one reflected index j with $a_j = N - c_j$ an active prime, hence an actual reflected CRD row.

Original motivation:

The Reflection Gate Lemma makes every c_j composite, and compositeness was read as if it also produced a parent row.

Status:

KILLED.

Minimal / strong counterexample:

$$N = 242, \quad x = 121, \quad p_0 = 127, \quad \text{previous prime } 113, \quad \text{so } h = 6 < G = 14.$$

Exact certificate:

The two internal offsets give $(c_1, a_1) = (125, 117)$ and $(c_2, a_2) = (123, 119)$, with $125 = 5^3$,

$117 = 3^2 \cdot 13$, $123 = 3 \cdot 41$, $119 = 7 \cdot 17$. Both reflections are composite, so no reflected row exists at all. Verified exactly.

Failed hypothesis or mechanism:

The gate was treated as a state activation statement.

Why the argument failed:

ARS-REFLECTION-GATE is an *if and only if*: the factorisation of c_j is a CRD row precisely when a_j is an active prime, and nothing in $h < G$ supplies that.

What survives:

Compositeness of every c_j , and the classification C0b-RD which does not require any a_j to be prime.

Safe weaker version:

State separately *boundary* $\rightarrow$ *active reflected state* as an open implication.

Current replacement:

Theory Core, ARS-REFLECTION-GATE; target BIA remains **OPEN**.

Do not repeat:

Do not hide a missing activation step under the word “reflection”.

Related current target:

T_I , $P1^{\text{br}}$.

Domain note:

$N = 242$ is not a failed canonical instance, so it does not refute BIA itself; it refutes any proof of BIA that uses only selector geometry before invoking failure.

FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001

FAIL-ID:

FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001.

Date / iteration:

vNEXT4.18.

Restriction:

The fixed-polylog multiset limit of the four-ary multiparent selector is a fundamental limit on repeated rows from one base.

Status:

KILLED.

Correction:

A greedy rough-support matching argument selects

$$\gg N^{1/4-\varepsilon}/(\log N)^2$$

rows from one already-constructed base fan while keeping unintended rough supports pairwise disjoint.

FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNAVOIDABLE-001

FAIL-ID:

FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNAVOIDABLE-001.

Date / iteration:

vNEXT4.18.

Claim:

Repeating a designated base necessarily incurs $\binom{\rho_q}{2} \log q$ collision even when rows are separated.

Status:

SCOPE-CORRECTED.

Correction:

The quadratic term is unavoidable inside one multiparent round. Separate singleton rounds with planned designated overlap and disjoint unintended rooted payment supports admit

exact single charging.

FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-STORAGE-001

FAIL-ID:

FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-STORAGE-001.

Status:

KILLED.

Rejected statement:

A fresh repeated hub of multiplicity M creates $\binom{M}{2} \log q$ units of horizontal factor-flow storage.

Certificate:

actual incremental J at that layer is $(M-1) \log q$; the exact difference $((M-1)(M-2)/2) \log q$ is pairwise curvature.

Do not repeat:

Pairwise-gcd statistics and incremental storage answer different questions.

FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-BINDING-WALL-001

FAIL-ID:

FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-BINDING-WALL-001.

Status:

SUPERSEDED.

Superseded description:

Analyse $k_i = (p_i - u_i)/r_i$ in isolation.

Replacement:

Use the full owner-return relation $a_i = q_0 c_i + k_i$ and compare the support of a_i with the owner support. Preserved support forces verticality; changed support gives a foreign sibling and hence reuse or renewed state population.

FAIL-RETURN-COFACTOR-IS-FREE-001

FAIL-ID:

FAIL-RETURN-COFACTOR-IS-FREE-001.

Date / iteration:

vNEXT4.6.

Claim:

In $N - w = bq$, the cofactor b can be discarded as an unstructured residual.

Status:

KILLED as a description.

Exact certificate:

For a block-derived low-core target,

$$\frac{10}{9}s < b < 2s, \quad (b, N) = 1, \quad \text{PF}(b) \subset R.$$

Hence b is a rooted cofactor packet of comparable scale and its support must be retained.

Do not repeat:

Do not project a quotient-return event to the base q alone.

FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001

FAIL-ID:

FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001.

Status:

KILLED.

Killed inference:

if $\ell \notin R$ is returned by an extension row, that occurrence immediately contributes $\log \ell$ to $\log(J_{R \cup A}/J_R)$.

Certificate:

when $\kappa_R(\ell) = 0$, the first occurrence is the baseline copy in the exact fresh factor $F_R(A)$. Only a second occurrence contributes full-weight incremental J .

What survives:

owner-anchored labels or later returns do enter incremental J exactly as recorded by the v32 second-occurrence theorem.

FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-001**FAIL-ID:**

FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-001.

Status:

SCOPE-CORRECTED.

Correction:

A returning basin may leave a functional evaluated on the fixed set R unchanged, but relative to the enlarged child-closed world it has the exact incremental TUJ/source-lock cost. A nonempty source-free BAD extension satisfies $T_S/T_R \geq N - 1$.

Do not repeat:

Distinguish fixed- R invisibility from zero incremental cost.

FAIL-REUSE-DOUBLE-CHARGE-001**FAIL-ID:**

FAIL-REUSE-DOUBLE-CHARGE-001.

Date / iteration:

v1.21 depletion audit, 1 September 2026.

Claim:

If one nonpositive clean round forces a rooted reuse payment $J_* \mid J_R$, then running clean selection again automatically forces a new, disjoint payment and therefore decreases the remaining rooted budget.

Status:

KILLED.

Exact logical obstruction:

The second “round” may select the identical family or may use complements sharing exactly the same rooted negative labels. Then the same set B_- and the same factor J_* are being observed again. Nothing implies $J_*^2 \mid J_R$ or any decrease of a residual quotient.

What survives:

If complete complement supports are pairwise disjoint across rounds, then the negative rooted label sets are disjoint and the product of the corresponding J_i divides J_R .

Current replacement:

Theory Core, G-MULTIROUND-EXTERNAL-SINGLE-CHARGE and
G-BAD-BASE-FORBIDDEN-SUPPORT-CLEAN-CONTINUATION.

Do not repeat:

Never infer depletion from repeated invocation of a selector alone. First prove cross-round support freshness or enter the explicit support-saturation alternative.

Related current target:

K1, P1BR, TF renewal termination.

FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RPD-001

FAIL-ID:

FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RPD-001.

Status:

SCOPE-CORRECTED.

Killed overreach:

a distribution estimate for $m_k(r) = \#\{i : r^k \mid N - q_i\}$ on the initial v4.41 targets automatically proves RPD for H_Q .

Certificate:

$H_Q = S_Q \cap (N^{1/3}, N/3)$ contains later high descendants of the stopped basins as well as the initial q_i .

What survives:

an initial-target estimate is a valid closure input only after an ancestry-localisation theorem shows that every uncharged full-scale deep obstruction can be moved to an equivalent designated generation.

FAIL-ROOTED-AMBIENT-001

FAIL-ID:

FAIL-ROOTED-AMBIENT-001.

Date / iteration:

Canonical bridge audit, 31 August 2026.

Claim:

After adjoining a selected set $\mathcal{P}$ of external active parents, the old rooted lock still controls the enlarged system, so T_R, U_R, J_R bound the promoted valuation mass.

Original motivation:

The rooted congruence was the strongest available global identity and was reused outside its domain.

Status:

KILLED.

Minimal / strong counterexample:

Structural, by G-MULTISOURCE-LOCK.

Exact certificate:

For child-closed S , $T_S \equiv (-1)^{|S|} \prod_{d \in D_S^{\text{src}}} d \pmod{N}$. Adjoining $\mathcal{P}$ and re-closing changes the source set from $\{p_0\}$ to a set containing $\mathcal{P}$, so the right-hand side changes. A large ambient J is consistent with a large source product.

Failed hypothesis or mechanism:

A one-source identity was applied to a multi-source system.

Why the argument failed:

The lock is an identity *about the sources*, not a bound on the collision channel.

What survives:

The exact multi-source congruence itself, and the promotion inequality $\log(J_{R \cup \mathcal{P}}/J_R) \leq K_{\times}(R, \mathcal{P}) + K(\mathcal{P})$.

Safe weaker version:

Any future capacity theorem must book the new sources explicitly.

Current replacement:

Theory Core, G-MULTISOURCE-LOCK, G-AMBIENT-PROMOTION, target K1.

Do not repeat:

Never carry a rooted bound across a change of source set.

Related current target:

K1, W1.

FAIL-ROUTING-INDIFFERENCE-SCOPE-001

FAIL-ID:

FAIL-ROUTING-INDIFFERENCE-SCOPE-001.

Date / iteration:

vNEXT4.1, Phase 0A.

Claim:

vNEXT4 (RI1), TD-UNIVERSAL-ROUTING $\not\Rightarrow$ P1BR.

Original motivation:

Under routing every fresh seed returns, and the return branch was shown to move no recorded quantity of R .

Status:

SCOPE-CORRECTED. The theorem survives; its reading does not.

Exact certificate:

The proof enumerates the *present* functionals $\kappa_R, \Gamma_N(R), \eta(R), J_R, U_R, T_R, \gamma, C, H, \tau, Q$ and observes that each is a function of the set R , which does not change when a closure returns to it. That establishes a non-implication *through those functionals*. It says nothing about a theorem that uses the arithmetic of the return path — the actual integers $N - p$ along it, the congruences placing a returning seed inside R , or $\gcd(N - p, N - u) \mid p - u$. Quantifying over all future theorems was never proved and cannot be proved this way.

What survives:

Theory Core, G-ROUTING-INDIFFERENCE-SCOPED: routing alone together with the currently recorded CRD ledgers does not close P1BR.

Safe weaker version:

the scoped form above, which is what is used.

Do not repeat:

Loss-control rule (LR5) is amended: routing may not be cited as a *sufficient* route to P1BR through the recorded ledgers; it remains admissible as an *ingredient* of an additive argument.

Related current target:

P1BR-ADDITIVE-COUPPLING, TD-UNIVERSAL-ROUTING.

FAIL-ROUTING-LOSES-SEED-WEIGHT-001

FAIL-ID:

FAIL-ROUTING-LOSES-SEED-WEIGHT-001.

Status:

KILLED.

Killed description:

after basin merging/returns, the original logarithmic weight of the external seed packet may disappear from all exact ledgers.

Certificate:

v4.38 proves

$$\prod_{\ell \in E} \ell \mid \Sigma_{\text{ext}}(S) F_R(A),$$

so

$$W(E) \leq \log \Sigma_{\text{ext}}(S) + \log F_R(A).$$

For $|E| \gg N^{1/3}$ one of the two reservoirs has ambient polynomial weight.

What survives:

neither reservoir yet has the required rooted upper consumer.

FAIL-ROUTING-RESCUES-P1BR-001

FAIL-ID:

FAIL-ROUTING-RESCUES-P1BR-001.

Date / iteration:

vNEXT4, Phase 6.

Claim:

That proving TD-UNIVERSAL-ROUTING — every BAD child-closed closure meets R — would remove the obstruction at P1BR-POSITIVE-ROUND-CONSEQUENCE, since it removes the disjoint-packet configuration “without requiring any new theory”.

Original motivation:

vNEXT3.1 recorded, correctly, that routing removes the disjoint-packet discussion, the recorded-ledger bridge theorem and the CRD-2 question. It was natural to expect that it also unblocked the positive round.

Status:

KILLED. The routing statement itself is untouched and remains **OPEN**; only its claimed relevance to P1BR is rejected.

Exact certificate:

Under routing, $B_{\text{ext}} = \emptyset$ and every fresh seed of a positive round returns. A returning seed r produces the enlarged world $R' = R \cup R_N(r)$, but κ_R , $\Gamma_N(R)$, $\eta(R)$, J_R , U_R , T_R , γ , $|R|$, C , H , τ , Q are functions of the set R alone and do not move; and the capacity bounds (FC7)–(FC10) do not transfer to R' , whose source set is not a single upper-half root with composite complement. See Theory Core, G-RETURN-BRANCH-NO-CANONICAL-CHARGE and G-ROUTING-INDIFFERENCE, (RI1).

Second, independent certificate:

The witness family of G-POSITIVE-ROUND-LEDGER-INDEPENDENCE is routing-compatible — every fresh state’s closure meets R_n — and still has $\eta(R_n)$ unbounded at fixed round datum.

Failed hypothesis or mechanism:

Confusing “the seed reaches R ” with “ R is charged”. Reaching a fixed set does not alter any functional of that set.

What survives:

Routing remains the right question for the bridge, the disjoint-packet configuration and the CRD-2 question, and the restricted form TD-POSITIVE-ROUND-ROUTING is recorded in the Core.

Do not repeat:

Loss-control rule (LR5): TD-UNIVERSAL-ROUTING may not be cited as a *sufficient* route to P1BR through the recorded ledgers. Scope-corrected at vNEXT4.1: see FAIL-ROUTING-INDIFFERENCE-SCOPE-001; routing remains admissible as an ingredient of an additive argument.

Related current target:

P1BR-ADDITIVE-COUPLING, TD-UNIVERSAL-ROUTING.

FAIL-SATURATION-DOUBLE-CHARGE-001**FAIL-ID:**

FAIL-SATURATION-DOUBLE-CHARGE-001.

Date / iteration:

v1.22 support-saturation continuation, 1 September 2026.

Claim:

If a current external BAD packet substantially overlaps the support of an earlier nonpositive globally clean family, then the pivot payment (SP3) is an additional disjoint factor which can be multiplied into the old v1.21 depletion product.

Status:

KILLED.

Why the inference fails:

The pivot uses a subfamily of the *same old rows*. Its negative rooted support can therefore be a subset of the already charged payment set of those rows. Relabelling changes the certificate

used to lower-bound the payment, not the underlying layer-profile balance.

What survives:

The overlap gives the quantitative localisation (SP1)–(SP5), and may reveal a positive signed subfamily. It becomes a new depletion charge only after a separate proof of support disjointness from all previous payment sets.

Current replacement:

Theory Core, G-SATURATED-BASE-RELABEL-PIVOT together with G-MULTIROUND-EXTERNAL-SINGLE-CHARGE.

Do not repeat:

Never add a pivot-derived rooted factor to $\prod_i J_i$ merely because the designated bases are new.

Related current target:

K1, globally clean renewal termination.

FAIL-SEALED-MULTISOURCE-LOCK-001

FAIL-ID:

FAIL-SEALED-MULTISOURCE-LOCK-001.

Date / iteration:

v1.6 sealed-source continuation, 1 September 2026.

Claim:

If many external upper-half bad parents have all children inside the canonical failed world R , adjoining them and applying G-MULTISOURCE-LOCK yields a new modular obstruction stronger than the rooted lock.

Original motivation:

The sealed branch creates genuine new sources, and the multi-source congruence explicitly records the source product.

Status:

KILLED.

Minimal / strong counterexample:

Structural; the proposed new congruence reduces identically to the old one.

Exact certificate:

Let $\mathcal{P}$ be finite, upper-half, external and bad with $\Gamma_N(\mathcal{P}) \subseteq R$, and put $S = R \cup \mathcal{P}$. Then

$$\text{Src}(S) = \{p_0\} \dot{\cup} \mathcal{P}, \quad T_S = T_R \prod_{p \in \mathcal{P}} (N - p).$$

The multi-source lock reads

$$T_R \prod_{p \in \mathcal{P}} (N - p) \equiv (-1)^{|R|+|\mathcal{P}|} p_0 \prod_{p \in \mathcal{P}} p \pmod{N}.$$

Using $N - p \equiv -p \pmod{N}$ and cancelling the active labels p , which are units modulo N , leaves exactly

$$T_R \equiv (-1)^{|R|} p_0 \pmod{N},$$

the original G-ROOT-LOCK congruence.

Failed hypothesis or mechanism:

A larger source set was assumed to create new modular information merely because the source product changed.

Why the argument failed:

The row factors contributed by the new sealed sources are congruent to the negatives of those same source labels, so their entire modular contribution cancels.

What survives:

G-SEALED-SOURCE-LOCK-REDUCTION is an exact **PROVED** structural statement. The sealed branch remains potentially useful through population size, valuation/reuse overload, or deeper topology, but not through the source-lock congruence alone.

Safe weaker version:

Use the sealed-source lift as bookkeeping only; seek a noncongruent invariant for the new sources.

Current replacement:

Theory Core v1.6, **G-SEALED-SOURCE-LOCK-REDUCTION**; see also the four-ary sign-overload disposition theorem in the Core.

Do not repeat:

Do not count new sealed sources as new modular constraints without simplifying the multi-source congruence first.

Related current target:

sealed-source branch, P1BR, K1.

FAIL-SELECTOR-CLOSES-CASCADE-001**FAIL-ID:**

FAIL-SELECTOR-CLOSES-CASCADE-001.

Date / iteration:

vNEXT3, Phase 4.

Claim:

Once **G-BAD-BASE-SQUAREFREE-POLYLOG-EXTENSION** is proved, the (FC10) clean-round route yields a contradiction.

Original motivation:

The selector was the last analytic blocker on that route.

Status:

KILLED. The selector is **PROVED**; the inference is not.

Exact certificate:

The selector supplies at most one clean row per designated base, and the bases lie in $[3, (\log N)^4]$, so $\sum_i M_i \log z_i \ll (\log N)^4 \log N$. (FC10) forces a positive signed round only once $\sum_i M_i \log z_i \geq \gamma \log(25N^2/18)$, i.e. once $\gamma \lesssim (\log N)^4$. The cascade therefore closes exactly on the subdomain $\gamma \leq (\log N)^4$ and on no larger one by this route.

Failed hypothesis or mechanism:

Removing an analytic obstruction was read as closing the chain; the counting obstruction downstream is untouched.

What survives:

G-CASCADE-GAMMA-TERMINUS, which states the subdomain exactly, and the selector itself.

Safe weaker version:

“Contradiction on $\gamma \leq (\log N)^4$.”

Current replacement:

Theory Core, the retired **G-CASCADE-GAMMA-TERMINUS**, whose proof body is kept at Subsection [fail:cascade-body] of this document and no longer in the Core.

Do not repeat:

Do not announce a cascade closure without counting the number of rounds the selector actually supplies against the capacity it must exhaust.

Related current target:

γ , K1, X1.

FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001**FAIL-ID:**

FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001.

Date / iteration:

vNEXT4.21.

Strategy:

Shorten the parent interval so the cofactor is smaller and use classical short-interval BV plus

an unweighted lower linear sieve to force P_3 .

Status:

KILLED relative to that stated analytic architecture.

Reason:

The loss in available distribution level dominates the gain in cofactor length. The successful vNEXT4.21 P_3 theorem instead uses weighted-sieve information at the full-interval BV level.

Scope guard:

This is not a universal impossibility result for stronger short-interval/switching technology.

FAIL-SIMPLE-STATE-001

FAIL-ID:

FAIL-SIMPLE-STATE-001.

Date / iteration:

v0.1 foundations review.

Claim:

The simple directed support graph on $\mathcal{A}_N$ is a faithful presentation of the CRD dynamics.

Original motivation:

Reachability questions are graph questions.

Status:

SUPERSEDED.

Minimal / strong counterexample:

Structural.

Exact certificate:

The support graph loses the valuations $\nu_N(p, q) = v_q(N - p)$, hence the prime-power layers $\mu_{\ell, k}$, hence the collision and cut functionals; and it carries no record of ambient external parents.

G-TUJ, G-FULLQ-LEDGER, G-COLLISION-CUT and G-AMBIENT-PROMOTION are all invisible to it.

Failed hypothesis or mechanism:

Edge existence was taken for the whole structure.

Why the argument failed:

$T = UJ$, residual production, prime-power collision and ambient promotion all depend on valuations, not on the mere existence of an edge.

What survives:

Graph-level facts: closure, terminal SCC, source deletion.

Safe weaker version:

Use the support graph only for reachability statements.

Current replacement:

Theory Core, the rooted ambient valuation-incidence object $\mathfrak{C}_N = (N, \mathcal{A}_N, p_0, \nu_N)$.

Do not repeat:

Do not state a valuation theorem in graph language.

Related current target:

L1, K1.

FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-VERTICAL-MIGRATION-001

FAIL-ID:

FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-VERTICAL-MIGRATION-001.

Status:

KILLED.

Killed inference:

a polynomial exact star on one base must first be extracted before diffuse vertical storage can be forced into another CRD ledger.

Certificate:

v4.37 k -power divisor packing sums over all powerful parts simultaneously and forces a D/J

lower bound without a common base.

What survives:

exact-star peeling is still useful when concentration is actually present.

FAIL-SINGLETON-STAR-LOCAL-WINDOW-001

FAIL-ID:

FAIL-SINGLETON-STAR-LOCAL-WINDOW-001.

Date / iteration:

vNEXT4.16.

Claim:

A fixed small prime r cannot divide parent cofactors for many high reflected targets because the legal coefficient interval is too short.

Status:

KILLED.

Exact obstruction:

For $c_j = sq$, the admissible cofactor interval has length $> 5s/9$. If $s > (18/5)r$, it already contains an odd multiple of r .

What remains:

Primality/rootedness, mirror feedback, or a signed multiplicity capacity must be used.

FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-001

FAIL-ID:

FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-001.

Status:

SCOPE-CORRECTED.

Killed inference:

derive $X \geq Y$ using only the same local size inequalities $x_i \leq y_i$ whose product already proves $X \leq Y$.

Reason:

this merely asks the local envelope to contradict itself.

What survives:

an independent global, canonical, high-gap, analytic or other nonlocal theorem may imply $X \geq Y$ on a special failed domain; together with the envelope this would prove that the domain is empty. Therefore the strong revised2 phrase “no global lower bound can ever contradict the envelope” is not an active prohibition.

FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVISIBLE-001

FAIL-ID:

FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVISIBLE-001.

Status:

KILLED.

Killed inference:

$\gamma < N^{1/3}/\log N$ makes the high-gap reflected block too small to yield a polynomial CRD object.

Certificate:

the whole-block disposition with any fixed $1/3 < \delta < 0.475$ forces

$$E \subset (\mathcal{A}_N \setminus R) \cap (0, N/5), \quad |E| \gg N^{1/3}.$$

What survives:

the packet is external and requires a source/fresh consumer; it is not automatically absorbed by R .

FAIL-STAR-SELF-REENTRY-001

FAIL-ID:

FAIL-STAR-SELF-REENTRY-001.

Status:

KILLED.

Killed inference:

the star $N^{1/3-1/(m+1)-o(1)}$ produced at fixed m can be fed back into the same power-threshold theorem to obtain a terminating descent.

Certificate:

the theorem requires a packet carrying defect $\gg N^{1/3}$, while the fixed- m star has an exponent deficit $1/(m+1) > 0$.

Scope guard:

v4.44 does not use such self-reentry; it chooses growing m before concentration and then uses simultaneous joint-profile freezing.

FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-ROUND-001

FAIL-ID:

FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-ROUND-001.

Source:

v4.52 scope correction and explicit Failure Log addition.

Status:

KILLED.

Killed inference:

remove the common r^e and then automatically apply **G-EXTERNAL-BASE-SIGNED-REUSE** as though the residuals were pairwise coprime and squarefree.

Reason:

the full complements are not pairwise coprime, and the stripped residuals b_p need not be pairwise coprime or squarefree.

Replacement:

use the exact r -stripped surplus ledger and split into paid residual collision/verticality or owned unique external sibling renewal. The stronger pairwise-coprime interface estimate remains conditional.

FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001

FAIL-ID:

FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001.

Date / iteration:

vNEXT4.21.

Killed inference:

Knowing $b \bmod q$ together with $\Omega(b) \leq 3$ forces some sibling $r \mid b$ to satisfy $r \equiv N \pmod{q}$.

Status:

KILLED for every prime $q \geq 5$.

Exact certificate:

With $G = (\mathbb{Z}/q\mathbb{Z})^\times$ and $a = N \bmod q$,

$$(G \setminus \{a\})^2 = G.$$

Thus two non-return sibling residues already realise every possible product residue.

Exceptional case:

$q = 3$ is different. If $N \equiv 2 \pmod{3}$ and $b \equiv 2 \pmod{3}$, at least one sibling returns to 3.

FAIL-TD-NAIVE-001

FAIL-ID:

FAIL-TD-NAIVE-001.

Date / iteration:

Interrupted T_D branch round, 31 August 2026.

Claim:

Every composite D -carrier a_j has an active prime factor, so every $\mathcal{D}$ index produces an actual CRD state.

Original motivation:

Fresh carriers do produce active states, and the smooth case was assumed to differ only quantitatively.

Status:

KILLED.

Minimal / strong counterexample:

$N = 1862$, $r = 1733$, $j = 2$.

Exact certificate:

$m = N - r = 129$ and $a_j = m + 2j = 133 = 7 \cdot 19$, while $7 \mid 1862$ and $19 \mid 1862$. Both prime factors of the carrier divide N , so neither is active and a_j contributes no CRD row. The reflected partner is $c_j = 1729 = 7 \cdot 13 \cdot 19$. Verified exactly.

Failed hypothesis or mechanism:

Compositeness of the carrier was confused with activity of its factors.

Why the argument failed:

Activity is a condition on N , not on a_j ; a carrier can be built entirely from primes dividing N .

What survives:

The smooth/fresh split of TD-CARRIER-SPLIT covers this instance instead of hiding it: a *fresh* carrier, that is one with $P^+(a_j) \geq h$, does give an active prime state $s_j < N/6$ by strict reflected-block nonreuse. The exception lives in $\mathcal{D}_{\text{sm}}$.

Safe weaker version:

Assert activity only for $j \in \mathcal{D}_{\text{fr}}$.

Current replacement:

Theory Core, TD-CARRIER-SPLIT and C-STRICT-NONREUSE.

Do not repeat:

Never promote “composite” to “has an active factor”.

Related current target:

T_D , $P1^{\text{br}}$.

Domain note:

$r = 1733$ is a *noncanonical* upper root; the canonical selector for $N = 1862$ is $p_0 = 937$. The certificate is an exact arithmetic fact and is domain-independent, but it must not be quoted as a canonical-world witness.

FAIL-TD-SMOOTH-INACTIVE-001

FAIL-ID:

FAIL-TD-SMOOTH-INACTIVE-001.

Date / iteration:

Phase-3 T_D second-layer attack, 31 August 2026.

Claim:

The canonical geometry alone — p_0 canonical, $h < G$, $a_j = m + 2j$ odd composite, c_j composite — forces a smooth reflected carrier to transport some cost: an active state, a prime-power layer, or a source term.

Original motivation:

TD-CARRIER-SPLIT gives $\Omega(a_j) > \log m / \log h$, a large factorisation mass, and mass was assumed to be transportable.

Status:

KILLED as a proof route resting on the canonical geometry. It is *not* a refutation of any statement conditioned on canonical failure; see the domain note below.

Minimal / strong counterexample:

$N = 168$, $x = 84$, $p_0 = 89$, $h = 5 < G = 6$, $m = 79$, $t = 2$, $j = 1$. Smallest instance with *composite* complement, hence with a genuine first front: $N = 2684$, $p_0 = 1361$, $h = 19 < G = 34$, $m = 1323 = 3^3 \cdot 7^2$, $j = 4$, $a_4 = 1331 = 11^3$, $c_4 = 1353 = 3 \cdot 11 \cdot 41$, $11 \mid 2684$.

Exact certificate:

$a_1 = m + 2 = 81 = 3^4$ and $c_1 = N - a_1 = 87 = 3 \cdot 29$. Since $3 \mid 168$, the only prime of a_1 divides N , so $3 \notin \mathcal{A}_N$. The carrier therefore contributes no active state, no prime-power layer of the CRD system, and no source. Its whole factorisation mass $\Omega(a_1) = 4$ is inert. Verified exactly.

Domain note — read before transferring this witness:

all six known instances are *successful* worlds. At $N = 168$ and $N = 492$ the complement m is prime, so p_0 is GOOD at depth zero and $R = \emptyset$; at $N = 2684$, 4380, 4386, 6260 the complement is composite but the traversal succeeds deeper. In none of them is R a failed world, so in none of them is the index j shown to lie in $\mathcal{D}$. By PROHIB-DOMAIN-TRANSFER this witness therefore kills only proofs that use the canonical geometry alone; the failure-conditioned form of the claim is untouched and remains open. This is the same shape of kill as FAIL-TF-LOCAL-SCOPE-001.

Further instances:

$(N, p_0, j, a_j) = (492, 251, 1, 3^5)$, $(4380, 2203, 5, 3^7)$, $(4386, 2203, 2, 3^7)$, $(6260, 3137, 1, 5^5)$. Over canonical $N < 1.2 \cdot 10^5$, 22.24% of composite reflected carriers have at least one prime factor dividing N .

Failed hypothesis or mechanism:

Ω -mass of a_j was identified with CRD mass. The two live in different objects: a_j is not a row, and its prime factors need not be states.

Why the argument failed:

Activity is a condition relative to N . A carrier assembled from primes dividing N is invisible to the CRD system.

Structural note:

such a carrier is necessarily *smooth*. If $\ell \mid a_j$ and $\ell \mid N$ then $\ell \mid N - a_j = c_j$, so ℓ divides two distinct members of the strict reflected block and C-STRICT-NONREUSE forces $\ell < h$. The fresh branch is therefore immune, which is exactly the content of TD-CARRIER-SPLIT: only $\mathcal{D}_{\text{sm}}$ can be inert.

What survives:

The smooth branch is bounded in *count* rather than transported in *mass*. TD-SMOOTH-CEILING gives $|\mathcal{D}_{\text{sm}}| \leq \sigma_N(h)$ unconditionally, and that ceiling is what the second-layer trichotomy consumes.

Safe weaker version:

Never assert that a smooth carrier produces a state. Use only its cardinality, via the ceiling.

Current replacement:

Theory Core, TD-SMOOTH-CEILING and TD-SECOND-LAYER.

Do not repeat:

Do not attempt to transport smooth carrier mass to actual layers. That direction is closed by this certificate; the productive direction is the ceiling.

Relation to FAIL-TD-NAIVE-001:

complementary, not stronger. FAIL-TD-NAIVE-001 uses a noncanonical root, $(N, r) = (1862, 1733)$ with canonical selector 937, and kills “every composite carrier has an active factor”. This entry uses canonical roots with $h < G$ and kills the same claim for any proof resting on that geometry, while strengthening the configuration from “some prime factor is inactive” to “the carrier is wholly inert”. Neither entry reaches the canonical *failed* domain,

because no canonical failed world is known.

Related current target:

$T_D, P1^{\text{br}}$.

Exact classification (Phase-3 continuation):

the failure mode is now completely understood. By TD-CARRIER-DICHOTOMY, a carrier produces no active state *precisely* when $\text{rad}(a_j) \mid h - 2j$. All six instances above are of that kind, and each realises the normal form $x = q^e + d$: $84 = 3^4 + 3$, $246 = 3^5 + 3$, $1342 = 11^3 + 11$, $2190 = 3^7 + 3$, $2193 = 3^7 + 6$, $3130 = 5^5 + 5$. The class is bounded by TD-DEADSMOOTH-CEILING and is empty whenever no prime below h divides x — 48.0% of canonical instances with $N < 3 \cdot 10^5$. Across that whole range only 19 dead-smooth carriers exist among 514 447, and no N carries two.

FAIL-TF-LOCAL-SCOPE-001

FAIL-ID:

FAIL-TF-LOCAL-SCOPE-001.

Date / iteration:

Interrupted T_D branch round, hostile scope witness, 31 August 2026.

Claim:

A fresh reflected state ℓ_j , certified locally as $\mathcal{F}$ -class together with $h < G$, routes to Good_N or into R .

Original motivation:

The recorded census showed $\mathcal{F}$ to be empirically rigid, and the local certificate looked self-sufficient.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

$N = 128$, $p_0 = 67$, $j = 1$, $c_1 = 65$, $\ell = 13$.

Exact certificate:

$x = 64$, $p_0 = 67$, previous prime 61, so $h = 3 < G = 6$; $t = 1$ and $c_1 = 65 = 5 \cdot 13$ with $P^+(c_1) = 13 \geq h$, so the local $\mathcal{F}$ certificate holds. The state 13 is active and not GOOD, since $128 - 13 = 115 = 5 \cdot 23$, and its stopped traversal fails with the separate closed world $\{3, 5, 7, 11, 13, 23, 29, 41\}$. Verified exactly.

Failed hypothesis or mechanism:

The certificate was evaluated without the hypothesis that defines R .

Why the argument failed:

Here $128 - 67 = 61$ is prime, so p_0 *succeeds* at depth zero and $R = R_N(p_0) = \emptyset$. The condition $\text{PF}(c_j) \cap R = \emptyset$ is satisfied vacuously. The instance is a *successful world*, and R is not even defined as a failed world.

What survives:

The canonical-failure-conditioned form of TF is untouched. What is killed is every proof of it that uses only the local $\mathcal{F}$ certificate together with $h < G$.

Safe weaker version:

Always carry the failure hypothesis into the $\mathcal{F}$ certificate; never evaluate $\mathcal{I}, \mathcal{D}, \mathcal{S}, \mathcal{F}$ at a root that succeeds.

Current replacement:

Theory Core, target TF.

Do not repeat:

A vacuous R is not an empty support intersection in the intended sense.

Related current target:

TF, T_D .

FAIL-TWOVARIABLE-FULL-LEVEL-001

FAIL-ID:

FAIL-TWOVARIABLE-FULL-LEVEL-001.

Date / iteration:

vNEXT4.12.

Claim:

The exact local density $(2p-1)/p^2 = 2/p - 1/p^2$ for the polynomial pair $q(N - kq^2)$ permits an elementary sieve level equal to a fixed power of N .

Status:

KILLED.

Exact certificate:

The coefficient coordinate has length only

$$K = N/h^2.$$

The elementary remainder term therefore restricts the second logarithmic saving to $\log K$, not $\log N$.

Do not repeat:

A local dimension-two identity is not a distribution level theorem.

FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-O(n)-PARENTS-001

FAIL-ID:

FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-O(n)-PARENTS-001.

Status:

KILLED on the high layer extracted in v4.40.

Killed inference:

even after extracting linearly many fresh load-one targets, they may all be fed by $o(n)$ parent rows.

Certificate:

after removing targets below $n/(\log N)^2$, one parent can feed at most three surviving targets because their product divides one complement $< N$. Therefore the parent population is at least $(1/27 - o(1))n$, and a greedy matching gives at least $(1/63 - o(1))n$ vertex-disjoint parent-target edges.

What survives:

the cofactor supports of those matched rows remain uncontrolled.

FAIL-UNIQUE-MISSING-DATUM-001

FAIL-ID:

FAIL-UNIQUE-MISSING-DATUM-001.

Date / iteration:

vNEXT3.1, Phase-5 reclassification.

Claim:

The vNEXT3 information-loss certificate named “a share of the ambient valuation tensor” as *the exact datum missing* for a coupling.

Original motivation:

It is the most natural non-additive quantity in sight.

Status:

SCOPE-CORRECTED.

Exact certificate:

What is proved is that all recorded packet functionals factor on disjoint child-closed unions, hence cannot express competition for an R -only capacity. What is *not* proved is that ambient tensor occupancy is the unique repair, that no other non-additive functional works, or that a

CRD-2 is necessary. No uniqueness argument was given, and none is available.

What survives:

The diagnosis, reclassified as FRONTIER CERTIFICATE / INFORMATION-LOSS DIAGNOSIS, together with four candidates recorded as CONJECTURAL THEORY-DESIGN DIRECTIONS: ambient tensor occupancy, cofactor provenance, parent-pair correlations, non-additive resource ownership.

Do not repeat:

Do not call a design hypothesis an “exact missing datum” without a uniqueness proof.

Related current target:

TD-UNIVERSAL-ROUTING, CRD-2 question.

FAIL-UNIT-COMPLEMENT-001

FAIL-ID:

FAIL-UNIT-COMPLEMENT-001.

Date / iteration:

Domain audit, frozen 31 August 2026.

Claim:

Every failed active root has composite complement.

Original motivation:

Failure has no prime-complement success, so the complement was informally treated as composite.

Status:

KILLED.

Minimal / strong counterexample:

$N = 20, r = 19$.

Exact certificate:

$19 \in \mathcal{A}_{20}$, $20 - 19 = 1$, and $1 \notin \mathbb{P}$. The row has no prime children, hence the stopped traversal fails, although the complement is not composite.

Failed hypothesis or mechanism:

The implication “nonprime $\Rightarrow$ composite” silently assumed the complement exceeds one.

Why the argument failed:

The unit row was omitted from the domain.

What survives:

If $N - r > 1$ and r is not GOOD, then $N - r$ is composite.

Safe weaker version:

State the composite-complement hypothesis explicitly whenever $Q = \text{PF}(N - r)$ or a root ledger is used.

Current replacement:

Theory Core, rooted algorithm and Theorems G-FULLQ-LEDGER/G-FULLQ-PRODUCTION on their stated domains.

Do not repeat:

Do not infer compositeness from failure before checking the unit row.

Related current target:

C0a and every generic failed-root lemma.

FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001

FAIL-ID:

FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001.

Date / iteration:

vNEXT3.1, Phase-4 scope correction.

Claim:

G-BRIDGE-ROUTING-EQUIVALENCE of vNEXT3, read as: no coupling of disjoint packets is

possible at all.

Original motivation:

(OD1)–(OD4) really do make every recorded functional additive, which felt exhaustive.

Status:

SCOPE-CORRECTED. The mathematics is correct; the quantifier was too large.

Exact certificate:

(OD1)–(OD4) quantify over the recorded ledger $T, J, U_\bullet, \eta, \nu, K, \text{Src}$. They say nothing about an arbitrary functional $\Psi_N(R, U)$, which need not be a disjoint-union homomorphism. Candidates not excluded: ambient valuation-tensor occupancy, parent or cofactor correlations, residue geometry, shared ambient prime supply, non-additive resource ownership.

What survives:

G-RECORDED-LEDGER-BRIDGE-ROUTING: no bridge built *solely from the recorded disjoint-union-homomorphic ledgers* can charge $\eta(U)$ against an R -only capacity, and any such bridge requires $U \cap R \neq \emptyset$.

Do not repeat:

Do not promote “every functional we have recorded” to “every functional”.

Related current target:

TD/TF routing, TD-UNIVERSAL-ROUTING.

FAIL-UPPERHALF-SCOPE-001

FAIL-ID:

FAIL-UPPERHALF-SCOPE-001.

Date / iteration:

v0.1 hostile mathematical audit §3.

Claim:

Upper-half source compression applies to a failed canonical world, so every active upper-half prime is a source of the bad graph.

Original motivation:

Both settings are called “failure”.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

Hypothesis mismatch; no finite witness is required.

Exact certificate:

The source theorem assumes $\text{Good}_N = \emptyset$, that is, that N is a Goldbach counterexample. Only then do stopped and raw reachability coincide and every active $N/2 < p < N$ become a source. Canonical p_0 -failure asserts nothing about other roots: the recorded corpus contains many failed worlds for arbitrary roots at values of N that have Goldbach representations.

Failed hypothesis or mechanism:

The global hypothesis was imported into the local one.

Why the argument failed:

Using a theorem whose hypothesis is global Goldbach failure in order to contradict p_0 -failure assumes what is to be proved for every other root.

What survives:

ARS-UPPERHALF-COMPRESSION on its own domain, and the direct-Goldbach-failure DAG in which it is legitimate.

Safe weaker version:

Keep two separate DAGs — universal p_0 -success, and direct contradiction under global failure — and never move a statement between them without annotation.

Current replacement:

Theory Core, **ARS-UPPERHALF-COMPRESSION** with its prohibition, and the exact Goldbach boundary section.

Do not repeat:

PROVED $\neq$ COMPUTATION $\neq$ OPEN $\neq$ REJECTED, and failure of the canonical traversal is not global Goldbach failure.

Related current target:

X1.

FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHT-001**FAIL-ID:**

FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHT-001.

Date / iteration:

vNEXT4.14 correction of vNEXT4.13.

Claim:

The full weighted mass $W_{\text{ext}} = \sum v_q(c_j) \log q$ can be passed unchanged into a family consumer that retains one seed per prime label.

Status:

SCOPE-CORRECTED.

Exact correction:

On $h > N^{1/3}$, high reflected valuations are at most two, so for the distinct external labels

$$W(E_{\text{ext}}) \geq W_{\text{ext}}/2.$$

What survives:

The natural-scale external branch survives with an absolute factor-two loss.

FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001**FAIL-ID:**

FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001.

Date / iteration:

vNEXT4.19 status repair of vNEXT4.18.

Killed inference:

A polynomial BAD source packet by itself is evidence specific to a Goldbach counterexample and should replace the mirror route as the primary proof target.

Status:

KILLED.

Reason:

The source-star selector is essentially a generic reduced residue-class/sieve construction once an active BAD base exists. Large source products are compatible with multisource defect identities and are lower obligations, not upper collision/source capacities.

What survives:

G-POLYNOMIAL-REPEATED-BASE-ROUGH-FAN and G-PLANNED-BASE-OVERLAP-SINGLE-CHARGE remain PROVED-PENDING-INDEPENDENT-AUDIT; C-POLYNOMIAL-SOURCE-STAR-CAPACITY becomes a secondary structural frontier.

FAIL-V410-ANONYMOUS-STAR-COFACTOR-001**FAIL-ID:**

FAIL-V410-ANONYMOUS-STAR-COFACTOR-001.

Date / iteration:

vNEXT4.11 correction of the vNEXT4.10 frontier.

Claim/strategy:

For a large inverse-sibling star, treat $(N - v)/\ell$ as an anonymous large quotient and split it into generic rough/smooth cases.

Status:

SUPERSEDED on the monomial branch.

Exact certificate:

Retained provenance gives the two-level factorisation

$$N - v = bw, \quad N - w = q^2, \quad w > \sqrt{N},$$

with the whole entry cofactor b already small. Repeated values of b produce the exact entry comb, including $\text{rad}(b)^{L-1} \mid J_R$.

Replacement:

Use the whole-entry-cofactor atom/comb, not an anonymous quotient split.

FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001**FAIL-ID:**

FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001.

Source target:

C-DEEP-RPD-ANCESTRY-LOCALISATION.

Status:

SCOPE-CORRECTED.

Overstrong formulation:

a full uncharged deep-RPD obstruction can be pigeonholed into one common global generation.

Why not proved:

generation-edge amortisation charges cross-generation edges through horizontal surplus or source destruction but does not bound the total number of generations strongly enough for that pigeonhole.

Safe v4.47 replacement:

polynomial exact deep layer, followed by original v4.41 submatching, large incremental J , or a local immediate-ancestry restart.

v4.48 refinement:

the primary replacement is further sharpened to C-DEEP-RPD-DESIGNATED-OWNER-LOCALISATION; total unique parenthood is not required.

FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001**FAIL-ID:**

FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001.

Alias:

v4.48 R48-4.

Status:

SCOPE-CORRECTED.

Killed automatic inference:

if an ancestor was born as a factor of a reflected carrier, a later descendant automatically divides that same carrier.

Certificate:

divisibility by the original carrier is not preserved by a generic edge $v_{j-1} = N - c_j v_j$.

What survives:

the v4.47 reflected congruences remain valid on the explicit intersection domain where a target really has both tags.

Replacement:

path-labelled affine ancestry transport, G-CRD-PATH-AFFINE-COFACTOR-NORMAL-FORM and G-DEEP-ANCESTRAL-AFFINE-CONGRUENCE.

FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001

FAIL-ID:

FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001.

Alias:

v4.48 R48-3.

Status:

SCOPE-CORRECTED.

Old scope:

treat the cube-root high-target row trichotomy as requiring global load one of the designated target.

Correction:

inspection of the proof uses only an actual BAD edge $u \rightarrow p$, $p > N^{1/3}$, and factorisation of $N - u$. The accepted replacement is **G-CUBEROOT-HIGH-TARGET-OWNER-TRICHOTOMY**.

FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001

FAIL-ID:

FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001.

Alias:

v4.48 R48-1.

Status:

SCOPE-CORRECTED.

Killed promotion:

from $|X| = N^{1/3-o(1)}$ and $M \gg N^{1/3}$, promote $\Omega(|X| \log N)$ automatically to an ambient/full- M -scale payment.

Correction:

retain only the explicit subpolynomial comparison delivered by the pigeonhole losses, recorded in v4.48 as $|X| \geq MN^{-o(1)}$. The local J -payment is valid; the stronger scale label was not justified.

FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001

FAIL-ID:

FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001.

Alias:

v4.48 R48-5.

Status:

KILLED as a direct domain application.

Killed transfer:

apply **C-LOWCHILD-J-OR-SMALLHUB** unchanged to the deep designated-owner family.

Domain mismatch:

the old theorem requires a family of size $\gg N^{1/3}$, whereas the v4.48 owner family is guaranteed only at $MN^{-o(1)} = N^{1/3-o(1)}$.

Replacement:

G-DESIGNATED-LOWCHILD-SCALE-DISPOSITION and the open consumer **C-DEEP-LOWCHILD-RENEWAL-CAPACITY**.

FAIL-V48-MONOMIAL-NOVELTY-001

FAIL-ID:

FAIL-V48-MONOMIAL-NOVELTY-001.

Date / iteration:

vNEXT4.9 audit of vNEXT4.8.

Claim:

C-MONOMIAL-TWO-EDGE-RIGIDITY introduced a new square-map theorem.

Status:

SCOPE-CORRECTED (provenance correction).

Exact certificate:

The injectivity/no-cycle/no-three-consecutive-square argument is already contained in the **PROVED** theorem TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY.

What survives:

The extracted standalone formulation and its use of component starts in vNEXT4.8 remain valid.

FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001**FAIL-ID:**

FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001.

Alias:

v4.48 R48-2.

Status:

SUPERSEDED as a necessary proof split.

Old design:

separate deep targets by total indegree and cash out the multi-parent part before constructing a restart matching.

Replacement:

choose one actual parent per target. Because one parent row cannot feed three distinct targets above $N^{1/3}$, the chosen-edge graph has bounded degree and yields a vertex-disjoint designated-owner matching directly.

FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001**FAIL-ID:**

FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001.

Alias:

v4.48 R48-7.

Status:

KILLED for immediate closure.

Killed inference:

once the deep semiprime owner family enters a positive clean signed round, P1BR follows.

Reason:

the established positive-round independence theorem remains binding.

New datum retained:

every designated target additionally satisfies the common deep-layer relation $Q \mid N - p_i$, which was not part of the old round datum.

Live replacement:

P1BR-DEEP-DESIGNATED-TARGET-COUPLING.

FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001**FAIL-ID:**

FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001.

Alias:

v4.48 R48-6.

Status:

SUPERSEDED.

Superseded idea:

restart the v4.43 fixed- m recurrence with growing $m \sim \log \log N$ as a new proof route.

Reason:

v4.44 already performed the variable-threshold correction and replaced the fixed- m recurrence

by simultaneous valuation freezing, including the near-critical polylog star boundary.

FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001

FAIL-ID:

FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001.

Status:

KILLED as a sufficient general bridge.

Certificate:

the cheap semiprime orthogonal-broom branch admits a large selected subfamily with the relevant owner-support / later-target gcds equal to 1.

Consequence:

the common deep modulus does not automatically couple the old owner round to the descendant target rows through cross-generation gcd.

FAIL-V49-DEEP-SEMIPRIME-AUTOMATIC-POSITIVE-ROUND-CLOSURE-001

FAIL-ID:

FAIL-V49-DEEP-SEMIPRIME-AUTOMATIC-POSITIVE-ROUND-CLOSURE-001.

Status:

SCOPE-CORRECTED.

Killed promotion:

treat the localised deep semiprime family as automatically in the former ambient positive-round closure after the $N^{-o(1)}$ localisation loss.

Reason:

the scale/domain transfer is not automatic, and positive signed production remains insufficient for P1BR without an independent additive/capacity coupling.

Replacement:

v4.49 orthogonal-broom analysis and the later local reflected-anchor frontier.

FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001

FAIL-ID:

FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001.

Status:

KILLED / **SCOPE-CORRECTED**.

Killed inference:

from $\ell \rightsquigarrow p$ and $\ell \mid c_j^{\text{ref}}$, infer $p \mid c_j^{\text{ref}}$.

Reason:

direct divisor incidence is not hereditary along a generic CRD path.

Replacement:

the exact path-labelled identity **G-REFLECTED-ANCESTRY-EXACT-IDENTITY** and branch application **C-DEEP-ANCESTRAL-REFLECTED-TAG-TRANSPORT**.

FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001

FAIL-ID:

FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001.

Status:

KILLED.

Killed description:

once direct factor inheritance is lost, reflected arithmetic is lost entirely.

Counterdatum:

exact ancestry retains

$$N(2s\Xi - 1) - 2d_j = 2(-1)^t sC_t(N - p).$$

Scope guard from v4.50:

the exact identity is useful data, but its bare reduction modulo the terminal deep modulus is automatic and is not by itself capacity.

FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLOSES-001**FAIL-ID:**

FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLOSES-001.

Status:

KILLED as an immediate contradiction route.

Reason:

the exact relative $T/U/J/F$ identities give lower reservoirs but no independent upper capacity for the external extension.

What survives:

the identities remain valid accounting tools.

FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-CAPACITY-001**FAIL-ID:**

FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-CAPACITY-001.

Alias:

v4.50 R50-2.

Status:

SCOPE-CORRECTED.

What remains proved:

the exact identity

$$N(2s\Xi - 1) - 2d_j = 2(-1)^t sC_t(N - p).$$

What is removed:

the claim that reducing this identity modulo $Q \mid N - p$ creates a new reflected path-profile capacity condition.

Current distinction:

direct reflected incidence is a capacity datum; bare remote reflected ancestry is not, at residue level.

FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001**FAIL-ID:**

FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001.

Alias:

v4.50 R50-5; consistent with FAIL-BOUNDED-ARITY-WEIGHT-001.

Status:

KILLED as a weighted-capacity inference.

Safe use:

branching may be combined with exact cofactor/path arithmetic or another single-charged quantity, but it is not a capacity theorem by itself.

FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-SELECTION-001**FAIL-ID:**

FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-SELECTION-001.

Alias:

v4.50 R50-3.

Status:

KILLED on the return-heavy cheap- J branch.

Certificate:

C-RETURNHEAVY-REFLECTED-CUBEROOT-LOADONE-MATCHING recovers a positive cube-root-scale vertex-disjoint matching whose targets retain direct reflected tags with distinct reflected indices and quotient-return normal forms.

FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001**FAIL-ID:**

FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001.

Alias:

v4.50 R50-4.

Status:

KILLED.

Killed inference:

if a reflected seed has some parent in the failed extension, then the parent lies in the same seed basin and closes a directed cycle.

Reason:

source retention supplies a parent somewhere in the basin union; it may belong to a different seed basin.

What survives:

when a genuine same-basin return or actual cycle is proved, G-CRD-CYCLE-COFACTOR-LOCK applies.

FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001**FAIL-ID:**

FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001.

Alias:

v4.50 R50-6.

Status:

KILLED.

What is proved:

a source-heavy packet gives a large external source product / T_U -type lower defect.

What is missing:

an independent source-sensitive upper capacity.

Consequence:

full C-REFLECTED-FIRST-OCCURRENCE-CONVERSION remains open; only the return-heavy half is proved pending audit.

FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-DEEP-POWER-001**FAIL-ID:**

FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-DEEP-POWER-001.

Source:

v4.51 R51-3.

Status:

KILLED.

Killed inference:

use the alternate-path cofactor congruence to obtain extra divisibility from the common factor r^{E-1} .

Reason:

$r \nmid N$, so the relevant factor is invertible modulo N and cancels.

What survives:

G-ALTERNATE-PATH-COFACTOR-LOCK remains **PROVED**.

FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001

FAIL-ID:

FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001.

Source:

v4.51 R51-5.

Status:

KILLED.

Killed description:

the inability of aggregate path products to see E means ancestry localisation itself failed.

Correction:

ancestry localisation remains useful and v4.51 strengthens it through G-DEEP-ANCHOR-FIBRE-COUNTING-TRADEOFF and C-DEEP-LINKED-REFLECTED-ANCHOR-RECOVERY. What fails is only an overly coarse path-product consumer.

FAIL-V51-CYCLE-LOCK-FALSE-001

FAIL-ID:

FAIL-V51-CYCLE-LOCK-FALSE-001.

Source:

v4.51 R51-4.

Status:

KILLED as a description of the theorem.

Correction:

G-CRD-CYCLE-COFACTOR-LOCK remains **PROVED** and is subsumed as a corollary of the stronger G-ALTERNATE-PATH-COFACTOR-LOCK.

FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARRIES-DEEP-MASS-001

FAIL-ID:

FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARRIES-DEEP-MASS-001.

Source:

v4.51 R51-1 / scope correction to v4.50.

Status:

SCOPE-CORRECTED.

Killed proof-DAG edge:

full cube-root reflected matching from v4.50 $\implies$ its descendant closure carries the current common-deep ob

Reason:

the deep obstruction was extracted from the descendant closure of the v4.41/v4.42 high matching, while the v4.50 reflected matching was re-extracted from the original external seed packet. A large deep obstruction may be concentrated in basins omitted by that unweighted matching.

What survives:

the v4.50 reflected matching theorem itself remains **PROVED-PENDING-INDEPENDENT-AUDIT**; v4.51 repairs the transfer by selecting reflected anchors after the actual deep rows are known.

FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001

FAIL-ID:

FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001.

Date / iteration:

vNEXT4.9.

Claim:

A dyadic band produced at $Y \geq h > N^{1/3}$ automatically lies in the range of G-DYADIC-RECIPROCAL-REENTRY.

Status:

KILLED.

Exact certificate:

The current analytic theorem requires

$$2Y^{3+\vartheta} \leq \frac{N^{1/2}}{(\log N)^{C_0}},$$

so its upper scale is only roughly $N^{1/(6+2\vartheta)}$, modulo logarithms. The window between that scale and the cubic range is not covered.

Do not repeat:

Keep deterministic inverse-scale transport and analytic BV re-entry as separate arrows in the proof DAG.

FAIL-VERTICAL-FC8-SEPARATION-001**FAIL-ID:**

FAIL-VERTICAL-FC8-SEPARATION-001.

Date / iteration:

vNEXT4.5.

Claim:

The nonroot vertical factor U_R is an essentially independent store which the exact FC8 low-core ledger does not see.

Status:

SCOPE-CORRECTED / **KILLED**.

Exact certificate:

In the proved $3 \nmid N$ fringe domain,

$$D_C J_R U_R^{\text{nr}} = W_C^{\text{nr}}.$$

Thus denominator, horizontal reuse, and nonroot vertical layers are exact factors of one conserved low-core mass. The root-only term $U_0 = m/\text{rad}(m)$ is separate but has $\log U_0 < \log N$.

What survives:

The partition among D, J, U can still be the obstruction; what is killed is treating U^{nr} as outside the same functional.

Do not repeat:

Preserve the full C -part of each row before projecting to J and U .

Related current target:

C-MONOMIAL-REFLECTED-PAIR-SPARSITY.

FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001**FAIL-ID:**

FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001.

Date / iteration:

vNEXT4.11.

Claim:

High-base block-derived vertical mass has no independent upper capacity and can only be controlled through J_R , $K(R)$, or a γ -dependent ledger.

Status:**KILLED**.**Exact certificate:**

For every $Y > N^{1/3}$, exact depth two and the square-parent slot interval give

$$m_q \leq 1 + \frac{N}{6q^2}, \quad V_{\geq Y}^{\text{nr}} \ll \frac{N}{Y}.$$

In particular $V_{\text{blk}} \ll N/h$ on the block-derived high-gap branch.

What survives:

This is an upper capacity on the vertical channel. It is not by itself a Goldbach contradiction and does not prove that the whole reflected obligation must enter that channel.

FAIL-VERTICAL-SQRTLOG-WALL-001**FAIL-ID:**

FAIL-VERTICAL-SQRTLOG-WALL-001.

Date / iteration:

vNEXT4.12.

Claim:

Under the fixed-proportion vertical-heavy hypothesis, the best available confinement is $h \ll \sqrt{N/\log N}$.

Status:

SUPERSEDED within that conditional branch.

Replacement:

The hybrid linear sieve gives

$$h \ll_{\delta} \sqrt{\frac{N}{\log N \log \log N}}$$

when $V_{\text{blk}} \geq \delta t \log h$.

Scope guard:

vNEXT4.13 shows that this remains a conditional vertical-heavy subroutine; it is not an unconditional ceiling on the whole canonical branch.

FAIL-VERTICAL-TO-ETA-LOSES-LOG-001**FAIL-ID:**

FAIL-VERTICAL-TO-ETA-LOSES-LOG-001.

Date / iteration:

vNEXT4.9.

Claim:

Routing a linear vertical family only to $\eta(R) \gg h$ irretrievably loses the $\log h$ factor and therefore cannot preserve a $t \log h$ -scale obligation.

Status:**KILLED**.**Exact certificate:**

The distinct parent rows are forced through a much smaller low-base sibling support. Cauchy/pigeonhole concentration yields

$$K_{<h}^{(1)}(R) \gg h \log h$$

for a linear family of distinct vertical bases, and the weighted theorem gives

$$V_{\text{blk}} \gg t \log h \implies \log J_R \gg V_{\text{blk}} \vee K_{<h}^{(1)}(R) \gg t \log h.$$

What survives:

Large $K(R)$ is still not a contradiction; the missing ingredient is a source-sensitive consumer/upper capacity.

FAIL-VERTICAL-UNLOCALISED-001**FAIL-ID:**

FAIL-VERTICAL-UNLOCALISED-001.

Date / iteration:

vNEXT4.6.

Claim:

Vertical storage may occur anywhere among block-derived rooted low-core factors.

Status:

KILLED as a description.

Exact certificate:

If $c_j = sq$ and $N - w = bq$, then

$$v_q(N - w) \geq 2 \iff q \mid b.$$

Since $b < 2s$ and $q > N/(2s)$, this forces

$$s > \sqrt{N}/2.$$

What survives:

The true vertical zone is the large-quotient region; below it every block-derived rooted factor is D/J-only.

FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001**FAIL-ID:**

FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001.

Status:

KILLED as a failure certificate; it must not be used.

Incorrect claim:

the v4.33 uniform residual floor was allegedly unproved for $q > 4h$.

Repair:

the already proved G-FULLQ-PRODUCTION gives $M_q > (5/3)q$, hence $q > 4h$ gives $M_q > 20h/3$. For $q \leq 4h$, the shadow bound works after explicitly excluding $q = 2h$ by parity. Thus the full sub-root floor survives.

Do not repeat:

before recording a gap for a quantity, search the whole Core for an older bound on that same quantity, not only the newest report block.

FAIL-WEIGHTED-REUSE-AMPLIFICATION-001**FAIL-ID:**

FAIL-WEIGHTED-REUSE-AMPLIFICATION-001.

Status:

SCOPE-CORRECTED.

Generic witness:

for $N = 2200$, root 1471 and $R = \{3, 13, 1471\}$, the complements are $3^6, 13^3, 3^7$ and $\log J_R = \log 3$; the vertical channel carries nearly all prime-power mass. This refutes a near-unit $\gamma \log N$ lower bound on $\log J_R$ over the full generic FC7/FC8 failed-root domain.

Critical scope:

the root is not canonical and lies outside the near-square-root subdomain. The witness

does *not* refute a theorem in which independent canonical/high-gap information forces an incompatible lower bound; such a theorem would prove that its failed domain is empty.

FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTED-PROVENANCE-001

FAIL-ID:

FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTED-PROVENANCE-001.

Status:

KILLED.

Killed inference:

projecting the reflected block to a coarse set of external prime factors destroys the generating reflected indices so severely that no polynomial-weight tagged family can be recovered.

Certificate:

v4.39 assigns each prime to one reflected index, notes that one index carries $< \log N$ assigned weight, and recovers an injectively tagged subfamily preserving an $\alpha - o(1)$ fraction of the weight for a $N^{\alpha-o(1)}$ packet.

What survives:

the recovered packet may still lack an immediate second occurrence / upper consumer.

FAIL-WORLD-SIZE-NAIVE-001

FAIL-ID:

FAIL-WORLD-SIZE-NAIVE-001.

Date / iteration:

vNEXT4, Phase 1b.

Claim:

For a failed world R with composite complement, $|R| \leq \eta(R) + 1$, on the ground that every row satisfies $\omega(N - u) \geq 2$ and that $\sum_{u \in R} \omega(N - u) = \eta(R) + |R| - 1$.

Original motivation:

Every state of a failed world is BAD, so its complement is composite; a composite number “has at least two prime factors”.

Status:

KILLED as stated.

Minimal / strong counterexample:

$N = 2200$, root $r_* = 1471$,

$$R = \{3, 13, 1471\}, \quad N - 1471 = 729 = 3^6, \quad N - 3 = 2197 = 13^3, \quad N - 13 = 2187 = 3^7.$$

Here $\Gamma_N(R) = \{3, 13\} = R \setminus \{r_*\}$, $\kappa_R(3) = 2$, $\kappa_R(13) = 1$, so $\eta(R) = 1$ while $|R| = 3 > \eta(R) + 1 = 2$.

Failed hypothesis or mechanism:

A composite number has at least two prime factors *counted with multiplicity*; it need not have two *distinct* ones. Every row of this world is a prime power, so $\omega = 1$ on all three.

Why the argument failed:

κ is layer-one incidence and therefore sees ω , not Ω . The step that requires $\omega \geq 2$ is the only place where badness was used, and badness gives $\Omega \geq 2$.

What survives:

The corrected count $|R| \leq \eta(R) + 1 + 2a_0 + a_1$ with $a_0 \leq 1$ and $a_1 \leq \gamma(\lfloor \log_2 N \rfloor - 1)$, hence $|R| < 4\gamma \log N$ under the (FC9) hypotheses: Theory Core, G-CANONICAL-WORLD-SIZE-CAP, (WS1)–(WS3). Zero violations over the recorded 301 worlds.

Safe weaker version:

The corrected form above; nothing weaker is needed.

Do not repeat:

Whenever a count over the rows of a failed world uses $\omega(N - u) \geq 2$, the prime-power rows must be excluded explicitly. Recorded as loss-control rule (LR6).

Related current target:P1BR-ADDITIVE-COUPPLING, γ .**FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001****FAIL-ID:**

FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001.

Status:**SUPERSEDED**.**Superseded description:**the useful unique-parent target floor is only $q > n/(\log N)^2 = N^{1/3-o(1)}$.**Certificate:**the total log mass of all primes $q \leq N^{1/3}$ is $O(N^{1/3}) = o(n \log N)$ on $n \gg N^{1/3}$. v4.41 therefore moves essentially all unique-parent mass to the strict floor $q > N^{1/3}$.**What survives:**

the v4.40 matching is valid but its constants/floor are superseded by the v4.41 cube-root matching.

FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001**FAIL-ID:**

FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001.

Status:**SCOPE-CORRECTED**.**Scope correction:**v4.41 correctly proves/uses the interval mechanism, but the statement that every monomial component has at most two edges was already **PROVED** in the Theory Core as C-MONOMIAL-TWO-EDGE-RIGIDITY.**New part retained:**v4.42 records the stronger fact that for fixed N at most one monomial component can contain two edges.**FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001****FAIL-ID:**

FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001.

Status:**KILLED**.**Killed inference:**fixing the integer cofactor c leaves no useful deterministic capacity on the owned high row-slot packet.**Replacement:**fixed c forces

$$2QFc \mid p_i - p_j,$$

and summing over $c \leq C$ gives the v4.57 harmonic row-slot capacity.**FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-C-001****FAIL-ID:**

FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-C-001.

Status:**KILLED**.**Killed inference:**summing fixed- c Selberg bounds over a polynomial cofactor range necessarily creates an $O(C)$ loss comparable to the whole packet.

Replacement:

choose

$$C_\varepsilon = MN^{-1/(\log \log N)^\varepsilon} = o(M).$$

Then the total $O(1)$ -per- c remainder is $o(M)$, while discarded cofactors still carry full logarithmic mass.

FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-ROUTING-SINK-001**FAIL-ID:**

FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-ROUTING-SINK-001.

Status:

KILLED as a routing diagnosis.

Reason:

exact target-row entry accounting shows that cheap oscillation can survive only by consuming fresh target labels. Repeated targets are paid; the unresolved issue is population capacity, not missing routing bookkeeping.

FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001**FAIL-ID:**

FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001.

Status:

KILLED.

Killed inference:

a renewed polynomial owned support population obtained from cofactor factorisation is already a contradiction.

Reason:

the standing renewal-is-not-divergence restriction remains binding; there is no proved global upper capacity for arbitrary fresh external population.

FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-ITSELF-A-CONTRADICTION-001**FAIL-ID:**

FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-ITSELF-A-CONTRADICTION-001.

Status:

KILLED.

Killed inference:

a polynomial family of distinct external high target rows, each used once, is already an upper-capacity contradiction.

Reason:

this is the standing “renewal is not divergence” restriction in the present deep-owned domain. A fresh external population is an obligation until a CRD-specific capacity theorem consumes it.

FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-THAN-SPACING-001**FAIL-ID:**

FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-THAN-SPACING-001.

Status:

KILLED.

Killed inference:

after v4.57 fixed- c spacing, primality of both endpoints cannot yield further capacity.

Replacement:

after parity parametrisation the endpoint primes are two explicit non-proportional affine forms with determinant $-2FN$, so the already audited classical dimension-two Selberg upper-bound

sieve gives logarithmic-square compression.

FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001

FAIL-ID:

FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001.

Status:

KILLED.

Killed inference:

the large-base hypothesis $r > Y$ by itself forces monotone r -adic dynamics.

Certificate:

the explicit v4.55 high-base diagnostic witness with nonmonotone level pattern. It does not satisfy the full deep exponent hypothesis and therefore does not rule out stronger theorems using deep/reflected provenance.

FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001

FAIL-ID:

FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001.

Status:

KILLED.

Killed inference:

large c_i may absorb the row-slot obligation without producing a trackable CRD resource.

Replacement:

large cofactors carry explicit logarithmic mass with exact support/horizontal/vertical factorisation; this yields genuine residual J/U or owned fresh-support renewal.

FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001

FAIL-ID:

FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001.

Status:

KILLED.

Killed inference:

the v4.58 full-scale large-cofactor renewal branch alone gives a global contradiction.

Reason:

renewed support is a real relation-preserving CRD obligation, but no proved capacity theorem yet bounds its total fresh external population.

FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001

FAIL-ID:

FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001.

Status:

KILLED.

Killed inference:

the v4.57 product-modulus upper bound already contradicts the deep lower bound on $Q = r^e$.

Reason:

the deep hypothesis permits $Q = N^{o(1)}$, so even a bound of scale $QF \leq N^{1-\alpha+o(1)}$ need not contradict the branch.

FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-DESCENT-001

FAIL-ID:

FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-DESCENT-001.

Status:**SCOPE-CORRECTED**.**Overstrong inference:**

organising threshold classes at $e, \lceil e/2 \rceil, \lceil e/4 \rceil, \dots$ automatically produces strict forward descent at every owned transition.

Correction:

threshold layers may remain useful organisationally, but an edge may jump upward to a larger exact valuation. Any finite-return argument must use an independent payment/capacity statement.

FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001**FAIL-ID:**

FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001.

Status:**KILLED**.**Killed inference:**

for an owned transition $p \rightarrow s$, the quantity $v_r(N - p)$ must decrease along the forward dynamics.

Certificate:

the exact v4.55 three-cycle template

$$N = r^{a+b+c} + 1, \quad N - p_0 = r^a p_1, \quad N - p_1 = r^b p_2, \quad N - p_2 = r^c p_0,$$

which realises arbitrary cyclic valuation pattern $a \rightarrow b \rightarrow c \rightarrow a$ whenever the states are active primes.

Safe replacement:

retain the exact adjacent-row transition identities and price actual repeated target rows rather than valuation-level recurrence.

FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001**FAIL-ID:**

FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001.

Status:**KILLED**.**Killed inference:**

a return to a previously used value of $v_r(N - s)$ must create horizontal reuse.

Certificate:

horizontal reuse is indexed by target prime labels; the v4.55 cycle realises level recurrence through distinct target rows.

Safe replacement:

after the first copy at one target row, every later occurrence at that same target row is fully priced by incremental J .

FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTART-001**FAIL-ID:**

FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTART-001.

Status:**KILLED** as a packet-design inference.**Killed inference:**

an owned edge landing on another high row with the same exact valuation e should be treated as a new external restart branch.

Replacement:

saturate the exact level in the stopped reflected-seed basin first; then every high exact- e target already belongs to the saturated layer.

FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001**FAIL-ID:**

FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001.

Status:

KILLED .

Killed inference:

the v4.58 log-saved bound

$$QF \ll_{\varepsilon} \frac{N(\log \log N)^{1+\varepsilon}}{M \log N}$$

is already incompatible with the current deep modulus.

Reason:

Q may still be only $N^{o(1)}$; the logarithmic saving is real but not a contradiction.

Chapter 13

Research Agenda

13.1 How to choose what to work on

Three rules for triage, in order of importance.

1. **Check the wall first.** If the proposed argument’s only canonical input is a finite transcript — a bounded number of shifts, congruences, valuations, divisibility certificates, bounded-depth rows, or neighbouring roots — it cannot work (Corollary 8.3). Stop before writing anything.
2. **Check the functional.** If the argument proves a lower bound in one functional and consumes it in another, it is the $P1^{br}/K1$ error. Write down both functionals explicitly *before* starting.
3. **Check the scope class.** If the argument uses a class-B lemma to prove a class-A statement, it proves nothing about p_0 . If it uses a class-D hypothesis under canonical failure, it is invalid.

13.1.1 Tiered targets

Tier	Target	Why it is in this tier
1	Prove or refute (N3): $\sum_q d_R(q) \geq R + 1$ (§5.7, Q8)	Small, self-contained, elementary in appearance, and it repairs the part of the corpus that used the false Euler normalisations. Zero counterexamples in 21 093 worlds. A model with a few hours should attempt this first.
1	Refute L1 by exhibiting $\nu(R) = O(1)$ with J_R unbounded (§9.5)	A <i>negative</i> result that would remove a whole family of capacity arguments. The data already point this way. Cheap to test computationally.
1	Extend the B_N^{low} census beyond $6 \cdot 10^4$	The coincidence of the three six-element lists (§10.5) is unexplained. A seventh N would be the single most informative new datum in the project.
2	TS: smooth branch to escape (§9.3)	The branch with the largest quantitative margin and the most elementary statement. Success would give Goldbach in a specific regime, which is a real theorem.
2	E1: escape-or-reuse, or the bounded-charging lemma (§9.6)	Overwhelming data, explicit statement, and the missing ingredient is named (bounded charging multiplicity). The most likely target to actually fall.
2	Invariants of the boundary filtration (Q7)	Purely combinatorial, independent of Goldbach, publishable on its own terms if anything nontrivial exists.

Tier	Target	Why it is in this tier
3	TD / C-SAMEFUNC-BRIDGE (§9.3)	The real bridge. Hard, and previous attempts all failed the same way. Attempt only with the same-functional constraint written down first.
3	RPD (§9.10)	The only open target carrying an explicit quantitative margin. Its original phrasing should be re-checked against the source before use.
4	x1 / (H- p_0)	Strictly stronger than Goldbach. Do not attack directly.
4	Anything requiring level of distribution $> 2/3$ or a binary prime–Liouville correlation	These are the external walls (§11.4). Working on them is working on a different, famous problem.

13.1.2 Experiments that would change the picture

1. B_N^{low} **census to** 10^6 . Currently $6 \cdot 10^4$. Cost: roughly quadratic in the limit as implemented; a smarter implementation should use the fact that B_0 can be built once per N from a shared sieve. A seventh nonempty B_N^{low} , or its absence at 10^6 , is decisive information about whether the six are an artefact of smallness.
2. **Full upper-root census beyond** $2 \cdot 10^4$. The 301 number is the project’s most-cited statistic and it is known only to $2 \cdot 10^4$. Extending it to 10^5 would test whether “failing roots cluster at $B_N^{\text{low}} \neq \emptyset$ numbers” persists.
3. J_R **versus** $\nu(R)$ **scatter**. Directly tests L1 and is a few lines on top of the existing world routine.
4. G_{esc} **to** 10^6 . Tests Q2 (does the margin grow?). The current dyadic minima 1, 1, 1, 1, 1, 2, 2, 3, 3, 3 are suggestive but the range is small.
5. **Phase-conditioned traversal statistics**. Does anything about the traversal, beyond immediate success, depend on θ ? Currently unknown, and ρ_0 is the only boundary invariant that varies between canonical worlds (Q1).

Chapter 14

Literature and Source Registry

14.1 Internal sources

Tag	Source	Role
[BIB]	the Goldbach Biblia (v7.5 and its post-v7.5 technical log)	Shadow, retention, path/winding invariants, prime-power witness, escape capacity, the 10^6 scan tables, the ten margin-one numbers.
[CORE]	CRD Core	Canonical geometry, C-GAP-EQUIVALENCE, C-STRICT-NONREUSE, the reflection gate, C0a, C0b-RD, the genealogical layer (V71.21–V71.44), the open-target registry, P0-CANONICITY-ABLATION.
[PCRD]	GOLDBACH_POST_CRD_RESEARCH_TALENTS	THIS FILE ALONE MAKES PRECEDENCE ON CONFLICTS. B_N^{low} fixed point and characterisation, the six N to $6 \cdot 10^4$, the successor-theory walls, finite affine canonical programmability, finite bad-tree universality, the canonical prime cell theorem, the H1 descent and the 2-core wall, the final verdict, its own theorem registry and 30-row log.
[FAIL]	the project failure log	All 239 live entries, reproduced in Chapter 12.
[LOG]	post-v7.5 technical log	D_q , M_q , the prime-power ledger, primitive residual normal form.
[ILA]	incidence-ledger audit	IL1–IL6, two reconstruction routes, 1 000 worlds.
[ORPH], [RT]	orphan census and red team	Escape-or-reuse data: 61 452 and 60 413 instances, 0 orphans.
[D4], [F3], [FRO], [FND], [V481], [NOV]	question-answer campaigns and the novelty note	Statistical ablations, structural twins, the rooted container identity, the prior-art boundary. Several of their numbers are <i>reported, not reproduced here</i> and are marked as such wherever they appear.

Warning 14.1 (Precedence). Where sources conflict, [PCRD] takes precedence, then [CORE], then [BIB]. Where this file’s own computation conflicts with any of them, this file states both and marks the discrepancy — it does not silently overwrite the source. The recorded discrepancies are: the Euler-defect normalisations (§5.7), the diagonal scope of the genealogical layer (Scope note 3.17), the lower bound in the ρ_0 formula (Warning 6.18), and the hypothesis of high-gap amplification (Warning 7.5).

14.2 Prior art and the novelty boundary

Warning 14.2. No claim of novelty is made anywhere in this file. Establishing novelty would require a separate literature review that has not been done. Any label of the form “derived here” means *derived from the corpus’s own definitions in this document*, not “new in number theory”.

The bibliography is a search record, not a citation list. It reproduces the Biblia’s full list of 92 sources plus three later additions. Many entries are *not* cited anywhere in the running text: they are there so that a future model can see what has already been looked at and does not repeat the search. An entry’s presence means “this was examined”; the disposition tables of §14.3–§14.3.3 say with what result. An entry’s absence is the useful signal — it means the project has probably not looked there.

14.2.1 What must *not* be attributed to this project

The relation $p \rightarrow q$ for $q \mid N - p$; the notion of a Goldbach factorization graph; recursive search through complement factors; valuation weights $v_q(N - p)$; reachability, source and sink components, autonomous closed components; the study of Goldbach pairs near $N/2$; elementary parent residue classes and gcd identities. [NOV]

Topic	Source	Boundary
Factorization graph and exceptional autonomous components	Božek [8]	The graph and the terminal pathology are not ours. Orientation matters: a source component in the paper’s orientation corresponds to terminal behaviour after reversal (Warning 3.23).
Recursive factor search from a given root	OEIS A306746	Search from an arbitrary root is prior art; the root condition there is <i>not</i> the unconditional least prime $\geq N/2$.
First upper prime already forming a pair	OEIS A078496	$a(n) = \min\{p > n : p, 2n - p \in \mathbb{P}\}$ — a selector <i>conditioned on the outcome</i> . This is not p_0 .
Nearest symmetric Goldbach pair	OEIS A082467	Goldbach near the midpoint is prior art.
Short intervals containing a prime	Baker–Harman–Pintz [13]	Calibrates $h \ll x^{0.525}$; it is not an escape lemma and is not used as one.
Verification of Goldbach	Oliveira e Silva et al. [12]	Says nothing about (H- p_0). Must never be cited as this project’s own computation.
Linear forms in logarithms	Matveev [79]	Astronomical constants; asymptotic content only. Used only for the 2-core wall.
Almost-prime representations $(1 + 1.9)$	Li–Liu [15]	External input for CAPT; does not give the last bit P_1 versus P_2 .
Exponent of distribution $5/8$	Pascadi [93]	External input for RSD; below the $2/3$ that RSD needs.
Chen-type lower bound	R. Li [94]	$D_{1,2}(N) \geq 1.9728 C(N)N/\log^2 N$.
Asymptotic sieve for primes	Friedlander–Iwaniec [38]	The template for the missing bilinear information.

Topic	Source	Boundary
Goldbach problem for the Liouville function	Mangerel [95]	Calibrates the centred-correlation wall.

14.2.2 The safe formulation

Building on earlier work on Goldbach factorization graphs and on recursive search through complement factors, we study the *unconditional* selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$ as a distinguished initial condition of the full dynamics. To the best of our knowledge we have not found earlier work using this selector — without requiring $N - p_0$ to be prime — as a distinguished root, and systematically comparing its reachable bad world with those of non-canonical roots. *This is a report on the extent of a literature search, not a claim of priority.*

14.3 Register of rejected literature applications

This section exists so that a future model does not re-import a tool the project has already tried and discarded. Each row records a specific *application* of a specific external result, and why it does not do the job here. A **KILLED** row means the application is dead, *not* that the cited theorem is wrong. An **OPEN** row means the tool may still be usable once a stated precondition is met.

14.3.1 First-front applications (Biblia v7.4)

Claim / tool	Status	Reason
$ Q \sim N/\log N$	KILLED	$Q = \text{PF}(m_0)$, so $ Q = \omega(m_0)$; a global density comparison does not apply to the first front.
SMOOTH $\Rightarrow$ huge powerful part	KILLED	A SMOOTH row can be squarefree with many distinct small prime factors (Warning 7.13).
Positive density of squarefree values $\Rightarrow$ positive fraction of Q	KILLED	A result for all arguments, or all primes, does not transfer to the <i>adaptive</i> set of prime divisors of a single m_0 .
Bombieri–Vinogradov on Q	KILLED	The theorem controls primes in arithmetic progressions <i>on average over moduli</i> , not the distribution of the prime divisors of one specific m_0 .
Filaseta–Trifonov as a uniform lower gap between consecutive powerful numbers	KILLED	The short-interval squarefull-counting literature does not supply the universal lower gap barrier that was assumed.
Darmon–Granville $\Rightarrow$ uniform bound for a varying family	KILLED	Coefficients, the difference and the curve itself vary with N ; finiteness for a fixed curve gives no uniformity.
Fixed-polynomial radical theorem with $F(X) = \prod_{q \in Q} (X - q)$	NEEDS VERIFICATION	Degree, height and the whole polynomial vary with N ; an explicit uniform dependence is required.
Pairwise classical <i>abc</i> closes SMOOTH	KILLED	The shift $ q - r $ can be of order N and consumes the radical; without extra global structure the required first-front lower bound does not arise.
Corvaja–Zannier greatest-prime-factor result	OPEN	Qualitatively supports escape, but gives no <i>moving</i> threshold $P^+(N - q) \geq m_0/q$.

Claim / tool	Status	Reason
Evertse–Schlickewei–Schmidt [18]	OPEN	Usable only after freezing a small alphabet/rank inside one counterexample. See also the vacuity result recorded in §9.2.4.
Gallagher’s larger sieve [41]	OPEN	Requires first showing that a large part of Q occupies few classes modulo many primes.
Multiplicity-order literature	OPEN	Known generic / almost-all results give no deterministic lower side for specific ports modulo a specific N .

14.3.2 Iteration-12 audit of 52 primary sources

An honest protocol instead of a fictitious “1000”. The scope and metadata of **52 distinct primary sources** were checked; for 18 of them the main theorems and the proof passages directly relevant to the project were read as well. This is *not* called “reading 1000 papers”: automatically fetching a thousand bibliographic records is not reading and does not raise mathematical credibility. Below, mode T means “theorems and relevant sections”, mode Z means “scope audit from abstract, introduction and statements of results”.

Sources	Mode	Conclusion for the project
Bhowmik–Grimmelt [36]; Li–Liu [15]	T	Goldbach still open. The first localises the barrier in minor arcs and exceptional sets; the second gives $(1 + 1.9)$, not an adaptive traversal path. Contributed calibration of the main gap.
Lichtman [26]; Ford [30]	T	Strong distribution and the anatomy of shifted primes work <i>on average</i> . They do not give a pointwise choice from a reachable parent set.
Grimmelt–Teräväinen [45]; Halupczok [46]	Z	Transference and Goldbach in progressions / short intervals still leave an exceptional set or averaged conditions. Useful as a weighting template, insufficient pointwise.
Friedlander–Iwaniec [38]; Polymath8 [39]	T	The most important analytic input: breaking parity needs Type-II bilinear information, not a linear sieve alone.
Green–Tao [40]; Bienvenu–Shao–Teräväinen [44]	T/Z	Transference needs a pseudorandom majorant and positive relative density for a <i>fixed</i> configuration. Those hypotheses are not available for adaptive reachability.
Ford [31, 47]; Assing–Blomer–Li [28]	T	Pratt-type closures and uniform Titchmarsh problems are the closest structural analogues, but the relation $q \mid N - p$ moves with N .
Matomäki–Radziwiłł [48]	T/Z	“Almost all” short intervals give no certificate for every adaptive set.
Helfgott [16, 17]; Oliveira e Silva et al. [12]; Huang–Li [25]; Li [14]	T/Z	Circle method, finite verification, EH and short intervals are important calibration, but none synchronises a prime with the basin of p_0 .
Evertse–Schlickewei–Schmidt [18]	T/Z	The number of S -unit solutions may depend only on dimension and rank, which is better than fixed- S constants. Application needs genuinely small rank; the known zero-wrap branch is empty.
Vaaler [49]	T/Z	Heights may give a short multiplicative dependence <i>after</i> an independent rank and height bound. They do not handle the present δ -forcing automatically.
Amoroso–Sombra–Zannier [50]	Z	Sparse-polynomial / unlikely-intersection results work for fixed families; Smith normal form is an excellent lattice algorithm but supplies no arithmetic contradiction by itself.

Sources	Mode	Conclusion for the project
Bugeaud [20]; Siksek–Stoll [21]; Bennett–Siksek [19, 5]	Z	Effective once a specific curve or support is frozen; no uniformity for a moving full SCC.
Bellman [51]; Karp [52]	T	The variational calculus of paths reduces to dynamic programming and a cycle criterion. Gave the cycle–potential duality, not prime selection.
Prajna–Jadbabaie [53]	Z	Barrier certificates and occupation measures are the right shape for reachability; on an SCC they need an arithmetic inequality that is missing.
Angeli–Philippe– Athanasopoulos–Jungers [54]	T/Z	A complete Lyapunov function is constant on recurrent classes, which justifies SCC compression. Multi-part versions do not evade the cycle barrier.
Friedman–Tillich [55]; Jiang– Lim–Yao–Ye [56]; De Leenheer [57]; Dodziuk [7]	T/Z	Calculus and Hodge theory on graphs formalise the gradient and the cyclic part. They contributed an exact no-go, not a primality theorem.
Bass [58]; Rangarajan [59]	Z	The graph zeta function and $\text{coker}(I - A)$ repackaged determinant/Smith data. No new information about the rows $N - p$.
Z-transform / Koopman resolvent literature	Z	Appropriate for a <i>fixed</i> operator; the adaptive graph is not an LTI system.
Discrete variational calculus lit- erature	Z	Gives stationarity equations, but for a sum of path costs it is not stronger than Bellman.

14.3.3 Frontier salvage audit (v4.81)

The v4.81 attack performed a targeted salvage audit of the older Biblia against the *current* frontier. Its verdict in one sentence: *the old corpus contributed exactly one genuinely missing tool — the gap-scale shadow identity — and one correct **shape** for the replacement frontier (the smoothness budget); everything else in the neighbourhood of the current obstruction is either already in the Core, or vacuous on the primitive branch, or domain-mismatched.*

Salvaged and used.

Old item	Restated form	Status and use
Prime-Gap Shadow Identity (Theorem 5.22)	$N = 2m + \delta, \delta = 2h; \text{ for } r \mid m:$ $r \mid N - y \iff r \mid \delta - y$	PROVED , re-proved from v4.80 definitions. Used to obtain the gap-scale form of primitive load-one (V81.11). <i>Not previously used in the v4.79/v4.80 chain.</i>
Retention–Escape (The- orem 5.24)	$\text{PF}(N - q) \cap Q = \text{PF}(d_q), d_q =$ $\text{gcd}(m, \delta - q)$	SCOPE-CORRECTED . Identical to $D_q = \text{gcd}(m, N - q)$, giving $D_q = \text{gcd}(m, 2h - q)$. The old “escape” reading (“all children are new”) must <i>not</i> be upgraded to success.
Gap/first-front disjoint- ness	$\text{gcd}(h, m) = 1, \text{PF}(h) \cap Q = \emptyset$	PROVED , re-proved; used for the block/ Q separation.
Multi-source product lock (Theorem 5.7)	$T_S \equiv (-1)^{ S } \prod_{\text{src}} d \pmod{N}$	Already in the Core; the <i>new</i> use is the selected-family instance (V81.13) and the residual instance $T_{R'} \equiv (-1)^{ R' } \Sigma'$.

Old item	Restated form	Status and use
Two-alphabet retention	$\gcd(N - q, mh) = \gcd(N - q, m) \gcd(N - q, h);$ $\gcd(E_q, E_r) \mid q - r$	PROVED (the split). On the primitive branch the two-alphabet retention of every first-front row collapses to the h -alphabet.
Smoothness budget (every row of a child-closed world is C -smooth)	The count of available C -smooth numbers in the row window is a necessary condition	SCOPE-CORRECTED to EMPIRICAL PATTERN : the old text carries it under an equidistribution assumption. Adopted here <i>only as the shape</i> of the new preferred frontier (§9.2.4). The equidistribution assumption is not imported.

Interesting but insufficient.

Old item	Content	Why insufficient at the frontier
Prime-power witness (Prop. 5.29)	$\text{PF}(N - q) \subseteq Q \Rightarrow \exists r \in Q : v_r(m) = v_r(\delta - q)$	On the primitive branch the hypothesis is never satisfied ($\text{PF}(N - q) \cap Q = \emptyset$): vacuous domain exactly where it is needed.
Shadow two-cycle (Prop. 5.28), $\delta = q + r + tqr$	Exact form of a Q -shadow two-cycle	The primitive branch makes the Q -shadow graph <i>empty</i> ; no cycle exists to exploit. Vacuous domain.
Quotient-rigidity master lemma	$ u - v < 1.1u/m^2,$ $\min(u, v) > 0.9m^2$ for row quotients	Proved inside a terminal-SCC band normalisation; its row-window and row-gap hypotheses are unavailable for primitive first-front rows. Domain mismatch.
One-escape residue lock	$g \equiv (-1)^{ W } p_0(U_W J_W)^{-1} \pmod{N}$	Requires a unique good child; in a failed world there is none. The useful residue content is exactly G-MULTISOURCE-LOCK . Retained as motivation for the secondary frontier.
C -smooth budget calibration on the six EAC	Empirical ratios 0.62–0.79	VERIFIED COMPUTATION only; the six EAC are terminal-SCC objects, <i>not</i> canonical failed worlds. Not usable as evidence for the canonical branch.

Rejected or already superseded.

Old item	Disposition
“Terminal bad SCC $\Rightarrow$ Goldbach contradiction” programme	Not revived. A terminal bad SCC is not by itself a Goldbach counterexample (Scope note 3.27). No lemma of that chain is used.
Exponent–determinant collapses	Domain is the two- or three-element terminal core with fixed Ω ; the primitive first front has neither. SUPERSEDED by the adaptive low-core theory.

Old item	Disposition
Fringe alphabet $\{\lambda, 2\lambda - 1\}$	FAIL-FRIDGE-ALPHABET-TRANSFER-001. Theorem 9.17 is <i>not</i> that claim: it derives the alphabet as the full prime range $\mathcal{L}_N \subseteq [\lambda, 2\lambda)$, which is precisely the “what survives” clause of that failure entry.
“Prime-free boundary activates a reflected parent”	FAIL-REFLECTION-ACTIVATION-001 ($N = 242$, $p_0 = 127$) is respected throughout; Theorem 9.10 is its sharpened form.
Evertse S -unit counting as a closure device	Vacuous, as the Biblia itself noted; recorded as FAIL-SUNIT-COUNT-BOUNDS-THE-LOWCORE-001.

14.4 Literature verdicts and named missing lemmas

Iteration-157 verdict. Checked: the Euclidean algorithm for positive 2×2 matrices, the free monoid of positive SL_2 , Conway–Coxeter friezes [86, 87], unit equations [88], graph toric ideals [81, 82], higher Lawrence configurations [89], cyclic congruence systems [90], and prime-escape theorems from subset products [91]. *None* supplies a dynamical transition preserving simultaneously N , the row identities, the full valuations and the self-label roles. The nearest missing theorem is named:

Matched Euclidean Gluing Lemma (**OPEN**). To the word or matrix of one packet one must assign an *actual* continuation to the next packet in the same system, preserving all roles, in such a way that a Euclidean measure strictly decreases or an escape is forced.

Without that lemma, word length, continued-fraction depth and matrix entry sums are only *static* coordinates. **They must not be used as Lyapunov functions for self-label dynamics.** Compare Remark 7.45, where the same continued-fraction structure appears exactly and still yields no descent.

The one external input whose shape is known. Across every audit the same conclusion recurs: what is missing is *Type-II bilinear information* in the sense of the asymptotic sieve for primes [38, 39]. Every internal construction that appeared to supply it turned out on inspection to be a Buchstab/sieve decomposition or an affine reparametrisation (§11.3.1). A future model should treat “does this really produce bilinear information, or is it an affine relabelling?” as the first question to ask of any new proposal.

Chapter 15

Historical and Superseded Results

Kept because the *history* matters: each item below was once load-bearing, and knowing why it stopped being load-bearing is what prevents its reconstruction under a new name. Nothing in this chapter may be used as a lemma. Where a superseded statement contains a correct sub-lemma, the sub-lemma appears in Chapter 4 under its own ID; the programme does not come back with it.

Item	Status	History and what replaced it
CRD1 as a proof engine	SUPERSEDED	Replaced by the fixed-point reformulation (Theorem 3.29), which is exact but is a computation, not a mechanism. What died: the “same container” argument, generic smooth counting, residue saturation, and the exact self-consistency of the fixed point.
TD-DENSITY	SUPERSEDED	The density formulation of TD required a bound of the same strength as its own conclusion. Replaced by the root-anchored form (§9.3).
p_0 as information selector	SUPERSEDED	Not refuted, but not established: every clause of the bundle-lift construction survives the ablation $p_0 \rightarrow p_1$ on rows with composite c'_j . The corpus verdict is “quantitatively useful, structurally inessential”. FAIL-P0-INFORMATION-SELECTOR-001
Euler defect in normalisations (N1), (N2)	KILLED	Refuted here computationally (§5.7). Any earlier argument using $\eta + \nu \geq S + \text{Src}(S) $ in those normalisations must be re-derived. The surviving forms are (N3) (VERIFIED COMPUTATION) and (N4) (trivial).
Prefix flux $B_k \equiv (-1)^k p_0$	AUDITED IDENTITY	Recorded in [BIB] (iterations 20, 22) with the flag “not auditable from packet”. It has never been used downstream and is kept only so that it is not rediscovered and trusted.
Rooted container identity	AUDITED IDENTITY	[1481]. Closed the size-based route — i.e. its role was to <i>end</i> a line of attack, not to open one.
Four-channel ledger on all of Q_0	KILLED	Replaced by IL1–IL6 on Q_{smooth} after $N = 20$ refuted the wider domain. [ILA]
Unconditional CAB	KILLED	$N = 322$ and $N = 858$. No repaired version exists; the capacity route now goes through K1, which is open.

Item	Status	History and what replaced it
Upper-half source compression	PROVED (class D)	Correct, but its hypothesis is $\text{Good}_N = \emptyset$. It was repeatedly invoked under canonical failure, which is invalid. FAIL-UPPERHALF-SCOPE-001
Symmetric fracture, forbidden offset band	PROVED (class D)	Same situation: correct under the minimal-counterexample hypothesis, unusable otherwise.
$h = 1$ local closure	KILLED	Finite bad-tree universality. Its successor, the elementary 2-core closure, is OPEN but reduces to Pillai/Bennett $a^\alpha - a = b^\beta - b$, i.e. to an external Diophantine wall. [PCRD]
CMFT, CGST, CSD, CAPT, RSD, rough- P_2	KILLED	§11.2. Each was an attempt to preserve information lost by its predecessor; each reached the finite-local wall or the parity barrier in a new coordinate system.
$q < m_0/3$	SCOPE-CORRECTED	Corrected to $q \leq m_0/3$; $N = 32$ attains equality. The consequence $q < N/6$ is unaffected.
$\rho_0 = \#\{D_s < 2h\}$	SCOPE-CORRECTED	Corrected here by adding the lower condition $D_s > h$ (Warning 6.18).
Genealogy stated for “every prime $r \geq x$ ”	SCOPE-CORRECTED	Corrected here to the non-diagonal domain; smallest witness $N = 14$ (Scope note 3.17).
High-gap amplification $h^2 > 4m_0$	SCOPE-CORRECTED	Hypothesis is Exact-D pair failure, not CRD failure (Warning 7.5).
COB-RD v1 (without $Q_\leq \neq \emptyset$)	PROVED WITH REPAIR	The missing hypothesis made the inequality read $1 < 1$ in 47 551 of 72 446 canonical instances with $h \geq 3$ and $N < 2 \cdot 10^5$. Repaired version is Theorem 8.6 (Scope note 8.7).

Chapter 16

Handoff to the Next Model

If you read only one chapter, read this one — then Chapter 2, then Chapter 9.

16.1 What not to try again

1. **Do not build a finite local certificate.** Any argument whose only canonical input is a bounded transcript — finitely many shifts, congruences, valuations, divisibility or compositeness certificates, bounded-depth rows, or neighbouring roots — is dead on arrival. Every finite transcript is realisable in infinitely many canonical worlds with $h = 1$ (Theorems 8.1, 8.2, Corollary 8.3). This kills CMFT, CGST, CSD, CAPT, RSD and the rough- P_2 matrix, and it will kill their successors.
2. **Do not restart the programme in new coordinates.** Five successor theories have already done exactly that (§11.2). A new coordinate system does not change what information is available.
3. **Do not prove a bound in one functional and use it in another.** This is the single defect blocking TD, $P1^{\text{br}}$ and K1 simultaneously. “Morally the same” and “dominates on average” are not the same functional.
4. **Do not treat canonical failure as Goldbach failure.** Class-D theorems (upper-half compression, symmetric fracture, forbidden offset band) require $\text{Good}_N = \emptyset$ and are unusable under canonical failure alone. FAIL-UPPERHALF-SCOPE-001
5. **Do not quote a count without its domain.** Which roots, which complements, unit rows included or not. FAIL-CENSUS-DOMAIN-001, FAIL-UNIT-COMPLEMENT-001
6. **Do not promote a scan.** 0 failures to 10^7 is not evidence for X1; it is the statement that X1 has no small counterexample.
7. **Do not use the mass of the boundary holes.** $O_k(\log N)$ against a block mass of order $t_k \log(x/h_k)$. Any argument routed through hole mass is dead before it starts. FAIL-P0-ZEROHOLE-ESSENTIAL-001
8. **Do not strengthen the reflection gate to activation.** $N = 242$ is final: both lower mirrors are composite and no active reflected row exists. The phase diagram explains why (Type I). FAIL-REFLECTION-ACTIVATION-001
9. **Do not widen the reflected block.** The endpoint pair has difference exactly $2h$ and is not covered by strict non-reuse. FAIL-BROADBLOCK-ENDPOINT
10. **Do not use the Euler defect in normalisation (N1) or (N2).** Refuted here: 6 044 and 3 804 violations respectively for $N \leq 6 \cdot 10^4$ (§5.7).
11. **Do not attack X1 or Goldbach directly.** X1 is strictly stronger than Goldbach for each N .
12. **Do not route around parity by enlarging the graph.** The obstruction lives in the sieve information, not in the graph (§11.4).
13. **Do not re-prove what is already PROVED.** Use it by ID from Chapter 4.
14. **Do not raise a status.** If you prove something, add a row; do not edit an old one.

16.2 What currently looks best

1. **C-PO-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPACITY** — the actual current frontier (§9.2.4). It is the *only* archimedean route the trichotomy leaves open, the high band is completely described, and a bound on $|C_\lambda|$ converts every existing product capacity into a finite quantity. It is also honestly hard: an S -smooth counting problem in a short interval with sparse, non-initial support, where $\Psi(x, y)$ technology does not apply and Evertse’s bound is vacuous.
2. **E1 (escape-or-reuse)**. The strongest data in the project (61 452 instances and an independent 60 413, zero orphans), a fully explicit statement, and a *named* missing ingredient: a bounded-charging lemma. If any frontier target falls, this is the one.
3. **The active-mass Euler bound (N3)**. Small, elementary in appearance, zero counterexamples in 21 093 worlds, and it repairs the part of the corpus that used the false normalisations. High value per unit effort.
4. **ts (the smooth branch)**. The largest quantitative margin of the four branch targets, and its success criterion — $G_{\text{esc}}(N) > 0$ in a specified regime — would be a genuine theorem about Goldbach in that regime.
5. **Refuting L1**. A negative result that would cleanly remove a whole family of capacity arguments. The measured data already lean this way.
6. **The boundary filtration as pure combinatorics (Q7)**. Independent of Goldbach, unexplored, and the only part of the project that is not blocked by any known wall.

Warning 16.1 (What does *not* look good, despite appearances). The genealogical layer is the most beautiful part of the project and the most likely to mislead. It is exact, it has genuine converses, and it does not imply anything dynamical — and Warning 6.7(2) raises the real possibility that it cannot, because “ $\kappa_{\text{gen}}(r) = 0$ ” may carry exactly the information “ r is the first upper prime” and nothing more. Before building on it, decide whether your argument would survive that equivalence.

16.3 The most important dependency theorems

If any of the following were found to be wrong, large parts of the file would fall. They are listed in order of how much depends on them.

ID	Name	What depends on it
C-GAP-EQ	Canonical gap equivalence (Thm. 3.10)	Everything canonical. It is the definition of the canonical input and the only class-A hypothesis anywhere.
ARS-CLOSED-WORLD	Closure and unique source (Lem. 3.26)	The whole failure analysis, the fixed point, the Euler layer, every capacity statement.
ARS-NORETURN	No-bad-return-to- p_0 (Thm. 5.33)	Uniqueness of the source, hence ARS-CLOSED-WORLD in the canonical case, hence the two-sheet budget.
CRD-FIXPOINT	Fixed-point characterisation (Thm. 3.29)	The entire computational programme; the six special N ; the tractability of the census.
C0b-RD	Reflection-defect ledger (Thm. 8.6)	All four branch targets TI, TD, TS, TF, hence P1 ^{br} , hence everything downstream. <i>Note its repaired hypothesis</i> (Scope note 8.7).
V71.21	Genealogical rank (Thm. 6.2)	Ledger completeness, uniqueness of the reflection-complete root, the lossless code, the ablation’s interpretation. <i>Note its domain</i> (Scope note 3.17).
C-STRICT-NONREUSE	Strict non-reuse (Lem. 3.13)	TI’s injectivity for large primes, W1’s collision-freeness, the $\mathcal{F}$ branch of C0b-RD.

ID	Name	What depends on it
S-SHADOW-ID	Shadow identity (Thm. 5.22)	Retention, D_q , M_q , cycle localisation, the prime-power witness — the whole multiplicative layer.
V-TUJ	$T_S = U_S J_S$ (Thm. 3.19)	Every ledger and capacity statement. Note that the <i>Euler defect</i> built on it does not inherit its reliability (§5.7).
G-MULTISOURCE-LOCK	Multi-source product lock (Thm. 5.7)	G-ROOT-LOCK, the selected residue lock (V81.13), and route (B) of the surviving-route trichotomy.
V81-CONTAINER	Rooted container identity (Thm. 9.7)	The whole current frontier: the slack bound, the death of the size route, and the trichotomy that names the only remaining routes.
V81-C5-ALPHABET	Adaptive fringe cofactor alphabet (Thm. 9.17)	The preferred target: it is what makes the high band completely known, so that the low core is the entire unknown.
FACP & FBTU	The wall (Thms. 8.1, 8.2)	Not a dependency but a constraint: they are what makes most of §16.1 binding. Their proofs are <i>not</i> reproduced in this file — they should be audited before the wall is treated as final.

16.4 The five most valuable next moves

1. **Attack C-PO-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPACITY** (§9.2.4). Bound $|C_\lambda|$ from above, or prove no such bound exists at this scale. This is the only archimedean route left after the container theorem, and both outcomes are decisive: a bound closes the primitive branch, its impossibility redirects the whole programme to routes (B) and (C). Read the honest difficulty statement and the recorded vacuity of the S -unit bound before starting.
2. **Prove or refute** $\sum_q d_R(q) \geq |R| + 1$ (normalisation (N3), §5.7). Elementary-looking, zero counterexamples in 21 093 canonical worlds, and it decides whether the Euler layer has any surviving content. Either outcome is worth having; a refutation would close the layer entirely.
3. **Prove the bounded-charging lemma for E1** (§9.6). Observed charging multiplicity is 1; the constant- $C = 1$ scheme is refuted with maximum 9.42243, so the target is an absolute bound, not the constant 1. This is the named missing ingredient of the best-supported open target in the project.
4. **Extend the B_N^{low} census from $6 \cdot 10^4$ to 10^6** (§13.1). Three different six-element lists coincide and nobody knows why. A seventh nonempty B_N^{low} — or a clean absence over a range sixteen times longer — is the single most informative new datum available.
5. **Test L1 directly: plot $\nu(R)$ against $\log J_R$** . A few lines on top of the existing world routine. If $\nu(R)$ stays bounded while J_R grows, L1 is dead and with it every capacity route through the layer part — which would be real progress by subtraction.
6. **Write down C-SAMEFUNC-BRIDGE explicitly and try to refute it** (§9.11). Before attempting TD, P1^{br} or K1 again, state the two functionals and look for a family where the production functional and the capacity functional provably diverge. Note that Corollary 9.9 has already done exactly this for the *selected-charge* instance, and the answer was negative — so the honest prior is that the same will happen again unless a genuinely new functional is introduced.

Warning 16.2 (Two moves demoted by the v4.81 result). Anything of the shape “find new selected storage disjoint from $\mathcal{C}_Q$ at fixed positive logarithmic scale” is now known to be *insufficient by construction*, not merely unproved (Corollary 9.9); and anything of the shape “exploit an overlap between the reflected block and the selected family” is known to be charge-free (Theorem 9.10). Do not spend a run on either.

Final orientation. Binary Goldbach is **OPEN** and is not the subject of this file. Universal canonical success (**H- p_0**) is **OPEN** and is strictly stronger. The p_0 -centred proof programme is **SUPERSEDED**. What the project actually owns is an exact structural theory of one distinguished vertex, a complete record of what fails around it, and a small number of sharply stated open problems that are interesting on their own terms. Treat it that way and it is a good research object. Treat it as a proof engine and it will consume arbitrarily many runs, as 239 log entries attest.

Appendix A

Reproduction Code

A.1 The data generator

The following script reproduces *every* number marked **VERIFIED COMPUTATION** in this file. It has no external dependencies (standard library, Python 3.9+). Blocks EXP1–EXP8 correspond to the identifiers in the provenance table of §10.1. Comments are the project’s own, transliterated to ASCII; the executable code is unchanged.

Listing A.1: `p0_experiments.py` — complete self-contained generator for every **VERIFIED COMPUTATION** number in this file.

```
1 #!/usr/bin/env python3
2 # -*- coding: utf-8 -*-
3 """
4 p0_experiments.py -- JEDEN samowystarczalny skrypt odtwarzajacy KAZDA liczbe
5 oznaczona COMPUTATIONAL w dossier p_0 oraz dane wszystkich rysunkow.
6
7 Bez zaleznosci zewnetrznych (tylko biblioteka standardowa Pythona 3.9+).
8
9     python3 p0_experiments.py --scan 10000000 --census 20000 --phase 200000 \
10         --fixedpoint 60000 --json dane.json
11
12 Domyślne, szybkie ustawienia (kilka minut):
13     python3 p0_experiments.py
14
15 Każdy blok EXP wypisuje dokładny zakres, który jest częścią twierdzenia
16 obliczeniowego -- zgodnie z reguła "nigdy nie cytuj liczby bez jej domeny".
17 """
18 from __future__ import annotations
19
20 import argparse
21 import json
22 from array import array
23 from collections import Counter, deque
24
25
26 # ===== sito
27 def build(limit):
28     spf = array('i', range(limit + 1))
29     i = 2
30     while i * i <= limit:
31         if spf[i] == i:
32             for j in range(i * i, limit + 1, i):
33                 if spf[j] == j:
34                     spf[j] = i
35             i += 1
36     isp = bytearray(limit + 1)
37     for n in range(2, limit + 1):
38         isp[n] = 1 if spf[n] == n else 0
39     return spf, isp
40
41
42 def pf(n, spf):
```

```

43 out = []
44 while n > 1:
45     q = spf[n]
46     out.append(q)
47     while n % q == 0:
48         n //= q
49 return out
50
51
52 def bigomega(n, spf):
53     c = 0
54     while n > 1:
55         q = spf[n]
56         while n % q == 0:
57             n //= q
58             c += 1
59     return c
60
61
62 class Env:
63     """Sito + selektory kanoniczne + dwa tryby przejścia."""
64
65     def __init__(self, limit):
66         self.L = limit
67         self.spf, self.isp = build(limit)
68
69     def nextp(self, n):
70         m = n + 1
71         while m <= self.L and not self.isp[m]:
72             m += 1
73         return m
74
75     def prevp(self, n):
76         m = n - 1
77         while m >= 2 and not self.isp[m]:
78             m -= 1
79         return m if m >= 2 else -1
80
81     def p0(self, N):
82         x = N // 2
83         return x if self.isp[x] else self.nextp(x)
84
85     def census(self, N, r):
86         """tryb census: (sukces, glebokosc, liczba wierzchołkow)"""
87         if 2 * r == N:
88             return True, 0, 1
89         q = deque([(r, 0)])
90         seen = {r}
91         nodes = 0
92         while q:
93             p, d = q.popleft()
94             nodes += 1
95             n = N - p
96             if n >= 2 and self.isp[n]:
97                 return True, d, nodes
98             while n > 1:
99                 f = self.spf[n]
100                 while n % f == 0:
101                     n //= f
102                 if N % f and f not in seen:
103                     seen.add(f)
104                     q.append((f, d + 1))
105         return False, -1, nodes
106
107     def world(self, N, r):
108         """tryb world: (sukces, glebokosc, |R|, liczba krawedzi GOOD)"""
109         if 2 * r == N or (N - r >= 2 and self.isp[N - r]):
110             return True, 0, 0, 0
111         q = deque([(r, 0)])
112         queued = {r}
113         R, esc, ok, dep = set(), 0, False, None
114         while q:
115             p, d = q.popleft()
116             if p in R:
117                 continue

```

```

118     n = N - p
119     if n >= 2 and self.isp[n]:
120         if not ok:
121             ok, dep = True, d
122         continue
123     R.add(p)
124     while n > 1:
125         f = self.spf[n]
126         while n % f == 0:
127             n //= f
128         if N % f:
129             if N - f >= 2 and self.isp[N - f]:
130                 esc += 1
131                 if not ok:
132                     ok, dep = True, d + 1
133                 elif f not in R and f not in queued:
134                     queued.add(f)
135                     q.append((f, d + 1))
136     return ok, dep, len(R), esc
137
138
139 # ===== EXP1: wyczerpujący skan p_0
140 def exp1_scan(limit):
141     """Kazde parzyste 4<=N<=limit, korzen kanoniczny, tryb census."""
142     e = Env(limit)
143     fails, tot, s = [], 0, 0
144     maxn, maxd, hist = (0, 0), (0, 0), Counter()
145     for N in range(4, limit + 1, 2):
146         r = e.p0(N)
147         ok, d, nodes = e.census(N, r)
148         tot += 1
149         s += nodes
150         if not ok:
151             fails.append((N, r))
152         else:
153             hist[d] += 1
154             if d > maxd[0]:
155                 maxd = (d, N)
156             if nodes > maxn[0]:
157                 maxn = (nodes, N)
158     return {"limit": limit, "cases": tot, "failures": len(fails),
159           "failure_list": fails[:20], "mean_nodes": s / tot,
160           "max_nodes": maxn, "max_depth": maxd,
161           "depth_hist": {str(k): hist[k] for k in sorted(hist)}}
162
163
164 # ===== EXP2: ablacja p_0 -> p_k (werdykt roota)
165 def exp2_ablation(limit, kmax=6):
166     e = Env(limit)
167     comp = [0] * (kmax + 1)
168     unit = [0] * (kmax + 1)
169     first = {}
170     for N in range(4, limit + 1, 2):
171         p = e.p0(N)
172         for k in range(kmax + 1):
173             if p >= N:
174                 break
175             ok, _, _ = e.census(N, p)
176             if not ok:
177                 if N - p == 1:
178                     unit[k] += 1
179                 else:
180                     comp[k] += 1
181                     first.setdefault(k, (N, p))
182         p = e.nextp(p)
183     return {"limit": limit, "kmax": kmax, "composite_complement_failures": comp,
184           "unit_row_failures": unit,
185           "first": {str(k): v for k, v in sorted(first.items())}}
186
187
188 # ===== EXP3: census WSZYSTKICH zawodzacych gornych korzeni
189 def exp3_upper_census(limit):
190     e = Env(limit)
191     rows, unit = [], 0
192     for N in range(4, limit + 1, 2):

```

```

193     r0 = e.p0(N)
194     p = r0
195     while p < N:
196         if e.isp[p] and N % p:
197             if N - p == 1:
198                 unit += 1
199             elif not e.census(N, p)[0]:
200                 rows.append((N, p, p == r0))
201             p += 1
202         while p < N and not e.isp[p]:
203             p += 1
204     Ns = sorted({r[0] for r in rows})
205     return {"limit": limit, "failed_roots": len(rows), "unit_rows": unit,
206           "N_values": Ns, "per_N": {str(n): sum(1 for r in rows if r[0] == n)
207                                     for n in Ns},
208           "canonical_among_them": sum(1 for r in rows if r[2])}
209
210
211 # ===== EXP4: margines uciezki G_esc
212 def exp4_margin(limit):
213     e = Env(limit)
214     ones, sizes = [], []
215     hist = Counter()
216     maxW = (0, 0)
217     dyad = {}
218     for N in range(4, limit + 1, 2):
219         r = e.p0(N)
220         ok, _, W, esc = e.world(N, r)
221         if W == 0:
222             continue
223         hist[esc] += 1
224         sizes.append(W)
225         if W > maxW[0]:
226             maxW = (W, N)
227         if esc == 1:
228             ones.append((N, r, W))
229         if W >= 3:
230             dyad.setdefault(N.bit_length(), []).append(esc)
231     return {"limit": limit, "bad_p0_worlds": len(sizes),
232           "G_esc_hist_small": {str(k): hist[k] for k in sorted(hist) if k <= 6},
233           "margin_one_all": [o[0] for o in ones],
234           "margin_one_W_ge_3": [(N, r, W) for N, r, W in ones if W >= 3],
235           "max_world": maxW,
236           "dyadic_min_G": {str(k): min(v) for k, v in sorted(dyad.items())}}
237
238
239 # ===== EXP5: trichotomia fazowa i dwukierunkowa ranga rho_0
240 def exp5_phase(limit):
241     e = Env(limit)
242     primes = [n for n in range(2, limit + 1) if e.isp[n]]
243     idx = {p: i for i, p in enumerate(primes)}
244     types = Counter()
245     imm = Counter()
246     viol = Counter()
247     rho_hist = Counter()
248     rho_theta = {}
249     theta_hist = Counter()
250     for N in range(6, limit + 1, 2):
251         x = N // 2
252         P = e.p0(N)
253         h, m0 = P - x, N - P
254         Pm = e.prevp(P)
255         G = P - Pm
256         if m0 - Pm != G - 2 * h:
257             viol["identity"] += 1
258         t = "diag" if h == 0 else ("I" if 2 * h < G else
259                                   ("II" if 2 * h == G else "III"))
260         types[t] += 1
261         theta_hist[int(20 * h / G)] += 1
262         if e.isp[m0] or m0 == 1:
263             imm[t] += 1
264             if t == "I":
265                 viol["typeI_prime_m0"] += 1
266         if t == "II" and m0 != Pm:
267             viol["typeII"] += 1

```



```

268     if t == "III":
269         js, tt = h - G // 2, (h - 1) // 2
270         if not (1 <= js <= tt and m0 + 2 * js == Pm):
271             viol["typeIII_mirror"] += 1
272         if N % Pm == 0:
273             viol["typeIII_inactive"] += 1
274     if h >= 3:
275         tt = (h - 1) // 2
276         direct = sum(1 for j in range(1, tt + 1) if e.isp[m0 + 2 * j])
277         i0, f, r = idx[P], 0, 1
278         while i0 - r >= 0:
279             D = P - primes[i0 - r]
280             if D >= 2 * h:
281                 break
282             if D > h:
283                 f += 1
284             r += 1
285             if direct != f:
286                 viol["rho0"] += 1
287             rho_hist[direct] += 1
288             rho_theta.setdefault(round(h / G, 2), []).append(direct)
289     return {"limit": limit, "type_counts": dict(types),
290           "immediate_success_by_type": dict(imm),
291           "violations": dict(viol),
292           "theta_hist_20": {str(k): theta_hist[k] for k in sorted(theta_hist)},
293           "rho_hist": {str(k): rho_hist[k] for k in sorted(rho_hist)},
294           "rho0_mean_by_theta": {str(k): round(sum(v) / len(v), 4)
295                                for k, v in sorted(rho_theta.items())
296                                if len(v) >= 50}}
297
298
299 # ===== EXP6: filtracja brzegu (embedding)
300 def exp6_filtration(limit, kmax=6):
301     e = Env(limit)
302     checked = viol = 0
303     for N in range(50, limit + 1, 2):
304         x = N // 2
305         roots, p = [], e.p0(N)
306         for _ in range(kmax + 1):
307             if p >= N:
308                 break
309             roots.append(p)
310             p = e.nextp(p)
311         for k in range(len(roots) - 1):
312             pk, pk1 = roots[k], roots[k + 1]
313             s = (pk1 - pk) // 2
314             tk, tk1 = (pk - x - 1) // 2, (pk1 - x - 1) // 2
315             if tk1 != tk + s:
316                 viol += 1
317             for i in range(1, tk + 1):
318                 checked += 1
319                 if pk1 - 2 * (i + s) != pk - 2 * i:
320                     viol += 1
321     return {"limit": limit, "checked": checked, "violations": viol}
322
323
324 # ===== EXP7: fixed point B_N^low
325 def exp7_fixedpoint(limit):
326     e = Env(limit)
327     out = []
328     for N in range(6, limit + 1, 2):
329         cap = N // 3
330         B, rows = set(), {}
331         for p in range(3, cap + 1, 2):
332             if not e.isp[p] or N % p == 0:
333                 continue
334             v = N - p
335             if v < 2 or e.isp[v]:
336                 continue
337             B.add(p)
338             rows[p] = pf(v, e.spf)
339         changed = True
340         while changed:
341             changed = False
342             for p in list(B):

```

```

343         for q in rows[p]:
344             if N % q and q not in B:
345                 B.discard(p)
346                 changed = True
347                 break
348     if B:
349         out.append((N, sorted(B)))
350     return {"limit": limit, "nonempty_count": len(out),
351            "nonempty": [(N, B) for N, B in out]}
352
353
354 # ===== EXP8: ablacja obserwabli p_0 vs p_1
355 def exp8_observables(limit):
356     e = Env(limit)
357     acc = {k: {n: [] for n in ("depth", "world", "esc", "Omega", "ret", "kap")}
358            | {"fail": 0, "n": 0, "pp": []} for k in (0, 1)}
359     for N in range(50, limit + 1, 2):
360         x = N // 2
361         r0 = e.p0(N)
362         r1 = e.nextp(r0)
363         for k, r in ((0, r0), (1, r1)):
364             if r >= N or N - r <= 1:
365                 continue
366             m = N - r
367             ok, dep, W, esc = e.world(N, r)
368             a = acc[k]
369             a["n"] += 1
370             if not ok:
371                 a["fail"] += 1
372             if dep is not None:
373                 a["depth"].append(dep)
374             a["world"].append(W)
375             a["esc"].append(esc)
376             a["Omega"].append(bigomega(m, e.spf))
377             Q = pf(m, e.spf)
378             Qs = set(Q)
379             a["pp"].append(max(Q) / m)
380             a["ret"].append(sum(1 for q in Q for z in pf(N - q, e.spf) if z in Qs))
381             hh = r - x
382             tt = (hh - 1) // 2
383             a["kap"].append(sum(1 for j in range(1, tt + 1) if e.isp[r - 2 * j]))
384
385     def st(v):
386         v = sorted(v)
387         return {"mean": round(sum(v) / len(v), 4), "median": v[len(v) // 2],
388                "max": v[-1]}
389     return {"limit": limit,
390            **{f"p{k}": {n: st(acc[k][n]) for n in
391                       ("depth", "world", "esc", "Omega", "ret", "kap")}
392               | {"fail": acc[k]["fail"], "n": acc[k]["n"],
393                  "Pplus_over_m_mean": round(sum(acc[k]["pp"]) / len(acc[k]["pp"]), 5),
394                  "Omega_hist": {str(a): b for a, b in
395                                sorted(Counter(acc[k]["Omega"]).items())},
396                  "depth_hist": {str(a): b for a, b in
397                                sorted(Counter(acc[k]["depth"]).items())}}
398               for k in (0, 1)}}
399
400
401 # ===== pojedynczy swiadkowie
402 def witnesses(limit=1_100_000):
403     """Wszyscy pojedynczy swiadkowie cytowani w dossier."""
404     e = Env(limit)
405
406     def fac(n):
407         c = Counter(pf(n, e.spf))
408         d = Counter()
409         m = n
410         for q in c:
411             while m % q == 0:
412                 m //= q
413                 d[q] += 1
414         return dict(sorted(d.items()))
415
416     w = {}
417     for N in (124, 242, 240, 2200, 103344, 364844, 14198, 11822):

```

```

418 P = e.p0(N)
419 Pm = e.prevp(P)
420 h, G, m0 = P - N // 2, P - Pm, N - P
421 ok, dep, W, esc = e.world(N, P)
422 t = (h - 1) // 2
423 w[str(N)] = {"p0": P, "h": h, "G": G, "prev_prime": Pm, "m0": m0,
424             "m0_fac": {str(a): b for a, b in fac(m0).items()},
425             "type": "diag" if h == 0 else
426                     ("I" if 2 * h < G else
427                      ("II" if 2 * h == G else "III")),
428             "kappa_gen": sum(1 for j in range(1, t + 1)
429                             if e.isp[P - 2 * j]),
430             "rho0": sum(1 for j in range(1, t + 1) if e.isp[m0 + 2 * j]),
431             "verdict": "SUCCESS" if ok else "FAILURE",
432             "depth": dep, "world": W, "escapes": esc}
433 w["2200_core"] = {"2200-3": 2200 - 3, "13^3": 13 ** 3,
434                  "2200-13": 2200 - 13, "3^7": 3 ** 7,
435                  "shift": 3 ** 7 - 3 == 13 ** 3 - 13 == 2184}
436 N = 1_000_000
437 Q, R = (61, 67, 73), (113, 179, 599)
438 w["K33"] = {"N": N, "Q": Q, "R": R,
439             "all_entries_prime": all(e.isp[N - q * r] for q in Q for r in R),
440             "all_factor_vertices_bad": all(not e.isp[N - v] for v in Q + R)}
441 w["cells"] = [{"P": P, "G": P - e.prevp(P),
442               "all_h_have_p0_eq_P": all(e.p0(2 * (P - h)) == P
443                                         for h in range(0, P - e.prevp(P)))}
444               for P in (67, 127, 1009, 104729)]
445 return w
446
447
448 # ===== driver
449 def main():
450     ap = argparse.ArgumentParser()
451     ap.add_argument("--scan", type=int, default=1_000_000)
452     ap.add_argument("--ablation", type=int, default=200_000)
453     ap.add_argument("--census", type=int, default=20_000)
454     ap.add_argument("--margin", type=int, default=200_000)
455     ap.add_argument("--phase", type=int, default=200_000)
456     ap.add_argument("--filtration", type=int, default=20_000)
457     ap.add_argument("--fixedpoint", type=int, default=20_000)
458     ap.add_argument("--observables", type=int, default=200_000)
459     ap.add_argument("--json", default="dane_p0.json")
460     a = ap.parse_args()
461     out = {
462         "EXP1_scan": exp1_scan(a.scan),
463         "EXP2_ablation": exp2_ablation(a.ablation),
464         "EXP3_upper_census": exp3_upper_census(a.census),
465         "EXP4_margin": exp4_margin(a.margin),
466         "EXP5_phase": exp5_phase(a.phase),
467         "EXP6_filtration": exp6_filtration(a.filtration),
468         "EXP7_fixedpoint": exp7_fixedpoint(a.fixedpoint),
469         "EXP8_observables": exp8_observables(a.observables),
470         "witnesses": witnesses(),
471     }
472     with open(a.json, "w") as f:
473         json.dump(out, f, indent=1)
474     print(json.dumps({k: (v if k == "witnesses" else
475                        {kk: vv for kk, vv in v.items()
476                          if not isinstance(vv, (list, dict)) or len(str(vv)) < 400})
477                       for k, v in out.items()}}, indent=1)[:6000])
478
479
480 if __name__ == "__main__":
481     main()

```

A.2 The Euler-defect check

This script produced the refutations of normalisations (N1) and (N2) and the zero-violation results for (N3) and (N4) in §5.7. It is new in this file.

Listing A.2: `euler_check.py` — tests the four candidate normalisations of the layered Euler defect on all canonical failed worlds.

```

1  #!/usr/bin/env python3
2  """Test the four candidate normalisations of the layered Euler defect
3  on every canonical failed world with  $50 \leq N \leq \text{LIMIT}$ ."""
4  from collections import deque
5
6  LIM = 200001
7  spf = list(range(LIM))
8  i = 2
9  while i * i < LIM:
10     if spf[i] == i:
11         for j in range(i * i, LIM, i):
12             if spf[j] == j:
13                 spf[j] = i
14         i += 1
15
16
17 def fac(n):
18     f = {}
19     while n > 1:
20         p = spf[n]
21         e = 0
22         while n % p == 0:
23             n //= p
24             e += 1
25         f[p] = e
26     return f
27
28
29 def isp(n):
30     return n >= 2 and spf[n] == n
31
32
33 def world(N):
34     """Canonical stopped traversal, world mode.
35     Returns (R, rows, depth, success, root) or None when the root wins
36     immediately (diagonal case or  $N - p_0$  prime)."""
37     x = N // 2
38     r = x if isp(x) else None
39     if r is None:
40         p = x + 1
41         while not isp(p):
42             p += 1
43         r = p
44     if 2 * r == N:
45         return None
46     if isp(N - r):
47         return None
48     q = deque([(r, 0)])
49     seen = {r}
50     R, rows, depth = set(), {}, {}
51     ok = False
52     while q:
53         p, d = q.popleft()
54         if p in R:
55             continue
56         if isp(N - p):
57             ok = True
58             continue
59         R.add(p)
60         rows[p] = fac(N - p)
61         depth[p] = d
62         for c in sorted(rows[p]):
63             if N % c == 0:
64                 continue
65             if isp(N - c):
66                 ok = True
67                 continue
68             if c not in R and c not in seen:
69                 seen.add(c)
70                 q.append((c, d + 1))
71     return (R, rows, depth, ok, r)
72
73

```

```

74 bad = [0, 0, 0, 0]
75 first = [None, None, None, None]
76 tested = 0
77 for N in range(50, 60001, 2):
78     w = world(N)
79     if w is None:
80         continue
81     R, rows, depth, ok, r = w
82     if not R:
83         continue
84     tested += 1
85     nu = len(set(depth.values()))
86     # (N1) row-excess form
87     eta1 = sum(sum(rows[p].values()) - len([c for c in rows[p] if c in R])
88               for p in R)
89     # (N2) valuation / reuse split
90     d = {}
91     kap = {}
92     for p in R:
93         for q_, e_ in rows[p].items():
94             if N % q_ == 0:
95                 continue
96             d[q_] = d.get(q_, 0) + e_
97             kap[q_] = kap.get(q_, 0) + 1
98     eta2 = sum(d[q_] - kap[q_] for q_ in d)
99     nu2 = sum(kap[q_] - 1 for q_ in d)
100    # (N3) active-mass form ; (N4) trivial form
101    mass = sum(d.values())
102    eta4 = sum(sum(rows[p].values()) - 1 for p in R)
103    target = len(R) + 1
104    for k, val in enumerate((eta1 + nu, eta2 + nu2, mass, eta4 + nu)):
105        if val < target:
106            bad[k] += 1
107            if first[k] is None:
108                first[k] = (N, len(R), val, target)
109    print("worlds tested:", tested)
110    for k, name in enumerate(("N1", "N2", "N3", "N4")):
111        print(name, "violations:", bad[k], "first:", first[k])

```

A.3 Auxiliary scripts not embedded here

The project repository also contains the following, which are *not* embedded because no number in this file depends on them.

File	Role
p0_algorithm.py	Listing 3.1: the self-contained reference algorithm (embedded, §3.4).
p0_experiments.py	Listing A.1: the data generator (embedded above).
p0_core.py	Extended engine: both traversal modes, canonical geometry, reflected block, genealogical rank, D_q/M_q , FRESH/SMOOTH split, $B_{\text{abs}}/B_{\text{re}}$, CLI.
p0_certificates.py	The 34 hostile certificates of §10.4: regressions 11 822 and 3 542 948, counterexamples 242, 128, 322, 858, 4736, 6142, 14198, 20, 1718, 51 032, the Exact-D models, the $K_{3,3}$ grid and the 2-core.
p0_bigscan.py, p0_phase.py, p0_figures.py, p0_upper_root_census.py, p0_escape_margin.py	Supporting scripts for individual experiments; all of their outputs are reproducible from p0_experiments.py.

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